

FARKALEET SERIES

***LINEAR ALGEBRA AND
ANALYTICAL GEOMETRY
(REVISED EDITION)***

AUTHOR:

Prof. (R) Muhammad Urs Shaikh

Prof. Dr. Feroz Shah

Dr. Sania Nizamani

CO-AUTHORS:

Muhammad Tahir Wahocho (Msc)

Edited By:

Prof. Dr. Asif Ali Shaikh

***JAHANGIR PUBLICATIONS, LARKANA
Cell: 0333-756-5595, 0304-453-2100***

PREFACE TO FIRST EDITION

About thirty years ago I wrote a series of books on different subjects of Engineering Mathematics. The series was named "FARKALEET SERIES". FARKELEET means MUHAMMAD (S.A.W.S). This name of our Prophet (S.A.W.S) is mentioned in the BIBLE.

The series though was hand written but became very much popular among my students at Mehran University of Engineering and Technology, Jamshoro, Pakistan. The series also gained popularity in the students of other universities and colleges of Sindh as well. Many teachers of different universities and colleges also appreciated such efforts of mine. The reason is that each topic was clearly explained and that every student as well as teacher enjoyed reading without any difficulty.

The book "LINEAR ALGEBRA AND ANALYTICAL GEOMETRY" is in your hands. Each topic is explained in well manner and in detail as well.

First chapter is devoted towards basic definitions and operations and rank of matrices. Second chapter is on the solutions of system of linear homogeneous and non-homogeneous equations and augmented matrix. Chapter three is on determinants, inverse of square matrix and Cramer's rule. Chapters four, five and six are on three dimensional analytical geometry including lines, planes and sphere. Chapter seven is on multiple integrals.

We shall be looking forward to hear from the readers any critic that will improve the standard of the book.

Prof. (Rtd) Muhammad Urs SHAIKH
Cell: 0333- 756-5595
email: ursshaikh_muet@yahoo.com

Co-Authors:
Asif Ali Shaikh
Muhammad Tahir Wahocho (M.Sc)
Ms Sania Qureshi

PREFACE TO SECOND EDITION

Tin the first edition, the book was containing some typo-graphical as well as some mathematical calculation mistakes. These mistakes have been now removed from the first edition. We hope that, students will be much comfortable with the new revised edition. However, if any mistake is found, we will appreciate if the same may be conveyed to the authors.

Prof. (R) Muhammad Urs Shaikh

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CHAPTER ONE

INTRODUCTION TO MATRICES

INTRODUCTION

Before we furnish a formal definition of a matrix, consider a situation where two groups A and B containing x and y persons respectively ask over each other the following:

People of `Group A` asks to `Group B` that if 2 persons from you join us we will be double to you.

People of `Group B` asks to `Group A` that if 2 persons from you join us we will be equal to you.

Now a question arises that how many people are in each group? Let us try to find the answer of this problem.

According to given conditions when `Group A` asks `Group B` that if 2 persons from you join us we will be double to you. This gives rise to an equation

$$x + 2 = 2(y - 2)$$

Similarly, when `Group B` asks `Group A` that if 2 persons from you join us we will be equal to you, this gives rise to an equation

$$y + 2 = x - 2$$

Putting these equations in systematic form, we get:

$$x - 2y = -6 \quad (1)$$

$$x - y = 4 \quad (2)$$

This system of linear equations can be expressed as

$$\begin{bmatrix} 1 & -2 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} -6 \\ 4 \end{bmatrix}$$

Solving equations (1) and (2), we get: $x = 14$ and $y = 10$. This means `Group A` contains 14 and `Group B` contains 10 people respectively.

Now consider a situation where six boys are asked to stand in queue(s). They may stand in one of the following form:

$$\text{i. } \begin{bmatrix} b_1 & b_2 & b_3 & b_4 & b_5 & b_6 \end{bmatrix} \quad \text{ii. } \begin{bmatrix} b_1 \\ b_2 \\ b_3 \\ b_4 \\ b_5 \\ b_6 \end{bmatrix} \quad \text{iii. } \begin{bmatrix} b_1 & b_2 \\ b_3 & b_4 \\ b_5 & b_6 \end{bmatrix} \quad \text{iv. } \begin{bmatrix} b_1 & b_2 & b_3 \\ b_4 & b_5 & b_6 \end{bmatrix}$$

Note: Write down different arrangements with nine boys using the above format.

In the first problem we observe that a system of linear equations can be arranged in some particular form and in the second problem six boys can be placed in systematic way in four different orders/arrangements. The horizontal arrangement is called row and the vertical one is called column. Such type of arrangements of elements is called a matrix.

Definition: A rectangular arrangement of elements in the following order is called a matrix.

The first suffix indicates the row and the second indicates the column in which the element lies.

This matrix has m rows and n columns. Hence the *order* of this matrix is $m \times n$ (read as m by n).

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} & \dots & a_{1n} \\ a_{21} & a_{22} & a_{23} & \dots & a_{2n} \\ \dots & \dots & \dots & \dots & \dots \\ a_{m1} & a_{m2} & a_{m3} & \dots & a_{mn} \end{bmatrix}$$

James Joseph Sylvester was the first mathematician who introduced the term “Matrix” in 1848. Cayley, Hamilton, Hermann Grossmann, Frobenius and Von Neumann were among the famous mathematicians who had worked on matrix theory.

Matrices (plural of matrix) have a long history of applications in solving linear equations. Leibniz, one of the two founders of Calculus, developed the theory of determinants in 1693. Cramer developed the theory further, presenting Cramer’s Rule in 1750. Carl Friedrich Gauss and Wilhelm Jordan developed Gauss-Jordan elimination method in 1800s.

Matrices are usually denoted by capital letters and its elements either by small case letters or numbers. For example:

$$A = \begin{bmatrix} 2 & 4 \\ 0 & -1 \end{bmatrix}, \quad B = \begin{bmatrix} a \\ b \\ c \end{bmatrix}, \quad C = [3 \quad -3 \quad 5 \quad 9], \quad D = \begin{bmatrix} 4 & \sqrt{6} \\ 3/4 & 9 \\ -1 & 0 \end{bmatrix}$$

If a matrix contains equal number of rows and columns it is known as a square matrix. If it contains one row only it is known as a row matrix/vector. Similarly, if matrix contains only one column it is known as column matrix/vector. If number of rows and columns is not same the matrix is called rectangular matrix.

Remark: In a square matrix, the elements $a_{11}, a_{22}, \dots, a_{nn}$ are called main or principal diagonal elements.

If A is a matrix of order $m \times n$ it can generally be written as

$$A = [a_{ij}], \quad i = 1, 2, \dots, m; \quad j = 1, 2, \dots, n$$

Matrices allow us to store huge data in a concise manner. For example, suppose that at a boarding house, there are 35 boys and 3 teachers who take breakfast, 65 boys and 7 teachers take lunch and 48 boys and 5 teachers take dinner. In matrix form this data can be placed in a concise form as

shown here.

	Boys	Teachers
Breakfast	35	3
Lunch	65	7
Dinner	48	5

Class Activity: Ali, Bilal and Mustafa appear in four subjects Mathematics, Physics, Chemistry and English. Their respective marks are: 66, 55, 77, 44; 55, 66, 45, 77 and 88, 77, 66, 55. Arrange this data in matrix form.

IMPORTANT MATRICES

In this section we list some important matrices which will be of great importance for the future work on matrix theory.

Zero Matrix

A matrix whose each element is zero regardless of its order is called a *zero matrix*. For example, the following matrices are all zero matrices.

$$\begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}, [0 \ 0 \ 0 \ 0], \begin{bmatrix} 0 & 0 \\ 0 & 0 \\ 0 & 0 \end{bmatrix}$$

Zero matrix is usually denoted by $O_{m,n}$ where m, n is its order. However, if O is a square matrix of order n , it is usually denoted by O_n . In the above example, 1st, 2nd, 3rd and 4th matrices are thus denoted by; O_2 , $O_{3,1}$, $O_{1,4}$ and $O_{3,2}$ respectively.

Diagonal Matrix

A square matrix whose each/some elements are zero with at least one non-zero diagonal element is called a *diagonal matrix*. For example, the following matrices are diagonal matrices.

$$\begin{bmatrix} 2 & 0 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & 7 \end{bmatrix}, \begin{bmatrix} 2 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 7 \end{bmatrix}, \begin{bmatrix} 1 & 0 & 0 \\ 0 & 5 & 0 \\ 0 & 0 & 0 \end{bmatrix}, \begin{bmatrix} 9 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

Scalar Matrix

A diagonal matrix in which all elements are equal is called a *scalar matrix*. For example, the following matrix is a scalar matrix.

$$\begin{bmatrix} 7 & 0 & 0 \\ 0 & 7 & 0 \\ 0 & 0 & 7 \end{bmatrix}$$

In general, if A is a scalar matrix then: $a_{ij} = \begin{cases} k, & \text{if } i = j \\ 0, & \text{if } i \neq j \end{cases}$, k being a scalar.

Identity Matrix

A scalar matrix in which each diagonal element is unity is called an *identity matrix*. For example, the following matrices are identity matrices of order 3, 2 and 1 respectively.

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \text{ and } [1]$$

An identity matrix of order n is denoted by I_n . When the order is immaterial, the identity matrix is usually denoted by I . In general, if I is an identity matrix then:

$$a_{ij} = \begin{cases} 1, & \text{if } i = j \\ 0, & \text{if } i \neq j \end{cases}$$

Unit Matrix

A matrix with all entries equal to 1 is called a *unit matrix*. A unit matrix may be square or rectangular. For example, the following matrices are unit matrices.

$$\begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}, \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$$

It may be noted that there is a great confusion that an identity and unit matrices are treated as same. But you have observed that this is not the case, that is; identity and unit matrices are different.

Upper and Lower Triangular Matrices

A square matrix in which each element below the principal diagonal is zero is, said to be *upper triangular matrix* and a square matrix in which every element above the principal diagonal is zero is, called *lower triangular matrix*. The following matrices are lower and upper triangular matrices respectively.

$$\begin{bmatrix} 5 & 0 & 0 & 0 \\ 1 & 6 & 0 & 0 \\ 9 & 5 & 2 & 0 \\ 4 & 8 & 7 & 1 \end{bmatrix}, \begin{bmatrix} 6 & 8 & 1 \\ 0 & 5 & 2 \\ 0 & 0 & 9 \end{bmatrix}$$

In general, upper and lower triangular matrices are defined as:

i. A is upper triangular matrix if: $a_{ij} = 0$ for all $i > j$

ii. A is lower triangular matrix if: $a_{ij} = 0$ for all $i < j$

Definition: A matrix is said to be a *triangular matrix* if it is either upper triangular or lower triangular matrix.

Complex Matrix

A matrix whose at least one element is complex number or purely imaginary number is called a *complex matrix*. For example, the following matrix is a complex matrix.

$$\begin{bmatrix} 2i & 6 & 4 \\ -1 & 4+3i & i \\ 3 & 2 & 7 \end{bmatrix}$$

Trace of a Matrix

The sum of diagonal elements of square matrix is called the *trace of a matrix*. For example, for the following matrix, $\text{Trace}(A) = 6 + 5 - 9 = 2$.

$$A = \begin{bmatrix} 6 & 8 & 1 \\ 0 & 5 & 2 \\ 7 & 1 & -9 \end{bmatrix}$$

Equality of Matrices

Two matrices A and B are said to be *equal* if and only if they have the same order and each element of one is equal to the corresponding element of the other. For example, the following

matrices are equal matrices: $A = \begin{bmatrix} 6 & 2 \\ 5 & 7 \end{bmatrix}$ and $B = \begin{bmatrix} \sqrt{36} & 1+1 \\ 3+2 & 9-2 \end{bmatrix}$

You may observe that the orders of matrices A and B are same and that $a_{ij} = b_{ij}$ for all positive integers i and j. We write this as $A = B$.

Example 01: Find the values of x, y, z and w which satisfy the matrix equation

$$\begin{bmatrix} x & 2y+x \\ z-1 & 4w-6 \end{bmatrix} = \begin{bmatrix} 0 & -7 \\ 3 & 2w \end{bmatrix}$$

Solution: By the definition of equality of matrices

$$x + 3 = 0$$

$$2y + x = -7$$

$$z - 1 = 3$$

$$4w - 6 = 2w$$

Solving these equations, we obtain: $x = -3$, $y = -2$, $z = 4$, $w = 3$.

OPERATIONS ON MATRICES

In this section, we shall define some fundamental operations on matrices.

Scalar Multiplication: If A is any matrix and k is a constant, then kA is a matrix whose elements are obtained by multiplying each element of matrix A by k . For example, if

$$A = \begin{bmatrix} 6 & 5 & 2 \\ 3 & 0 & -7 \end{bmatrix} \Rightarrow 3A = \begin{bmatrix} 18 & 15 & 6 \\ 9 & 0 & -21 \end{bmatrix}$$

Addition and Subtraction of Matrices

If A and B are two matrices of same order $m \times n$, then their sum $A + B$ is a matrix C of same order $m \times n$ where each element of matrix C is the sum of the corresponding elements of matrices A and B . For example, if

$$A = \begin{bmatrix} 6 & 5 & 4 \\ -2 & 3 & 7 \end{bmatrix} \text{ and } B = \begin{bmatrix} 4 & 1 & -5 \\ 3 & 9 & 2 \end{bmatrix} \text{ then}$$

$$C = A + B = \begin{bmatrix} 6 & 5 & 4 \\ -2 & 3 & 7 \end{bmatrix} + \begin{bmatrix} 4 & 1 & -5 \\ 3 & 9 & 2 \end{bmatrix} = \begin{bmatrix} 6+4 & 5+1 & 4-5 \\ -2+3 & 3+9 & 7+2 \end{bmatrix} = \begin{bmatrix} 10 & 6 & -1 \\ 1 & 12 & 9 \end{bmatrix}$$

Similarly, the difference $A - B$ of matrices A and B is a matrix each element of which is obtained by subtracting the elements of matrix B from the corresponding elements of matrix A . For example, if

$$A = \begin{bmatrix} 6 & 2 \\ 7 & -5 \end{bmatrix} \text{ and } B = \begin{bmatrix} 8 & 1 \\ 3 & -5 \end{bmatrix} \text{ then}$$

$$D = A - B = \begin{bmatrix} 6 & 2 \\ 7 & -5 \end{bmatrix} - \begin{bmatrix} 8 & 1 \\ 3 & -5 \end{bmatrix} = \begin{bmatrix} 6-8 & 2-1 \\ 7-3 & -5-(-5) \end{bmatrix} = \begin{bmatrix} -2 & 1 \\ 4 & 0 \end{bmatrix}$$

Using double subscript notations, we say that if the matrices A and B are confirmable for addition and subtraction and that $C = A + B$ and $D = A - B$ then,

$$c_{ij} = a_{ij} + b_{ij}, \forall i, j \quad \text{and} \quad d_{ij} = a_{ij} - b_{ij}, \forall i, j$$

Remarks: (i) Two matrices are said to be confirmable for addition or subtraction if and only if their order is same.

(ii) If the matrices A , B and C are confirmable for addition/subtraction, then

- a. $A + A = 2A$ b. $A - A = O$ c. $A + A + \dots + A = nA$
d. $A + B = B + A$ e. $A - B \neq B - A$ f. $A + (B + C) = (A + B) + C$

From (d) and (e) we observe that matrix addition is commutative but same is not true for matrix subtraction. In fact, matrix subtraction is anti-commutative. Property (f) is known as “Associative law of addition”.

Remark: A set of all matrices ‘ M ’ confirmable under addition form an “Abelian Group”. For if $A, B, C \in M$, then

$A + B \in M$	[Closure property under addition]
$A + (B + C) = (A + B) + C$	[Existence of associate law under addition]
$A + O = O + A = A$, O being zero matrix	[Existence of identity matrix under addition]
$A + (-A) = (-A) + A = O$, where $O \in M$	[Existence of inverse matrix under addition]
$A + B = B + A$	[Existence of commutative law for addition]

Multiplication of Matrices

Let $A = (a_{ij})$ be an $m \times p$ matrix and $B = (b_{ij})$ be $r \times n$ matrix, then the product $A.B$ in this order, is defined only when $p = r$ that is; number of columns in matrix A equal to number of rows in

matrix B. In this case, the product matrix $C = A.B$ is an $m \times n$ matrix. The following example illustrates how two matrices are multiplied.

Example 02: Find AB if $A = \begin{bmatrix} 1 & 0 & -1 \\ 2 & 4 & 7 \\ 5 & 3 & 0 \end{bmatrix}$ and $B = \begin{bmatrix} 6 & 1 \\ 0 & 4 \\ -2 & 3 \end{bmatrix}$

Solution: The above matrices are confirmable for multiplication because the number of columns in matrix A is equal to the number of rows in matrix B.

Now,

$$AB = \begin{bmatrix} 1 & 0 & -1 \\ 2 & 4 & 7 \\ 5 & 3 & 0 \end{bmatrix} \begin{bmatrix} 6 & 1 \\ 0 & 4 \\ -2 & 3 \end{bmatrix}$$

$$= \begin{bmatrix} 1 \times 6 + 0 \times 0 - 1 \times -2 & 1 \times 1 + 0 \times 4 - 1 \times 3 \\ 2 \times 6 + 4 \times 0 + 7 \times -2 & 2 \times 1 + 4 \times 4 + 7 \times 3 \\ 5 \times 6 + 3 \times 0 + 0 \times -2 & 5 \times 1 + 3 \times 4 + 0 \times 3 \end{bmatrix} = \begin{bmatrix} 8 & -2 \\ -2 & 39 \\ 30 & 17 \end{bmatrix}$$

Can you compute B.A? The answer is simply no because the number of columns of B is 2 and number of rows of A is 3. Hence, B.A is not confirmable for multiplication.

Remarks: (i) A matrix can be multiplied by itself, if it is a square matrix. The product $A.A = A^2$. Similarly $A.A^2 = A^3$ and $A^2.A^2 = A^4$, etc.

(ii) For matrix multiplication A.B matrix A is post-multiplied by matrix B and matrix B is pre-multiplied by matrix A.

(iii) If $A.B = O$ it is not necessary that A or B is a zero/null matrix. For instance, if

$$A = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} \text{ and } B = \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}$$

Then,

$$A.B = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} = O$$

The product AB is a zero matrix but neither A nor B is zero matrix.

It may be noted that if $A.B = O$ then:

- $|A| = 0$ or $|B| = 0$, where $|A|$ denotes the determinant of A.
- Matrices A and B are called “Divisors of zero”.
- $AB = BA$. Readers may compute B.A and confirm that $B.A = O$.

(iv) In number theory, we know that if $a.b = a.c$ then $b = c$ provided $a \neq 0$.

This is known as cancellation law. Now, let us see this in matrices.

If A, B and C are three matrices and $AB = AC$, it is not necessary that $B = C$. For example,

Let $A = \begin{bmatrix} 1 & 2 & 0 \\ 1 & 1 & 0 \\ -1 & 4 & 0 \end{bmatrix}$, $B = \begin{bmatrix} 1 & 2 & 3 \\ 1 & 1 & -1 \\ 2 & 2 & 2 \end{bmatrix}$ and $C = \begin{bmatrix} 1 & 2 & 3 \\ 1 & 1 & -1 \\ 3 & 3 & 3 \end{bmatrix}$

Then, $AB = \begin{bmatrix} 3 & 4 & 1 \\ 2 & 3 & 2 \\ -3 & 2 & -7 \end{bmatrix}$ and $AC = \begin{bmatrix} 3 & 4 & 1 \\ 2 & 3 & 2 \\ -3 & 2 & -7 \end{bmatrix}$ (Readers may verify this)

You may observe that $AB = AC$ but $B \neq C$. Thus, cancellation law does not hold for matrix multiplication. You may observe that, however, in such case $|A| = 0$.

(v) If A is a square matrix then $(A^m)^n = (A^n)^m = A^{mn}$ where m and n being positive integers.

(vi) If A is a square matrix then $A^0 = I$.

(vii) If A and B are confirmable for multiplication as A.B and B.A then it is not necessary that

A.B and B.A are equal. For instance, let $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 1 \\ 0 & 2 \end{bmatrix}$

Then, $AB = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 0 & 2 \end{bmatrix} = \begin{bmatrix} 1.1+1.0 & 1.1+2.2 \\ 3.1+4.0 & 3.1+4.2 \end{bmatrix} = \begin{bmatrix} 1 & 5 \\ 6 & 11 \end{bmatrix}$

$$AB = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 0 & 2 \end{bmatrix} = \begin{bmatrix} 1.1+1.0 & 1.1+2.2 \\ 3.1+4.0 & 3.1+4.2 \end{bmatrix} = \begin{bmatrix} 1 & 5 \\ 6 & 11 \end{bmatrix}$$

$$BA = \begin{bmatrix} 1 & 1 \\ 0 & 2 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} = \begin{bmatrix} 1.1+1.3 & 1.2+1.4 \\ 0.1+2.3 & 0.2+2.4 \end{bmatrix} = \begin{bmatrix} 4 & 6 \\ 6 & 8 \end{bmatrix}$$

We may observe that $AB \neq BA$. This shows that in matrix theory, generally, commutative law under multiplication does not hold. Using this property, we may prove that in general

Theorem: Prove that (a) $A^2 - B^2 \neq (A - B)(A + B)$ (b) $(A \pm B)^2 \neq A^2 \pm 2AB + B^2$

Proof: (a) Consider $(A - B)(A + B) = A(A + B) - B(A + B) = A^2 + AB - BA - B^2$.

Since $AB \neq BA$, hence $AB - BA \neq 0$. This implies that in general,

$$A^2 - B^2 \neq (A - B)(A + B).$$

This property is true only when $AB = BA$.

(b) Left as an exercise.

Other Matrix Multiplication Properties

- $A(BC) = (AB)C$ (Associative law under multiplication)
- $A(B + C) = AB + AC$ (Left distributive law)
- $(A + B)C = AC + BC$ (Right distributive law)
- $k(AB) = (kA)B = A(kB)$, (k being some scalar)

Class Activity: Students are advised to take any three square matrices of order 2 and $k = 3$ and prove the above four results.

The following practical problem makes use of matrix multiplication that helps readers to understand the importance of matrix multiplication.

Example 03: A company produces products x_1 , x_2 , and x_3 at plants p_1 , p_2 , and p_3 , using raw materials a, b, c and d. The number of units of raw materials required to produce one unit of each product is given in the first table. The cost per unit of raw materials at each plant is given in the second table.

Raw Material					Cost per unit		
	A	B	C	D		p_1	p_2
x_1	11	20	5	6	A	3	1
x_2	2	1	18	6	B	2	2
x_3	1	2	5	0	C	1	2
					D	1	1

What is the cost of producing each product at each plant?

Solution: Let $A = \begin{bmatrix} 11 & 20 & 5 & 6 \\ 2 & 1 & 18 & 6 \\ 1 & 2 & 5 & 0 \end{bmatrix}$ and $B = \begin{bmatrix} 3 & 1 \\ 2 & 2 \\ 1 & 2 \\ 1 & 1 \end{bmatrix}$

The cost of producing one unit of x_1 at plant p_1 is: $11 \times 3 + 20 \times 2 + 5 \times 1 + 6 \times 1 = 84$ because it requires 11 units of raw material 'a' at \$3 per unit, 20 units of 'b' at \$2 per unit, 5 units of 'c' at \$1 per unit and 6 units of 'd' at \$1 per unit. This calculation is just the sum of the product of row 1 of A (corresponding to x_1) and column 1 of B (corresponding to p_1). Similarly other values may be computed which are obtained simply multiplying matrix A with matrix B. The result is shown as under:

	p_1	p_2
x_1	84	67
x_2	32	46
x_3	12	15

At the cross section of say x_1 and p_1 the result 84 shows that cost of producing x_1 items at plant p_1 is \$84, etc.

Example 04: Evaluate the matrix multiplication $\begin{bmatrix} x & y & z \end{bmatrix} \begin{bmatrix} a & h & g \\ h & b & f \\ g & f & c \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix}$

Solution: $\begin{bmatrix} x & y & z \end{bmatrix} \begin{bmatrix} a & h & g \\ h & b & f \\ g & f & c \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix}$

$$= \begin{bmatrix} x \times a + y \times h + z \times g & x \times h + y \times b + z \times f & x \times g + y \times f + z \times c \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix}$$

$$= \begin{bmatrix} ax + hy + gz & hx + by + fz & gx + fy + cz \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix}$$

$$= \left[(ax + hy + gz) \times x + (hx + by + fz) \times y + (gx + fy + cz) \times z \right]$$

$$= \left[ax^2 + by^2 + cz^2 + 2hxy + 2fyz + 2gxz \right]$$

Example 05: Let $A = \begin{bmatrix} 2 & -3 & -5 \\ -1 & 4 & 5 \\ 1 & -3 & -4 \end{bmatrix}$ and $B = \begin{bmatrix} -1 & 3 & 5 \\ 1 & -3 & -5 \\ -1 & 3 & 5 \end{bmatrix}$

Find (i) $A + B$ (ii) $A - B$ (iii) $2A + 3B$ (iv) $3A - 5B$ (v) AB (vi) BA .

Solution: (i) $A + B = \begin{bmatrix} 2 & -3 & -5 \\ -1 & 4 & 5 \\ 1 & -3 & -4 \end{bmatrix} + \begin{bmatrix} -1 & 3 & 5 \\ 1 & -3 & -5 \\ -1 & 3 & 5 \end{bmatrix} = \begin{bmatrix} 2-1 & -3+3 & -5+5 \\ -1+1 & 4-3 & 5-5 \\ 1-1 & -3+3 & -4+5 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

(ii) $A - B = \begin{bmatrix} 2 & -3 & -5 \\ -1 & 4 & 5 \\ 1 & -3 & -4 \end{bmatrix} - \begin{bmatrix} -1 & 3 & 5 \\ 1 & -3 & -5 \\ -1 & 3 & 5 \end{bmatrix} = \begin{bmatrix} 2+1 & -3-3 & -5-5 \\ -1-1 & 4+3 & 5+5 \\ 1+1 & -3-3 & -4-5 \end{bmatrix} = \begin{bmatrix} 3 & -6 & -10 \\ -2 & 7 & 10 \\ 2 & -6 & -9 \end{bmatrix}$

$$(iii) \quad 2A + 3B = 2 \begin{bmatrix} 2 & -3 & -5 \\ -1 & 4 & 5 \\ 1 & -3 & -4 \end{bmatrix} + 3 \begin{bmatrix} -1 & 3 & 5 \\ 1 & -3 & -5 \\ -1 & 3 & 5 \end{bmatrix}$$

$$= \begin{bmatrix} 4 & -6 & -10 \\ -2 & 8 & 10 \\ 2 & -6 & -8 \end{bmatrix} + \begin{bmatrix} -3 & 9 & 15 \\ 3 & -9 & -15 \\ -3 & 9 & 15 \end{bmatrix} = \begin{bmatrix} 1 & 3 & 5 \\ 1 & -1 & -5 \\ -1 & 3 & 7 \end{bmatrix}$$

$$(iv) \quad 3A - 5B = 3 \begin{bmatrix} 2 & -3 & -5 \\ -1 & 4 & 5 \\ 1 & -3 & -4 \end{bmatrix} - 5 \begin{bmatrix} -1 & 3 & 5 \\ 1 & -3 & -5 \\ -1 & 3 & 5 \end{bmatrix}$$

$$= \begin{bmatrix} 6 & -9 & -15 \\ -3 & 12 & 15 \\ 3 & -9 & -12 \end{bmatrix} - \begin{bmatrix} -5 & 15 & 25 \\ 5 & -15 & -25 \\ -5 & 15 & 25 \end{bmatrix} = \begin{bmatrix} 11 & -24 & -40 \\ -8 & 27 & 40 \\ 8 & -24 & -37 \end{bmatrix}$$

(v)

$$AB = \begin{bmatrix} 2 & -3 & -5 \\ -1 & 4 & 5 \\ 1 & -3 & -4 \end{bmatrix} \begin{bmatrix} -1 & 3 & 5 \\ 1 & -3 & -5 \\ -1 & 3 & 5 \end{bmatrix} = \begin{bmatrix} -2-3+5 & 6+9-15 & 10+15-25 \\ 1+4-5 & -3-12+15 & -5-20+25 \\ -1-3+4 & 3+9-12 & 5+15-20 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}.$$

(vi) Similarly, $BA = O$

Example 06: Evaluate $\begin{bmatrix} 1 & 2 \\ 4 & -3 \end{bmatrix}^3$

$$\begin{aligned} \text{Solution: } \begin{bmatrix} 1 & 2 \\ 4 & -3 \end{bmatrix}^3 &= \begin{bmatrix} 1 & 2 \\ 4 & -3 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 4 & -3 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 4 & -3 \end{bmatrix} = \begin{bmatrix} 1+8 & 2-6 \\ 4-12 & 8+9 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 4 & -3 \end{bmatrix} \\ &= \begin{bmatrix} 9 & -4 \\ -8 & 17 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 4 & -3 \end{bmatrix} = \begin{bmatrix} 9-16 & 18+12 \\ -8+68 & -16-51 \end{bmatrix} = \begin{bmatrix} -7 & 30 \\ 60 & -67 \end{bmatrix} \end{aligned}$$

Example 07: Prove that if the product of matrices

$$A = \begin{bmatrix} \cos^2 \theta & \cos \theta \sin \theta \\ \cos \theta \sin \theta & \sin^2 \theta \end{bmatrix} \text{ and } B = \begin{bmatrix} \cos^2 \phi & \cos \phi \sin \phi \\ \cos \phi \sin \phi & \sin^2 \phi \end{bmatrix}$$

is the zero matrix then θ and ϕ differ by an odd multiple of $\pi/2$.

$$\begin{aligned} \text{Solution: Consider } AB &= \begin{bmatrix} \cos^2 \theta & \cos \theta \sin \theta \\ \cos \theta \sin \theta & \sin^2 \theta \end{bmatrix} \begin{bmatrix} \cos^2 \phi & \cos \phi \sin \phi \\ \cos \phi \sin \phi & \sin^2 \phi \end{bmatrix} \\ &= \begin{bmatrix} \cos^2 \theta \cos^2 \phi + \cos \theta \sin \theta \cos \phi \sin \phi & \cos^2 \theta \cos \phi \sin \phi + \cos \theta \sin \theta \sin^2 \phi \\ \cos \theta \sin \theta \cos^2 \phi + \sin^2 \theta \cos \phi \sin \phi & \cos \theta \sin \theta \cos \phi \sin \phi + \sin^2 \theta \sin^2 \phi \end{bmatrix} \\ &= \begin{bmatrix} \cos \theta \cos \phi (\cos \theta \cos \phi + \sin \theta \sin \phi) & \cos \theta \sin \phi (\cos \theta \cos \phi + \sin \theta \sin \phi) \\ \sin \theta \cos \phi (\cos \theta \cos \phi + \sin \theta \sin \phi) & \sin \theta \sin \phi (\cos \theta \cos \phi + \sin \theta \sin \phi) \end{bmatrix} \\ &= \begin{bmatrix} \cos \theta \cos \phi \cos(\theta - \phi) & \cos \theta \sin \phi \cos(\theta - \phi) \\ \sin \theta \cos \phi \cos(\theta - \phi) & \sin \theta \sin \phi \cos(\theta - \phi) \end{bmatrix} \\ &= \cos(\theta - \phi) \begin{bmatrix} \cos \theta \cos \phi & \cos \theta \sin \phi \\ \sin \theta \cos \phi & \sin \theta \sin \phi \end{bmatrix} \end{aligned}$$

Now given that $AB = O$ which, implies that $\cos(\theta - \phi) = 0$ because the matrix on right is not a zero matrix. Hence, if $\cos(\theta - \phi) = 0$ then $(\theta - \phi) = (2n - 1)\pi/2$. This proves that the difference between θ and ϕ is odd multiple of $\pi/2$.

Example 08: If $AB = BA$ then show that $(AB)^n = A^n \cdot B^n$.

Solution: (Proof by Mathematical Induction)

Given that $AB = BA$ then $(AB)^1 = (BA)^1$.

$$(AB)^2 = (AB)(AB) = A(BB)A = A(B^2A) = A(AB^2) = A^2B^2.$$

Thus proposition is true for $n = 1$ and $n = 2$. Let us assume that proposition is true for $n = k$, that is, $(AB)^k = A^k B^k$. Then,

$$\begin{aligned}(AB)^k(AB) &= A^k B^k (AB) = A^k B^k (BA) = A^k (B^k B)A = A^k (B^{k+1} A) = A^k A B^{k+1} \\ &= A^{k+1} B^{k+1}.\end{aligned}$$

Thus proposition is true for $n = k + 1$. Thus by Mathematical Induction, given proposition is true for all positive integer n .

Example 09: If $A = \begin{bmatrix} \cos a & \sin a \\ -\sin a & \cos a \end{bmatrix}$ then show that $A^n = \begin{bmatrix} \cos na & \sin na \\ -\sin na & \cos na \end{bmatrix}$

Solution: By Mathematical Induction put $n = 1$, then $A^1 = \begin{bmatrix} \cos 1a & \sin 1a \\ -\sin 1a & \cos 1a \end{bmatrix} = \begin{bmatrix} \cos a & \sin a \\ -\sin a & \cos a \end{bmatrix}$

Put $n = 2$, then

$$\begin{aligned}A^2 &= \begin{bmatrix} \cos a & \sin a \\ -\sin a & \cos a \end{bmatrix} \begin{bmatrix} \cos a & \sin a \\ -\sin a & \cos a \end{bmatrix} = \begin{bmatrix} \cos^2 a - \sin^2 a & \sin a \cos a + \sin a \cos a \\ -\sin a \cos a - \cos a \sin a & \cos^2 a - \sin^2 a \end{bmatrix} \\ &= \begin{bmatrix} \cos 2a & \sin 2a \\ -\sin 2a & \cos 2a \end{bmatrix}\end{aligned}$$

Let us assume that proposition is true for $n = k$, that is,

$$A^k = \begin{bmatrix} \cos ka & \sin ka \\ -\sin ka & \cos ka \end{bmatrix}. \text{ Now multiplying both sides by } A = \begin{bmatrix} \cos a & \sin a \\ -\sin a & \cos a \end{bmatrix}$$

$$\begin{aligned}\text{or } A.A^k &= A^{k+1} = \begin{bmatrix} \cos a & \sin a \\ -\sin a & \cos a \end{bmatrix} \begin{bmatrix} \cos ka & \sin ka \\ -\sin ka & \cos ka \end{bmatrix} \\ &= \begin{bmatrix} \cos ka \cos a - \sin ka \sin a & \sin ka \cos a + \cos ka \sin a \\ -\sin ka \cos a - \cos ka \sin a & \cos ka \cos a - \sin ka \sin a \end{bmatrix} = \begin{bmatrix} \cos(k+1)a & \sin(k+1)a \\ -\sin(k+1)a & \cos(k+1)a \end{bmatrix}^R\end{aligned}$$

Replacing $(k + 1)$ by n , we get: $A^n = \begin{bmatrix} \cos na & \sin na \\ -\sin na & \cos na \end{bmatrix}$. This is required proposition. Hence by mathematical induction given proposition is true for all natural numbers n .

Example 10: If $A = \begin{bmatrix} 3 & -4 \\ 1 & -1 \end{bmatrix}$ then show that $A^n = \begin{bmatrix} 1+2n & -4n \\ n & 1-2n \end{bmatrix}$

$$\text{Solution: Consider, } A^2 = \begin{bmatrix} 3 & -4 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} 3 & -4 \\ 1 & -1 \end{bmatrix} = \begin{bmatrix} 9-4 & -12+4 \\ 3-1 & -4+1 \end{bmatrix} = \begin{bmatrix} 5 & -8 \\ 2 & -3 \end{bmatrix} = \begin{bmatrix} 1+2.2 & -4.2 \\ 2 & 1-2.2 \end{bmatrix}$$

$$A^3 = A.A^2 = \begin{bmatrix} 3 & -4 \\ 1 & -1 \end{bmatrix} \cdot \begin{bmatrix} 5 & -8 \\ 2 & -3 \end{bmatrix} = \begin{bmatrix} 15-8 & -24+12 \\ 5-2 & -8+3 \end{bmatrix} = \begin{bmatrix} 7 & -12 \\ 3 & -5 \end{bmatrix} = \begin{bmatrix} 1+3.2 & -4.3 \\ 3 & 1-2.3 \end{bmatrix}$$

Continue this way n time, we get: $A^n = \begin{bmatrix} 1+2n & -4n \\ n & 1-2n \end{bmatrix}$

Example 11: If $A = \begin{bmatrix} 1 & 2 \\ 1 & 3 \end{bmatrix}$ and $f(x) = x^2 + 2x - 2$, find $f(A)$.

Solution: In matrix algebra, $f(A) = A^2 + 2A - 2I$. Now

$$A^2 = \begin{bmatrix} 1 & 2 \\ 1 & 3 \end{bmatrix}^2 = \begin{bmatrix} 1 & 2 \\ 1 & 3 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 1 & 3 \end{bmatrix} = \begin{bmatrix} 3 & 8 \\ 4 & 11 \end{bmatrix} \text{ and } 2A = 2 \begin{bmatrix} 1 & 2 \\ 1 & 3 \end{bmatrix} = \begin{bmatrix} 2 & 4 \\ 2 & 6 \end{bmatrix}. \text{ Thus}$$

$$f(A) = A^2 + 2A - 2I = \begin{bmatrix} 3 & 8 \\ 4 & 11 \end{bmatrix} + \begin{bmatrix} 2 & 4 \\ 2 & 6 \end{bmatrix} - \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix} = \begin{bmatrix} 3 & 12 \\ 6 & 15 \end{bmatrix}$$

INVERSE OF A MATRIX

We know that zero is an identity element under addition of a set real numbers for $a + 0 = 0 + a = a$, a being real. Similarly, one is an identity element under multiplication for a set of real numbers for $a \cdot 1 = 1 \cdot a = a$, a being a real.

In matrix algebra, a zero matrix is an additive identity and identity matrix is multiplicative identity for $A + O = O + A = A$ and $A \cdot I = I \cdot A = A$.

Definition: Let A and B be two square matrices of same order and $AB = BA = I$ then matrices A and B are the multiplicative inverse of each other and we write $A = B^{-1}$ and $B = A^{-1}$.

Similarly, let A and B be two matrices of same order and $A + B = O$, then B is called additive inverse of the matrix A and vice versa and we write $B = -A$ or $A = -B$. In this section we shall discuss multiplicative inverse of a square matrix. It may be noted that

- (i) Only the square matrices may possess multiplicative inverses.
- (ii) The symbol A^{-1} (read as A -inverse) does not mean $1/A$ or I/A . The symbol is just the notation for the inverse of a matrix A because there is no matrix division exists.
- (iii) The order of multiplicative inverse of square matrix (if it exists) will be same as that of the matrix A .

Example 01: Show that following matrices are inverses of each other.

$$A = \begin{bmatrix} 1 & 0 & 3 \\ 2 & 4 & 1 \\ 1 & 3 & 0 \end{bmatrix}, \quad B = \begin{bmatrix} -1 & 3 & -4 \\ 1/3 & -1 & 5/3 \\ 2/3 & -1 & 4/3 \end{bmatrix}$$

Solution: Consider

$$\begin{aligned} AB &= \begin{bmatrix} 1 & 0 & 3 \\ 2 & 4 & 1 \\ 1 & 3 & 0 \end{bmatrix} \begin{bmatrix} -1 & 3 & -4 \\ 1/3 & -1 & 5/3 \\ 2/3 & -1 & 4/3 \end{bmatrix} = \begin{bmatrix} -1+0+2 & 3+0-3 & -4+0+4 \\ -2+4/3+2/3 & 6-4-1 & -8+20/3+4/3 \\ -1+1+0 & 3-3+0 & -4+5+0 \end{bmatrix} \\ &= \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = I \end{aligned}$$

Similarly, $BA = I$ (left as exercise). Since $AB = BA = I$, this implies that A and B are multiplicative inverses of each other.

Theorem: Prove that if inverse of a square matrix exists then it will be unique.

Proof: Suppose that matrix A possesses two inverses, say B and C . Then, by definition,

$$AB = BA = I \text{ and } AC = CA = I.$$

Now consider any one above equations say, $AB = I$. Pre-multiplying by C , we get

$$C(AB) = CI \text{ or } (CA)B = C \text{ or } IB = C. \text{ Thus, } B = C$$

Similarly, let us consider, $CA = I$. Post-multiplying by B , we get:

$$(CA)B = IB \text{ or } C(AB) = B \text{ or } CI = B \text{ which implies that } C = B.$$

This gives contradiction to our assumption. Hence if the inverse of a square matrix A exists, it is always unique.

Definition: The square matrix, whose inverse exists, is called *non-singular or regular or invertible matrix*. The square matrix whose inverse does not exist is called a *singular matrix*.

Theorem: If the matrices A and B are invertible matrices, so is the matrix AB and that $(AB)^{-1} = B^{-1}A^{-1}$.

Proof: Consider, $(AB)(AB)^{-1} = (AB)(B^{-1}A^{-1}) = A(BB^{-1})A^{-1} = AI A^{-1} = AA^{-1} = I$

Similarly, $(AB)^{-1}(AB) = (B^{-1}A^{-1})(AB) = B^{-1}(A^{-1}A)B = B^{-1}IB = BB^{-1} = I$.

This proves that $(AB)^{-1} = B^{-1}A^{-1}$.

TYPICAL SQUARE MATRICES

In this section we shall discuss some important square matrices.

Periodic Matrix

A square matrix A is said to be *periodic* if $A^{k+1} = A$. If k is the least positive integer for which this property holds then A is called periodic matrix with period k .

Example 02: Show that the matrix $A = \begin{bmatrix} 1 & -2 & -6 \\ -3 & 2 & 9 \\ 2 & 0 & -3 \end{bmatrix}$ is a periodic matrix with period 2.

Solution: Consider $A^2 = \begin{bmatrix} 1 & -2 & -6 \\ -3 & 2 & 9 \\ 2 & 0 & -3 \end{bmatrix} \begin{bmatrix} 1 & -2 & -6 \\ -3 & 2 & 9 \\ 2 & 0 & -3 \end{bmatrix}$

$$= \begin{bmatrix} 1+6-12 & -2-4+0 & -6-18+18 \\ -3-6+18 & 6+4+0 & 18+18-27 \\ 2-0-6 & -4+0-0 & -12+0+9 \end{bmatrix} = \begin{bmatrix} -5 & -6 & -6 \\ 9 & 10 & 9 \\ -4 & -4 & -3 \end{bmatrix}$$

Now consider, $A^3 = A^2.A = \begin{bmatrix} -5 & -6 & -6 \\ 9 & 10 & 9 \\ -4 & -4 & -3 \end{bmatrix} \begin{bmatrix} 1 & -2 & -6 \\ -3 & 2 & 9 \\ 2 & 0 & -3 \end{bmatrix}$

$$= \begin{bmatrix} -5+18-12 & 10-12-0 & 30-54+18 \\ 9-30+18 & -18+20+0 & -54+90-27 \\ -4+12-6 & 8-8-0 & 24-36+9 \end{bmatrix} = \begin{bmatrix} 1 & -2 & -6 \\ -3 & 2 & 9 \\ 2 & 0 & -3 \end{bmatrix} = A$$

Since $A^3 = A$ hence A is a periodic matrix with period 2.

Idempotent Matrix

A square matrix A is said to be *idempotent* if $A^2 = A$.

Example 03: Show that $A = \begin{bmatrix} 2 & -2 & -4 \\ -1 & 3 & 4 \\ 1 & -2 & -3 \end{bmatrix}$ is an idempotent matrix.

Solution: Consider $A^2 = \begin{bmatrix} 2 & -2 & -4 \\ -1 & 3 & 4 \\ 1 & -2 & -3 \end{bmatrix} \begin{bmatrix} 2 & -2 & -4 \\ -1 & 3 & 4 \\ 1 & -2 & -3 \end{bmatrix} = \begin{bmatrix} 4+2-4 & -4-6+8 & -8-8+12 \\ -2-3+4 & 2+9-8 & 4+12-12 \\ 2+2-3 & -2-6+6 & -4-8+9 \end{bmatrix}$

$$= \begin{bmatrix} 2 & -2 & -4 \\ -1 & 3 & 4 \\ 1 & -2 & -3 \end{bmatrix} = A$$

Since $A^2 = A$, the given matrix A is an idempotent matrix.

Remarks: (i) Identity matrix is always periodic matrix with period one. It is also an Idempotent matrix because $I^2 = I$.

(ii) It may also be noted that an idempotent matrix is periodic matrix with period 1.

Nilpotent Matrix

A square matrix A is said to be *nilpotent* if $A^n = O$ where n is positive integer. If n is the least positive integer for which this property holds then A is called nilpotent matrix of index n .

Example 04: Show that $\begin{bmatrix} 1 & -3 & -4 \\ -1 & 3 & 4 \\ 1 & -3 & -4 \end{bmatrix}$ is nilpotent.

Solution: Let $A = \begin{bmatrix} 1 & -3 & -4 \\ -1 & 3 & 4 \\ 1 & -3 & -4 \end{bmatrix}$ then $A^2 = A \cdot A = \begin{bmatrix} 1 & -3 & -4 \\ -1 & 3 & 4 \\ 1 & -3 & -4 \end{bmatrix} \begin{bmatrix} 1 & -3 & -4 \\ -1 & 3 & 4 \\ 1 & -3 & -4 \end{bmatrix}$

$$= \begin{bmatrix} 1+3-4 & -3-9+12 & -4-12+16 \\ -1-3+4 & 3+9-12 & 4+12-16 \\ 1+3-4 & -3-9+12 & -4-12+16 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} = O$$

Thus given matrix is nilpotent of index 2.

Remark: If A is a nilpotent matrix then its determinant is zero, that is $|A| = 0$.

Involutory Matrix

A square matrix A is said to be involutory matrix if $A^2 = I$.

Example 05: Show that $A = \begin{bmatrix} 0 & 1 & -1 \\ 4 & -3 & 4 \\ 3 & -3 & 4 \end{bmatrix}$ is involutory.

Solution: Consider

$$A^2 = \begin{bmatrix} 0 & 1 & -1 \\ 4 & -3 & 4 \\ 3 & -3 & 4 \end{bmatrix} \begin{bmatrix} 0 & 1 & -1 \\ 4 & -3 & 4 \\ 3 & -3 & 4 \end{bmatrix} = \begin{bmatrix} 0+4-3 & 0-3+3 & 0+4-4 \\ 0-12+12 & 4+9-12 & -4-12+16 \\ 0-12+12 & 3+9-12 & -3-12+16 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = I$$

Since, $A^2 = I$, hence A is an involutory matrix.

Remark: If A is involutory then $A^2 = I$ or $A A = I$. Pre-multiplying both sides by A^{-1} , we obtain, $A^{-1}(A A) = A^{-1} \cdot I$ or $(A^{-1} A) \cdot A = A^{-1}$ or $IA = A^{-1}$. Thus, $A = A^{-1}$.

This shows that if A is an involutory matrix, it is equal to its inverse.

Transpose of a Matrix

Let A be any matrix of order $m \times n$. The matrix obtained by interchanging rows and columns of A is called the *transpose* of a matrix. The transpose of matrix is denoted by A^t . For example, if

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 0 & 5 & 7 \end{bmatrix} \text{ then } A^t = \begin{bmatrix} 1 & 0 \\ 2 & 5 \\ 3 & 7 \end{bmatrix}$$

Thus by definition, if matrix $A = [a_{ij}]$, then $A^t = [a_{ji}]$

Properties of Transpose of Matrices

If matrices A and B are conformable for the sum $A \pm B$ and the product AB , then

$$\begin{aligned} \text{(i)} (A \pm B)^t &= A^t \pm B^t & \text{(ii)} ((A)^t)^t &= A \\ \text{(iii)} (kA^t) &= k A^t, k \text{ being scalar} & \text{(iv)} (AB)^t &= B^t \cdot A^t \end{aligned}$$

Proofs are beyond the scope of this book. Readers are suggested to see any text book on matrix theory.

Corollary: $(ABC)^t = C^t B^t A^t$

Symmetric and Skew-Symmetric Matrices

Let A be a square matrix. If $A^t = A$ then A is called symmetric matrix and if $A^t = -A$ then A is called skew symmetric matrix.

For example, the matrices $\begin{bmatrix} 4 & 9 & 12 \\ 9 & 6 & 13 \\ 12 & 13 & 10 \end{bmatrix}$ and $\begin{bmatrix} 0 & 7 & -2 \\ -7 & 0 & -5 \\ 2 & 5 & 0 \end{bmatrix}$ respectively are symmetric and

skew-symmetric matrices.

Theorem: If A is square matrix, prove that $A + A^t$ and $A - A^t$ are symmetric and is a skew-symmetric matrix respectively.

Proof: Let $C = A + A^t$ then, $C^t = (A + A^t)^t = A^t + (A^t)^t = A^t + A = A + A^t = C$.

Since $C^t = C$, hence C is symmetric. But $C = A + A^t$, hence $A + A^t$ is symmetric matrix.

Now let $D = A - A^t$ then, $D^t = (A - A^t)^t = A^t - (A^t)^t = A^t - A = -(A - A^t) = -D$.

Since $D^t = -D$, hence D is skew-symmetric. But $D = A - A^t$, hence $A - A^t$ is skew-symmetric.

Theorem: If A is skew-symmetric matrix, then its diagonal elements are zero.

Proof: Given that A is a skew-symmetric matrix, hence $A^t = -A$. This implies that

$$a_{ij} = -a_{ji} \text{ for all } i, j.$$

This must be true for the diagonal elements as well. Thus,

$$a_{ii} = -a_{ii} \text{ or } a_{ii} + a_{ii} = 0 \text{ or } 2a_{ii} = 0 \text{ or } a_{ii} = 0.$$

This proves that for a skew-symmetric matrix A the diagonal elements are zero.

Theorem: If A and B are two symmetric matrices then $(AB)^t = A^t B^t = AB$ if and only if they commute.

Proof: Given that A and B are symmetric matrices, that is, $A^t = A$ and $B^t = B$.

Now let us assume that A and B commute, that is;

$$AB = BA \quad (1)$$

Then by definition, $(AB)^t = B^t A^t = BA = AB$ [Using eq. 1 & notice that $A^t = A$ and $B^t = B$]

Thus, $(AB)^t = AB$

This proves that AB is symmetric.

Now let us assume that AB is symmetric, i.e. $(AB)^t = AB$.

This implies that, $B^t A^t = AB$

Or, $BA = AB$ [Note: $A^t = A$ and $B^t = B$]

This shows that A and B commute.

Example 06: If matrix A is give as under then show that $A + A^t$ is symmetric and $A - A^t$ is skew

symmetric where $A = \begin{bmatrix} 2 & 8 & 5 \\ 1 & 3 & 4 \\ 7 & 9 & 5 \end{bmatrix}$

Solution: Given that $A = \begin{bmatrix} 2 & 8 & 5 \\ 1 & 3 & 4 \\ 7 & 9 & 5 \end{bmatrix}$ then $A^t = \begin{bmatrix} 2 & 1 & 7 \\ 8 & 3 & 9 \\ 5 & 4 & 5 \end{bmatrix}$

$$\therefore A + A^t = \begin{bmatrix} 4 & 9 & 12 \\ 9 & 6 & 13 \\ 12 & 13 & 10 \end{bmatrix} \quad \text{and} \quad A - A^t = \begin{bmatrix} 0 & 7 & -2 \\ -7 & 0 & -5 \\ 2 & 5 & 0 \end{bmatrix}$$

We see that $A + A^t$ is a symmetric matrix and $A - A^t$ is skew-symmetric matrix. We also see that all the diagonal elements of a skew-symmetric matrix are zero.

Example 06: Write the matrix $A = \begin{bmatrix} 1 & 2 & 4 \\ -2 & 5 & 3 \\ -1 & 6 & 3 \end{bmatrix}$ as sum of symmetric and skew-symmetric

matrices.

Solution: Given $A = \begin{bmatrix} 1 & 2 & 4 \\ -2 & 5 & 3 \\ -1 & 6 & 3 \end{bmatrix}$ or $A^t = \begin{bmatrix} 1 & -2 & -1 \\ 2 & 5 & 6 \\ 4 & 3 & 3 \end{bmatrix}$

Thus, $A + A^t = \begin{bmatrix} 2 & 0 & 3 \\ 0 & 10 & 9 \\ 3 & 9 & 6 \end{bmatrix}$ (1) and $A - A^t = \begin{bmatrix} 0 & 4 & 5 \\ -4 & 0 & -3 \\ -5 & 3 & 0 \end{bmatrix}$ (2)

Adding (1) and (2), we get: $2A = \begin{bmatrix} 2 & 0 & 3 \\ 0 & 10 & 9 \\ 3 & 9 & 6 \end{bmatrix} + \begin{bmatrix} 0 & 4 & 5 \\ -4 & 0 & -3 \\ -5 & 3 & 0 \end{bmatrix}$

or $A = \begin{bmatrix} 1 & 0 & 3/2 \\ 0 & 5 & 9/2 \\ 3/2 & 9/2 & 3 \end{bmatrix} + \begin{bmatrix} 0 & 2 & 5/2 \\ -2 & 0 & -3/2 \\ -5/2 & 3/2 & 0 \end{bmatrix}$
 Symmetric matrix + Skew – symmetric matrix

Conjugate of a Matrix

A complex matrix obtained by replacing its complex elements by their corresponding complex conjugates is called the *conjugate* and is denoted by $\bar{A}$. For example, if

$A = \begin{bmatrix} 2+3i & 4 & 5i \\ 0 & 4i & 8 \\ 4-3i & 0 & 7+3i \end{bmatrix}$ then $\bar{A} = \begin{bmatrix} 2-3i & 4 & -5i \\ 0 & -4i & 8 \\ 4+3i & 0 & 7-3i \end{bmatrix}$ is the conjugate matrix of A.

Tranjugate or Transpose of a Conjugate Matrix

The transpose of the conjugate of a matrix A is called the *tranjugate* or *transposed conjugate* of A. It is denoted by $(\bar{A})^t$. For example, if

$$A = \begin{bmatrix} 2+3i & 4 & 5i \\ 0 & 4i & 8 \\ 4-3i & 0 & 7+3i \end{bmatrix} \text{ then } (\bar{A})^t = \begin{bmatrix} 2-3i & 0 & 4+3i \\ 4 & -4i & 0 \\ -5i & 8 & 7-3i \end{bmatrix}$$

Properties of Complex Matrix

If A and B are two complex matrices confirmable for addition and multiplication, then

$$(i) \overline{A \pm B} = \bar{A} \pm \bar{B} \quad (ii) (\bar{A})^t = \overline{(A^t)} \quad (iii) \overline{A \cdot B} = \bar{A} \cdot \bar{B} \quad (iv) \overline{kA} = \bar{k}\bar{A},$$

$$(v) \overline{\left(\overline{\left(\overline{A} \right)} \right)} = A \quad (vi) A + \bar{A} \text{ is real matrix} \quad (vii) A - \bar{A} \text{ is purely imaginary matrix. Here, } k$$

is some constant.

Hermitian Matrix

A square matrix A for which $(\bar{A})^t = A$ is called *Hermitian matrix*. For example, the matrix

$$A = \begin{bmatrix} -1 & i-1 & 2i+3 \\ -i-1 & -2 & i \\ -2i+3 & -i & 0 \end{bmatrix}$$

is a Hermitian matrix because,

$$\bar{A} = \begin{bmatrix} -1 & i-1 & 2i+3 \\ -i-1 & -2 & i \\ -2i+3 & -i & 0 \end{bmatrix} \text{ then } (\bar{A})^t = \begin{bmatrix} -1 & i-1 & 2i+3 \\ -i-1 & -2 & i \\ -2i+3 & -i & 0 \end{bmatrix} = A.$$

Hence A is Hermitian matrix.

Skew-Hermitian Matrix

A square matrix A for which $(\bar{A})^t = -A$ is called a *skew-Hermitian matrix*. For example, if

$$A = \begin{bmatrix} i & 1-i & 2 \\ -1-i & 3i & i \\ -2 & i & 0 \end{bmatrix}, \text{ then } \bar{A} = \begin{bmatrix} -i & 1+i & 2 \\ -1+i & -3i & -i \\ -2 & -i & 0 \end{bmatrix}$$

$$\text{Then, } (\bar{A})^t = \begin{bmatrix} -i & -1+i & -2 \\ 1+i & -3i & -i \\ 2 & -i & 0 \end{bmatrix} = - \begin{bmatrix} i & 1-i & 2 \\ -1-i & 3i & i \\ -2 & i & 0 \end{bmatrix} = -A.$$

Theorem: Prove that diagonal elements of Hermitian matrix are real and that of skew-Hermitian matrix are zero or purely imaginary.

Proof: Let A be a Hermitian matrix then $(\bar{A})^t = A$. Now let its diagonal element be $a_{ii} = \alpha + i\beta$. This implies that $\overline{a_{ii}} = \alpha - i\beta$. But $a_{ii} = \overline{a_{ii}}$ which implies that $\alpha + i\beta = \alpha - i\beta$ or $2i\beta = 0$ giving $\beta = 0$. Thus $a_{ii} = \alpha$ which is a real number. Thus diagonal elements of Hermitian matrix are real.

Now, let A be a Skew-Hermitian matrix then $(\bar{A})^t = -A$. Now let its diagonal element be $a_{ii} = \alpha + i\beta$. This implies that $\overline{a_{ii}} = -a_{ii}$ or $\alpha - i\beta = -(\alpha + i\beta)$ or $\alpha - i\beta = -\alpha - i\beta$ giving $2\alpha = 0$ or $\alpha = 0$. Thus, $a_{ii} = i\beta$ which is purely imaginary. If $\beta = 0$ then $a_{ii} = 0$. Thus, for Skew-Hermitian matrix, its diagonal elements are purely imaginary or zero.

Orthogonal Matrix

A square matrix A is said to be *orthogonal* if $A^t A = I = A A^t$.

Example 07: Show that the matrix $\begin{bmatrix} \cos \theta & 0 & \sin \theta \\ 0 & 1 & 0 \\ \sin \theta & 0 & \cos \theta \end{bmatrix}$ is orthogonal.

Solution: By definition consider,

$$\begin{aligned} A A^t &= \begin{bmatrix} \cos \theta & 0 & \sin \theta \\ 0 & 1 & 0 \\ \sin \theta & 0 & \cos \theta \end{bmatrix} \begin{bmatrix} \cos \theta & 0 & \sin \theta \\ 0 & 1 & 0 \\ -\sin \theta & 0 & \cos \theta \end{bmatrix} \\ &= \begin{bmatrix} \cos^2 \theta + \sin^2 \theta & 0 & \cos \theta \sin \theta - \cos \theta \sin \theta \\ 0 & 1 & 0 \\ \cos \theta \sin \theta - \cos \theta \sin \theta & 0 & \cos^2 \theta + \sin^2 \theta \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = I \end{aligned}$$

Similarly we can show that $A^t A = I$. Hence given matrix is orthogonal.

Example 08: Prove that $A = \frac{1}{3} \begin{bmatrix} -1 & 2 & 2 \\ 2 & -1 & 2 \\ 2 & 2 & -1 \end{bmatrix}$ is orthogonal.

Solution: $A = \frac{1}{3} \begin{bmatrix} -1 & 2 & 2 \\ 2 & -1 & 2 \\ 2 & 2 & -1 \end{bmatrix}$ or $A^t = \frac{1}{3} \begin{bmatrix} -1 & 2 & 2 \\ 2 & -1 & 2 \\ 2 & 2 & -1 \end{bmatrix}$. Then

$$A^t A = \frac{1}{9} \begin{bmatrix} -1 & 2 & 2 \\ 2 & -1 & 2 \\ 2 & 2 & -1 \end{bmatrix} \begin{bmatrix} -1 & 2 & 2 \\ 2 & -1 & 2 \\ 2 & 2 & -1 \end{bmatrix} = \frac{1}{9} \begin{bmatrix} 9 & 0 & 0 \\ 0 & 9 & 0 \\ 0 & 0 & 9 \end{bmatrix} = \frac{9}{9} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = I.$$

Since, $A^t A = I$, therefore, matrix A is orthogonal.

Theorem: If A and B are orthogonal matrices, then AB and BA are also orthogonal matrices.

Proof: Since A and B are orthogonal matrices, we have, $AA^t = I$ and $BB^t = I$. We have to prove that AB and BA are also orthogonal.

Consider, $(AB)(AB)^t = AB \cdot B^t A^t = A(B B^t)A^t = A(I)A^t = A A^t = I$.

This proves that AB is orthogonal. Now, consider

$$(BA)(BA)^t = BA \cdot A^t B^t = B(A A^t)B^t = B(I)B^t = B B^t = I.$$

This shows that BA is orthogonal.

Example 09: If the matrix $A = \begin{bmatrix} 0 & 2b & c \\ a & b & -c \\ a & -b & c \end{bmatrix}$ is orthogonal find a , b and c .

Solution: Since A is orthogonal hence $A A^t = I$. Thus,

$$A A^t = \begin{bmatrix} 0 & 2b & c \\ a & b & -c \\ a & -b & c \end{bmatrix} \begin{bmatrix} 0 & a & a \\ 2b & b & -b \\ c & -c & c \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 4b^2 + c^2 & 2b^2 - c^2 & -2b^2 + c^2 \\ 2b^2 - c^2 & a^2 + b^2 + c^2 & a^2 - b^2 - c^2 \\ -2b^2 + c^2 & a^2 - b^2 - c^2 & a^2 + b^2 + c^2 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Equating the corresponding entries, we obtain:

$4b^2 + c^2 = 1$ and $2b^2 - c^2 = 0$. Adding, we get : $b = \pm 1/\sqrt{6}$ and $c = \pm 1/\sqrt{3}$

Also, $a^2 + b^2 + c^2 = 1$. Putting the values of b and c we get, $a = \pm 1/\sqrt{2}$

Unary Matrix

Square matrix A is said to be unitary if $(\bar{A})^t A = I$

Example 10: Show that $A = \begin{bmatrix} \frac{1+i}{2} & \frac{-1+i}{2} \\ \frac{1+i}{2} & \frac{1-i}{2} \end{bmatrix}$ is a unitary matrix.

Solution: $A = \begin{bmatrix} \frac{1+i}{2} & \frac{-1+i}{2} \\ \frac{1+i}{2} & \frac{1-i}{2} \end{bmatrix}$ then $A^t = \begin{bmatrix} \frac{1+i}{2} & \frac{1+i}{2} \\ \frac{-1+i}{2} & \frac{1-i}{2} \end{bmatrix}$ and $(\bar{A}^t) = \begin{bmatrix} \frac{1-i}{2} & \frac{1-i}{2} \\ \frac{-1-i}{2} & \frac{1+i}{2} \end{bmatrix}$. Now,

$$\begin{aligned} (\bar{A}^t)A &= \begin{bmatrix} \frac{1-i}{2} & \frac{1-i}{2} \\ \frac{-1-i}{2} & \frac{1+i}{2} \end{bmatrix} \begin{bmatrix} \frac{1+i}{2} & \frac{-1+i}{2} \\ \frac{1+i}{2} & \frac{1-i}{2} \end{bmatrix} = \left(\frac{1}{2}\right)\left(\frac{1}{2}\right) \begin{bmatrix} 1-i & 1-i \\ -(1+i) & 1+i \end{bmatrix} \begin{bmatrix} 1+i & -1+i \\ 1+i & 1-i \end{bmatrix} \\ &= \frac{1}{4} \begin{bmatrix} (1-i^2) + (1-i^2) & -(1-i)^2 + (1-i)^2 \\ -(1+i) + (1+i)^2 & (1-i^2) + (1-i^2) \end{bmatrix} = \frac{1}{4} \begin{bmatrix} 4 & 0 \\ 0 & 4 \end{bmatrix} = \frac{4}{4} \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = I. \end{aligned}$$

Hence A is a unitary matrix.

Theorem: If A is a square complex matrix, then $A + (\bar{A})^t$ is Hermitian and $A - (\bar{A})^t$ is skew-Hermitian matrix.

Proof: Let

$$C = A + (\bar{A})^t \text{ then } \bar{C} = \overline{A + (\bar{A})^t} = \bar{A} + \overline{(\bar{A})^t} = \bar{A} + (\bar{\bar{A}})^t = \bar{A} + A^t \quad \left[\because \overline{(\bar{A})^t} = A^t \right]$$

$$\text{Thus, } (\bar{C})^t = (\bar{\bar{A} + A^t})^t = (\bar{\bar{A}})^t + (\bar{A^t})^t = (\bar{A})^t + A = C.$$

Since, $(\bar{C})^t = C$ hence C is a Hermitian matrix. Now let,

$$D = A - (\bar{A})^t \text{ or } \bar{D} = \overline{A - (\bar{A})^t} = \bar{A} - \overline{(\bar{A})^t} = \bar{A} - (\bar{\bar{A}})^t = \bar{A} - A^t \quad \left[\because \overline{(\bar{A})^t} = A^t \right]$$

$$\text{Thus, } (\bar{D})^t = (\bar{\bar{A} - A^t})^t = (\bar{\bar{A}})^t - (\bar{A^t})^t = (\bar{A})^t - A = -(A - (\bar{A})^t) = -D.$$

Since, $(\bar{D})^t = -D$ hence D is Skew Hermitian matrix.

Example 11: Express $\begin{bmatrix} i & 2-3i & 4+5i \\ 6+i & 0 & 4-5i \\ -i & 2-i & 2+i \end{bmatrix}$ as a sum of Hermitian and Skew – Hermitian

matrices.

Solution: Given

$$A = \begin{bmatrix} 1+i & 2 & 5-5i \\ 2i & 2+i & 4+2i \\ -1+i & -4 & 7 \end{bmatrix} \text{ so } \bar{A} = \begin{bmatrix} 1-i & 2 & 5+5i \\ -2i & 2-i & 4-2i \\ -1-i & -4 & 7 \end{bmatrix} \text{ or } (\bar{A})^t = \begin{bmatrix} 1-i & -2i & -1-i \\ 2 & 2-i & -4 \\ 5+5i & 4-2i & 7 \end{bmatrix}$$

$$\text{Thus, } A + (\bar{A})^t = \begin{bmatrix} 1+i & 2 & 5-5i \\ 2i & 2+i & 4+2i \\ -1+i & -4 & 7 \end{bmatrix} + \begin{bmatrix} 1-i & -2i & -1-i \\ 2 & 2-i & -4 \\ 5+5i & 4-2i & 7 \end{bmatrix} = \begin{bmatrix} 2 & 2-2i & 4-6i \\ 2+2i & 4 & 2i \\ 4+6i & -2i & 14 \end{bmatrix}$$

$$\text{Thus, } \frac{1}{2} \{A + (\bar{A})^t\} = \begin{bmatrix} 1 & 1-i & 2-3i \\ 1+i & 2 & i \\ 2+3i & -i & 7 \end{bmatrix} \quad (1)$$

$$\text{Similarly, } A - (\bar{A})^t = \begin{bmatrix} 1+i & 2 & 5-5i \\ 2i & 2+i & 4+2i \\ -1+i & -4 & 7 \end{bmatrix} - \begin{bmatrix} 1-i & -2i & -1-i \\ 2 & 2-i & -4 \\ 5+5i & 4-2i & 7 \end{bmatrix} = \begin{bmatrix} 2i & 2+2i & 6-4i \\ -2+2i & 2i & 8+2i \\ -6-4i & -8+2i & 0 \end{bmatrix}$$

$$\text{Thus, } \frac{1}{2} \{A - (\bar{A})^t\} = \begin{bmatrix} i & 1+i & 3-2i \\ -1+i & i & 4+i \\ -3-2i & -4+i & 0 \end{bmatrix} \quad (2)$$

Adding (1) and (2), we have

$$\therefore \frac{1}{2} \{A + (\bar{A})^t\} + \frac{1}{2} \{A - (\bar{A})^t\} = A = \begin{bmatrix} 1 & 1-i & 2-3i \\ 1+i & 2 & i \\ 2+3i & -i & 7 \end{bmatrix} + \begin{bmatrix} i & 1+i & 3-2i \\ -1+i & i & 4+i \\ -3-2i & -4+i & 0 \end{bmatrix}.$$

Hermitian Matrix + Skew – Hermitian Matrix

ELEMENTARY ROW OPERATIONS (ERO)

Many applications in matrix theory make an extensive use of elementary row operations. These elementary row operations are of three types and are presented here.

- (i) Multiplying a given row by a non-zero number. This is usually denoted by $k.R_i$ which means multiply row R_i by a constant k .
- (ii) Interchanging any two rows of a matrix. This is usually denoted by R_{ij} which means interchange row R_i with R_j .
- (iii) Addition of any multiple of one row to another row. This means multiply any row of a matrix by a non-zero number and the result so obtained may be added to any other row. This is usually denoted by $k.R_i + R_j$. This means multiply row R_i by a non-zero number k and result so obtained is added in R_j . Some teachers use this operation as: $R_j + k.R_i$.

Row Equivalent Matrices

Two matrices A and B are called row equivalent matrices, written as $A \sim B$, if one can be obtained from the other by performing a finite sequence of elementary row operations. For example, let

$$A = \begin{bmatrix} 1 & 5 & 2 & 3 \\ 3 & 1 & 8 & -1 \\ 2 & 5 & 1 & 6 \end{bmatrix}$$

Then, if we multiply R_2 by 3, (denoted by $3R_2$) we get a new matrix say B, given by:

$$B = \begin{bmatrix} 1 & 5 & 2 & 3 \\ 9 & 3 & 24 & -3 \\ 2 & 5 & 1 & 6 \end{bmatrix}$$

We say that “matrix B is Row Equivalent to matrix A” and write $B \sim A$.

Similarly, if we interchange R_1 and R_3 of matrix A, (denoted by R_{13}) we get:

$$C = \begin{bmatrix} 2 & 5 & 1 & 6 \\ 3 & 1 & 8 & -1 \\ 1 & 5 & 2 & 3 \end{bmatrix}$$

We say that “matrix C is Row Equivalent to matrix A” and write $C \sim A$.

Finally, if R_1 is multiplied by say 4 and the result is added to R_2 , (denoted by $R_2 + 4R_1$), we get

$$D = \begin{bmatrix} 1 & 5 & 2 & 3 \\ 7 & 21 & 16 & 11 \\ 2 & 5 & 1 & 6 \end{bmatrix}$$

You may notice that row R_1 remains same while the change occurs in row R_2 . We say that “matrix D is Row Equivalent to matrix A” and write $D \sim A$.

Echelon and Reduced Echelon Matrices

A row of any matrix is called zero if all its entries are zeros. Now a matrix is in echelon form if

(i) All zero rows come below the non-zero rows. (ii) The first nonzero entry in each nonzero row is 1 and occurs in a column to the right of the leading 1 in the row above it.

For example,

$$A = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}, B = \begin{bmatrix} 1 & 3 & -1 \\ 0 & 1 & 3 \\ 0 & 0 & 0 \end{bmatrix}, C = \begin{bmatrix} 1 & 3 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix} \text{ and } D = \begin{bmatrix} 1 & 0 & 3 & 4 \\ 0 & 0 & 1 & 4 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

are echelon matrices.

If an echelon matrix has the additional property that in each row of the column containing leading 1 has all its other entries zero the matrix is called *Reduced Echelon Matrix*. For Example, the following matrices are reduced echelon matrices.

$$\begin{bmatrix} 1 & 3 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}, \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 2 \end{bmatrix} \text{ and } \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

Example 01: Transform the matrix $A = \begin{bmatrix} 1 & 4 & -2 \\ 2 & 7 & -1 \\ 2 & 9 & -7 \end{bmatrix}$ into an echelon matrix and then to the

reduced echelon matrix.

Solution: Given $A = \begin{bmatrix} 1 & 4 & -2 \\ 2 & 7 & -1 \\ 2 & 9 & -7 \end{bmatrix} \begin{matrix} R_2 - 2R_1 \\ R_3 - 2R_1 \end{matrix}$

$$A \approx \begin{bmatrix} 1 & 4 & -2 \\ 0 & -1 & 3 \\ 0 & 1 & -3 \end{bmatrix} \xrightarrow{R_3 + R_2} \begin{bmatrix} 1 & 4 & -2 \\ 0 & -1 & 3 \\ 0 & 0 & 0 \end{bmatrix} \xrightarrow{(-1)R_2} \begin{bmatrix} 1 & 4 & -2 \\ 0 & 1 & -3 \\ 0 & 0 & 0 \end{bmatrix}$$

which is in echelon/reduced echelon form.

Example 02: Change the matrix $A = \begin{bmatrix} -1 & 3 & 0 & 1 & 4 \\ 1 & -3 & 0 & 0 & -1 \\ 2 & -6 & 2 & 4 & 0 \\ 0 & 0 & 1 & 3 & -4 \end{bmatrix}$ into reduced echelon matrix.

Solution: Given matrix is $A = \begin{bmatrix} -1 & 3 & 0 & 1 & 4 \\ 1 & -3 & 0 & 0 & -1 \\ 2 & -6 & 2 & 4 & 0 \\ 0 & 0 & 1 & 3 & -4 \end{bmatrix} \xrightarrow{\left(\frac{1}{2}\right)R_3}$

$$\begin{aligned} &\approx \begin{bmatrix} -1 & 3 & 0 & 1 & 4 \\ 1 & -3 & 0 & 0 & -1 \\ 1 & -3 & 1 & 2 & 0 \\ 0 & 0 & 1 & 3 & -4 \end{bmatrix} \xrightarrow{R_{12}} \begin{bmatrix} 1 & -3 & 0 & 0 & -1 \\ 0 & 0 & 1 & 3 & -4 \\ 0 & 0 & 1 & 2 & 1 \\ 0 & 0 & 1 & 3 & -4 \end{bmatrix} \xrightarrow{R_{24}} \begin{bmatrix} 1 & -3 & 0 & 0 & -1 \\ 0 & 0 & 1 & 3 & -4 \\ 0 & 0 & 1 & 2 & 1 \\ 0 & 0 & 0 & 1 & 3 \end{bmatrix} \\ &\approx \begin{bmatrix} 1 & -3 & 0 & 0 & -1 \\ 0 & 0 & 1 & 3 & -4 \\ 0 & 0 & 0 & -1 & 5 \\ 0 & 0 & 0 & 1 & 3 \end{bmatrix} \xrightarrow{R_4 + R_3} \begin{bmatrix} 1 & -3 & 0 & 0 & -1 \\ 0 & 0 & 1 & 3 & -4 \\ 0 & 0 & 0 & -1 & 5 \\ 0 & 0 & 0 & 0 & 8 \end{bmatrix} \xrightarrow{\begin{matrix} R_2 + 3R_3 \\ \frac{1}{8}R_4 \end{matrix}} \begin{bmatrix} 1 & -3 & 0 & 0 & -1 \\ 0 & 0 & 1 & 0 & 11 \\ 0 & 0 & 0 & -1 & 5 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix} \\ &\approx \begin{bmatrix} 1 & -3 & 0 & 0 & -1 \\ 0 & 0 & 1 & 0 & 11 \\ 0 & 0 & 0 & -1 & 5 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix} \xrightarrow{(-1)R_3} \begin{bmatrix} 1 & -3 & 0 & 0 & -1 \\ 0 & 0 & 1 & 0 & 11 \\ 0 & 0 & 0 & 1 & -5 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix} \xrightarrow{\begin{matrix} R_1 + R_4 \\ R_2 - 11R_4 \\ R_3 + 5R_4 \end{matrix}} \begin{bmatrix} 1 & -3 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix} \\ &\approx \begin{bmatrix} 1 & -3 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix} \end{aligned}$$

The matrix is now in reduced echelon form.

Example 03: Transform the following matrices into (i) echelon form and (ii) reduced echelon form.

$$(i) \ A = \begin{bmatrix} 1 & 2 & 8 \\ 1 & 0 & 2 \\ 0 & 0 & 0 \end{bmatrix} \quad (ii) \ B = \begin{bmatrix} 1 & -2 & 3 & 4 \\ 3 & 2 & -1 & -3 \\ 4 & 1 & 1 & 4 \end{bmatrix} \quad (iii) \ C = \begin{bmatrix} 1 & 3 & 0 & 2 & 5 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 2 & 2 & 6 \end{bmatrix}$$

Solution: (i) Given matrix is $A = \begin{bmatrix} 1 & 2 & 8 \\ 1 & 0 & 2 \\ 0 & 0 & 0 \end{bmatrix} \xrightarrow{R_2 - R_1}$

$$\approx \begin{bmatrix} 1 & 2 & 8 \\ 0 & -2 & -6 \\ 0 & 0 & 0 \end{bmatrix} \xrightarrow{(-1/2)R_2} \begin{bmatrix} 1 & 2 & 8 \\ 0 & 1 & 3 \\ 0 & 0 & 0 \end{bmatrix}. \text{ This is an Echelon form.}$$

$$\text{Furthermore, } \begin{bmatrix} 1 & 2 & 8 \\ 0 & 1 & 3 \\ 0 & 0 & 0 \end{bmatrix} \xrightarrow{R_1 - 2R_2} \begin{bmatrix} 1 & 0 & 2 \\ 0 & 1 & 3 \\ 0 & 0 & 0 \end{bmatrix}. \text{ This is in the reduced echelon form.}$$

$$\begin{aligned} \text{(ii) } B &= \begin{bmatrix} 1 & -2 & 3 & 4 \\ 3 & 2 & -1 & -3 \\ 4 & 1 & 1 & 4 \end{bmatrix} \begin{matrix} R_2 - 3R_1 \\ R_3 - 4R_1 \end{matrix} \\ &\approx \begin{bmatrix} 1 & -2 & 3 & 4 \\ 0 & 8 & -10 & -15 \\ 0 & 9 & -11 & -12 \end{bmatrix} \begin{matrix} R_3 + (-1)R_2 \end{matrix} \approx \begin{bmatrix} 1 & -2 & 3 & 4 \\ 0 & 8 & -10 & -15 \\ 0 & 1 & -1 & 3 \end{bmatrix} R_{23} \\ &\approx \begin{bmatrix} 1 & -2 & 3 & 4 \\ 0 & 1 & -1 & 3 \\ 0 & 8 & -10 & -15 \end{bmatrix} \begin{matrix} R_3 + (-8)R_2 \end{matrix} \approx \begin{bmatrix} 1 & -2 & 3 & 4 \\ 0 & 1 & -1 & 3 \\ 0 & 0 & -2 & -39 \end{bmatrix} \begin{matrix} (-1/2)R_3 \end{matrix} \\ &\approx \begin{bmatrix} 1 & -2 & 3 & 4 \\ 0 & 1 & -1 & 3 \\ 0 & 0 & 1 & 39/2 \end{bmatrix}. \text{ This is in echelon form.} \end{aligned}$$

$$\text{Furthermore, } \begin{bmatrix} 1 & 0 & 1 & 10 \\ 0 & 1 & -1 & 3 \\ 0 & 0 & 1 & 39/2 \end{bmatrix} \begin{matrix} R_1 - R_3 \\ R_2 + R_3 \end{matrix} \approx \begin{bmatrix} 1 & 0 & 0 & -19/2 \\ 0 & 1 & 0 & 45/2 \\ 0 & 0 & 1 & 39/2 \end{bmatrix}$$

This is in Reduced Echelon Form.

$$\begin{aligned} \text{(iii) } C &= \begin{bmatrix} 1 & 3 & 0 & 2 & 5 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 2 & 2 & 6 \end{bmatrix} R_{23} \approx \begin{bmatrix} 1 & 3 & 0 & 2 & 5 \\ 0 & 0 & 2 & 2 & 6 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix} (1/2)R_2 \\ &\approx \begin{bmatrix} 1 & 3 & 0 & 2 & 5 \\ 0 & 0 & 1 & 1 & 3 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}. \text{ This is in echelon as well as in reduced echelon form.} \end{aligned}$$

Example 04: Show that $A = \begin{bmatrix} 1 & 2 & 1 \\ 3 & 1 & 2 \\ 0 & 1 & 2 \end{bmatrix} \approx I_3$

$$\begin{aligned} \text{Solution: We have } A &= \begin{bmatrix} 1 & 2 & 1 \\ 3 & 1 & 2 \\ 0 & 1 & 2 \end{bmatrix} \xrightarrow{R_2 + (-3)R_1} \begin{bmatrix} 1 & 2 & 1 \\ 0 & -5 & -1 \\ 0 & 1 & 2 \end{bmatrix} R_{23} \\ &\approx \begin{bmatrix} 1 & 2 & 1 \\ 0 & 1 & 2 \\ 0 & -5 & -1 \end{bmatrix} \begin{matrix} R_1 + (-2)R_2, R_3 + 5R_2 \end{matrix} \approx \begin{bmatrix} 1 & 0 & -3 \\ 0 & 1 & 2 \\ 0 & 0 & 9 \end{bmatrix} \left(\frac{1}{9}\right)R_3 \\ &\approx \begin{bmatrix} 1 & 0 & -3 \\ 0 & 1 & 2 \\ 0 & 0 & 1 \end{bmatrix} \begin{matrix} R_1 + 3R_3 \\ R_2 + (-2)R_3 \end{matrix} \approx \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = I_3 \end{aligned}$$

Rank of a Matrix

If any matrix is reduced to an echelon form or reduced echelon form, the number of non-zero rows is called the rank of that matrix and is usually denoted by $\rho(A)$. It may be noted that the non-zero rows of echelon matrix form a set of linearly independent vectors. You will learn this in coming sections.

Remarks: (i) Zero matrix has zero rank. (ii) Every non-zero matrix has at least rank 1.

Example 05: Find the rank of matrix $A = \begin{bmatrix} 5 & 9 & 3 \\ -3 & 5 & 6 \\ -1 & -5 & -3 \end{bmatrix}$

Solution: Consider

$$\begin{aligned}
 A &= \begin{bmatrix} 5 & 9 & 3 \\ -3 & 5 & 6 \\ -1 & -5 & -3 \end{bmatrix} \xrightarrow{R_{13}} \begin{bmatrix} -1 & -5 & -3 \\ -3 & 5 & 6 \\ 5 & 9 & 3 \end{bmatrix} \xrightarrow{(-1)R_1} \begin{bmatrix} 1 & 5 & 3 \\ -3 & 5 & 6 \\ 5 & 9 & 3 \end{bmatrix} \xrightarrow{\substack{R_2+3R_1 \\ R_3-5R_1}} \\
 &\approx \begin{bmatrix} 1 & 5 & 3 \\ 0 & 20 & 15 \\ 0 & -16 & -12 \end{bmatrix} \xrightarrow{(1/4)R_3} \begin{bmatrix} 1 & 5 & 3 \\ 0 & 20 & 15 \\ 0 & 4 & 3 \end{bmatrix} \xrightarrow{\frac{1}{20}R_2} \begin{bmatrix} 1 & 5 & 3 \\ 0 & 1 & 3/4 \\ 0 & 4 & 3 \end{bmatrix} \xrightarrow{R_3+(-4)R_2} \\
 &\approx \begin{bmatrix} 1 & 5 & 3 \\ 0 & 1 & 3/4 \\ 0 & 0 & 0 \end{bmatrix}. \text{ This is in echelon form and number of non-zero rows are } 2 \Rightarrow \rho(A) = 2
 \end{aligned}$$

Example 06: Find the rank of the matrix $A = \begin{bmatrix} 1 & 3 & 1 & -2 & -3 \\ 1 & 4 & 3 & -1 & -4 \\ 2 & 3 & -4 & -7 & -3 \\ 3 & 8 & 1 & -7 & -8 \end{bmatrix}$

Solution: Consider $A = \begin{bmatrix} 1 & 3 & 1 & -2 & -3 \\ 1 & 4 & 3 & -1 & -4 \\ 2 & 3 & -4 & -7 & -3 \\ 3 & 8 & 1 & -7 & -8 \end{bmatrix} \xrightarrow{\substack{R_2+(-1)R_1 \\ R_3+(-2)R_1 \\ R_4+(-3)R_1}}$

$$\begin{aligned}
 &\approx \begin{bmatrix} 1 & 3 & 1 & -2 & -3 \\ 0 & 1 & 2 & 1 & -1 \\ 0 & -3 & -6 & -3 & 1 \\ 0 & -1 & -2 & -1 & 3 \end{bmatrix} \xrightarrow{\substack{\frac{1}{3}R_3 \\ (-1)R_4}} \begin{bmatrix} 1 & 3 & 1 & -2 & -3 \\ 0 & 1 & 2 & 1 & -1 \\ 0 & 1 & 2 & 1 & -1 \\ 0 & 1 & 2 & 1 & -1 \end{bmatrix} \xrightarrow{\substack{R_3+(-1)R_2 \\ R_4+(-1)R_2}} \begin{bmatrix} 1 & 3 & 1 & -2 & -3 \\ 0 & 1 & 2 & 1 & -1 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}
 \end{aligned}$$

This is the echelon form of matrix A. The number of non-zero rows is two, hence the rank of matrix A is 2.

An Alternative Method to find the Rank of a Matrix

In the previous section the rank of a matrix A was found by reducing the matrix to an echelon form. This involves many row operations and the process becomes much difficult if fractions occur in computation.

A new method is very elegant and well-designed which involves no fractions and also yields an echelon matrix that is row equivalent to A.

This method involves evolution of determinants of order 2. An example is produced below for those who have no idea about the determinants of order 2.

Definition: Let A be a square matrix of order 2. We define determinant of matrix A as under:

$$\det(A) = |A| = \begin{vmatrix} a & b \\ c & d \end{vmatrix} = ad - bc$$

The alternative method of finding the rank of matrix A is presented in the following example.

Example 07: Find the rank of the matrix $A = \begin{bmatrix} -3 & 5 & 1 & 2 \\ 7 & 2 & 0 & -4 \\ -8 & 3 & 1 & 6 \end{bmatrix}$

Solution: Since matrix A is a non-zero matrix of order 3×4 , hence its rank must be at least 1. [Readers are advised to see the process carefully. They may ask their tutors to help them to understand the technique presented here].

$$\text{Rank } A = 1 + \text{Rank} \begin{bmatrix} \begin{vmatrix} -3 & 5 \\ 7 & 2 \end{vmatrix} & \begin{vmatrix} -3 & 1 \\ 7 & 0 \end{vmatrix} & \begin{vmatrix} -3 & 2 \\ 7 & -4 \end{vmatrix} \\ \begin{vmatrix} -3 & 5 \\ -8 & 3 \end{vmatrix} & \begin{vmatrix} -3 & 1 \\ -8 & 1 \end{vmatrix} & \begin{vmatrix} -3 & 2 \\ -8 & 6 \end{vmatrix} \end{bmatrix}$$

$$\text{Rank } A = 1 + \text{Rank} \begin{bmatrix} -41 & 7 & -2 \\ 31 & 5 & -2 \end{bmatrix}$$

This reduced matrix is now of order 2×3 and is a non-zero matrix. Therefore,

$$\text{Rank } A = 1 + 1 + \text{Rank} \begin{bmatrix} \begin{vmatrix} -41 & 7 \\ 31 & 5 \end{vmatrix} & \begin{vmatrix} -41 & -2 \\ 31 & -2 \end{vmatrix} \end{bmatrix} = 2 + \text{Rank} [12 \quad 144]$$

The last matrix is of order 1×2 and is a non-zero matrix. Hence,

$\text{Rank } A = 2 + 1 = 3$. An echelon matrix equivalent to A is

$$\begin{bmatrix} -3 & 5 & 1 & 2 \\ 0 & -41 & 7 & -2 \\ 0 & 0 & 12 & 144 \end{bmatrix}$$

Remark: Here, we have taken the first row of matrix A and subsequently, the first row of each reduced matrix together with zeros to form the echelon matrix equivalent to A.

Example 08: Find the rank of the matrix $A = \begin{bmatrix} 1 & -2 & 1 & 0 & 5 \\ -1 & 0 & 1 & -2 & 2 \\ 1 & -6 & 3 & -2 & 12 \\ 2 & -3 & 0 & 2 & 3 \end{bmatrix}$

Also write an echelon matrix row equivalent to A.

Solution: Since A is a non-zero matrix of order 4×5 , hence its rank must be at least one.

$$\text{Rank } A = 1 + \text{Rank} \begin{bmatrix} \begin{vmatrix} 1 & -2 \\ -1 & 0 \end{vmatrix} & \begin{vmatrix} 1 & 1 \\ -1 & 1 \end{vmatrix} & \begin{vmatrix} 1 & 0 \\ -1 & -2 \end{vmatrix} & \begin{vmatrix} 1 & 5 \\ -1 & 2 \end{vmatrix} \\ \begin{vmatrix} 1 & -2 \\ 1 & -6 \end{vmatrix} & \begin{vmatrix} 1 & 1 \\ 1 & 3 \end{vmatrix} & \begin{vmatrix} 1 & 0 \\ 1 & -2 \end{vmatrix} & \begin{vmatrix} 1 & 5 \\ 1 & 12 \end{vmatrix} \\ \begin{vmatrix} 1 & -2 \\ 2 & -3 \end{vmatrix} & \begin{vmatrix} 1 & 1 \\ 2 & 0 \end{vmatrix} & \begin{vmatrix} 1 & 0 \\ 2 & 2 \end{vmatrix} & \begin{vmatrix} 1 & 5 \\ 2 & 3 \end{vmatrix} \end{bmatrix}$$

$$\text{Rank } A = 1 + \text{Rank} \begin{bmatrix} -2 & 2 & -2 & 7 \\ -4 & 2 & -2 & 7 \\ 1 & -2 & 2 & -7 \end{bmatrix}$$

This reduced non – zero matrix is of order 3×4 . Hence,

$$\text{Rank } A = 1 + 1 + \text{Rank} \begin{bmatrix} \begin{bmatrix} -2 & 2 \\ -4 & 2 \end{bmatrix} & \begin{bmatrix} -2 & -2 \\ -4 & -2 \end{bmatrix} & \begin{bmatrix} -2 & 7 \\ -4 & 7 \end{bmatrix} \\ \begin{bmatrix} -2 & 2 \\ 1 & -2 \end{bmatrix} & \begin{bmatrix} -2 & -2 \\ 1 & 2 \end{bmatrix} & \begin{bmatrix} -2 & 7 \\ 1 & -7 \end{bmatrix} \end{bmatrix}$$

$$\text{Rank } A = 2 + \text{Rank} \begin{bmatrix} 4 & -4 & 14 \\ 2 & -2 & 7 \end{bmatrix}. \text{ This reduced non – zero matrix is of order } 2 \times 3.$$

$$\text{Thus Rank } A = 3 + \text{Rank} \begin{bmatrix} \begin{bmatrix} 4 & -4 \\ 2 & -2 \end{bmatrix} & \begin{bmatrix} 4 & 14 \\ 2 & 7 \end{bmatrix} \end{bmatrix}. \text{ Rank } A = 3 + \text{Rank} \begin{bmatrix} 0 & 0 \end{bmatrix} = 3 + 0 = 3$$

$$\text{An echelon matrix equivalent to } A = \begin{bmatrix} 1 & -2 & 1 & 0 & 5 \\ 0 & -2 & 2 & -2 & 7 \\ 0 & 0 & 4 & -4 & 14 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

Inverse of Matrix by Row Operation Method

In the above sections we have discussed what an identity matrix is. In this section we shall learn how to compute the inverse of square matrix by using EROs.

We state the theorem that helps to find the inverse of square matrix. The proof may be found in any text book on Linear Algebra. It may be noted that

- (i) Only inverse of a square matrix exists.
- (ii) It is not necessary that inverse of every square matrix exists.

Theorem: If a square matrix A is reduced to an identity matrix I by a sequence of elementary row operations, the same operation applied on I will produce the inverse of matrix A .

Let us see how the matrix inverse is found by using above theorem.

Example 09: Find the inverse of the following matrices by using elementary row operations.

$$\begin{aligned} \text{(i)} \begin{bmatrix} 1 & 0 & 3 \\ 2 & 4 & 1 \\ 1 & 3 & 0 \end{bmatrix} & \text{(ii)} \begin{bmatrix} -1 & 2 & -3 \\ 2 & 1 & 0 \\ 4 & -2 & 5 \end{bmatrix} & \text{(iii)} \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 3 & 7 \end{bmatrix} & \text{(iv)} \begin{bmatrix} 2 & 1 & -1 \\ 0 & 2 & 1 \\ 5 & 2 & -3 \end{bmatrix} \end{aligned}$$

Solution:

$$\begin{aligned} & \begin{array}{cc} \text{Given matrix} & \text{Identity matrix } (I_3) \end{array} \\ \text{(i)} & \begin{bmatrix} 1 & 0 & 3 \\ 2 & 4 & 1 \\ 1 & 3 & 0 \end{bmatrix} \quad \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{array}{l} R_2 + (-2)R_1 \\ R_3 + (-1)R_1 \end{array} \\ & = \begin{bmatrix} 1 & 0 & 3 \\ 0 & 4 & -5 \\ 0 & 3 & -3 \end{bmatrix} \quad \begin{bmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ -1 & 0 & 1 \end{bmatrix} \begin{array}{l} R_2 + (-1)R_3 \left(-\frac{1}{3}\right)R_3 \\ (-1/3)R_3 \end{array} \end{aligned}$$

$$\begin{aligned}
&= \begin{bmatrix} 1 & 0 & 3 \\ 0 & 1 & -2 \\ 0 & -1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ -1 & 1 & -1 \\ 1/3 & 0 & -1/3 \end{bmatrix} \begin{matrix} \\ (-1)R_3 \\ \end{matrix} \\
&= \begin{bmatrix} 1 & 0 & 3 \\ 0 & 1 & -2 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ -1 & 1 & -1 \\ 2/3 & -1 & 4/3 \end{bmatrix} \begin{matrix} R_1 + (-3)R_3 \\ R_2 + 2R_3 \\ \end{matrix} \\
&= \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} -1 & 3 & -4 \\ 1/3 & -1 & 5/3 \\ 2/3 & -1 & 4/3 \end{bmatrix}
\end{aligned}$$

Hence the inverse of the given matrix is $A^{-1} = \begin{bmatrix} -1 & 3 & -4 \\ 1/3 & -1 & 5/3 \\ 2/3 & -1 & 4/3 \end{bmatrix}$.

(ii) Given matrix Identity matrix (I_3)

$$\begin{aligned}
&\begin{bmatrix} -1 & 2 & -3 \\ 2 & 1 & 0 \\ 4 & -2 & 5 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{matrix} \\ R_2 + 2R_1 \\ R_3 + 4R_1 \end{matrix} \\
&= \begin{bmatrix} -1 & 2 & -3 \\ 0 & 5 & -6 \\ 0 & 6 & -7 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 4 & 0 & 1 \end{bmatrix} \begin{matrix} (-1)R_1 \\ \\ R_3 + (-1)R_2 \end{matrix} \\
&= \begin{bmatrix} 1 & -2 & 3 \\ 0 & 5 & -6 \\ 0 & 1 & -1 \end{bmatrix} \begin{bmatrix} -1 & 0 & 0 \\ 2 & 1 & 0 \\ 2 & -1 & 1 \end{bmatrix} \begin{matrix} R_{23} \\ \\ \end{matrix} \\
&= \begin{bmatrix} 1 & -2 & 3 \\ 0 & 1 & -1 \\ 0 & 5 & -6 \end{bmatrix} \begin{bmatrix} -1 & 0 & 0 \\ 2 & -1 & 1 \\ 2 & 1 & 0 \end{bmatrix} \begin{matrix} R_1 + 2R_2 \\ \\ R_3 + (-5)R_2 \end{matrix} \\
&= \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & -1 \\ 0 & 0 & -1 \end{bmatrix} \begin{bmatrix} 3 & -2 & 2 \\ 2 & -1 & 1 \\ -8 & 6 & -5 \end{bmatrix} \begin{matrix} R_1 + R_3 \\ R_2 + (-1)R_3 \\ (-1)R_3 \end{matrix} \\
&= \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} -5 & 4 & -3 \\ 10 & -7 & 6 \\ 8 & -6 & 5 \end{bmatrix}. \text{ Thus, } A^{-1} = \begin{bmatrix} -5 & 4 & -3 \\ 10 & -7 & 6 \\ 8 & -6 & 5 \end{bmatrix}
\end{aligned}$$

Students are advised to verify that $A A^{-1} = A^{-1} A = I$.

(iii) Given matrix Identity matrix

$$\begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 3 & 7 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix} \begin{matrix} \\ \\ R_5 + (-3)R_4 \\ \end{matrix}$$

$$\begin{aligned}
 &= \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix} \quad \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -3 & 1 \end{bmatrix} R_4 + (-2)R_5 \\
 &= \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix} \quad \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 7 & -2 \\ 0 & 0 & 0 & -3 & 1 \end{bmatrix}
 \end{aligned}$$

Hence the inverse of the given matrix is $A^{-1} = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 7 & -2 \\ 0 & 0 & 0 & -3 & 1 \end{bmatrix}.$

(iv)

Given Matrix

Identity Matrix

$$\begin{aligned}
 &\begin{bmatrix} 2 & 1 & -1 \\ 0 & 2 & 1 \\ 5 & 2 & -3 \end{bmatrix} \quad \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} R_3 + (-2)R_1 \\
 &= \begin{bmatrix} 2 & 1 & -1 \\ 0 & 2 & 1 \\ 1 & 0 & -1 \end{bmatrix} \quad \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -2 & 0 & 1 \end{bmatrix} R_{13} \\
 &= \begin{bmatrix} 1 & 0 & -1 \\ 0 & 2 & 1 \\ 2 & 1 & -1 \end{bmatrix} \quad \begin{bmatrix} -2 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix} R_3 + (-2)R_1 \\
 &= \begin{bmatrix} 1 & 0 & -1 \\ 0 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix} \quad \begin{bmatrix} -2 & 0 & 1 \\ -5 & 1 & 2 \\ 5 & 0 & -2 \end{bmatrix} R_3 + (-1)R_2 \\
 &= \begin{bmatrix} 1 & 0 & -1 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad \begin{bmatrix} -2 & 0 & 1 \\ -5 & 1 & 2 \\ 10 & -1 & -4 \end{bmatrix} R_2 + R_3 \\
 &= \begin{bmatrix} 1 & 0 & -1 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad \begin{bmatrix} 8 & -1 & -3 \\ -5 & 1 & 2 \\ 10 & -1 & -4 \end{bmatrix}
 \end{aligned}$$

Hence the inverse of the given matrix is $\begin{bmatrix} 8 & -1 & -3 \\ -5 & 1 & 2 \\ 10 & -1 & -4 \end{bmatrix}.$

WORKSHEET 01

1. If $A = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$ find a and b such that $(aI + bA)^2 = A$

2. If $A = \begin{bmatrix} 2 \\ 0 \end{bmatrix}$, $B = \begin{bmatrix} -1 & 1 & 1 \end{bmatrix}$, and $C = \begin{bmatrix} 3 & 0 \\ -2 & 1 \\ 0 & 1 \end{bmatrix}$, show that $(ABC)^t = C^t B^t A^t$.

3. Write the following matrix as sum of symmetric and skew-symmetric matrices.

$$A = \begin{bmatrix} 4 & 5 & 2 \\ -2 & 7 & 2 \\ 1 & 1 & 8 \end{bmatrix}$$

4. Find the inverse of matrix $A = \begin{bmatrix} 3 & -3 & 4 \\ 2 & -3 & 4 \\ 0 & -1 & 1 \end{bmatrix}$ and then show that $A^3 = A^{-1}$.

5. Show that the matrices $A = \frac{1}{3} \begin{bmatrix} 2 & 2 & 1 \\ 2 & -1 & 2 \\ -1 & 2 & 2 \end{bmatrix}$ and $B = \frac{1}{3} \begin{bmatrix} 1 & -2 & 2 \\ -2 & 1 & 2 \\ -2 & -2 & -1 \end{bmatrix}$ are orthogonal matrices.

6. Show that matrix $A = \begin{bmatrix} \cos a & 0 & \sin a \\ 0 & 1 & 0 \\ -\sin a & 0 & \cos a \end{bmatrix}$ is orthogonal matrix.

7. Find the values of unknowns in each of the following matrices.

(i) $\begin{bmatrix} x-3 & y+7 \\ 4 & z-1 \end{bmatrix} = \begin{bmatrix} -1 & 2 \\ 4 & -3 \end{bmatrix}$ (ii) $\begin{bmatrix} x-4 & y+1 \\ 5 & -2 \end{bmatrix} = \begin{bmatrix} 5 & -3 \\ 5 & z \end{bmatrix}$ (iii) $\begin{bmatrix} 2 & 3 \\ 0 & a \end{bmatrix} = \begin{bmatrix} r+2 & s-10 \\ 0 & 12 \end{bmatrix}$

8. Let $A = \begin{bmatrix} 1 & 3 & 0 \\ -1 & 2 & 1 \\ 0 & 0 & 2 \end{bmatrix}$ and $B = \begin{bmatrix} 2 & 3 & 4 \\ 1 & 2 & 3 \\ -1 & 1 & 2 \end{bmatrix}$. Show that $AB \neq BA$.

9. If $A = \begin{bmatrix} 1 & 2 & 2 \\ 2 & 1 & 2 \\ 2 & 2 & 1 \end{bmatrix}$, show that $A^2 - 4A - 5I = 0$, where I and O are the unit matrix and null

matrix of order 3 respectively.

10. Let $A = \begin{bmatrix} 1 & 2 & -1 \\ 3 & 0 & 2 \\ 4 & 5 & 0 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 0 & 1 & 3 \end{bmatrix}$. Verify that $(AB)^t = A^t B^t$.

11. Prove that the matrix $\begin{bmatrix} 1 & 1+i & 2 \\ 1-i & 3 & i \\ 2 & -i & 0 \end{bmatrix}$ is Hermitian.

12. Show that $\begin{bmatrix} -i & 3+2i & -2+i \\ -3+2i & 0 & 3-4i \\ 2-i & -3-4i & -2i \end{bmatrix}$ skew – Hermitian.

13. Prove that the matrix $A = \frac{1}{\sqrt{3}} \begin{bmatrix} 1 & 1+i \\ 1-i & -1 \end{bmatrix}$ is unitary.

14. Show that the matrices $\begin{bmatrix} -1 & -1 & -1 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ and $\begin{bmatrix} 4 & 3 & 3 \\ -1 & 0 & -1 \\ -4 & 4 & -3 \end{bmatrix}$ are involutory.

15. Prove that the matrices $\begin{bmatrix} -1 & -1 & -1 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ and $\begin{bmatrix} 2 & -3 & -5 \\ -1 & 4 & 5 \\ 1 & -3 & -4 \end{bmatrix}$ are idempotent.

16. Show that the matrix $A = \begin{bmatrix} 1 & 1 & 3 \\ 5 & 2 & 6 \\ -2 & -1 & -3 \end{bmatrix}$ is nilpotent of index 3.

17. Express $\begin{bmatrix} i & 2-3i & 4+5i \\ 6+i & 0 & 4-5i \\ -i & 2-i & 2+i \end{bmatrix}$ as the sum of a Hermitian and a skew – Hermitian

matrices. Also express $B = \begin{bmatrix} 2 & 3 & 4 \\ 1 & 2 & 3 \\ -1 & 1 & 2 \end{bmatrix}$ as sum of symmetric and skew-symmetric matrices.

18. Reduce each of the following matrices into the indicated form. Also find their ranks.

(i) $\begin{bmatrix} 1 & 2 & 1 & -1 & 1 \\ 1 & 2 & -1 & 1 & 5 \\ -1 & 1 & 2 & 0 & 0 \end{bmatrix}$ echelon form (ii) $\begin{bmatrix} 1 & 1 & 2 & 9 \\ 2 & 4 & -3 & 1 \\ 3 & 6 & -5 & 0 \end{bmatrix}$ reduced echelon form

(iii) $\begin{bmatrix} 1 & 1 & 2 & 8 \\ -1 & -2 & 3 & 1 \\ 3 & -7 & 4 & 10 \end{bmatrix}$ reduced echelon form (iv) $\begin{bmatrix} 4 & -8 & 3 & 16 \\ -1 & 2 & -5 & -21 \\ 3 & -6 & 1 & 7 \end{bmatrix}$ echelon form

19. Find the inverse of the following matrices using elementary row operations:

$$\begin{bmatrix} 3 & -3 & 4 \\ 2 & -3 & 4 \\ 0 & -1 & 1 \end{bmatrix}, \begin{bmatrix} 1 & 3 & 3 \\ 1 & 4 & 3 \\ 1 & 3 & 4 \end{bmatrix}, \begin{bmatrix} 7 & 6 & 2 \\ -1 & 2 & 4 \\ 3 & 6 & 8 \end{bmatrix}, \begin{bmatrix} 2 & -6 & -2 & -3 \\ 5 & -13 & -4 & -7 \\ -1 & 4 & 1 & 2 \\ 0 & 1 & 0 & 1 \end{bmatrix}$$

CHAPTER Two

SYSTEM OF LINEAR EQUATIONS

INTRODUCTION

Before we discuss the system of linear equations and their solutions, let us consider some of the problems that will help us to understand where such systems exist and why is it necessary to study them.

Problem Statement 1: Consider a situation where two groups A and B containing x and y persons respectively ask over each other the following:

People of 'Group A' asks to 'Group B' that if 2 persons from you join us we will be double to you.

People of 'Group B' asks to 'Group A' that if 2 persons from you join us we will be equal to you. How many people are in each group?

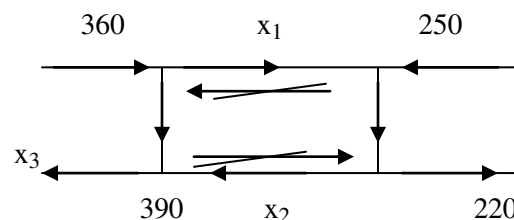
Problem Statement 2: The housing construction company plans to undertake four housing projects and lists material requirements for each house in each of the project as follows:

	Project1	Project2	Project3	Project4
Paint (in 100 gallons)	1	2	1	1.5
Wood (in 10,000 cu.ft.)	3	4	2.5	2.5
Bricks (in millions)	1	2	1.5	1
Labour (in 1000 hours)	10	10	9	8

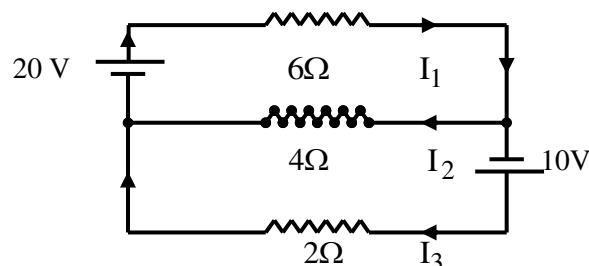
If the supplier delivers 6800 gallons of paint, 1,420,000 cubic feet of wood, 67 million bricks and 4, 48, 000 hours of labor, find the number of houses built for each project.

Problem Statement 3: A soap manufacturer decides to spend 600, 000 rupees on radio, magazine and TV advertising. If he spends as much on TV advertising as on magazines and radio together, and the amount spent on magazines and TV combined equals five times that spent on radio, what is the amount to be spent on each type of advertising?

Problem Statement 4: In a certain part of the city two sets of one way streets intersect as shown in the following figure. The average hourly volume of traffic entering and leaving this part during such hours is given in the figure. Determine the amount of traffic between each of the four junctions.



Problem Statement 5: Find the unknown currents in each branch of the circuit shown in the following figure.



All these examples give rise to a set of linear equations known as '*System of Linear Equations*'. If we try to find out the solution of all these problems we may not be able to find them by using an ordinary arithmetic.

In the first problem statement we get two equations in two variables. This has been shown at the beginning of first chapter. There are many situations where the number of equations is large. Moreover, the number of equations may not be equal to number of variables/unknowns. In addition to this, some systems are homogeneous where the right sides of these equations are zero. In all these situations we must have a universal way to find the solution of the system of linear equations particularly when the number of equations is greater than three.

Before we present these methods let us generalize the situation. The system of m linear equations in n variables/unknowns is given by

$$\begin{aligned} a_{11} x_1 + a_{12} x_2 + \dots + a_{1n} x_n &= b_1 \\ a_{21} x_1 + a_{22} x_2 + \dots + a_{2n} x_n &= b_2 \\ \dots & \\ \dots & \\ a_{m1} x_1 + a_{m2} x_2 + \dots + a_{mn} x_n &= b_m \end{aligned} \quad (1)$$

Here a_{ij} and b_j for $i = 1, 2, \dots, m$ and $j = 1, 2, \dots, n$ are constants. The numbers a_{ij} are called the *system coefficients*. The system (1) is known as *Non-homogeneous System of Linear Equations*. If each b_j in (1) is zero, the system becomes

[illegible]

This is known as *Homogeneous System of Linear Equations*.

A sequence of n numbers $s_1, s_2, s_3, \dots, s_n$, for which (1) is satisfied when we substitute $x_1 = s_1, x_2 = s_2, \dots, x_n = s_n$, is called a *solution* of (1). The set of all such solutions is called *solution set* or the *general solution* of (1).

A system of equations that has no solution is said to be *inconsistent*. If there is at least one solution of the system, the system is called *consistent system*.

The system (2) always possesses one solution, namely the *zero solution* that is $x_1 = 0, x_2 = 0, \dots, x_n = 0$. The zero solution is also known as *trivial solution*. Any other solution, if it exists, is called a *non-zero* or *non-trivial solution*. In coming sections we shall learn under what conditions the

solution of systems (1) and (2) exist. The system of equations (1) and (2) can be put in matrix form as:

$$\begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \dots & \dots & \dots & \dots \\ a_{m1} & a_{m2} & \dots & a_{mn} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_n \end{bmatrix} \quad (1a)$$

$$\begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \dots & \dots & \dots & \dots \\ a_{m1} & a_{m2} & \dots & a_{mn} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \end{bmatrix} \quad (1b)$$

Remarks: (i) In place of variables $x_1, x_2, \dots, x_n$ if we use other variables say $u_1, u_2, \dots, u_n$ there will be no change in the solution. These variables are dummy variables and are immaterial. However, if coefficients a_{ij} and/or b_i are changed the solution of the system will be different. (ii) If $m = n$ the system is called square system.

METHODS OF SOLVING SYSTEM OF LINEAR EQUATIONS

In this section we shall discuss the two important methods of solving non-homogeneous equations $A\mathbf{x} = \mathbf{b}$ and homogeneous system of equations $A\mathbf{x} = \mathbf{0}$. They are

- (i) Gauss Elimination Method
- (ii) Gauss-Jordan Method.

Gauss Elimination Method

Gauss elimination method is one of the several methods employed to solve a system of m linear equations in n variables. The name Gauss was given to it after its inventor, the famous German mathematician Carl Friedrich Gauss (1777-1855).

Let us consider the linear system (1). In matrix form it can be expressed as $A\mathbf{x} = \mathbf{b}$, where A is the matrix of coefficients of order $m \times n$, $\mathbf{x}$ and $\mathbf{b}$ are column vectors of order $n \times 1$ and $m \times 1$ respectively. The following four steps are applied to solve a system using Gauss's elimination method.

Step1: Change the system of linear equations to form $A\mathbf{x} = \mathbf{b}$.

Step2: Form the augmented coefficient matrix A_b by including the elements of $\mathbf{b}$ as an extra column in the matrix A .

Step3: Convert the augmented matrix into echelon form by using elementary row operations.

Step4: Convert the echelon matrix into matrix form and find $\mathbf{x}$ by using "Backward Substitution".

Remarks: A system $A\mathbf{x} = \mathbf{b}$ is called *over-determined* if it has more equations than unknowns i.e. $m > n$. It is called *determined* if the number of equations is equal to the number of unknowns, i.e. $m = n$ and *underdetermined* if $m < n$.

Example 01: Use Gauss's elimination method to solve the system of linear equations ($m = n$).

$$x_1 + 5x_2 + 2x_3 = 9, \quad x_1 + x_2 + 7x_3 = 6, \quad -3x_2 + 4x_3 = -2$$

Solution: Step1: We change the system of linear equations in matrix form $A\mathbf{x} = \mathbf{b}$.

$$\begin{bmatrix} 1 & 5 & 2 \\ 1 & 1 & 7 \\ 0 & -3 & 4 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 9 \\ 6 \\ -2 \end{bmatrix}$$

Step2: The augmented coefficient matrix A_b is:

$$A_b = \begin{bmatrix} 1 & 5 & 2 & 9 \\ 1 & 1 & 7 & 6 \\ 0 & -3 & 4 & -2 \end{bmatrix}$$

Step3: Now convert this augmented coefficient matrix into an echelon form using elementary row operations as follows:

$$\begin{aligned} A_b &= \begin{bmatrix} 1 & 5 & 2 & 9 \\ 1 & 1 & 7 & 6 \\ 0 & -3 & 4 & -2 \end{bmatrix} \xrightarrow{R_2 + (-1)R_1} \begin{bmatrix} 1 & 5 & 2 & 9 \\ 0 & -4 & 5 & -3 \\ 0 & -3 & 4 & -2 \end{bmatrix} \xrightarrow{R_3 + (-1)R_2} \\ &= \begin{bmatrix} 1 & 5 & 2 & 9 \\ 0 & -4 & 5 & -3 \\ 0 & 1 & -1 & 1 \end{bmatrix} \xrightarrow{R_2 + 4R_3} \begin{bmatrix} 1 & 5 & 2 & 9 \\ 0 & 0 & 1 & 1 \\ 0 & 1 & -1 & 1 \end{bmatrix} \xrightarrow{R_{23}} \begin{bmatrix} 1 & 5 & 2 & 9 \\ 0 & 1 & -1 & 1 \\ 0 & 0 & 1 & 1 \end{bmatrix} \end{aligned}$$

Step4: Convert the echelon matrix into equation form

$$\begin{bmatrix} 1 & 5 & 2 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 9 \\ 1 \\ 1 \end{bmatrix}$$

This is equivalent to: $x_1 + 5x_2 + 2x_3 = 9$ (i)

$x_2 - x_3 = 1$ (ii)

$x_3 = 1$ (iii)

From (iii), we get $x_3 = 1$. Substituting $x_3 = 1$ into (ii), we get, $x_2 = 2$. Now putting $x_3 = 1$ and $x_2 = 2$ in (i), we get, $x_1 = -3$. Thus, required solution of given system of equations is

$x_1 = -3$, $x_2 = 2$, $x_3 = 1$ or $\mathbf{x} = [-3, 2, 1]^t$

Gauss-Jordan Method

The Gauss's method is based upon transferring the augmented matrix into an *echelon* form where backward substitution takes more operations (addition and multiplications). In Gauss-Jordan's method we reduce the augmented matrix into *Reduced Echelon* form where we do not need the process of backward substitution. This saves a lot of manual or computing time. Let us see how this method works.

Step1: Change the system of linear equations to the form $A\mathbf{x} = \mathbf{b}$.

Step2: Form the augmented coefficient matrix A_b by including the column vector $\mathbf{b}$.

Step3: Convert the augmented matrix into *Reduced Echelon* form by using elementary row operations.

Step4: Convert the reduced echelon matrix into system of equations and find $\mathbf{x}$ directly without using backward substitution.

Example 02: Use Gauss-Jordan method to solve the system of linear equations.

$$x_1 + 5x_2 + 2x_3 = 9, \quad x_1 + x_2 + 7x_3 = 6, \quad -3x_2 + 4x_3 = -2$$

Solution: **Step1:** Change the system of linear equations in matrix form $A\mathbf{x} = \mathbf{b}$

$$\begin{bmatrix} 1 & 5 & 2 \\ 1 & 1 & 7 \\ 0 & -3 & 4 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 9 \\ 6 \\ -2 \end{bmatrix}$$

Step2: The augmented matrix is

$$A_b = \begin{bmatrix} 1 & 5 & 2 & 9 \\ 1 & 1 & 7 & 6 \\ 0 & -3 & 4 & -2 \end{bmatrix}$$

Step3: Change the augmented coefficient matrix into *reduced echelon* form by using elementary row operations.

$$\begin{aligned} A_b &= \begin{bmatrix} 1 & 5 & 2 & 9 \\ 1 & 1 & 7 & 6 \\ 0 & -3 & 4 & -2 \end{bmatrix} \xrightarrow{R_2 + (-1)R_1} \begin{bmatrix} 1 & 5 & 2 & 9 \\ 0 & -4 & 5 & -3 \\ 0 & -3 & 4 & -2 \end{bmatrix} \xrightarrow{R_3 + (-1)R_2} \\ &= \begin{bmatrix} 1 & 5 & 2 & 9 \\ 0 & -4 & 5 & -3 \\ 0 & 1 & -1 & 1 \end{bmatrix} \xrightarrow{R_2 + 4R_3} \begin{bmatrix} 1 & 5 & 2 & 9 \\ 0 & 0 & 1 & 1 \\ 0 & 1 & -1 & 1 \end{bmatrix} \xrightarrow{R_{23}} \\ &= \begin{bmatrix} 1 & 5 & 2 & 9 \\ 0 & 1 & -1 & 1 \\ 0 & 0 & 1 & 1 \end{bmatrix} \xrightarrow{R_1 + (-5)R_2} \begin{bmatrix} 1 & 0 & 7 & 4 \\ 0 & 1 & -1 & 1 \\ 0 & 0 & 1 & 1 \end{bmatrix} \xrightarrow{\begin{matrix} R_1 + (-7)R_3 \\ R_2 + R_3 \end{matrix}} \\ &= \begin{bmatrix} 1 & 0 & 0 & -3 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & 1 & 1 \end{bmatrix} \end{aligned}$$

Step4: Writing this matrix into equation form, we have:

$$x_1 + 0 \cdot x_2 + 0 \cdot x_3 = -3 \quad (i)$$

$$0 \cdot x_1 + x_2 + 0 \cdot x_3 = 2 \quad (ii)$$

$$0 \cdot x_1 + 0 \cdot x_2 + x_3 = 1 \quad (iii)$$

From (i), (ii) and (iii) we have: $x_1 = -3$, $x_2 = 2$, $x_3 = 1$. This is same solution as was obtained in Example 1 above.

Example 03: Use Gauss-Jordan method to solve the system of linear equations

$$x_1 + x_2 + x_3 = a$$

$$x_1 + (1+a)x_2 + x_3 = 2a$$

$$x_1 + x_2 + (1+a)x_3 = 3a$$

Solution: Step1: Change the system of linear equations in matrix form $Ax = b$

$$\begin{bmatrix} 1 & 1 & 1 \\ 1 & 1+a & 1 \\ 1 & 1 & 1+a \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} a \\ 2a \\ 3a \end{bmatrix}$$

Step2: The augmented coefficient matrix A_b is

$$A_b = \begin{bmatrix} 1 & 1 & 1 & a \\ 1 & 1+a & 1 & 2a \\ 1 & 1 & 1+a & 3a \end{bmatrix}$$

Step3: Change the augmented coefficient matrix into reduced echelon form using

elementary row operations: $A_b = \begin{bmatrix} 1 & 1 & 1 & a \\ 1 & 1+a & 1 & 2a \\ 1 & 1 & 1+a & 3a \end{bmatrix} \xrightarrow{\begin{matrix} R_2 + (-1)R_1 \\ R_3 + (-1)R_1 \end{matrix}}$

$$\begin{aligned}
&\approx \begin{bmatrix} 1 & 1 & 1 & a \\ 0 & a & 0 & a \\ 0 & 0 & a & 2a \end{bmatrix} \left(\frac{1}{a} \right) R_1, \left(\frac{1}{a} \right) R_3 \approx \begin{bmatrix} 1 & 1 & 1 & a \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 2 \end{bmatrix} R_1 + (-1)R_2 \\
&\approx \begin{bmatrix} 1 & 0 & 1 & a-1 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 2 \end{bmatrix} R_1 + (-1)R_3 \approx \begin{bmatrix} 1 & 0 & 0 & a-3 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 2 \end{bmatrix}
\end{aligned}$$

Step4: Writing the above matrix in equation form, we have:

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} a-3 \\ 1 \\ 2 \end{bmatrix}.$$

This gives

$$x_1 + 0 \cdot x_2 + 0 \cdot x_3 = a-3 \quad (i)$$

$$0 \cdot x_1 + x_2 + 0 \cdot x_3 = 1 \quad (ii)$$

$$0 \cdot x_1 + 0 \cdot x_2 + x_3 = 2 \quad (iii)$$

From (i), (ii) and (iii) we get the solution of the given system of linear equations:

$$x_1 = a-3, x_2 = 1, x_3 = 2 \text{ or } \mathbf{x} = [a-3, 1, 2]^t$$

Homogeneous System of Linear Equations

Consider a system of m homogeneous equations in n unknowns $x_1, x_2, \dots, x_n$.

$$\left. \begin{aligned} a_{11}x_1 + a_{12}x_2 + \dots + a_{1n}x_n &= 0 \\ a_{21}x_1 + a_{22}x_2 + \dots + a_{2n}x_n &= 0 \\ \dots &\dots \\ a_{m1}x_1 + a_{m2}x_2 + \dots + a_{mn}x_n &= 0 \end{aligned} \right\}$$

In matrix form the above system of equations may be expressed as:

$$\mathbf{A} \mathbf{x} = \mathbf{O} \quad (1)$$

$$\text{where, } \mathbf{A} = \begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \dots & \dots & \dots & \dots \\ a_{m1} & a_{m2} & \dots & a_{mn} \end{bmatrix}, \mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ \dots \\ x_n \end{bmatrix} \text{ and } \mathbf{O} = \begin{bmatrix} 0 \\ 0 \\ \dots \\ 0 \end{bmatrix}.$$

You may observe that $x_1 = x_2 = \dots = x_n = 0$ form a solution of (1). This is the *trivial/zero solution*. All other solutions (if they exist) are known as *non-trivial solutions* of (1).

Example 04: Solve the following system of homogeneous equations ($m = n$)

$$x + y + z = 0, \quad 4x + 5y + 2z = 0, \quad 2x + 3y + 3z = 0$$

Solution: Change the above system of equations to the matrix form $\mathbf{Ax} = \mathbf{O}$.

$$\text{Or } \begin{bmatrix} 1 & 1 & 1 \\ 4 & 5 & 2 \\ 2 & 3 & 3 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}. \text{ Now consider,}$$

$$\mathbf{A} = \begin{bmatrix} 1 & 1 & 1 \\ 4 & 5 & 2 \\ 2 & 3 & 3 \end{bmatrix} R_2 + (-4)R_1 = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & -2 \\ 0 & 1 & 1 \end{bmatrix} R_3 + (-1)R_2 = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & -2 \\ 0 & 0 & 3 \end{bmatrix}$$

This last matrix implies that

$$\begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & -2 \\ 0 & 0 & 3 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}.$$

$$x + y + z = 0 \quad (i)$$

This gives

$$y - 2z = 0 \quad (ii)$$

$$3z = 0 \quad (iii)$$

From equation (iii) we get $z = 0$. Putting $z = 0$ in (ii) we get $y = 0$ and then from (i), we have $x = 0$. This shows that given system has only zero/trivial solution.

Example 05: Solve the following system of homogeneous equations ($m = n$)

$$x + y + z = 0$$

$$4x + 5y + 2z = 0$$

$$2x + 3y = 0$$

Solution: We change the above system of equations to the matrix form $Ax = O$.

Or

$$\begin{bmatrix} 1 & 1 & 1 \\ 4 & 5 & 2 \\ 2 & 3 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}.$$

Now consider, $A = \begin{bmatrix} 1 & 1 & 1 \\ 4 & 5 & 2 \\ 2 & 3 & 0 \end{bmatrix} \begin{matrix} R_2 + (-4)R_1 \\ R_3 + (-2)R_1 \end{matrix} = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & -2 \\ 0 & 1 & -2 \end{bmatrix} \begin{matrix} \\ R_3 + (-1)R_2 \end{matrix} = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & -2 \\ 0 & 0 & 0 \end{bmatrix}$

This last matrix implies that

$$\begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & -2 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$x + y + z = 0 \quad (i)$$

This gives

$$y - 2z = 0 \quad (ii)$$

From equation (ii) we get $y = 2z$. Putting $y = 2z$ in (i) we get, $x + 2z + z = 0$. Now, let $z = k$, then $x = -3k$, $y = 2k$, and $z = k$. Here k is known as a “free parameter” or “an arbitrary constant”; so it can hold any value. Thus for different values of k , there are *infinite solutions* of the above homogeneous system of equations. For example,

If $k = 1$, then $x = -3$, $y = 2$ and $z = 1$.

If $k = 2$, then $x = -6$, $y = 4$ and $z = 2$, etc.

Example 06: Solve the following linear homogeneous equations ($m < n$)

$$2w + 3x - y - z = 0$$

$$4w - 6x - 2y + 2z = 0$$

$$-6w + 12x + 3y - 4z = 0$$

Solution: The matrix form of the given system is:

$$\begin{bmatrix} 2 & 3 & -1 & -1 \\ 4 & -6 & -2 & 2 \\ -6 & 12 & 3 & -4 \end{bmatrix} \begin{bmatrix} w \\ x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}.$$

We use the Gauss-Jordan method and have

$$A = \begin{bmatrix} 2 & 3 & -1 & -1 \\ 4 & -6 & -2 & 2 \\ -6 & 12 & 3 & -4 \end{bmatrix} \begin{matrix} R_2 + (-2)R_1 \\ R_3 + 3R_1 \end{matrix} = \begin{bmatrix} 2 & 3 & -1 & -1 \\ 0 & -12 & 0 & 4 \\ 0 & 21 & 0 & -7 \end{bmatrix} \begin{matrix} (1/4)R_2 \\ (-1/7)R_3 \end{matrix}$$

$$= \begin{bmatrix} 2 & 3 & -1 & -1 \\ 0 & -3 & 0 & 1 \\ 0 & -3 & 0 & 1 \end{bmatrix} \begin{matrix} R_3 - R_2 \end{matrix} = \begin{bmatrix} 2 & 3 & -1 & -1 \\ 0 & -3 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

Thus, the system becomes $\begin{bmatrix} 2 & 3 & -1 & -1 \\ 0 & -3 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} w \\ x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$

Writing in equation form, we get

This gives $2w + 3x - y - z = 0$ (i)

$-3x + z = 0$ (ii)

From (ii) we get $x = z/3$. Putting this into (i) we get

$$2w + 3\left(\frac{z}{3}\right) - y - z = 0 \quad \text{or} \quad 2w + z - y - z = 0 \quad \text{or} \quad 2w - y = 0, \text{ thus } y = 2w.$$

Thus, $x = z/3$ and $y = 2w$. Take $z = k_1$ and $w = k_2$, we get $x = k_1/3$ and $y = 2k_2$. (It may be noted that there are two free parameters k_1 and k_2). Hence, $x = k_1/3$, $y = 2k_2$, $z = k_1$ and $w = k_2$ constitute the general solution where k_1 and k_2 are arbitrary constants. Thus, there are infinite solutions of a given system.

CONSISTENCY CRITERIAN

Consider the system of linear equations: $x + 2y = 4$ and $3x + 2y = 2$.

Solving, we get $x = -1$ and $y = 5/2$ (a unique solution).

Now consider the system of equations: $x + 2y = 4$ and $3x + 6y = 12$

We observe that second equation is multiple of the first equation and hence there is one free parameter. From first equation, we have $x = -2y$. If $y = k$ then $x = -2k$. Here k is a free parameter hence there are infinite solutions of this system.

Finally consider the system of equations: $x + 2y = 4$ and $3x + 6y = 5$

Multiply first equation by 3 and subtracting second equation from it, we get, $0 = 7$. This is not possible hence the given system has no solution.

From the above discussion, we observe that a system of linear equations may have

- (i) Unique solution (ii) Infinite solutions (iii) No solution.

Thus, it becomes necessary to discuss that under what conditions the system of linear equations (either non-homogeneous or homogeneous) will possess the solution. This phenomenon is called consistency criterion.

Consistency Criteria

(a) When the system of linear equations $A \mathbf{x} = \mathbf{b}$ is a square system and

- i. The Rank of $A = \text{Rank of } A_b = n$ (where n is number of unknowns or equations) then the system has unique solution.

- ii. The Rank of $A = \text{Rank } A_b < n$ then the system has infinite solutions.
 - iii. The Rank $A \neq \text{Rank } A_b$ the system is said to be inconsistent and hence has no solution.
- (b) When the system of linear equations $A \mathbf{x} = \mathbf{b}$ is rectangular system and
- i. The Rank of $A = \text{Rank of } A_b$, then the system has an infinite number of solutions.
 - ii. The Rank $A \neq \text{Rank } A_b$, then the system is said to be inconsistent and hence has no solution.
- (c) When the homogeneous system $A \mathbf{x} = \mathbf{0}$ is a square system and
- i. The Rank of matrix $A = n$, the system has only a trivial/zero solution.
 - ii. The Rank $A < n$ then the system has infinite many solutions.
- (d) When the homogeneous system $A \mathbf{x} = \mathbf{0}$ is a rectangular system and
- i. The Rank of $A = n$, where n is the number of unknowns or variables, then the system has an infinite solutions.

Example 01: Is the system of equations, $2x + 6y = -11$, $6x + 20y - 6z = -3$ and $6y - 18z = -1$ consistent?

Solution: It may be noted that here $m = n$. The augmented matrix of system of equations is

$$A_b = \begin{bmatrix} 2 & 6 & 0 & -11 \\ 6 & 20 & -6 & -3 \\ 0 & 6 & -18 & -1 \end{bmatrix}$$

Now we reduce the matrix into echelon form using elementary row operations:

$$\begin{aligned} A_b &= \begin{bmatrix} 2 & 6 & 0 & -11 \\ 6 & 20 & -6 & -3 \\ 0 & 6 & -18 & -1 \end{bmatrix} \xrightarrow{R_2 + (-3)R_1} \begin{bmatrix} 2 & 6 & 0 & -11 \\ 0 & 2 & -6 & 30 \\ 0 & 6 & -18 & -1 \end{bmatrix} \xrightarrow{(1/2)R_2} \\ &\approx \begin{bmatrix} 2 & 6 & 0 & -11 \\ 0 & 1 & -3 & 15 \\ 0 & 6 & -18 & -1 \end{bmatrix} \xrightarrow{R_3 + (-6)R_2} \begin{bmatrix} 2 & 6 & 0 & -11 \\ 0 & 1 & -3 & 15 \\ 0 & 0 & 0 & -91 \end{bmatrix} \end{aligned}$$

The last matrix is echelon form of A_b . We observe that $\text{Rank}(A) = 2$ and $\text{Rank}(A_b) = 3$. Since, $\text{Rank}(A) \neq \text{Rank}(A_b)$, therefore, given system of equations is *inconsistent*. Observe that last row of A_b in equation form is $0x + 0y + 0z = -91$, which implies that $0 = -91$ and it is not true. Hence given system has no solution.

Example 02: Discuss the consistency of the following system of equations

$$2x + 3y + 4z = 11, \quad x + 5y + 7z = 15, \quad 3x + 11y + 13z = 25$$

In case the system is consistent, solve it.

Solution: The augmented matrix is $A_b = \begin{bmatrix} 2 & 3 & 4 & 11 \\ 1 & 5 & 7 & 15 \\ 3 & 11 & 13 & 25 \end{bmatrix}$

Reduce the matrix into echelon form using elementary row operations

$$A_b = \begin{bmatrix} 2 & 3 & 4 & 11 \\ 1 & 5 & 7 & 15 \\ 3 & 11 & 13 & 25 \end{bmatrix} \xrightarrow{R_{12}} \begin{bmatrix} 1 & 5 & 7 & 15 \\ 2 & 3 & 4 & 11 \\ 3 & 11 & 13 & 25 \end{bmatrix} \xrightarrow{\begin{matrix} R_2 + (-2)R_1 \\ R_3 + (-3)R_1 \end{matrix}}$$

$$\begin{aligned}
&= \begin{bmatrix} 1 & 5 & 7 & 15 \\ 0 & -7 & -10 & -19 \\ 0 & -4 & -8 & -20 \end{bmatrix} \begin{matrix} (-1)R_2 \\ -(1/4)R_3 \end{matrix} = \begin{bmatrix} 1 & 5 & 7 & 15 \\ 0 & 7 & 10 & 19 \\ 0 & 1 & 2 & 5 \end{bmatrix} R_{23} \\
&= \begin{bmatrix} 1 & 5 & 7 & 15 \\ 0 & 1 & 2 & 5 \\ 0 & 7 & 10 & 19 \end{bmatrix} R_3 + (-7)R_2 = \begin{bmatrix} 1 & 5 & 7 & 15 \\ 0 & 1 & 2 & 5 \\ 0 & 0 & -4 & -16 \end{bmatrix} \left(-\frac{1}{4}\right)R_3 = \begin{bmatrix} 1 & 5 & 7 & 15 \\ 0 & 1 & 2 & 5 \\ 0 & 0 & 1 & 4 \end{bmatrix}
\end{aligned}$$

From the last matrix we can easily observe that $\text{Rank}(A) = \text{Rank}(A_b) = 3$. Therefore, the given system is consistent and has a unique solution. Now to find the solution, we write the reduced system in equation form

$$x + 5y + 7z = 15 \quad (\text{i})$$

$$y + 2z = 5 \quad (\text{ii})$$

$$z = 4 \quad (\text{iii})$$

Solving, we have: $x = 2$, $y = -3$, $z = 4$. This is the solution of the given system.

Example 03: Test for consistency and solve the following system of linear equations

$$5x + 3y + 7z = 4, \quad 3x + 26y + 2z = 9, \quad 7x + 2y + 10z = 5$$

Solution: The augmented for the above system of equations is

$$A_b = \begin{bmatrix} 5 & 3 & 7 & 4 \\ 3 & 26 & 2 & 9 \\ 7 & 2 & 10 & 5 \end{bmatrix}$$

Now we reduce the matrix into echelon form using elementary row operations:

$$\begin{aligned}
A_b &= \begin{bmatrix} 5 & 3 & 7 & 4 \\ 3 & 26 & 2 & 9 \\ 7 & 2 & 10 & 5 \end{bmatrix} R_3 + (-2)R_2 = \begin{bmatrix} 5 & 3 & 7 & 4 \\ 3 & 26 & 2 & 9 \\ 1 & -50 & 6 & -13 \end{bmatrix} R_{13} \\
&\approx \begin{bmatrix} 1 & -50 & 6 & -13 \\ 3 & 26 & 2 & 9 \\ 5 & 3 & 7 & 4 \end{bmatrix} \begin{matrix} R_2 + (-3)R_1 \\ R_3 + (-5)R_1 \end{matrix} = \begin{bmatrix} 1 & -50 & 6 & -13 \\ 0 & 176 & -16 & 48 \\ 0 & 253 & -23 & 69 \end{bmatrix} \begin{matrix} (1/16) R_2 \\ (1/23) R_3 \end{matrix} \\
&\approx \begin{bmatrix} 1 & -50 & 6 & -13 \\ 0 & 11 & -1 & 3 \\ 0 & 11 & -1 & 3 \end{bmatrix} R_3 + (-1)R_2 = \begin{bmatrix} 1 & -50 & 6 & -13 \\ 0 & 11 & -1 & 3 \\ 0 & 0 & 0 & 0 \end{bmatrix}
\end{aligned}$$

From the last matrix we can easily observe that

$$\text{Rank } A = \text{Rank } A_b = 2 < 3 \quad (3 \text{ is number of unknowns})$$

Thus given system is *consistent* but having infinite solutions. From the last matrix we have

$$x - 50y + 6z = -13 \quad (\text{i})$$

$$11y - z = 3 \quad (\text{ii})$$

From (ii) we have $y = (3 + z)/11$. Substituting it into (i), we get

$$x - \frac{50}{11}(3 + z) + 6z = -13 \quad \text{or} \quad x - \frac{50}{11}z - \frac{150}{11} + 6z = -13$$

$$x + \left(\frac{-50 + 66}{11}\right)z = \frac{150}{11} - 13 \quad \text{or} \quad x + \frac{16}{11}z = \frac{7}{11} \quad \text{or} \quad x = \frac{1}{11}(7 - 16z)$$

Let $z = k$ then, $x = (7 - 16k)/11$, $y = (3 + k)/11$. As k is free parameter so given system has infinite solutions.

Example 04: Examine the following system for a non-trivial solution ($m < n$)

$$x_1 - x_2 + 2x_3 + x_4 = 0, \quad 3x_1 + 2x_2 + x_4 = 0, \quad 4x_1 + x_2 + 2x_3 + 2x_4 = 0$$

Solution: The matrix of the coefficients is: $A = \begin{bmatrix} 1 & -1 & 2 & 1 \\ 3 & 2 & 0 & 1 \\ 4 & 1 & 2 & 2 \end{bmatrix}$.

Now we reduce the matrix into echelon form using elementary row operations:

$$\begin{aligned} A &= \begin{bmatrix} 1 & -1 & 2 & 1 \\ 3 & 2 & 0 & 1 \\ 4 & 1 & 2 & 2 \end{bmatrix} \xrightarrow{\substack{R_2 + (-3)R_1 \\ R_3 + (-4)R_1}} \begin{bmatrix} 1 & -1 & 2 & 1 \\ 0 & 5 & -6 & -2 \\ 0 & 5 & -6 & -2 \end{bmatrix} \xrightarrow{R_3 + (-1)R_2} \\ &= \begin{bmatrix} 1 & -1 & 2 & 1 \\ 0 & 5 & -6 & -2 \\ 0 & 0 & 0 & 0 \end{bmatrix} \xrightarrow{\left(\frac{1}{5}\right)R_2} \begin{bmatrix} 1 & -1 & 2 & 1 \\ 0 & 1 & -6/5 & -2/5 \\ 0 & 0 & 0 & 0 \end{bmatrix} \xrightarrow{R_1 + R_2} \begin{bmatrix} 1 & 0 & 4/5 & 3/5 \\ 0 & 1 & -6/5 & -2/5 \\ 0 & 0 & 0 & 0 \end{bmatrix} \end{aligned}$$

The rank of A is 2 which is less than 4 (the number of equations), therefore, the given system of equations is consistent having infinite and non-trivial solutions.

The first two rows of the last matrix give the following relations

$$x_1 + \frac{4}{5}x_3 + \frac{3}{5}x_4 = 0 \quad (i)$$

$$x_2 - \frac{6}{5}x_3 - \frac{2}{5}x_4 = 0 \quad (ii)$$

From (i) and (ii), we have: $x_1 = -\left(\frac{4}{5}x_3 + \frac{3}{5}x_4\right)$ and $x_2 = \frac{6}{5}x_3 + \frac{2}{5}x_4$

Let $x_3 = k_1$ and $x_4 = k_2$ then: $x_1 = -\left(\frac{4}{5}k_1 + \frac{3}{5}k_2\right)$ and $x_2 = \frac{6}{5}k_1 + \frac{2}{5}k_2$.

Hence, we have: $x_1 = -\left(\frac{4}{5}k_1 + \frac{3}{5}k_2\right)$, $x_2 = \frac{6}{5}k_1 + \frac{2}{5}k_2$, $x_3 = k_1$ and $x_4 = k_2$

We can get an infinite number of solutions by giving different values to k_1 and k_2 .

Example 05: Show that the system of equations $2x - y + 3z = a$, $3x + y - 5z = b$, $-5x - 5y + 21z = c$ is consistent if $c = 2a - 3b$.

Solution: Consider, $A_b = \begin{bmatrix} 2 & -1 & 3 & a \\ 3 & 1 & -5 & b \\ -5 & -5 & 21 & c \end{bmatrix}$. Using elementary row operations:

$$\begin{aligned} A_b &= \begin{bmatrix} 2 & -1 & 3 & a \\ 3 & 1 & -5 & b \\ -5 & -5 & 21 & c \end{bmatrix} \xrightarrow{R_2 + (-1)R_1} \begin{bmatrix} 2 & -1 & 3 & a \\ 1 & 2 & -8 & b-a \\ -5 & -5 & 21 & c \end{bmatrix} \xrightarrow{R_{12}} \\ &\approx \begin{bmatrix} 1 & 2 & -8 & b-a \\ 2 & -1 & 3 & a \\ -5 & -5 & 21 & c \end{bmatrix} \xrightarrow{\substack{R_2 + (-2)R_1 \\ R_3 + 5R_1}} \begin{bmatrix} 1 & 2 & -8 & b-a \\ 0 & -5 & 19 & 3a-2b \\ 0 & 5 & -19 & c+5b-5a \end{bmatrix} \xrightarrow{R_3 + R_2} \\ &\approx \begin{bmatrix} 1 & 2 & -8 & b-a \\ 0 & -5 & 19 & 3a-2b \\ 0 & 0 & 0 & c+3b-2a \end{bmatrix} \end{aligned}$$

The last row reveals that : $0x + 0y + 0z = c + 3b - 2a$

We clearly observe that the system is inconsistent if $c + 3b - 2a \neq 0$. Thus given system is consistent if $c + 3b - 2a = 0$ or $c = 2a - 3b$.

Example 06: Find the values of k such that the system of equations

$$x + ky + 3z = 0, \quad 4x + 3y + kz = 0, \quad 2x + y + 2z = 0 \text{ have a non-trivial solution.}$$

Solution: The matrix of coefficients is

$$A = \begin{bmatrix} 1 & k & 3 \\ 4 & 3 & k \\ 2 & 1 & 2 \end{bmatrix}$$

Now we reduce the matrix into echelon form using elementary row operations:

$$\begin{aligned} A &= \begin{bmatrix} 1 & k & 3 \\ 4 & 3 & k \\ 2 & 1 & 2 \end{bmatrix} \begin{matrix} R_2 + (-4)R_1 \\ R_3 + (-2)R_1 \end{matrix} = \begin{bmatrix} 1 & k & 3 \\ 0 & 3-4k & k-12 \\ 0 & 1-2k & -4 \end{bmatrix} \begin{matrix} R_2 + (-2)R_3 \end{matrix} \\ &\approx \begin{bmatrix} 1 & k & 3 \\ 0 & 1 & k-4 \\ 0 & 1-2k & -4 \end{bmatrix} \begin{matrix} R_3 + \{-(1-2k)\}R_2 \end{matrix} \approx \begin{bmatrix} 1 & k & 3 \\ 0 & 1 & k-4 \\ 0 & 0 & 2k^2-9k \end{bmatrix} \end{aligned}$$

For non-trivial solution $2k^2 - 9k = 0$ or $k(2k - 9) = 0$ or $k = 0$ or $k = 9/2$.

Example 07: Find the value of k such that the system of equations $2x + 3y - 2z = 0$, $3x - y + 3z = 0$, $7x + ky - z = 0$ have non-trivial solutions. Also find the solution.

Solution: The matrix of coefficients for the above system is:

$$A = \begin{bmatrix} 2 & 3 & -2 \\ 3 & -1 & 3 \\ 7 & k & -1 \end{bmatrix}$$

Now we reduce the matrix into echelon form using elementary row operations:

$$\begin{aligned} A &= \begin{bmatrix} 2 & 3 & -2 \\ 3 & -1 & 3 \\ 7 & k & -1 \end{bmatrix} \begin{matrix} R_2 + (-1)R_1 \\ R_3 + (-7)R_1 \end{matrix} = \begin{bmatrix} 2 & 3 & -2 \\ 1 & -4 & 5 \\ 7 & k & -1 \end{bmatrix} \begin{matrix} R_{12} \end{matrix} \\ &= \begin{bmatrix} 1 & -4 & 5 \\ 2 & 3 & -2 \\ 7 & k & -1 \end{bmatrix} \begin{matrix} R_2 + (-2)R_1 \\ R_3 + (-7)R_1 \end{matrix} = \begin{bmatrix} 1 & -4 & 5 \\ 0 & 11 & -12 \\ 0 & k+28 & -36 \end{bmatrix} \begin{matrix} \left(\frac{1}{11}\right)R_2 \end{matrix} \\ &= \begin{bmatrix} 1 & -4 & 5 \\ 0 & 1 & -\frac{12}{11} \\ 0 & k+28 & -36 \end{bmatrix} \begin{matrix} R_3 + \{-(k+28)\}R_2 \end{matrix} = \begin{bmatrix} 1 & -4 & 5 \\ 0 & 1 & -11/12 \\ 0 & 0 & -36+12(k+28)/11 \end{bmatrix} \end{aligned}$$

For non-trivial solution we must have:

$$-36 + \frac{12}{11}(k+28) = 0 \text{ or } -396 + 12k + 308 = 0 \text{ or } 12k - 60 = 0 \text{ or } k = 5.$$

Putting this value of k in the above equations, we get:

$$x - 4y + 5z = 0 \quad (i)$$

$$y - 12z/11 = 0 \quad (ii)$$

From (ii) we have $y = 12z/11$. Substituting it into (i), we get

$$x - 4\left(\frac{12}{11}\right)z + 5z = 0 \quad \text{or} \quad x - \frac{48}{11}z + 5z = 0 \quad \text{or} \quad x = -\frac{7}{11}z.$$

Let $z = a$, then $x = -7a/11$, $y = 12a/11$ which is the solution of the given system. Since 'a' is a free parameter hence, the given system has infinitely many solutions for $k = 5$.

Example 08: Determine for what values of λ and μ the following system of equations: $x + y + z = 6$, $x + 2y + 3z = 10$, $x + 2y + \lambda z = \mu$ has (i) no solution (ii) a unique solution (iii) infinite number of solutions.

Solution: The augmented matrix for the above system of equations is

$$A_b = \begin{bmatrix} 1 & 1 & 1 & 6 \\ 1 & 2 & 3 & 10 \\ 1 & 2 & \lambda & \mu \end{bmatrix}.$$

Reducing into echelon form using elementary row operations

$$\begin{aligned} A_b &= \begin{bmatrix} 1 & 1 & 1 & 6 \\ 1 & 2 & 3 & 10 \\ 1 & 2 & \lambda & \mu \end{bmatrix} \begin{matrix} \\ R_2 + (-1)R_1 \\ R_3 + (-1)R_1 \end{matrix} \\ &= \begin{bmatrix} 1 & 1 & 1 & 6 \\ 0 & 1 & 2 & 4 \\ 0 & 1 & \lambda - 1 & \mu - 6 \end{bmatrix} \begin{matrix} \\ \\ R_3 + (-2)R_2 \end{matrix} = \begin{bmatrix} 1 & 1 & 1 & 6 \\ 0 & 1 & 2 & 4 \\ 0 & 0 & \lambda - 3 & \mu - 10 \end{bmatrix} \end{aligned}$$

- (i) There is no solution if $\text{Rank}(A) \neq \text{Rank}(A_b)$.
i.e. $\lambda - 3 = 0$ or $\lambda = 3$ and $\mu - 10 \neq 0$ or $\mu \neq 10$.
- (ii) There is a unique solution if $\text{Rank}(A) = \text{Rank}(A_b) = 3$.
i.e. $\lambda - 3 \neq 0$ or $\lambda \neq 3$ and μ may have any value.
- (iii) There are infinite solutions if $\text{Rank}(A) = \text{Rank}(A_b) < 3$.
i.e. $\lambda - 3 = 0$ or $\lambda = 3$ and $\mu - 10 = 0$ or $\mu = 10$.

Example 09: For what value of λ the following system of linear equations

$$(5 - \lambda)x_1 + 4x_2 + 2x_3 = 0$$

$$4x_1 + (5 - \lambda)x_2 + 2x_3 = 0$$

$$2x_1 + 2x_2 + (2 - \lambda)x_3 = 0$$

have a non-trivial solutions. Find these solutions.

Solution: The matrix of coefficients is

$$A = \begin{bmatrix} 5 - \lambda & 4 & 2 \\ 4 & 5 - \lambda & 2 \\ 2 & 2 & 2 - \lambda \end{bmatrix}$$

We reduce this matrix to the echelon form by applying elementary row operations

$$\begin{aligned} A &= \begin{bmatrix} 5 - \lambda & 4 & 2 \\ 4 & 5 - \lambda & 2 \\ 2 & 2 & 2 - \lambda \end{bmatrix} R_{13} \approx \begin{bmatrix} 2 & 2 & 2 - \lambda \\ 4 & 5 - \lambda & 2 \\ 5 - \lambda & 4 & 2 \end{bmatrix} \left(\frac{1}{2}\right)R_1 \\ &= \begin{bmatrix} 1 & 1 & 1 - \lambda/2 \\ 4 & 5 - \lambda & 2 \\ 5 - \lambda & 4 & 2 \end{bmatrix} \begin{matrix} \\ R_2 + (-4)R_1 \\ R_3 + \{-(5 - \lambda)\}R_1 \end{matrix} = \begin{bmatrix} 1 & 1 & 1 - \lambda/2 \\ 0 & 1 - \lambda & -2 + 2\lambda \\ 0 & \lambda - 1 & (-6 + 7\lambda - \lambda^2)/2 \end{bmatrix} \end{aligned}$$

$$\approx \begin{bmatrix} 1 & 1 & 1-\lambda/2 \\ 0 & 1-\lambda & -2(1-\lambda) \\ 0 & -(1-\lambda) & -(1-\lambda)(6-\lambda)/2 \end{bmatrix} \left(\frac{1}{1-\lambda} \right) R_2, \left(\frac{1}{1-\lambda} \right) R_3 ; (\text{assuming } 1-\lambda \neq 0)$$

$$= \begin{bmatrix} 1 & 1 & 1-\lambda/2 \\ 0 & 1 & -2 \\ 0 & -1 & (\lambda-6)/2 \end{bmatrix} \begin{matrix} \\ R_1 + (-1)R_2 \\ R_3 + R_2 \end{matrix} \quad \text{or} \quad A = \begin{bmatrix} 1 & 0 & 3-\lambda/2 \\ 0 & 1 & -2 \\ 0 & 0 & (\lambda-10)/2 \end{bmatrix}$$

For the non-trivial solution, rank of $A < 3$, so we must have $\frac{\lambda-10}{2} = 0$ or $\lambda = 10$. This implies

$$\text{that } x_1 + \left(3 - \frac{10}{2}\right)x_3 = 0 \quad \text{or} \quad x_1 - 2x_3 = 0 \quad \text{or} \quad x_1 = 2x_3 \quad \text{and} \quad x_2 - 2x_3 = 0 \quad \text{or} \quad x_2 = 2x_3$$

where x_3 is arbitrary. If $1-\lambda=0$ or $\lambda=1$, then above system becomes: $\begin{bmatrix} 1 & 1 & 1/2 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$

This gives $x_1 + x_2 + \frac{1}{2}x_3 = 0$ or $x_1 = -\left(x_2 + \frac{1}{2}x_3\right)$, where x_2 and x_3 are arbitrary constants.

Example 10: For what value of λ the system of linear equations

$$(3-\lambda)x_1 - x_2 + x_3 = 0$$

$$x_1 - (1-\lambda)x_2 + x_3 = 0 \quad \text{have a non-trivial solutions. Find these solutions.}$$

$$x_1 - x_2 + (1-\lambda)x_3 = 0$$

Solution: The matrix of the coefficients is

$$A = \begin{bmatrix} (3-\lambda) & -1 & 1 \\ 1 & -(1-\lambda) & 1 \\ 1 & -1 & (1-\lambda) \end{bmatrix}.$$

We reduce this matrix to the echelon form by applying elementary row operations:

$$A = \begin{bmatrix} (3-\lambda) & -1 & 1 \\ 1 & -(1-\lambda) & 1 \\ 1 & -1 & (1-\lambda) \end{bmatrix} \xrightarrow{R_{12}} \begin{bmatrix} (3-\lambda) & -1 & 1 \\ 1 & -(1-\lambda) & 1 \\ 1 & -1 & (1-\lambda) \end{bmatrix}$$

$$\approx \begin{bmatrix} 1 & -(1-\lambda) & 1 \\ (3-\lambda) & -1 & 1 \\ 1 & -1 & (1-\lambda) \end{bmatrix} \begin{matrix} \\ R_2 + \{-(3-\lambda)\}R_1 \\ R_3 + (-1)R_1 \end{matrix}$$

$$= \begin{bmatrix} 1 & -(1-\lambda) & 1 \\ 0 & 2-4\lambda+\lambda^2 & \lambda-2 \\ 0 & -\lambda & -\lambda \end{bmatrix} \xrightarrow{(-1)R_3} \begin{bmatrix} 1 & -(1-\lambda) & 1 \\ 0 & 2-4\lambda+\lambda^2 & \lambda-2 \\ 0 & 1 & 1 \end{bmatrix} \xrightarrow{R_{23}} \begin{bmatrix} 1 & -(1-\lambda) & 1 \\ 0 & 2-4\lambda+\lambda^2 & \lambda-2 \\ 0 & 1 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & \lambda-1 & 1 \\ 0 & 1 & 1 \\ 0 & 2-4\lambda+\lambda^2 & \lambda-2 \end{bmatrix} \begin{matrix} \\ R_1 + \{-(\lambda-1)\}R_2 \\ R_3 + \{-(2-4\lambda+\lambda^2)\}R_2 \end{matrix} = \begin{bmatrix} 1 & 0 & 2-\lambda \\ 0 & 1 & 1 \\ 0 & 0 & -4+5\lambda-\lambda^2 \end{bmatrix}$$

For non-trivial solutions rank of $A < 3$ (the number of unknowns), so we must have

$$-4+5\lambda-\lambda^2 = 0 \quad \text{or} \quad \lambda^2 - 5\lambda + 4 = 0 \quad \text{or} \quad \lambda = 1, 4.$$

Now when $\lambda = 1$, then from the last matrix, we have

$$x_1 + x_3 = 0 \quad \text{or} \quad x_1 = -x_3 \quad \text{and} \quad x_2 + x_3 = 0 \quad \text{or} \quad x_2 = -x_3$$

$$\text{When } \lambda = 4, \text{ then } x_1 - 2x_3 = 0 \quad \text{or} \quad x_1 = 2x_3 \quad \text{and} \quad x_2 + x_3 = 0 \quad \text{or} \quad x_2 = -x_3$$

$$\text{Hence, for } \lambda = 1, x_1 = x_2 = -x_3 \quad : \quad \text{for } \lambda = 4, x_1 = 2x_3, x_2 = -x_3$$

APPLICATIONS OF SYSTEM OF LINEAR EQUATIONS

In the introductory section, we gave many examples to show our students some practical problems where the systems of linear equations occur. In this section, we shall readdress these problems and few others and try to solve them in order to bring some satisfaction to our students who want to know how such problems are mathematically modeled and then their solutions are obtained.

Example 01: If 20 pounds of rice and 10 pounds of potatoes cost \$16.20 and 30 pounds of rice and 12 pounds of potatoes cost \$23.04, how much will 10 pounds of rice and 50 pounds of potatoes cost?

Solution: Let the cost of 1 pound of rice be \$ x and the cost of 1 pound of potatoes be \$ y . Then according to the question

$$20x + 10y = 16.20 \quad (i) \quad 30x + 12y = 23.04 \quad (ii)$$

Multiply (i) by 6 and (ii) by 4, we have

$$120x + 60y = 97.2 \quad (iii) \quad 120x + 48y = 92.16 \quad (iv)$$

Subtracting (iv) from (iii), we get: $12y = 5.04$ or $y = 0.42$

Substituting this value of y into (i), we obtain $x = 0.6$.

Thus, 10 pounds of rice cost $10 \times 0.6 = \$6$ and 50 pounds of potatoes cost $50 \times 0.42 = \$21$.

Example 2.5.2 A yarn merchant sells brands A, B, C of yarn each of which is a blend of Pakistani, Egyptian and American cotton in the ratio 1 : 2 : 1, 2 : 1 : 1 and 2 : 0 : 2 respectively. If one kilogram of A, B, C costs 40, 50 and 60 rupees respectively, find the cost of a kilogram of cotton of each country.

Solution: Let the cost of each kilogram of Pakistani, Egyptian and American cotton be x_1 , x_2 and x_3 rupees respectively. Then the above-mentioned problem is formulated into the following mathematical model involving equations:

$$\frac{1}{4}x_1 + \frac{2}{4}x_2 + \frac{1}{4}x_3 = 40, \quad \frac{2}{4}x_1 + \frac{1}{4}x_2 + \frac{1}{4}x_3 = 50, \quad \frac{2}{4}x_1 + \frac{2}{4}x_3 = 60,$$

Multiplying each equation by 4, we get

$$x_1 + 2x_2 + x_3 = 160, \quad 2x_1 + x_2 + x_3 = 200, \quad 2x_1 + 2x_3 = 240$$

Step1. Change the system of linear equations in matrix form $A \mathbf{x} = \mathbf{b}$.

$$A = \begin{bmatrix} 1 & 2 & 1 \\ 2 & 1 & 1 \\ 2 & 0 & 2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 160 \\ 200 \\ 240 \end{bmatrix}$$

Step2. Now form the augmented coefficient matrix A_b .

$$A_b = \begin{bmatrix} 1 & 2 & 1 & 160 \\ 2 & 1 & 1 & 200 \\ 2 & 0 & 2 & 240 \end{bmatrix}.$$

Step3. Convert this augmented coefficient matrix into an echelon form using elementary row operations.

$$\begin{aligned}
 A_b &= \begin{bmatrix} 1 & 2 & 1 & 160 \\ 2 & 1 & 1 & 200 \\ 2 & 0 & 2 & 240 \end{bmatrix} \begin{array}{l} R_2 + (-2)R_1 \\ R_3 + (-2)R_1 \end{array} \\
 &= \begin{bmatrix} 1 & 2 & 1 & 160 \\ 0 & -3 & -1 & -120 \\ 0 & -4 & 0 & -80 \end{bmatrix} \begin{array}{l} \\ (-1) R_2 \\ (1/4)R_3 \end{array} \\
 &= \begin{bmatrix} 1 & 2 & 1 & 160 \\ 0 & 3 & 1 & 120 \\ 0 & 1 & 0 & 20 \end{bmatrix} R_{23} = \begin{bmatrix} 1 & 2 & 1 & 160 \\ 0 & 1 & 0 & 20 \\ 0 & 3 & 1 & 120 \end{bmatrix} \begin{array}{l} \\ R_3 + (-3)R_2 \\ \end{array} = \begin{bmatrix} 1 & 2 & 1 & 160 \\ 0 & 1 & 0 & 20 \\ 0 & 0 & 1 & 60 \end{bmatrix}.
 \end{aligned}$$

Step4. Writing this echelon matrix into equation form, we get

$$x_1 + 2x_2 + x_3 = 160 \quad (i)$$

$$0 \cdot x_1 + x_2 + 0 \cdot x_3 = 20 \quad (ii)$$

$$0 \cdot x_1 + 0 \cdot x_2 + x_3 = 60 \quad (iii)$$

Solving equations (i) through (iii) using backward substitution, we get

$x_1 = 60$, $x_2 = 20$ and $x_3 = 60$. Thus, the cost of each kilogram of Pakistani, Egyptian and American cotton is Rs. 60/, Rs. 20/ and Rs. 60/ respectively.

Example 03: A housing construction company plans to undertake four housing projects and lists material requirements for each house in each of the project as follows:

	Project1	Project2	Project3	Project4
Paint (in 100 gallons)	1	2	1	1.5
Wood (in 10,000 cu.ft.)	3	4	2.5	2.5
Bricks (in millions)	1	2	1.5	1
Labour (in 1000 hours)	10	10	9	8

If the supplier delivers 6800 gallons of paint, 1,420,000 cubic feet of wood, 64 million bricks and 448,000 hours of labor, find the number of houses built for each project.

Solution: Let x_1 , x_2 , x_3 and x_4 be the numbers of houses built under Project-1, Project-2, Project-3 and Project-4 respectively. Then, according to given conditions, we have

$$x_1 + 2x_2 + x_3 + 1.5x_4 = 68 \quad (6800 \div 100 = 68)$$

$$3x_1 + 4x_2 + 2.5x_3 + 2.5x_4 = 142 \quad (1,420,000 \div 10,000 = 142)$$

$$x_1 + 2x_2 + 1.5x_3 + x_4 = 64$$

$$10x_1 + 10x_2 + 9x_3 + 8x_4 = 448 \quad (448,000 \div 1000 = 448)$$

The augmented matrix is:

$$A_b = \begin{bmatrix} 1 & 2 & 1 & 1.5 & 68 \\ 3 & 4 & 2.5 & 2.5 & 142 \\ 1 & 2 & 1.5 & 1 & 64 \\ 10 & 10 & 9 & 8 & 448 \end{bmatrix}.$$

Now convert this augmented coefficient matrix into reduced echelon form using elementary row operations:

$$A_b = \begin{bmatrix} 1 & 2 & 1 & 1.5 & 68 \\ 3 & 4 & 2.5 & 2.5 & 142 \\ 1 & 2 & 1.5 & 1 & 64 \\ 10 & 10 & 9 & 8 & 448 \end{bmatrix} \begin{array}{l} \\ R_2 + (-3)R_1 \\ R_3 + (-1)R_1 \\ R_4 + (-10)R_1 \end{array}$$

$$\begin{aligned}
&= \begin{bmatrix} 1 & 2 & 1 & 1.5 & 68 \\ 0 & -2 & -0.5 & -2 & -62 \\ 0 & 0 & 0.5 & -0.5 & -4 \\ 0 & -10 & -1 & -7 & -232 \end{bmatrix} \begin{matrix} R_1 + R_2 \\ (1/2)R_2 \\ 2R_3 \\ (-1)R_4 \end{matrix} = \begin{bmatrix} 1 & 0 & 0.5 & -0.5 & 6 \\ 0 & 1 & 0.25 & 1 & 31 \\ 0 & 0 & 1 & -1 & -8 \\ 0 & 10 & 1 & 7 & 232 \end{bmatrix} \begin{matrix} \\ \\ \\ R_4 + (-10)R_2 \end{matrix} \\
&\approx \begin{bmatrix} 1 & 0 & 0.5 & -0.5 & 6 \\ 0 & 1 & 0.25 & 1 & 31 \\ 0 & 0 & 1 & -1 & -8 \\ 0 & 0 & -1.5 & -3 & -78 \end{bmatrix} \begin{matrix} R_1 + (-0.5)R_3 \\ R_2 + (-0.25)R_3 \\ \\ R_4 + 1.5R_3 \end{matrix} = \begin{bmatrix} 1 & 0 & 0 & 0 & 10 \\ 0 & 1 & 0 & 1.25 & 33 \\ 0 & 0 & 1 & -1 & -8 \\ 0 & 0 & 0 & -4.5 & -90 \end{bmatrix} \begin{matrix} \\ \\ \\ -1/(4/5)R_4 \end{matrix} \\
&\approx \begin{bmatrix} 1 & 0 & 0 & 0 & 10 \\ 0 & 1 & 0 & 1.25 & 33 \\ 0 & 0 & 1 & -1 & -8 \\ 0 & 0 & 0 & 1 & 20 \end{bmatrix} \begin{matrix} \\ R_2 + -1.25R_4 \\ R_3 + R_4 \\ \end{matrix} \approx \begin{bmatrix} 1 & 0 & 0 & 0 & 10 \\ 0 & 1 & 0 & 0 & 8 \\ 0 & 0 & 1 & 0 & 12 \\ 0 & 0 & 0 & 1 & 20 \end{bmatrix}
\end{aligned}$$

Solving, we obtain:

$$x_1 = 10, x_2 = 8, x_3 = 12, x_4 = 20.$$

This means that with the available material, 10 houses can be built in Project-1, 8 houses in Project-2, 12 houses in Project-3 and 20 houses in Project-4.

Example 04: A soap manufacturer decides to spend 600, 000 rupees on radio, magazine and TV advertising. If he spends as much on TV advertising as on magazines and radio together, and the amount spent on magazines and TV combined equals five times that spent on radio, what is the amount to be spent on each type of advertising?

Solution: Let x , y , z be the amounts in rupees spent on radio, magazines and TV advertising respectively. Then, according to the given conditions, we have we get

$$x + y + z = 600, 000, x + y = z \text{ and } y + z = 5x. \text{ Re-arranging}$$

$$x + y + z = 600,000 \quad x + y - z = 0 \quad 5x - y - z = 0$$

The augmented matrix is:

$$\begin{aligned}
A_b &= \begin{bmatrix} 1 & 1 & 1 & 600,000 \\ 1 & 1 & -1 & 0 \\ 5 & -1 & -1 & 0 \end{bmatrix} \begin{matrix} \\ R_2 + (-1)R_1 \\ R_3 + (-5)R_1 \end{matrix} = \begin{bmatrix} 1 & 1 & 1 & 600,000 \\ 0 & 0 & -2 & -600,000 \\ 0 & -6 & -6 & -3,000,000 \end{bmatrix} \begin{matrix} \\ (-1/2)R_2 \\ (-1/6)R_3 \end{matrix} \\
&= \begin{bmatrix} 1 & 1 & 1 & 600,000 \\ 0 & 0 & 1 & 300,000 \\ 0 & 1 & 1 & 500,000 \end{bmatrix} R_{23} = \begin{bmatrix} 1 & 1 & 1 & 600,000 \\ 0 & 1 & 1 & 500,000 \\ 0 & 0 & 1 & 300,000 \end{bmatrix} \begin{matrix} R_1 + (-1)R_3 \\ R_2 + (-1)R_3 \\ \end{matrix} = \begin{bmatrix} 1 & 0 & 0 & 100,000 \\ 0 & 1 & 0 & 200,000 \\ 0 & 0 & 1 & 300,000 \end{bmatrix}
\end{aligned}$$

This gives: $x = 100, 000$, $y = 200, 000$, $z = 300, 000$. That is Rs.100, 000 is spent on radio, Rs.200, 000 is spent on magazines and Rs.300, 000 is spent on TV advertising. The sum of this amount is Rs. 600, 000.

Example 05: Find the three numbers whose sum is 34 where the sum of the first and second is 7 and the sum of the second and third is 22.

Solution: Let x , y and z be the required three numbers. Then according to given conditions:

$$x + y + z = 34, \quad x + y = 7, \quad y + z = 22$$

To solve the above system of equations, we use the Gauss-Jordan method and have

$$\begin{aligned}
 A_b &= \begin{bmatrix} 1 & 1 & 1 & 34 \\ 1 & 1 & 0 & 7 \\ 0 & 1 & 1 & 22 \end{bmatrix} R_2 + (-1)R_1 = \begin{bmatrix} 1 & 1 & 1 & 34 \\ 0 & 0 & -1 & -27 \\ 0 & 1 & 1 & 22 \end{bmatrix} (-1)R_2 \\
 &= \begin{bmatrix} 1 & 1 & 1 & 34 \\ 0 & 0 & 1 & 27 \\ 0 & 1 & 1 & 22 \end{bmatrix} R_{23} = \begin{bmatrix} 1 & 0 & 0 & 12 \\ 0 & 1 & 1 & 22 \\ 0 & 0 & 1 & 27 \end{bmatrix} R_2 + (-1)R_3 = \begin{bmatrix} 1 & 0 & 0 & 12 \\ 0 & 1 & 0 & -5 \\ 0 & 0 & 1 & 27 \end{bmatrix}
 \end{aligned}$$

This gives $x = 12$, $y = -5$ and $z = 27$. Thus, required numbers are -5 , 12 and 27 which satisfy the given conditions.

Example 06: Three brands of fertilizer are available that provide nitrogen, phosphoric acid, and soluble potash to the soil. One bag of each brand provides the following units of each nutrient.

Brand	Nitrogen	Phosphoric Acid	Potash
A	1	3	2
B	2	1	0
C	3	2	1

For ideal growth, the soil in a certain country needs 18 units of nitrogen, 23 units of phosphoric acid, and 13 units of potash per acre. How many bags of each brand of fertilizer should be used per acre for ideal growth?

Solution: Let x , y and z be the number of bags of each brand of fertilizer s should be used per acre for ideal growth respectively. According to the question,

$$x + 2y + 3z = 18,$$

$$3x + y + 2z = 23,$$

$$2x + z = 13$$

To solve the above system of equations, we use the Gauss-Jordan method and have

$$\begin{aligned}
 A_b &= \begin{bmatrix} 1 & 2 & 3 & 18 \\ 3 & 1 & 2 & 23 \\ 2 & 0 & 1 & 13 \end{bmatrix} R_2 + (-3)R_1 = \begin{bmatrix} 1 & 2 & 3 & 18 \\ 0 & -5 & -7 & -31 \\ 0 & -4 & -5 & -23 \end{bmatrix} (-1)R_2 = \begin{bmatrix} 1 & 2 & 3 & 18 \\ 0 & 5 & 7 & 31 \\ 0 & -4 & -5 & -23 \end{bmatrix} R_2 + (-1)R_3 \\
 &= \begin{bmatrix} 1 & 2 & 3 & 18 \\ 0 & 1 & 2 & 8 \\ 0 & 4 & 5 & 23 \end{bmatrix} R_1 + (-2)R_2 = \begin{bmatrix} 1 & 0 & -1 & 2 \\ 0 & 1 & 2 & 8 \\ 0 & 0 & -3 & -9 \end{bmatrix} (-1/9)R_3 \\
 &= \begin{bmatrix} 1 & 0 & -1 & 2 \\ 0 & 1 & 2 & 8 \\ 0 & 0 & 1 & 3 \end{bmatrix} R_1 + R_3 = \begin{bmatrix} 1 & 0 & 0 & 5 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & 1 & 3 \end{bmatrix}
 \end{aligned}$$

Solving, we obtain $x = 5$, $y = 2$ and $z = 3$.

This shows that 5 bags of brand 'A' fertilizer, 2 bags of brand 'B', and 3 bags of brand 'C' should be used per acre for ideal growth.

Example 07: A zoo charges \$6 for adults, \$3 for students and \$0.50 for children. One morning 79 people entered and paid a total of \$207. Determine the possible numbers of adults, students, and children.

Solution: Let A denote the number of adults, S be the number of students and C the number of children. Then according to the question,

$$A + S + C = 79 \quad (\text{Number of people})$$

$$6A + 3S + 0.5C = 207 \quad (\text{Total Amount})$$

To solve the above system of equations, we use the Gauss-Jordan method and have

$$\begin{aligned}
 A_b &= \begin{bmatrix} 1 & 1 & 1 & 79 \\ 6 & 3 & 0.5 & 207 \end{bmatrix} 2R_2 = \begin{bmatrix} 1 & 1 & 1 & 79 \\ 12 & 6 & 1 & 414 \end{bmatrix} R_2 + (-12)R_1 \\
 &= \begin{bmatrix} 1 & 1 & 1 & 79 \\ 0 & -6 & -11 & -534 \end{bmatrix} (-1/6)R_2 = \begin{bmatrix} 1 & 1 & 1 & 79 \\ 0 & 1 & 11/6 & 89 \end{bmatrix} R_1 + (-1)R_2 \\
 &= \begin{bmatrix} 1 & 0 & -5/6 & -10 \\ 0 & 1 & 11/6 & 89 \end{bmatrix}
 \end{aligned}$$

From the last matrix, we have

$$A - 5C/6 = -10 \text{ or } A = 5C/6 - 10 \text{ and } S + 11C/6 = 89 \text{ or } S = 11C/6 + 89.$$

Possible values for C are C = 12, 18, 24, 30, 36, 42 and 48 because for other values of C, the result will be either a fraction or a negative number. Hence the possible number of adults, students and children are shown in the following table.

Children	12	18	24	30	36	42	48
Adults	0	5	10	15	20	25	30
Students	67	56	45	34	23	12	1

Example 08: Three species of bacteria co-exist in a test tube and feed on three foods F_1 , F_2 and F_3 . Suppose that a bacterium of the i^{th} species consumes on the average an amount a_{ij} of the j^{th} food per day where

$$\begin{array}{lll}
 a_{11} = 1 & a_{12} = 1 & a_{13} = 1 \\
 a_{21} = 1 & a_{22} = 2 & a_{23} = 3 \\
 a_{31} = 1 & a_{32} = 3 & a_{33} = 5
 \end{array}$$

Further suppose that there are 15000, 30000, 45000 units of food of types F_1 , F_2 , F_3 respectively. Assuming that all food is consumed, what are the populations of the three species that can co-exist in this environment?

Solution: Let the population of the three species of bacteria be x_1 , x_2 and x_3 . Since the i^{th} species consumes a_{ij} units of food of type j we have

$$\begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \\ 1 & 3 & 5 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 15000 \\ 30000 \\ 45000 \end{bmatrix} \Rightarrow \begin{array}{ll} x_1 + x_2 + x_3 = 15000 & \text{(i)} \\ x_1 + 2x_2 + 3x_3 = 30000 & \text{(ii)} \\ x_1 + 3x_2 + 5x_3 = 45000 & \text{(iii)} \end{array}$$

The augmented matrix is

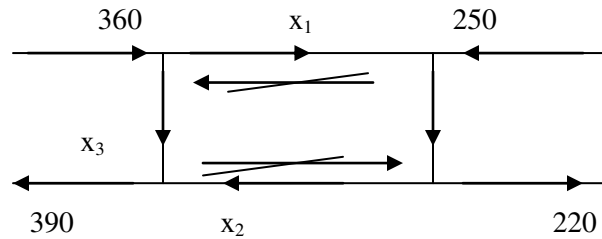
$$\begin{aligned}
 A_b &= \begin{bmatrix} 1 & 1 & 1 & 15000 \\ 1 & 2 & 3 & 30000 \\ 1 & 3 & 5 & 45000 \end{bmatrix} \begin{array}{l} R_2 + (-1)R_1 \\ R_3 + (-1)R_1 \end{array} = \begin{bmatrix} 1 & 1 & 1 & 15000 \\ 0 & 1 & 2 & 15000 \\ 0 & 2 & 4 & 30000 \end{bmatrix} \begin{array}{l} \\ R_3 + (-2)R_2 \end{array} \\
 &= \begin{bmatrix} 1 & 1 & 1 & 15000 \\ 0 & 1 & 2 & 15000 \\ 0 & 0 & 0 & 0 \end{bmatrix} \begin{array}{l} \\ R_1 + (-1)R_2 \end{array} = \begin{bmatrix} 1 & 0 & -1 & 0 \\ 0 & 1 & 2 & 15000 \\ 0 & 0 & 0 & 0 \end{bmatrix}
 \end{aligned}$$

$$x_1 - x_3 = 0 \text{ or } x_1 = x_3 \text{ and } x_2 + 2x_3 = 15000 \text{ or } x_2 = 15000 - 2x_3, \text{ where } x_3 \text{ is free parameter.}$$

From (i) we see that the total populations of the three species that can co-exist is 15000.

If we take $x_3 = 3000$ then, $x_1 = 3000$ and $x_2 = 9000$ and the population of the three species that can co-exist in this environment is 3000, 9000, and 3000.

Example 09: In a certain part of the city two sets of one way streets intersect as shown in the following figure. The average hourly volume of traffic entering and leaving this part during such hours is given in the figure. Determine the amount of traffic between each of the four junctions.



Solution: Suppose that x_1 , x_2 and x_3 denote the number of vehicles going along AB, CD and AD respectively. Equating the incoming traffic to the outgoing traffic at each intersection, we have the following mathematical model:

$$x_1 + x_3 = 360 \quad (i)$$

$$x_2 + x_3 = 390 \quad (ii)$$

$$x_1 + 250 = x_2 + 220 \text{ or } x_1 - x_2 = -30 \quad (iii)$$

Taking the augmented matrix and reducing it to the echelon form, we have:

$$A_b = \begin{bmatrix} 1 & 0 & 1 & 360 \\ 0 & 1 & 1 & 390 \\ 1 & -1 & 0 & -30 \end{bmatrix} \xrightarrow{R_3 + (-1)R_1} \begin{bmatrix} 1 & 0 & 1 & 360 \\ 0 & 1 & 1 & 390 \\ 0 & -1 & -1 & -390 \end{bmatrix} \xrightarrow{R_3 + R_2} \begin{bmatrix} 1 & 0 & 1 & 360 \\ 0 & 1 & 1 & 390 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

This gives, $x_1 + x_3 = 360$ or $x_1 = 360 - x_3$ and $x_2 + x_3 = 390$ or $x_2 = 390 - x_3$. Thus, x_1 and x_2 can be expressed in terms of x_3 where x_3 is a free parameter. We know that whenever a free parameter exists in a system of linear equation we always get an infinite set of solutions. But here x_3 must satisfy the condition that $0 \leq x_3 \leq 360$ because the maximum traffic flow at the first intersection is 360 and minimum value is 0. Thus, we have 361 different solutions for different values of x_3 .

Example 10: The flow of traffic flow per hour through a network of streets is shown in the adjacent figure. Solve the system for x_i , $i = 1, 2, 3, 4$. Also show the traffic flow when (a) $x_4 = 0$ (b) $x_4 = 100$.

Solution: Using the fact that flow into the junction is equal to flow out, we obtain

$$x_2 + 200 = x_1, \quad x_1 + 100 = x_3, \quad x_3 = 200 + x_4,$$

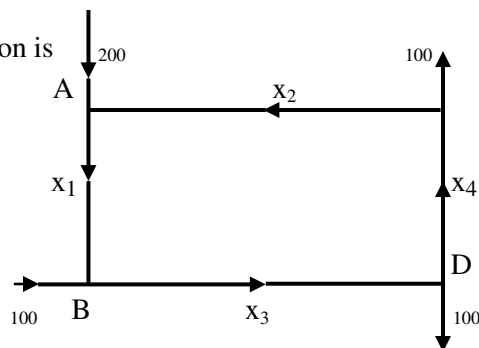
$$x_4 = 100 + x_2.$$

These equations can be written in the standard form as

$$x_1 + x_2 = 200, \quad x_1 - x_3 = -100,$$

$$x_3 - x_4 = 200, \quad x_2 - x_4 = -100$$

The augmented matrix for the above system of equations can be transformed into the reduced row echelon form as follows:



$$\begin{aligned}
 A_b &= \begin{bmatrix} 1 & -1 & 0 & 0 & 200 \\ 1 & 0 & -1 & 0 & -100 \\ 0 & 0 & 1 & -1 & 200 \\ 0 & 1 & 0 & -1 & -100 \end{bmatrix} \begin{matrix} R_2 + (-1)R_1 \\ \\ \\ \end{matrix} = \begin{bmatrix} 1 & -1 & 0 & 0 & 200 \\ 0 & 1 & -1 & 0 & -300 \\ 0 & 0 & 1 & -1 & 200 \\ 0 & 1 & 0 & -1 & -100 \end{bmatrix} \begin{matrix} R_1 + R_2 \\ \\ R_4 + (-1)R_2 \\ \end{matrix} \\
 &\approx \begin{bmatrix} 1 & 0 & -1 & 0 & -100 \\ 0 & 1 & -1 & 0 & -300 \\ 0 & 0 & 1 & -1 & 200 \\ 0 & 0 & 1 & -1 & 200 \end{bmatrix} \begin{matrix} R_1 + R_3 \\ R_2 + R_3 \\ \\ R_4 + (-1)R_3 \end{matrix} = \begin{bmatrix} 1 & 0 & 0 & -1 & 100 \\ 0 & 1 & 0 & -1 & -100 \\ 0 & 0 & 1 & -1 & 200 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}
 \end{aligned}$$

From the last matrix, the equivalent system of equations is:

$$x_1 - x_4 = 100, x_2 - x_4 = -100, x_3 - x_4 = 200$$

Here, x_4 is a free variable (parameter). With $x_4 = k$ we obtain

$$x_1 = 100 + k, x_2 = -100 + k \text{ and } x_3 = 200 + k$$

The solution is not unique. This means that there are a large number of possible traffic flows.

(a) When $x_4 = 0$, that is; no flow along the street CD, we obtain the unique solution

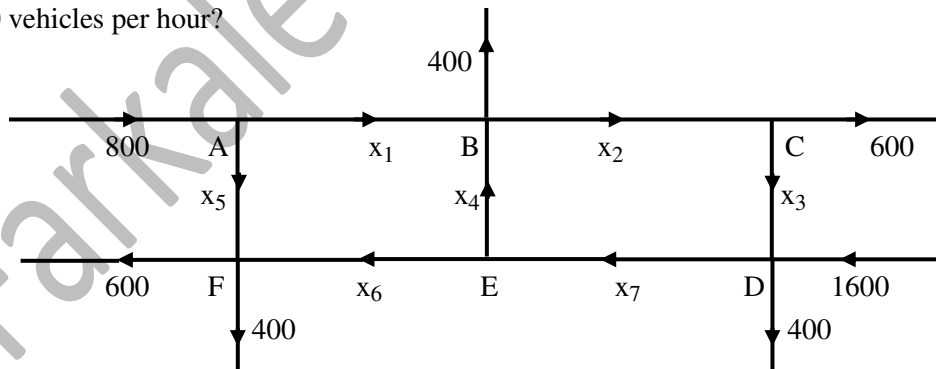
$$x_1 = 100, x_2 = -100 \text{ and } x_3 = 200$$

(b) When $x_4 = 100$, we have the unique solution

$$x_1 = 200, x_2 = 0 \text{ and } x_3 = 300$$

Remark: We normally want the flows in a network to be nonnegative. For instance, in the above example part (a) $x_2 = -100$ indicates that traffic was flowing from A to D rather than in the prescribed direction from D to A.

Example 11: Set up a system of equations that represents traffic flow for the network shown in the following figure. (The numbers represent the average flows into and out of the network at peak traffic hours). Solve the system of equations. What is the traffic flow if $x_6 = 300$ and $x_7 = 1300$ vehicles per hour?



Solution: Since the flow in at any node is equal to the flow out, we obtain the following system of equations:

$$\begin{aligned}
 800 &= x_1 + x_5 & (\text{Node A}), & & x_1 + x_4 &= x_2 + 400 & (\text{Node B}) \\
 x_2 &= x_3 + 600 & (\text{Node C}), & & x_3 + 1600 &= x_7 + 400 & (\text{Node D}) \\
 x_7 &= x_4 + x_6 & (\text{Node E}), & & x_5 + x_6 &= 400 + 600 & (\text{Node F})
 \end{aligned}$$

The augmented matrix for the above system of equations can be transformed into reduced row echelon form as follows:

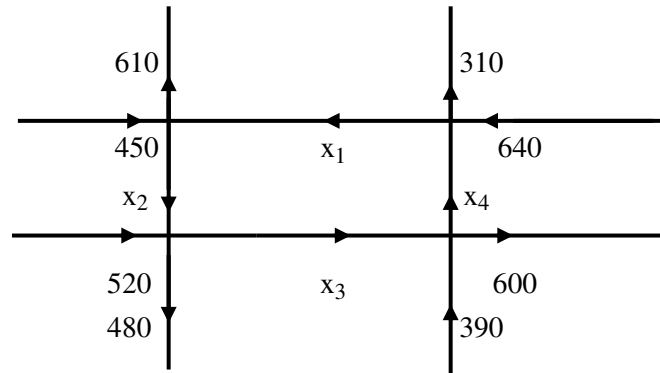
$$\begin{aligned}
 Ab &= \begin{bmatrix} 1 & 0 & 0 & 0 & 1 & 0 & 0 & 800 \\ 1 & -1 & 0 & 1 & 0 & 0 & 0 & 400 \\ 0 & 1 & -1 & 0 & 0 & 0 & 0 & 600 \\ 0 & 0 & 1 & 0 & 0 & 0 & -1 & -1200 \\ 0 & 0 & 0 & 1 & 0 & 1 & -1 & 0 \\ 0 & 0 & 0 & 0 & 1 & 1 & 0 & 1000 \end{bmatrix} \quad R_2 + (-1)R_1 \\
 &= \begin{bmatrix} 1 & 0 & 0 & 0 & 1 & 0 & 0 & 800 \\ 0 & -1 & 0 & 1 & -1 & 0 & 0 & -400 \\ 0 & 1 & -1 & 0 & 0 & 0 & 0 & 600 \\ 0 & 0 & 1 & 0 & 0 & 0 & -1 & -1200 \\ 0 & 0 & 0 & 1 & 0 & 1 & -1 & 0 \\ 0 & 0 & 0 & 0 & 1 & 1 & 0 & 1000 \end{bmatrix} \quad (-1)R_2 \\
 &= \begin{bmatrix} 1 & 0 & 0 & 0 & 1 & 0 & 0 & 800 \\ 0 & 1 & 0 & -1 & 1 & 0 & 0 & 400 \\ 0 & 1 & -1 & 0 & 0 & 0 & 0 & 600 \\ 0 & 0 & 1 & 0 & 0 & 0 & -1 & -1200 \\ 0 & 0 & 0 & 1 & 0 & 1 & -1 & 0 \\ 0 & 0 & 0 & 0 & 1 & 1 & 0 & 1000 \end{bmatrix} \quad R_3 + (-1)R_2 \\
 &= \begin{bmatrix} 1 & 0 & 0 & 0 & 1 & 0 & 0 & 800 \\ 0 & 1 & 0 & -1 & 1 & 0 & 0 & 400 \\ 0 & 0 & -1 & 1 & -1 & 0 & 0 & 200 \\ 0 & 0 & 1 & 0 & 0 & 0 & -1 & -1200 \\ 0 & 0 & 0 & 1 & 0 & 1 & -1 & 0 \\ 0 & 0 & 0 & 0 & 1 & 1 & 0 & 1000 \end{bmatrix} \quad (-1)R_3 \\
 &= \begin{bmatrix} 1 & 0 & 0 & 0 & 1 & 0 & 0 & 800 \\ 0 & 1 & 0 & -1 & 1 & 0 & 0 & 400 \\ 0 & 0 & 1 & -1 & 1 & 0 & 0 & -200 \\ 0 & 0 & 1 & 0 & 0 & 0 & -1 & -1200 \\ 0 & 0 & 0 & 1 & 0 & 1 & -1 & 0 \\ 0 & 0 & 0 & 0 & 1 & 1 & 0 & 1000 \end{bmatrix} \quad R_4 + (-1)R_3 \\
 &= \begin{bmatrix} 1 & 0 & 0 & 0 & 1 & 0 & 0 & 800 \\ 0 & 1 & 0 & 0 & 0 & 0 & -1 & -600 \\ 0 & 0 & 1 & 0 & 0 & 0 & -1 & -1200 \\ 0 & 0 & 0 & 1 & -1 & 0 & -1 & -1000 \\ 0 & 0 & 0 & 0 & 1 & 1 & 0 & 1000 \\ 0 & 0 & 0 & 0 & 1 & 1 & 0 & 1000 \end{bmatrix} \quad \begin{matrix} R_1 + (-1)R_5 \\ R_4 + R_5 \\ R_6 + (-1)R_5 \end{matrix} = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & -1 & 0 & -200 \\ 0 & 1 & 0 & 0 & 0 & 0 & -1 & -600 \\ 0 & 0 & 1 & 0 & 0 & 0 & -1 & -1200 \\ 0 & 0 & 0 & 1 & 0 & 1 & -1 & 0 \\ 0 & 0 & 0 & 0 & 1 & 1 & 0 & 1000 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}
 \end{aligned}$$

From the last matrix, the equivalent system of equations is:

$$\begin{aligned}
 x_1 - x_6 &= -200 & \text{or } x_1 &= x_6 - 200 & \text{and} & & x_2 - x_7 &= -600 & x_2 &= x_7 - 600 \\
 x_3 - x_7 &= -1200 & \text{or } x_3 &= x_7 - 1200 & \text{and} & & x_4 + x_6 - x_7 &= 0 & \text{or } x_4 &= x_7 - x_6 \\
 x_5 + x_6 &= 1000 & \text{or } x_5 &= 1000 - x_6
 \end{aligned}$$

Now, if $x_6 = 300$ and $x_7 = 1300$ then $x_1 = 100$, $x_2 = 700$, $x_3 = 100$, $x_4 = 1000$, $x_5 = 700$

Example 12: In a certain part of the city two sets of one way streets intersect as shown in the following figure:



The average hourly volume of traffic entering and leaving this part during such hours is given in the figure. Determine the amount of traffic between each of the four junctions.

Solution: We denote the number of automobiles between junctions by x_1, x_2, x_3 and x_4 . Using the fact that the number of automobiles entering a given junction equal that leaving the same, we obtain the following equations for the unknowns x_1, x_2, x_3 and x_4

$$\begin{cases} x_1 + 450 = x_2 + 610 \\ x_2 + 520 = x_3 + 480 \\ x_3 + 390 = x_4 + 600 \\ x_4 + 640 = x_1 + 310 \end{cases}$$

These equations can be written in the standard form as

$$\begin{cases} x_1 - x_2 = 160 \\ x_2 - x_3 = -40 \\ x_3 - x_4 = 210 \\ x_1 - x_4 = 330 \end{cases}$$

The augmented matrix for the above system of equations can be transformed into the reduced row echelon form as follows:

$$A_b = \begin{bmatrix} 1 & -1 & 0 & 0 & 160 \\ 0 & 1 & -1 & 0 & -40 \\ 0 & 0 & 1 & -1 & 210 \\ 1 & 0 & 0 & -1 & 330 \end{bmatrix} \begin{matrix} \\ \\ R_4 + (-1)R_1 \\ \end{matrix} = \begin{bmatrix} 1 & -1 & 0 & 0 & 160 \\ 0 & 1 & -1 & 0 & -40 \\ 0 & 0 & 1 & -1 & 210 \\ 0 & 1 & 0 & -1 & 170 \end{bmatrix} \begin{matrix} R_1 + R_2 \\ \\ R_4 + (-1)R_2 \\ \end{matrix}$$

$$\approx \begin{bmatrix} 1 & 0 & -1 & 0 & 120 \\ 0 & 1 & -1 & 0 & -40 \\ 0 & 0 & 1 & -1 & 210 \\ 0 & 0 & 1 & -1 & 210 \end{bmatrix} \begin{matrix} R_1 + R_3 \\ R_2 + R_3 \\ R_4 + (-1)R_3 \\ \end{matrix} = \begin{bmatrix} 1 & 0 & 0 & -1 & 330 \\ 0 & 1 & 0 & -1 & 170 \\ 0 & 0 & 1 & -1 & 210 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

From the last matrix, the equivalent system of equations is:

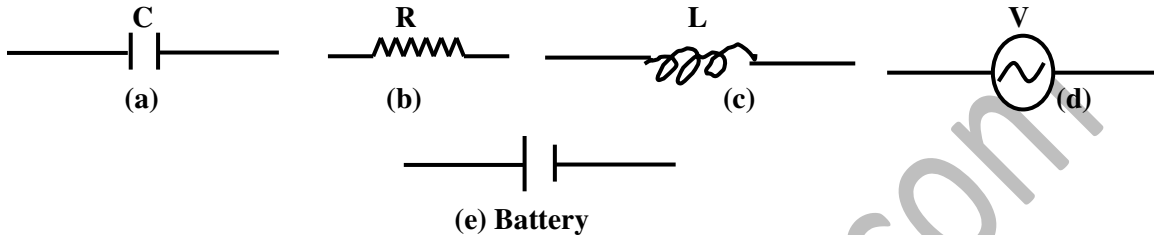
$$x_1 - x_4 = 330, x_2 - x_4 = 170, x_3 - x_4 = 210.$$

Here x_4 is a free variable/parameter hence, there are many possible solutions.

Electrical Networks

An electrical system may be in the form of a circuit or a network. The essential components of such a system are resistors, capacitors (or a condensers), an inductors and batteries or any other source of voltage including an electromotive force (e.m.f.).

A resistor uses electrical energy and converts it into heat. An electric light bulb, electric tube and electric heater are examples of resistors. A capacitor stores electrical energy and an inductor oppose a change in electrical current. The electromotive force (supplied by a battery or generator) is the cause of flow of current. Following symbols are used in electrical network.



In a circuit:

- (i) A capacitor is denoted by C , its capacitance is measured in farads (F) and its graphical representation is shown in figure (a).
- (ii) A resistor is denoted by R , its resistance is measured in ohms (Ω) and its graphical representation is displayed in figure (b).
- (iii) An inductor is denoted by L , its conductance is measured in henries (H) and its graphical representation is displayed in figure (c).
- (iv) The electromotive force is measured in volts (V). The e.m.f. of a battery is fixed and may be denoted by E . The e.m.f. of a generator varies periodically and may be denoted by $E_0 \sin \omega t$ where ω is the angular frequency. Graphical representations of a battery and generator are displayed in figure d and (e).

Electrical Quantities

The common electrical quantities are charge (quantity of electricity) denoted by q ; current i.e. the amount of charge flowing per second through the cross section of a wire, denoted by $I = dq/dt$ where the time t is measured in seconds. The charge is measured in coulombs and the current in amperes.

The flow of current in a circuit is governed by Ohm's law and Kirchoff's laws which can be stated as follows:

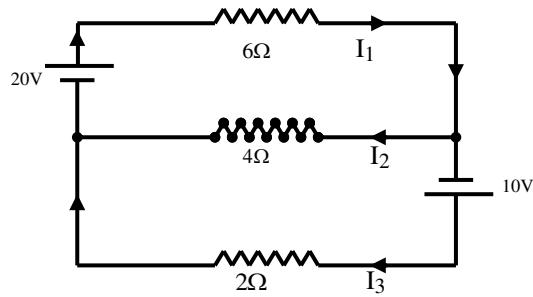
Ohm's Law: The voltage drop across a resistor is the product of the current and the resistance.

Kirchoff's First Law (Current Rule): The sum of the currents flowing into a node is equal to the sum of the currents flowing out.

Kirchoff's Second Law (Voltage Rule): The algebraic sum of the voltage drops around a closed circuit is equal to the total voltage in the circuit.

Remark: When applying the current rule, select a direction (clockwise or counterclockwise) around the closed circuit and then consider all voltages and currents positive or negative according to as the flow is in the same or opposite direction. This is why the term *algebraic sum* is used in the current rule.

Example.13: Find currents in each branch of the circuit shown in the following figure.



Solution: Using Kirchoff's first law (junction rule) at nodes A and B, we have

$$I_2 + I_3 = I_1 \quad \text{or} \quad -I_1 + I_2 + I_3 = 0 \quad (\text{i})$$

and by Kirchoff's second law (current rule) for the circuits ABCD and ABEF, we have

$$-4I_2 + 2I_3 = 10 \quad (\text{ii})$$

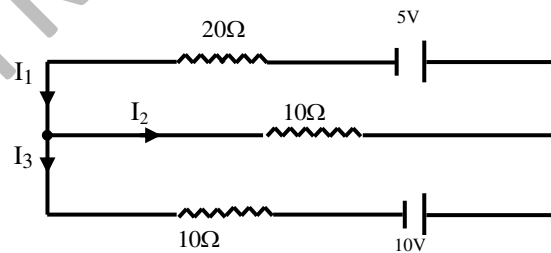
$$4I_2 + 6I_1 = 20 \quad (\text{iii})$$

The augmented matrix arises from above equations

$$\begin{aligned}
 A_b &= \begin{bmatrix} 1 & -1 & -1 & 0 \\ 0 & -4 & 2 & 10 \\ 6 & 4 & 0 & 20 \end{bmatrix} \begin{matrix} \\ \\ R_3 + (-6)R_1 \end{matrix} \\
 &\approx \begin{bmatrix} 1 & -1 & -1 & 0 \\ 0 & 2 & -1 & -5 \\ 0 & 5 & 3 & 10 \end{bmatrix} \begin{matrix} \\ \\ R_3 + (-2)R_2 \end{matrix} \\
 &= \begin{bmatrix} 1 & -1 & -1 & 0 \\ 0 & 1 & 5 & 20 \\ 0 & 2 & -1 & -5 \end{bmatrix} \begin{matrix} R_1 + R_2 \\ \\ R_3 + (-2)R_2 \end{matrix} \\
 &= \begin{bmatrix} 1 & 0 & 4 & 20 \\ 0 & 1 & 5 & 20 \\ 0 & 0 & 1 & 45/11 \end{bmatrix} \begin{matrix} R_1 + (-4)R_3 \\ R_2 + (-5)R_3 \\ \end{matrix} \\
 &= \begin{bmatrix} 1 & -1 & -1 & 0 \\ 0 & -4 & 2 & 10 \\ 0 & 10 & 6 & 20 \end{bmatrix} \begin{matrix} (-1/2) R_2 \\ (1/2) R_3 \\ \end{matrix} \\
 &= \begin{bmatrix} 1 & -1 & -1 & 0 \\ 0 & 2 & -1 & -5 \\ 0 & 1 & 5 & 20 \end{bmatrix} \begin{matrix} \\ R_{23} \\ \end{matrix} \\
 &= \begin{bmatrix} 1 & 0 & 4 & 20 \\ 0 & 1 & 5 & 20 \\ 0 & 0 & -11 & -45 \end{bmatrix} \begin{matrix} \\ \\ (-1/11)R_3 \end{matrix} \\
 &= \begin{bmatrix} 1 & 0 & 0 & 40/11 \\ 0 & 1 & 0 & -5/11 \\ 0 & 0 & 1 & 45/11 \end{bmatrix}
 \end{aligned}$$

From the last matrix, we see that $I_1 = 40/11 \Omega$, $I_2 = -5/11 \Omega$, $I_3 = 45/11 \Omega$.

Example 14: Determine the currents I_1 , I_2 and I_3 for the electrical network shown in the following figure.



Solution: Applying Kirchoff's second law to the loops, we obtain the equations:

$$-10I_2 + 10I_3 = 10$$

$$20I_1 + 10I_2 = 5$$

Applying Kirchoff's first law, we have $I_1 = I_2 + I_3$. Thus the above equations can be written in

$$\begin{aligned}
 \text{the standard form as:} \quad & \begin{matrix} I_1 & - & I_2 & - & I_3 & = & 0 \\ & - & 10I_2 & + & 10I_3 & = & 10 \\ & 20I_1 & + & 10I_2 & & = & 5 \end{matrix}
 \end{aligned}$$

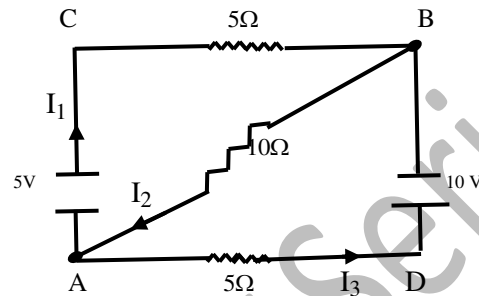
The corresponding augmented matrix can be reduced as follows:

$$\begin{aligned}
 A_b &= \begin{bmatrix} 1 & -1 & -1 & 0 \\ 0 & -10 & 10 & 10 \\ 20 & 10 & 0 & 5 \end{bmatrix} \begin{matrix} (-1/10)R_2 \\ (1/5)R_3 \end{matrix} &= \begin{bmatrix} 1 & -1 & -1 & 0 \\ 0 & 1 & -1 & -1 \\ 4 & 2 & 0 & 1 \end{bmatrix} \begin{matrix} R_3 + (-4)R_1 \\ \\ \end{matrix} \\
 &= \begin{bmatrix} 1 & -1 & -1 & 0 \\ 0 & 1 & -1 & -1 \\ 0 & 6 & 4 & 1 \end{bmatrix} \begin{matrix} R_1 + R_2 \\ \\ R_3 + (-6)R_2 \end{matrix} &= \begin{bmatrix} 1 & 0 & -2 & -1 \\ 0 & 1 & -1 & -1 \\ 0 & 0 & 10 & 7 \end{bmatrix} \begin{matrix} \\ \\ (1/10)R_3 \end{matrix} \\
 &\approx \begin{bmatrix} 1 & 0 & -2 & -1 \\ 0 & 1 & -1 & -1 \\ 0 & 0 & 1 & 0.7 \end{bmatrix} \begin{matrix} R_1 + 2R_3 \\ R_2 + R_3 \\ \end{matrix} &= \begin{bmatrix} 1 & 0 & 0 & 0.4 \\ 0 & 1 & 0 & -0.3 \\ 0 & 0 & 1 & 0.7 \end{bmatrix}
 \end{aligned}$$

Therefore, the currents are: $I_1 = 0.4\Omega$, $I_2 = -0.3\Omega$, $I_3 = 0.7\Omega$

Since I_2 is negative, the current flow is from C to B rather than from B to C, as tentatively assigned in the given figure.

Example 15: Find currents in the circuit shown in the following figure



Solution: Kirchhoff's first law applied at junction A gives

$$I_2 = I_1 + I_3 \Rightarrow -I_1 + I_2 - I_3 = 0$$

Similarly, Kirchhoff's second law applied to circuits ABC and ABD gives

$$10I_2 + 5I_1 = 5 \text{ and } -10I_2 - 5I_3 = -10 \text{ and}$$

Thus the above equations can be written in the standard form as:

$$\begin{aligned}
 I_1 - I_2 + I_3 &= 0 \\
 5I_1 + 10I_2 &= 5 \\
 10I_2 + 5I_3 &= 10
 \end{aligned}$$

The corresponding augmented matrix can be reduced as follows:

$$\begin{aligned}
 A_b &= \begin{bmatrix} 1 & -1 & 1 & 0 \\ 5 & 10 & 0 & 5 \\ 0 & 10 & 5 & 10 \end{bmatrix} \begin{matrix} (1/5)R_2 \\ (1/3)R_3 \end{matrix} &= \begin{bmatrix} 1 & -1 & 1 & 0 \\ 1 & 2 & 0 & 1 \\ 0 & 2 & 1 & 2 \end{bmatrix} \begin{matrix} R_2 + (-1)R_1 \\ \\ \end{matrix} \\
 &= \begin{bmatrix} 1 & -1 & 1 & 0 \\ 0 & 3 & -1 & 1 \\ 0 & 2 & 1 & 2 \end{bmatrix} \begin{matrix} \\ R_2 + (-1)R_3 \\ R_3 + (-2)R_2 \end{matrix} &= \begin{bmatrix} 1 & -1 & 1 & 0 \\ 0 & 1 & -2 & -1 \\ 0 & 2 & 1 & 2 \end{bmatrix} \begin{matrix} R_1 + R_2 \\ \\ R_3 + (-2)R_2 \end{matrix} \\
 &= \begin{bmatrix} 1 & 0 & -1 & -1 \\ 0 & 1 & -2 & -1 \\ 0 & 0 & 5 & 4 \end{bmatrix} \begin{matrix} (1/5)R_3 \\ \\ \end{matrix} &= \begin{bmatrix} 1 & 0 & -1 & -1 \\ 0 & 1 & -2 & -1 \\ 0 & 0 & 1 & 4/5 \end{bmatrix} \begin{matrix} R_1 + R_3 \\ R_2 + 2R_3 \\ \end{matrix} = \begin{bmatrix} 1 & 0 & 0 & -1/5 \\ 0 & 1 & 0 & 3/5 \\ 0 & 0 & 1 & 4/5 \end{bmatrix}
 \end{aligned}$$

This implies that $I_1 = -1/5\Omega$, $I_2 = 3/5\Omega$, $I_3 = 4/5\Omega$.

WORKSHEET 02

1. Solve the following systems of linear equations:

$$\begin{array}{ll}
 x + y + z = 3 & x + 2y + 3z = 1 \\
 \text{(i)} \quad x + 2y + 3z = 4 & \text{(ii)} \quad 3x - y - z = 2 \\
 x + 4y + 9z = 6 & 4x + 2y + z = 3 \\
 x - y + 2z = 3 & x_1 + 2x_2 - x_3 = 1 \\
 \text{(iii)} \quad x + 2y + 3z = 5 & \text{(iv)} \quad 3x_1 - 2x_2 + 2x_3 = 2 \\
 3x - 4y - 5z = -13 & 7x_1 - 2x_2 + 3x_3 = 5 \\
 3x_1 + 2x_3 + 2x_4 = 0 & 3x_1 + 2x_3 + 2x_4 = 0 \\
 \text{(v)} \quad -x_1 + 7x_2 + 4x_3 + 9x_4 = 0 & \text{(v)} \quad -x_1 + 7x_2 + 4x_3 + 9x_4 = 0 \\
 7x_1 - 7x_2 - 5x_4 = 0 & 7x_1 - 7x_2 - 5x_4 = 0 \\
 2x_1 + x_2 + 2x_3 + x_4 = 6 & \\
 \text{(vi)} \quad 3x_1 - 6x_2 + 6x_3 + 12x_4 = 36 & \\
 4x_1 + 3x_2 + 3x_3 - 3x_4 = -1 & \\
 2x_1 + 2x_2 - x_3 + x_4 = 10 &
 \end{array}$$

2. Test the consistency of the following equations and solve them if possible:

$$\begin{array}{ll}
 3x + 3y + 2z = 1 & t - w + x - y + z = 1 \\
 \text{(i)} \quad x + 2y = 4 & \text{(ii)} \quad 2t - w + 3x + 4z = 2 \\
 10y + 3z = -2 & 3t - 2w + 2x + y + z = 1 \\
 2x - 3y - z = 5 & t + x + y + z = 0 \\
 3w - 6x - y - z = 0 & 4x - 5y + z = 2 \\
 \text{(iii)} \quad w - 2x + 5y - 3z = 0 & \text{(iv)} \quad 3x + y - 2z = 9 \\
 2w - 4x + 3y - z = 3 & x + 4y + z = 5
 \end{array}$$

3. Find the value of k for which the following system of equations is consistent:

$$\begin{array}{l}
 3x - 2y + 2z = 3 \\
 x + ky - 3z = 0 \\
 4x + y + 2z = 7
 \end{array}$$

4. Find the value of λ for which the system of equations

$$\begin{array}{l}
 x + y + 4z = 1 \\
 x + 2y - 2z = 1 \\
 \lambda x + y + z = 1
 \end{array}$$

will have a unique solution.

5. For what values of λ and μ , the following system of equations

$$\begin{array}{l}
 2x + 3y + 5z = 9 \\
 7x + 3y - 2z = 8 \\
 2x + 3y + \lambda z = \mu
 \end{array}$$

will have (i) unique solution (ii) no solution.

6. Determine the values of a and b for which the system

$$3x - 2y + z = b; \quad 5x - 8y + 9z = 3; \quad 2x + y + az = -1$$

(i) unique solution (ii) has no solution and, (iii) has infinitely many solutions.

7. Choose λ that makes the following system of linear equations consistent and find the general solution of the system for this value of λ .

$$\begin{aligned}x + y - z + t &= 2 \\2y + 4z + 2t &= 3 \\x + 2y + z + 2t &= \lambda\end{aligned}$$

8. Show that the system of equations

$$\begin{aligned}3x + 4y + 5z &= a \\4x + 5y + 6z &= b \\5x + 6y + 7z &= c\end{aligned}$$

doesn't have a solution unless $a + c = 2b$. Solve the equations when $a = b = c = 1$.

9. Find the condition on λ for which the system of equations

$$\begin{aligned}3x - y + 4z &= 3 \\x + 2y - 3z &= -2 \\6x + 5y + \lambda z &= -3\end{aligned}$$

has a unique solution. Find the solution for $\lambda = -5$.

10. Find the value of k such that the system of equations

$$\begin{aligned}2x + 3y - 2z &= 0 \\3x - y + 3z &= 0 \\7x + ky - z &= 0\end{aligned}$$

has non-trivial solutions. Find the solutions.

11. Find the values of k , such that the system of equations

$$\begin{aligned}4x_1 + 9x_2 + x_3 &= 0 \\kx_1 + 3x_2 + kx_3 &= 0 \\x_1 + 4x_2 + 2x_3 &= 0\end{aligned}$$

has non-trivial solutions. Hence, find the solution of the system.

12. Find values of λ for which the following system of equations has a non-trivial solution.

$$\begin{aligned}3x_1 + x_2 - \lambda x_3 &= 0 \\2x_1 + 4x_2 + \lambda x_3 &= 0 \\8x_1 - 4x_2 - 6x_3 &= 0\end{aligned}$$

13. Find the value of k for which the following system of equations

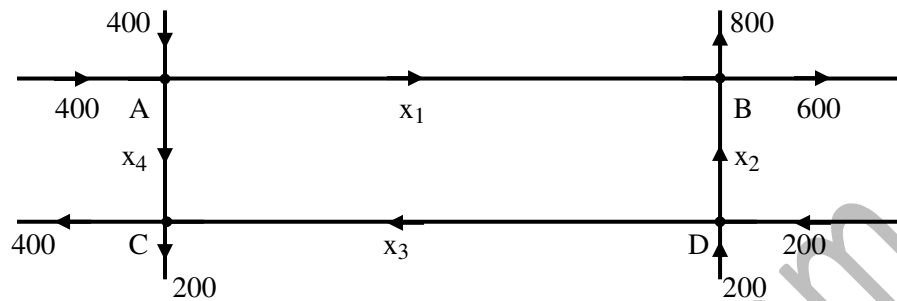
$$\begin{aligned}4x + 2y - 5z &= 0 \\x + ky + 2z &= 0 \\2x + y - z &= 0\end{aligned}$$

has a non-trivial solution.

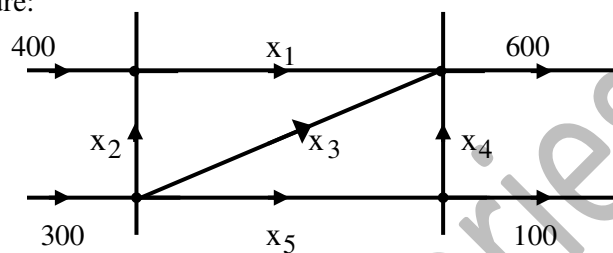
14. Determine the non-trivial solution (if any) of the following homogeneous system of linear equations:

$x_1 + 2x_2 - 3x_3 = 0$	$x_1 + x_2 - x_3 = 0$
(i) $2x_1 + 5x_2 + 2x_3 = 0$	(ii) $2x_1 - 3x_2 + x_3 = 0$
$3x_1 - x_2 - 4x_3 = 0$	$x_1 - 4x_2 + 2x_3 = 0$
$x_1 - 2x_2 + 3x_3 - 2x_4 = 0$	$x_1 + x_2 - x_3 = 0$
(iii) $3x_1 - 7x_2 - 2x_3 + 4x_4 = 0$	(iv) $2x_1 + 4x_2 - x_3 = 0$
$4x_1 + 3x_2 + 5x_3 + 2x_4 = 0$	$3x_1 + 2x_2 + 2x_3 = 0$

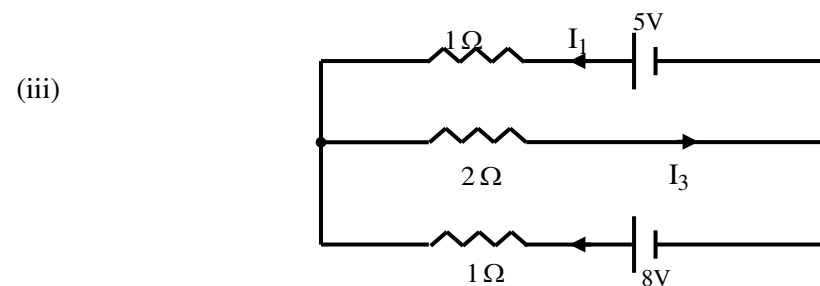
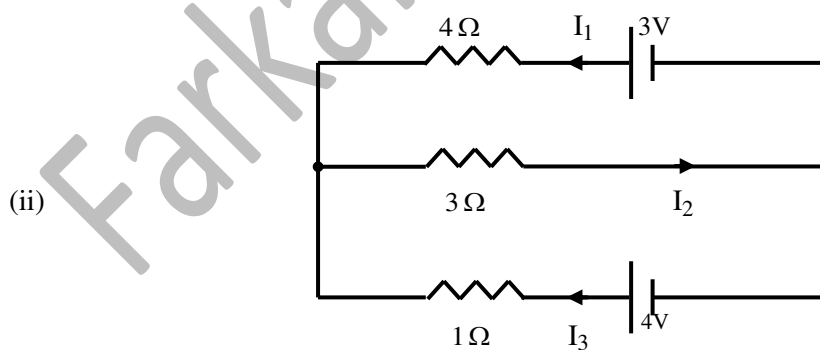
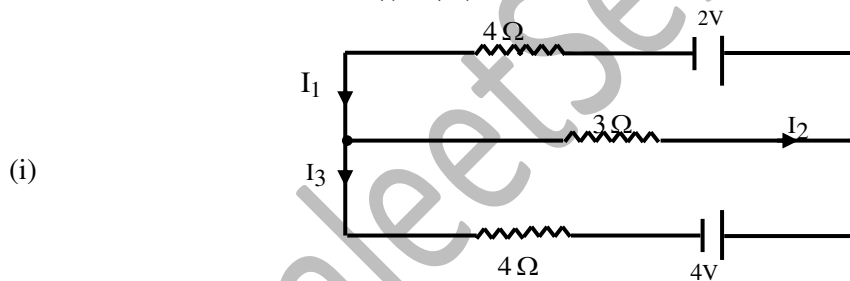
15. For the following network, (a) set up the system of equations that describes traffic flow; (b) determine the flows x_1, x_2 and x_3 if $x_4 = 100$; and (c) determine the maximum and minimum values for x_1 if all the flows are constrained to be nonnegative.



16. The flow of traffic (number of vehicles per hour) through a network of streets is shown in the following figure:



17. For the electrical networks (i) – (iii), determine the currents in the various branches:



CHAPTER THREE

DETERMINANTS

INTRODUCTION

Determinant is an important concept in matrix algebra. If a matrix is square, its elements are confined to compute a real or a complex number called the *determinant*.

Determinants have their origin in the solution of system of linear equations with two or more unknowns when $m = n$.

If a matrix $A = [a]$ is 1×1 matrix then its determinant is $|A| = a$.

If A is matrix of order 2, that is, $A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}$ then its determinant is defined as:

$$\det(A) = |A| = \begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix} = a_{11}a_{22} - a_{12}a_{21}$$

Similarly, if $A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$ is a matrix of order 3×3 then, its determinant is defined

$$\begin{aligned} \text{as, } \det(A) = |A| &= \begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix} = a_{11} \begin{vmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{vmatrix} - a_{12} \begin{vmatrix} a_{21} & a_{23} \\ a_{31} & a_{33} \end{vmatrix} + a_{13} \begin{vmatrix} a_{21} & a_{22} \\ a_{31} & a_{32} \end{vmatrix} \\ &= a_{11}(a_{22}a_{33} - a_{23}a_{32}) - a_{12}(a_{21}a_{33} - a_{23}a_{31}) + a_{13}(a_{21}a_{32} - a_{22}a_{31}) \end{aligned}$$

Example 01: Find the determinant of each of the following matrices

$$A = \begin{bmatrix} 3 & 5 \\ 6 & 12 \end{bmatrix}, \quad B = \begin{bmatrix} -4 & -3 \\ 2 & 6 \end{bmatrix}, \quad C = \begin{bmatrix} 3/2 & -4 \\ 2 & 8 \end{bmatrix}.$$

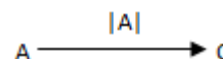
Solution: $|A| = \begin{vmatrix} 3 & 5 \\ 6 & 12 \end{vmatrix} = (3)(12) - (5)(6) = 36 - 30 = 6$

$$|B| = \begin{vmatrix} -4 & -3 \\ 2 & 6 \end{vmatrix} = (-4)(6) - (-3)(2) = -24 + 6 = -18$$

$$|C| = \begin{vmatrix} 3/2 & -4 \\ 2 & 8 \end{vmatrix} = \left(\frac{3}{2}\right)(8) - (-4)(2) = 12 + 8 = 20$$

Definition: The determinant of a square matrix A of order n (with real or complex elements) is a real or complex number associated with matrix A .

Definition: A determinant is a function whose domain is the set of all square matrices of order n and whose range is a subset of real or complex numbers say C . This is shown above.



Here D is a determinant/function, M is a square matrix whose elements are real/complex numbers and k is a constant (real or complex).

The converse is not true, that is, there are many matrices whose determinant may be same. For example,

$$\begin{bmatrix} 1 & 2 \\ 3 & 6 \end{bmatrix} = \begin{bmatrix} 1 & 3 \\ 2 & 6 \end{bmatrix} = \begin{bmatrix} 1 & 6 \\ 1 & 6 \end{bmatrix} = \begin{bmatrix} 6 & 1 \\ 6 & 1 \end{bmatrix} = \begin{bmatrix} 2 & 1 \\ 6 & 3 \end{bmatrix} \text{ etc,}$$

are different matrices but their determinant is zero. Thus determinant is many to one function.

Difference between Matrix and Determinant

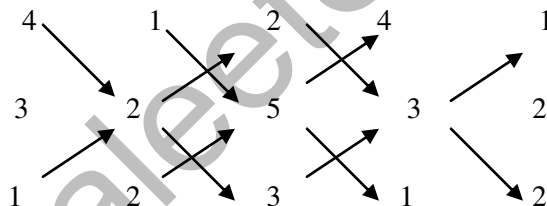
- (i) A matrix cannot be reduced to a single number whereas a determinant can be reduced to a single number.
- (ii) In a matrix the number of rows may or may not be equal to the number of columns. In a determinant the number of rows must be equal to the number of columns.
- (iii) An interchange of rows (columns) gives a different matrix. An interchange of rows (columns) gives the same determinant with opposite sign.

Example 02: If $A = \begin{bmatrix} 4 & 1 & 2 \\ 3 & 2 & 5 \\ 1 & 2 & 3 \end{bmatrix}$, then find $|A|$.

Solution: Method-I Expanding by first row, we get

$$|A| = 4 \begin{vmatrix} 2 & 5 \\ 2 & 3 \end{vmatrix} - 1 \begin{vmatrix} 3 & 5 \\ 1 & 3 \end{vmatrix} + 2 \begin{vmatrix} 3 & 2 \\ 1 & 2 \end{vmatrix} = 4(6 - 10) - 1(9 - 5) + 2(6 - 2) = -16 - 4 + 8 = -12.$$

Method-II We write the columns of A and adjoin to these the first two columns as below:



Write the six products of numbers on the directed lines taking plus sign with those products on arrows pointing downwards and negative sign with products on those arrows pointing upwards.

$$\text{Thus } |A| = [4 \cdot 2 \cdot 3 + 1 \cdot 5 \cdot 1 + 2 \cdot 3 \cdot 2] - [1 \cdot 2 \cdot 2 + 2 \cdot 5 \cdot 4 + 3 \cdot 3 \cdot 1] = (24 + 5 + 12) - (4 + 40 + 9) = 41 - 53 = -12$$

Remarks:

- (i) Method-II is useful only to evaluate a determinant of a matrix of order 3.
- (ii) Method-I is tedious and time consuming if the order of matrix is greater than 3.

Example 03: If $A = \begin{bmatrix} 3 & 2 & 1 & -4 \\ 4 & 5 & 1 & 2 \\ -2 & 3 & 0 & 1 \\ 2 & 1 & 3 & 5 \end{bmatrix}$, find $|A|$

Solution: Expanding by first row, we get

$$|A| = 3 \begin{vmatrix} 5 & 1 & 2 \\ 3 & 0 & 1 \\ 1 & 3 & 5 \end{vmatrix} - 2 \begin{vmatrix} 4 & 1 & 2 \\ -2 & 0 & 1 \\ 2 & 3 & 5 \end{vmatrix} + 1 \begin{vmatrix} 4 & 5 & 2 \\ -2 & 3 & 1 \\ 2 & 1 & 5 \end{vmatrix} - (-4) \begin{vmatrix} 4 & 5 & 1 \\ -2 & 3 & 0 \\ 2 & 1 & 3 \end{vmatrix}$$

$$= 3[5(0-3) - 1(15-1) + 2(9-0)] - 2[4(0-3) - 1(-10-2) + 2(-10-0)] \\ + [4(15-1) - 5(-10-2) + 2(-2-6)] + 4[4(9-0) - 5(-6-0) + 1(-2-6)] = 323$$

Properties of Determinants

Finding the value of a determinant by expanding is called *evaluation* of determinant. As mentioned above that evaluation of determinant of matrix of order greater than 3 is a tedious task. However, the use of certain properties of determinants makes the calculation much easier. These properties are presented below.

Property-I: If A is a square matrix, then $\det(A) = \det(A^t)$. This means that determinant of a square matrix has same value as that of its transpose.

For, if $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ then $A^t = \begin{bmatrix} a & c \\ b & d \end{bmatrix}$. or $|A| = ad - bc = |A^t|$

Property-II: If matrix B is obtained by interchanging any two rows of the square matrix A, then $\det(B) = -\det(A)$. Similarly, if a matrix C is obtained by interchanging two columns of matrix A, then $\det(C) = -\det(A)$.

For example, let, $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$, $B = \begin{bmatrix} c & d \\ a & b \end{bmatrix}$ and $C = \begin{bmatrix} b & a \\ d & c \end{bmatrix}$

Here matrix B is obtained by interchanging the 1st and 2nd rows of the matrix A. Similarly, matrix C is obtained by interchanging the 1st and 2nd columns of the matrix A.

Now, $\det(A) = ad - bc$, then $\det(B) = \det(C) = bc - ad = -(ad - bc) = -A$.

Property-III: If two rows or two columns of a square matrix A are identical, then $\det(A) = 0$.

For, let $A = \begin{bmatrix} a & b \\ a & b \end{bmatrix}$ then $|A| = ab - ab = 0$. [Here two rows of matrix A are equal]

Similarly, let $A = \begin{bmatrix} a & a \\ b & b \end{bmatrix}$ then $|A| = ab - ab = 0$. [Here two columns of matrix A are equal]

Property-IV: If all entries of a row or a column of square matrix are zero, then $\det(A) = 0$

For, let $A = \begin{bmatrix} 0 & 0 \\ a & b \end{bmatrix}$ then $|A| = 0.b - 0.a = 0$, $B = \begin{bmatrix} 0 & a \\ 0 & b \end{bmatrix}$ then $|B| = 0$

Property-V: If A is a matrix of order n and k is any scalar, then $\det(kA) = k^n \det(A)$.

For, let $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ then $kA = \begin{bmatrix} ka & kb \\ kc & kd \end{bmatrix}$ or $\det(kA) = k^2 ad - k^2 bc = k^2(ad - bc) = k^2 \det(A)$

[Note: Matrix A is of order 2]

Property-IV: Let B be a square matrix obtained by multiplying any row or column of matrix A by a non-zero number k, then $\det(B) = k \det(A)$.

For example, let $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ and $B = \begin{bmatrix} ka & kb \\ c & d \end{bmatrix}$ [B is obtained by multiplying 1st row of A by k]

Then, $\det(A) = ad - bc$ and $\det(B) = kad - kbc = k(ad - bc) = k \det(A)$

Property-VII: If each element in a row or column of a matrix is sum of two elements then its determinant can be expressed as the sum of two determinants. This is shown as under.

Let $A = \begin{bmatrix} a+w & b \\ c+z & d \end{bmatrix}$

Then,
$$\det(A) = \det \begin{bmatrix} a+w & b \\ c+z & d \end{bmatrix} = \det \begin{bmatrix} a & b \\ c & d \end{bmatrix} + \det \begin{bmatrix} w & b \\ z & d \end{bmatrix}$$

$$\text{LHS} = d(a+w) - b(c+z) = ad - bc + dw - bz$$

$$\text{RHS} = ad + dw - bc - bz = ad - bc + dw - bz$$

Property-VIII: If A is a diagonal or triangular matrix, then determinant of A is simply the product of diagonal elements. For example, if

$$A = \begin{bmatrix} a & 0 \\ c & d \end{bmatrix}, B = \begin{bmatrix} a & b \\ 0 & d \end{bmatrix}, C = \begin{bmatrix} a & 0 \\ 0 & d \end{bmatrix} \text{ then, } \det(A) = \det(B) = \det(C) = ad$$

Property-IX: The determinant of an identity matrix is 1. For example, if

$$A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \text{ then } |A| = 1.1.1 = 1.$$

Remark: The reason that an Identity matrix is called a unit matrix by some mathematicians because its determinant is unity (one). But we have said in chapter 2 that identity and unit matrices are two different concepts.

Property-X: If B is a square matrix obtained by multiplying any row/column of square matrix A and the result is added into another row/column, then $\det(B) = \det(A)$.

For example, let $A = \begin{bmatrix} 1 & 4 \\ 6 & 7 \end{bmatrix}$ or $\det(A) = 7 - 24 = -17$.

Now multiply 1st row of A by 3 and add the result in 2nd row, we get

$$B = \begin{bmatrix} 1 & 4 \\ 9 & 19 \end{bmatrix} \text{ or } \det(B) = 19 - 36 = -17, \text{ or } \det(A) = \det(B).$$

Remark: We shall present couple of examples that will make you to understand that how we use the above properties and evaluate the determinant of a square matrix.

Example 04: Compute the determinant of matrix by applying property X.

$$A = \begin{bmatrix} 5 & -3 & 2 & 3 \\ -4 & -6 & 7 & -5 \\ 3 & 5 & 6 & 9 \\ 2 & -3 & 5 & 4 \end{bmatrix}$$

Solution: $|A| = \begin{vmatrix} 5 & -3 & 2 & 3 \\ -4 & -6 & 7 & -5 \\ 3 & 5 & 6 & 9 \\ 2 & -3 & 5 & 4 \end{vmatrix} \xrightarrow{R_1+R_2} \begin{vmatrix} 1 & -9 & 9 & -2 \\ -4 & -6 & 7 & -5 \\ 3 & 5 & 6 & 9 \\ 2 & -3 & 5 & 4 \end{vmatrix} \xrightarrow{\begin{matrix} R_2+4R_1 \\ R_3-3R_1 \\ R_4-2R_1 \end{matrix}}$

$$= \begin{vmatrix} 1 & -9 & 9 & -2 \\ 0 & -42 & 43 & -13 \\ 0 & 32 & -21 & 15 \\ 0 & 15 & -13 & 8 \end{vmatrix} = \begin{vmatrix} -42 & 43 & -13 \\ 32 & -21 & 15 \\ 15 & -13 & 8 \end{vmatrix} \text{ (Expanding by } C_1)$$

$$= \begin{vmatrix} 1 & 43 & -13 \\ 11 & -21 & 15 \\ 2 & -13 & 8 \end{vmatrix} \xrightarrow{\begin{matrix} C_2-11C_1 \\ C_3-2C_1 \end{matrix}} \begin{vmatrix} 1 & 43 & -13 \\ 0 & -494 & 158 \\ 0 & -99 & 34 \end{vmatrix} \text{ Expanding by } C_1, \text{ we get}$$

$$= \begin{vmatrix} -494 & 158 \\ -99 & 34 \end{vmatrix} = (-494)(34) - (158)(-99) = -16796 + 15642 = -1154$$

Example 05: Prove that $\begin{vmatrix} a-b & b-c & c-a \\ b-c & c-a & a-b \\ c-a & a-b & b-c \end{vmatrix} = 0$

Solution: Adding second row into first row, we have

$$|A| = \begin{vmatrix} a-b+b-c & b-c+c-a & c-a+a-b \\ b-c & c-a & a-b \\ c-a & a-b & b-c \end{vmatrix} = \begin{vmatrix} a-c & b-a & c-b \\ b-c & c-a & a-b \\ c-a & a-b & b-c \end{vmatrix}$$

Adding third row into first row, we have

$$= \begin{vmatrix} a-c+c-a & b-a+a-b & c-b+b-c \\ b-c & c-a & a-b \\ c-a & a-b & b-c \end{vmatrix} = \begin{vmatrix} 0 & 0 & 0 \\ b-c & c-a & a-b \\ c-a & a-b & b-c \end{vmatrix} = 0 \quad [\cdot R_1 = 0]$$

Example 06: Prove that $|A| = \begin{vmatrix} 1 & \cos\alpha & \cos\beta \\ \cos\alpha & 1 & \cos(\alpha+\beta) \\ \cos\beta & \cos(\alpha+\beta) & 1 \end{vmatrix} = 0$

Solution: Using the trigonometric formula, we have

$$|A| = \begin{vmatrix} 1 & \cos\alpha & \cos\beta \\ \cos\alpha & 1 & \cos\alpha\cos\beta - \sin\alpha\sin\beta \\ \cos\beta & \cos\alpha\cos\beta - \sin\alpha\sin\beta & 1 \end{vmatrix}$$

Now multiply C_1 by $-\cos\alpha$ and add the result in C_2 , we have

$$= \begin{vmatrix} 1 & \cos\alpha - \cos\alpha & \cos\beta \\ \cos\alpha & 1 - \cos^2\alpha & \cos\alpha\cos\beta - \sin\alpha\sin\beta \\ \cos\beta & \cos\alpha\cos\beta - \sin\alpha\sin\beta - \cos\alpha\cos\beta & 1 \end{vmatrix}$$

$$= \begin{vmatrix} 1 & 0 & \cos\beta \\ \cos\alpha & \sin^2\alpha & \cos\alpha\cos\beta - \sin\alpha\sin\beta \\ \cos\beta & -\sin\alpha\sin\beta & 1 \end{vmatrix} \quad C_3 - \cos\beta C_1$$

$$= \begin{vmatrix} 1 & 0 & \cos\beta - \cos\beta \\ \cos\alpha & \sin^2\alpha & \cos\alpha\cos\beta - \sin\alpha\sin\beta - \cos\alpha\cos\beta \\ \cos\beta & -\sin\alpha\sin\beta & 1 - \cos^2\beta \end{vmatrix} = \begin{vmatrix} 1 & 0 & 0 \\ \cos\alpha & \sin^2\alpha & -\sin\alpha\sin\beta \\ \cos\beta & -\sin\alpha\sin\beta & \sin^2\beta \end{vmatrix}$$

Taking common from second and third column respectively, we have

$$= (\sin\alpha)(-\sin\beta) \begin{vmatrix} 1 & 0 & 0 \\ \cos\alpha & \sin\alpha & \sin\alpha \\ \cos\beta & -\sin\beta & -\sin\beta \end{vmatrix} = 0 \quad (\text{since } C_2 = C_3)$$

Example 07: Prove that $|A| = \begin{vmatrix} 1/2! & 1 & 0 \\ 1/3! & 1/2! & 1 \\ 1/4! & 1/3! & 1/2! \end{vmatrix} = 0$

Solution: Replacing factorials by their values, we have

$$= \begin{vmatrix} 1/2 & 1 & 0 \\ 1/6 & 1/2 & 1 \\ 1/24 & 1/6 & 1/2 \end{vmatrix}. \text{ Multiplying first, second and third row by 2, 6 and 24 respectively and}$$

dividing by the factor $(1/(2.6.24))$, we get

$$|A| = \frac{1}{2 \cdot 6 \cdot 24} \begin{vmatrix} 1 & 2 & 0 \\ 1 & 3 & 6 \\ 1 & 4 & 12 \end{vmatrix} = \frac{1}{2 \cdot 6 \cdot 24} \begin{vmatrix} 1 & 2 & 0 \\ 0 & 1 & 6 \\ 0 & 2 & 12 \end{vmatrix} \begin{matrix} R_2 - R_1 \\ R_3 - R_1 \end{matrix}$$

$$\text{Taking 2 common from third row, we have } |A| = \frac{2}{2 \cdot 6 \cdot 24} \begin{vmatrix} 1 & 2 & 0 \\ 0 & 1 & 6 \\ 0 & 1 & 6 \end{vmatrix} = \frac{1}{6 \cdot 24} (0) = 0 \quad [\because R_2 = R_3]$$

Example 3.1.8 Prove that $|A| = \begin{vmatrix} x & a & a & a \\ a & x & a & a \\ a & a & x & a \\ a & a & a & x \end{vmatrix} = (x-a)^3 (x+3a).$

Solution: By adding second, third and fourth rows into the first row, we have

$$|A| = \begin{vmatrix} x+3a & x+3a & x+3a & x+3a \\ a & x & a & a \\ a & a & x & a \\ a & a & a & x \end{vmatrix}$$

Taking $(x+3a)$ common from first row, we have

$$|A| = (x+3a) \begin{vmatrix} 1 & 1 & 1 & 1 \\ a & x & a & a \\ a & a & x & a \\ a & a & a & x \end{vmatrix} = (x+3a) \begin{vmatrix} 1 & 0 & 0 & 0 \\ a & x-a & 0 & 0 \\ a & 0 & x-a & 0 \\ a & 0 & 0 & x-a \end{vmatrix} \begin{matrix} C_2 - C_1 \\ C_3 - C_1 \\ C_4 - C_1 \end{matrix}$$

$$\text{Expanding by first row, we get } |A| = (x+3a) \begin{vmatrix} x-a & 0 & 0 \\ 0 & x-a & 0 \\ 0 & 0 & x-a \end{vmatrix}$$

Since the determinant is diagonal, therefore its value will be equal to the product of its diagonal elements. It implies that: $|A| = (x+3a)(x-a)(x-a)(x-a) = (x-a)^3 (x+3a) = \text{R.H.S.}$

Example 09: Prove that $\begin{vmatrix} 1+x & 1 & 1 & 1 \\ 1 & 1-x & 1 & 1 \\ 1 & 1 & 1+y & 1 \\ 1 & 1 & 1 & 1-y \end{vmatrix} = x^2 y^2$

Solution: Taking left hand side, we have

$$|A| = \begin{vmatrix} 1+x & 1 & 1 & 1 \\ 1 & 1-x & 1 & 1 \\ 1 & 1 & 1+y & 1 \\ 1 & 1 & 1 & 1-y \end{vmatrix} \begin{matrix} C_1 - C_4 \\ C_2 - C_4 \\ C_3 - C_4 \end{matrix} = \begin{vmatrix} x & 0 & 0 & 1 \\ 0 & -x & 0 & 1 \\ 0 & 0 & y & 1 \\ y & y & y & 1-y \end{vmatrix} = x \begin{vmatrix} -x & 0 & 1 \\ 0 & y & 1 \\ y & y & 1-y \end{vmatrix} - 1 \begin{vmatrix} 0 & -x & 0 \\ 0 & 0 & y \\ y & y & y \end{vmatrix}$$

(Expanding w.r.t R_1)

Now expanding these determinants, we obtain:

$$|A| = \{x[-x(y-y^2) - 0 + 1(0-y^2)]\} - \{0 + x(0-y^2) + 0\} = -x^2 y + x^2 y^2 + xy^2 = x^2 y^2.$$

Example 10: Show that
$$\begin{vmatrix} 1 & w & w^2 & w^3 \\ 1 & w^2 & w^4 & w \\ 1 & w^3 & w & w^4 \\ 1 & w^4 & w^3 & w^2 \end{vmatrix}^2 = 125, \text{ where } w \text{ is a fifth root of } 1.$$

Solution: Taking left hand side, we have

$$\begin{aligned} |A| &= \begin{vmatrix} 1 & w & w^2 & w^3 \\ 1 & w^2 & w^4 & w \\ 1 & w^3 & w & w^4 \\ 1 & w^4 & w^3 & w^2 \end{vmatrix}^2 = \begin{vmatrix} 1 & w & w^2 & w^3 \\ 1 & w^2 & w^4 & w \\ 1 & w^3 & w & w^4 \\ 1 & w^4 & w^3 & w^2 \end{vmatrix} \begin{vmatrix} 1 & w & w^2 & w^3 \\ 1 & w^2 & w^4 & w \\ 1 & w^3 & w & w^4 \\ 1 & w^4 & w^3 & w^2 \end{vmatrix} \\ &= \begin{vmatrix} 1+w+w^2+w^3 & w+w^3+1+w^2 & w^2+1+w^3+w & w^3+w^2+w+1 \\ 1+w^2+w^4+w & w+w^4+w^2+1 & w^2+w+1+w^4 & w^3+w^3+w^3+w^3 \\ 1+w^3+w+w^4 & w+1+w^4+w^3 & w^2+w^2+w^2+w^2 & w^3+w^4+1+w \\ 1+w^4+w^3+w^2 & w+w+w+w & w^2+w^3+w^4+1 & w^3+1+w^2+w^4 \end{vmatrix} \\ &= \begin{vmatrix} -w^4 & -w^4 & -w^4 & -w^4 \\ -w^3 & -w^3 & -w^3 & 4w^3 \\ -w^2 & -w^2 & 4w^2 & -w^2 \\ -w & 4w & -w & -w \end{vmatrix}, \end{aligned}$$

using the fact that sum of five 5th roots of unity is zero and their product is 1

that is, $w^5 = 1$ and $1 + w + w^2 + w^3 + w^4 = 0$.

$$|A| = (-w^4)(-w^3)(-w^2)(-w) \begin{vmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & -4 \\ 1 & 1 & -4 & 1 \\ 1 & -4 & 1 & 1 \end{vmatrix} = w^{10} \begin{vmatrix} 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & -5 \\ 1 & 0 & -5 & 0 \\ 1 & -5 & 0 & 0 \end{vmatrix} \begin{matrix} C_2 - C_1 \\ C_3 - C_1 \\ C_4 - C_1 \end{matrix}$$

Expanding by first row, we have: $|A| = w^5 w^5 \begin{vmatrix} 0 & 0 & -5 \\ 0 & -5 & 0 \\ -5 & 0 & 0 \end{vmatrix} \quad [\text{Note: Here, } w^5 = 1]$

Taking (-5) common from each row, we get: $= (1)(1)(-5)(-5)(-5) \begin{vmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{vmatrix}$

Interchanging 1st & 3rd row, we have: $= -(-125) \begin{vmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{vmatrix} = 125(1) = 125 \quad [\text{Note: } |I| = 1]$

Example 11: If 35,282; 44,759; 58,916; 80,652; and 92,469 are multiples of 13, show that

$$\begin{vmatrix} 3 & 5 & 2 & 8 & 2 \\ 4 & 4 & 7 & 5 & 9 \\ 5 & 8 & 9 & 1 & 6 \\ 8 & 0 & 6 & 5 & 2 \\ 9 & 2 & 4 & 6 & 9 \end{vmatrix} \text{ is also multiple of } 13.$$

Solution: Let, $|A| = \begin{vmatrix} 3 & 5 & 2 & 8 & 2 \\ 4 & 4 & 7 & 5 & 9 \\ 5 & 8 & 9 & 1 & 6 \\ 8 & 0 & 6 & 5 & 2 \\ 9 & 2 & 4 & 6 & 9 \end{vmatrix}$

Multiply C_4 by 10, C_3 by 100, C_2 by 1000, and adding the results in C_5 , we get

$$|A| = \begin{vmatrix} 3 & 5 & 2 & 8 & 35,282 \\ 4 & 4 & 7 & 5 & 44,759 \\ 5 & 8 & 9 & 1 & 58,916 \\ 8 & 0 & 6 & 5 & 80,652 \\ 9 & 2 & 4 & 6 & 92,469 \end{vmatrix} \text{ . Taking 13 common from } C_5 \text{ we get } |A| = 13 \begin{vmatrix} 3 & 5 & 2 & 8 & 2714 \\ 4 & 4 & 7 & 5 & 3443 \\ 5 & 8 & 9 & 1 & 4532 \\ 8 & 0 & 6 & 5 & 6204 \\ 9 & 2 & 4 & 6 & 7113 \end{vmatrix}$$

As the last determinant is multiple of 13 hence the given determinant is also multiple of 13.

Example 12: Without expanding evaluate the following determinants.

(i) $\begin{vmatrix} 1 & a & b+c \\ 1 & b & c+a \\ 1 & c & a+b \end{vmatrix}$ (ii) $\begin{vmatrix} 1 & a & a^2-bc \\ 1 & b & b^2-ca \\ 1 & c & c^2-ab \end{vmatrix}$ (iii) $\begin{vmatrix} bc & ca & ab \\ 1/a & 1/b & 1/c \\ a^2 & b^2 & c^2 \end{vmatrix}$ (iv) $\begin{vmatrix} a & a^2 & a/bc \\ b & b^2 & b/ca \\ c & c^2 & c/ab \end{vmatrix}$

Solution: (i) $\begin{vmatrix} 1 & a & b+c \\ 1 & b & c+a \\ 1 & c & a+b \end{vmatrix} C_2 + C_3 = \begin{vmatrix} 1 & a+b+c & b+c \\ 1 & a+b+c & c+a \\ 1 & a+b+c & a+b \end{vmatrix} = (a+b+c) \begin{vmatrix} 1 & 1 & b+c \\ 1 & 1 & c+a \\ 1 & 1 & a+b \end{vmatrix} = 0$

[Taking $a + b + c$ common from C_2 then see that C_1 and C_2 are equal so the value of determinant is zero]

(ii) Consider $\begin{vmatrix} 1 & a & a^2-bc \\ 1 & b & b^2-ca \\ 1 & c & c^2-ab \end{vmatrix} R_2 - R_1, R_3 - R_1 = \begin{vmatrix} 1 & a & a^2-bc \\ 0 & b-a & b^2-a^2+bc-ca \\ 0 & c-a & c^2-a^2+bc-ab \end{vmatrix}$

$$= \begin{vmatrix} 1 & a & a^2-bc \\ 1 & b & b^2-ca \\ 1 & c & c^2-ab \end{vmatrix} R_2 - R_1, R_3 - R_1 = \begin{vmatrix} 1 & a & a^2-bc \\ 0 & b-a & (b^2-a^2)+(bc-ca) \\ 0 & c-a & (c^2-a^2)+(bc-ab) \end{vmatrix}$$

$$= \begin{vmatrix} 1 & a & a^2-bc \\ 0 & b-a & (b-a)(b+a)+c(b-a) \\ 0 & c-a & (c-a)(c+a)+b(c-a) \end{vmatrix} = (b-a)(c-a) \begin{vmatrix} 1 & a & a^2-bc \\ 0 & 1 & a+b+c \\ 0 & 1 & a+b+c \end{vmatrix} = 0 \quad [\because R_2 = R_3]$$

(iii) Consider $\begin{vmatrix} bc & ca & ab \\ 1/a & 1/b & 1/c \\ a^2 & b^2 & c^2 \end{vmatrix}$. Multiply R_2 by abc and also divide by abc , we obtain:

$$= \frac{1}{abc} \begin{vmatrix} bc & ca & ab \\ bc & ca & ab \\ a^2 & b^2 & c^2 \end{vmatrix} = 0 \quad \because R_1 = R_2$$

(iv) Consider $\begin{vmatrix} a & a^2 & a/bc \\ b & b^2 & b/ca \\ c & c^2 & c/ab \end{vmatrix} = \frac{1}{abc} \begin{vmatrix} a & a^2 & a^2 \\ b & b^2 & b^2 \\ c & c^2 & c^2 \end{vmatrix} = 0 \quad [\because C_2 = C_3]$

[Here, multiply C_3 by abc and also divide by abc]

Example 13: Show that: (i)
$$\begin{bmatrix} (a+b)^2 & a^2+b^2 & ab \\ (c+d)^2 & c^2+d^2 & cd \\ (g+h)^2 & g^2+h^2 & gh \end{bmatrix} = 0$$
 (ii)
$$\begin{vmatrix} a & b & c & d & 1 \\ b & c & d & a & 1 \\ c & d & a & b & 1 \\ d & a & b & c & 1 \\ b & a & d & c & 1 \end{vmatrix} = 0$$

(iii)
$$\begin{bmatrix} a^2 & (a+1)^2 & (a+2)^2 & (a+3)^2 \\ b^2 & (b+1)^2 & (b+2)^2 & (b+3)^2 \\ c^2 & (c+1)^2 & (c+2)^2 & (c+3)^2 \\ d^2 & (d+1)^2 & (d+2)^2 & (d+3)^2 \end{bmatrix} = 0$$

Solution: (i) Consider.
$$\begin{bmatrix} (a+b)^2 & a^2+b^2 & ab \\ (c+d)^2 & c^2+d^2 & cd \\ (g+h)^2 & g^2+h^2 & gh \end{bmatrix} \quad [\text{Multiply } C_3 \text{ by 2 and add it to } C_2. \text{ Divide by 2 also}]$$

$$= \frac{1}{2} \begin{bmatrix} (a+b)^2 & a^2+b^2 & ab \\ (c+d)^2 & c^2+d^2 & cd \\ (g+h)^2 & g^2+h^2 & gh \end{bmatrix} = \begin{bmatrix} (a+b)^2 & (a+b)^2 & ab \\ (c+d)^2 & (c+d)^2 & cd \\ (g+h)^2 & (g+h)^2 & gh \end{bmatrix} = 0 \quad [\because C_1 = C_2]$$

[Note: $a^2 + 2ab + b^2 = (a+b)^2$]

(ii) Consider
$$\begin{vmatrix} a & b & c & d & 1 \\ b & c & d & a & 1 \\ c & d & a & b & 1 \\ d & a & b & c & 1 \\ b & a & d & c & 1 \end{vmatrix}$$
. Adding C_2, C_3 and C_4 in C_1 , we obtain:

$$\begin{vmatrix} a+b+c+d & b & c & d & 1 \\ a+b+c+d & c & d & a & 1 \\ a+b+c+d & d & a & b & 1 \\ a+b+c+d & a & b & c & 1 \\ a+b+c+d & a & d & c & 1 \end{vmatrix} = (a+b+c+d) \begin{vmatrix} 1 & b & c & d & 1 \\ 1 & c & d & a & 1 \\ 1 & d & a & b & 1 \\ 1 & a & b & c & 1 \\ 1 & a & d & c & 1 \end{vmatrix} = 0 \quad [\because C_1 = C_5]. \text{ Take } a+b+c+d \text{ common from } C_1$$

(iii) Consider
$$\begin{bmatrix} a^2 & (a+1)^2 & (a+2)^2 & (a+3)^2 \\ b^2 & (b+1)^2 & (b+2)^2 & (b+3)^2 \\ c^2 & (c+1)^2 & (c+2)^2 & (c+3)^2 \\ d^2 & (d+1)^2 & (d+2)^2 & (d+3)^2 \end{bmatrix}$$
. Subtract C_1 from C_2, C_3, C_4

$$\begin{bmatrix} a^2 & 2a+1 & 4a+4 & 6a+9 \\ b^2 & 2b+1 & 4b+4 & 6b+9 \\ c^2 & 2c+1 & 4c+4 & 6c+9 \\ d^2 & 2d+1 & 4d+4 & 6d+9 \end{bmatrix} = \begin{bmatrix} a^2 & 2a+1 & 4a+4 & 4 \\ b^2 & 2b+1 & 4b+4 & 4 \\ c^2 & 2c+1 & 4c+4 & 4 \\ d^2 & 2d+1 & 4d+4 & 4 \end{bmatrix}$$

Hint : Subtract C_2 and C_3 from C_4 in 1st determinant
Subtract C_4 from C_3 in 2nd determinant
& taking 4 common from C_4

$$= 4 \begin{vmatrix} a^2 & 2a+1 & 4a & 1 \\ b^2 & 2b+1 & 4b & 1 \\ c^2 & 2c+1 & 4c & 1 \\ d^2 & 2d+1 & 4d & 1 \end{vmatrix}. \text{ Now subtract } C_4 \text{ from } C_2, \text{ we get}$$

$$= 4 \begin{vmatrix} a^2 & 2a & 4a & 1 \\ b^2 & 2b & 4b & 1 \\ c^2 & 2c & 4c & 1 \\ d^2 & 2d & 4d & 1 \end{vmatrix}. \text{ Taking 2 and 4 common from } C_2 \text{ and } C_4 \text{ respectively.}$$

$$= 4.4.2 \begin{vmatrix} a^2 & a & a & 1 \\ b^2 & b & b & 1 \\ c^2 & c & c & 1 \\ d^2 & d & d & 1 \end{vmatrix} = 0 \quad [\because C_2 = C_3]$$

Example 15: Show that

$$(i) \begin{vmatrix} a & b & c & d \\ -b & a & -d & c \\ -c & d & a & -b \\ -d & -c & b & a \end{vmatrix} = (a^2 + b^2 + c^2 + d^2)^2 \quad (ii) \begin{vmatrix} (b+c)^2 & a^2 & a^2 \\ b^2 & (c+a)^2 & b^2 \\ c^2 & c^2 & (a+b)^2 \end{vmatrix} = 2abc(a+b+c)^3$$

$$(iii) \begin{vmatrix} (a^2+b^2)/c & c & c \\ a & (b^2+c^2)/a & a \\ b & b & (c^2+a^2)/b \end{vmatrix} = 4abc \quad (iv) \begin{vmatrix} 1 & 1 & 1 \\ \alpha & \beta & \gamma \\ \beta\gamma & \gamma\alpha & \alpha\beta \end{vmatrix} = (\alpha-\beta)(\beta-\gamma)(\gamma-\alpha)$$

Solution: (i) Consider $\begin{vmatrix} a & b & c & d \\ -b & a & -d & c \\ -c & d & a & -b \\ -d & -c & b & a \end{vmatrix}$ Multiply C_1, C_2, C_3 and C_4 by a, b, c and d and also dividing by $abcd$.

$$= \frac{1}{abcd} \begin{vmatrix} a^2 & b^2 & c^2 & d^2 \\ -ab & ab & -cd & cd \\ -ac & bd & ac & -bd \\ -ad & -bc & bc & ad \end{vmatrix}. \text{ Adding } C_2, C_3 \text{ and } C_4 \text{ in } C_1$$

$$= \frac{1}{abcd} \begin{vmatrix} a^2 + b^2 + c^2 + d^2 & b^2 & c^2 & d^2 \\ 0 & ab & -cd & cd \\ 0 & bd & ac & -bd \\ 0 & -bc & bc & ad \end{vmatrix}. \text{ Expanding by first column, we get}$$

$$= \frac{a^2 + b^2 + c^2 + d^2}{abcd} \begin{vmatrix} ab & -cd & cd \\ bd & ac & -bd \\ -bc & bc & ad \end{vmatrix}. \text{ Taking } a, b \text{ and } c \text{ common from } C_1, C_2, C_3$$

$$= bcd \left(\frac{a^2 + b^2 + c^2 + d^2}{abcd} \right) \begin{vmatrix} a & -d & c \\ b & a & -b \\ -b & b & a \end{vmatrix} = \left(\frac{a^2 + b^2 + c^2 + d^2}{a} \right) \begin{vmatrix} a & -d & c \\ b & a & -b \\ -b & b & a \end{vmatrix}$$

Expanding now, we get:

$$= \left(\frac{a^2 + b^2 + c^2 + d^2}{a} \right) (a^3 + ab^2 + ac^2 + ad^2) = a \left(\frac{a^2 + b^2 + c^2 + d^2}{a} \right) (a^2 + b^2 + c^2 + d^2) = (a^2 + b^2 + c^2 + d^2)^2$$

$$(ii) \begin{vmatrix} (b+c)^2 & a^2 & a^2 \\ b^2 & (c+a)^2 & b^2 \\ c^2 & c^2 & (a+b)^2 \end{vmatrix}. \text{ Subtract } C_3 \text{ from } C_1 \text{ and } C_2, \text{ we get:}$$

$$= \begin{vmatrix} (b+c)^2 - a^2 & 0 & a^2 \\ 0 & (c+a)^2 - b^2 & b^2 \\ c^2 - (a+b)^2 & c^2 - (a+b)^2 & (a+b)^2 \end{vmatrix} = \begin{vmatrix} (b+c-a)(b+c+a) & 0 & a^2 \\ 0 & (c+a-b)(c+a+b) & b^2 \\ (c-a-b)(c+a+b) & (c-a-b)(c+a+b) & (a+b)^2 \end{vmatrix}$$

Taking $(a+b+c)$ common from C_1 and C_2 , we obtain:

$$= (a+b+c)^2 \begin{vmatrix} (b+c-a) & 0 & a^2 \\ 0 & (c+a-b) & b^2 \\ (c-a-b) & (c-a-b) & (a+b)^2 \end{vmatrix}. \text{ Now expanding, we obtain}$$

$$= (a+b+c)^2 (2a^2bc + 2ab^2c + 2abc^2) = 2abc(a+b+c)^2(a+b+c) = 2abc(a+b+c)^3$$

$$\therefore (iii) \text{ Consider } \begin{vmatrix} (a^2 + b^2)/c & c & c \\ a & (b^2 + c^2)/a & a \\ b & b & (c^2 + a^2)/b \end{vmatrix} = 4abc$$

Multiply R_1 , R_2 and R_3 by c , a and b respectively and also dividing by abc .

$$= \frac{1}{abc} \begin{vmatrix} a^2 + b^2 & c^2 & c^2 \\ a^2 & b^2 + c^2 & a^2 \\ b^2 & b^2 & c^2 + a^2 \end{vmatrix}. \text{ Subtracting } C_3 \text{ from } C_2, \text{ we get}$$

$$= \frac{1}{abc} \begin{vmatrix} a^2 + b^2 & c^2 & c^2 \\ a^2 & b^2 + c^2 & a^2 \\ b^2 & b^2 & c^2 + a^2 \end{vmatrix}. \text{ Expanding, we get}$$

$$= \frac{1}{abc} \left\{ \begin{aligned} & \left[(a^2 + b^2) \left[(b^2 + c^2)(c^2 + a^2) - a^2b^2 \right] - c^2 \left[a^2(c^2 + a^2) - a^2b^2 \right] \right] \\ & + c^2 \left[a^2b^2 - b^2(b^2 + c^2) \right] \end{aligned} \right\}$$

$$= \frac{1}{abc} \left\{ \begin{aligned} & \left[(a^2 + b^2) (c^2b^2 + a^2b^2 + c^4 + c^2a^2 - a^2b^2) - c^2 (c^2a^2 + a^2 - a^2b^2) \right] \\ & + c^2 (a^2b^2 - b^4 - b^2c^2) \end{aligned} \right\}$$

$$= \frac{1}{abc} \left\{ \begin{aligned} & a^2b^2c^2 + a^4b^2 + a^2c^4 + a^4c^2 - a^4b^2 + b^4c^2 + a^2b^4 + b^2c^4 + a^2b^2c^2 + a^2b^4 \\ & - a^2c^4 - a^4c^2 + a^2b^2c^2 + a^2b^2c^2 - b^4c^2 - b^2c^4 \end{aligned} \right\}$$

$$= \frac{1}{abc} (4a^2b^2c^2) = 4abc$$

Example 16: Prove that

$$(i) \begin{vmatrix} 1 & 1 & 1 \\ \alpha & \beta & \gamma \\ \alpha^3 & \beta^3 & \gamma^3 \end{vmatrix} = (\beta - \gamma)(\gamma - \alpha)(\alpha - \beta)(\alpha + \beta + \gamma) \quad (ii) \begin{vmatrix} a^3 & 3a^2 & 3a & 1 \\ a^2 & a^2 + 2a & 2a + 1 & 1 \\ a & 2a + 1 & a + 2 & 1 \\ 1 & 3 & 3 & 1 \end{vmatrix} = (a - 1)^6$$

$$(iii) \begin{vmatrix} 1+a & 1 & 1 & 1 \\ 1 & 1+b & 1 & 1 \\ 1 & 1 & 1+c & 1 \\ 1 & 1 & 1 & 1+d \end{vmatrix} = abcd \left(1 + \frac{1}{a} + \frac{1}{b} + \frac{1}{c} + \frac{1}{d} \right)$$

$$(iv) \begin{vmatrix} a^2 & a & 1 & bcd \\ b^2 & b & 1 & acd \\ c^2 & c & 1 & abd \\ d^2 & d & 1 & abc \end{vmatrix} = (a-b)(a-c)(a-d)(b-c)(b-d)(c-d)$$

Solution: (i) Consider $\begin{vmatrix} 1 & 1 & 1 \\ \alpha & \beta & \gamma \\ \alpha^3 & \beta^3 & \gamma^3 \end{vmatrix}$ Subtract C_1 from C_2 and C_3

$$= \begin{vmatrix} 1 & 0 & 0 \\ \alpha & \beta - \alpha & \gamma - \alpha \\ \alpha^3 & \beta^3 - \alpha^3 & \gamma^3 - \alpha^3 \end{vmatrix} = \begin{vmatrix} & \beta - \alpha & \gamma - \alpha \\ (\beta - \alpha)(\beta^2 + \alpha\beta + \alpha^2) & (\gamma - \alpha)(\gamma^2 + \alpha\gamma + \alpha^2) \end{vmatrix} \text{Expand about } C_1$$

Taking $(\beta - \alpha)$ and $(\gamma - \alpha)$ common from C_1 and C_2 respectively, we obtain:

$$\begin{aligned} &= (\beta - \alpha)(\gamma - \alpha) \begin{vmatrix} 1 & 1 \\ \beta^2 + \alpha\beta + \alpha^2 & \gamma^2 + \alpha\gamma + \alpha^2 \end{vmatrix} \\ &= (\beta - \alpha)(\gamma - \alpha) [\gamma^2 + \alpha\gamma + \alpha^2 - \beta^2 - \alpha\beta - \alpha^2] = (\beta - \alpha)(\gamma - \alpha) [\gamma^2 + \alpha\gamma - \beta^2 - \alpha\beta] \\ &= (\beta - \alpha)(\gamma - \alpha) [(\gamma^2 - \beta^2) + \alpha(\gamma - \beta)] = (\beta - \alpha)(\gamma - \alpha) [(\gamma - \beta)(\gamma + \beta) + \alpha(\gamma - \beta)] \\ &= (\beta - \alpha)(\gamma - \alpha)(\gamma - \beta)[\gamma + \beta + \alpha] = [-(\alpha - \beta)][-(\beta - \gamma)][(\gamma - \alpha)](\gamma + \beta + \alpha) \\ &= (\alpha - \beta)(\beta - \gamma)(\gamma - \alpha)(\gamma + \beta + \alpha) \end{aligned}$$

$$(iii) \text{ Consider } \begin{vmatrix} a^3 & 3a^2 & 3a & 1 \\ a^2 & a^2 + 2a & 2a + 1 & 1 \\ a & 2a + 1 & a + 2 & 1 \\ 1 & 3 & 3 & 1 \end{vmatrix} \begin{matrix} R_1 - R_4 \\ R_2 - R_4 \\ R_3 - R_4 \end{matrix} = \begin{vmatrix} a^3 - 1 & 3(a^2 - 1) & 3(a - 1) & 0 \\ a^2 - 1 & a^2 + 2a - 3 & 2a - 2 & 0 \\ a - 1 & 2a - 2 & a - 1 & 0 \\ 1 & 3 & 3 & 1 \end{vmatrix}$$

$$= \begin{vmatrix} (a-1)(a^2 + a + 1) & 3(a-1)(a+1) & 3(a-1) & 0 \\ (a-1)(a+1) & (a+3)(a-1) & 2(a-1) & 0 \\ a-1 & 2(a-1) & a-1 & 0 \\ 1 & 3 & 3 & 1 \end{vmatrix} = (a-1)^3 \begin{vmatrix} a^2 + a + 1 & 3(a+1) & 3 & 0 \\ (a+1) & (a+3) & 2 & 0 \\ 1 & 2 & 1 & 0 \\ 1 & 3 & 3 & 1 \end{vmatrix}$$

$$= (a-1)^3 \begin{bmatrix} a^2 + a + 1 & 3(a+1) & 3 \\ a+1 & a+3 & 2 \\ 1 & 2 & 1 \end{bmatrix} \text{expanding by } C_4$$

$$= (a-1)^3 \{ (a^2 + a + 1)(a + 3 - 4) - 3(a+1)(a + 1 - 2) + 3(2a + 2 - a - 3) \}$$

$$\begin{aligned}
 &= (a-1)^3 \{(a^2 + a + 1)(a-1) - 3(a+1)(a-1) + 3(a-1)\} \\
 &= (a-1)^3 \{a^3 + a^2 + a - a^2 - a - 1 - 3a^2 + 3 + 3a - 3\} \\
 &= (a-1)^3 (a^3 - 3a^2 + 3a - 1) = (a-1)^3 (a-1)^3 = (a-1)^6
 \end{aligned}$$

$$\text{(iii) Consider } \begin{vmatrix} 1+a & 1 & 1 & 1 \\ 1 & 1+b & 1 & 1 \\ 1 & 1 & 1+c & 1 \\ 1 & 1 & 1 & 1+d \end{vmatrix} \begin{matrix} C_1 - C_4 \\ C_2 - C_4 \\ C_3 - C_4 \\ \end{matrix} = \begin{vmatrix} a & 0 & 0 & 1 \\ 0 & b & 0 & 1 \\ 0 & 0 & c & 1 \\ -d & -d & -d & 1+d \end{vmatrix} \text{ Expanding, we get}$$

$$\begin{aligned}
 &= a \begin{vmatrix} b & 0 & 1 \\ 0 & c & 1 \\ -d & -d & 1+d \end{vmatrix} - 1 \begin{vmatrix} 0 & b & 0 \\ 0 & 0 & c \\ -d & -d & -d \end{vmatrix} \\
 &= a\{b(c+cd+d) - 0 + 1(0+cd)\} - 1\{0 - b(0+cd) + 0\}
 \end{aligned}$$

$$= abc + abcd + abd + acd + bcd = abcd \left[1 + \frac{1}{a} + \frac{1}{b} + \frac{1}{c} + \frac{1}{d} \right]$$

$$\text{(iv) Consider } \begin{vmatrix} a^2 & a & 1 & bcd \\ b^2 & b & 1 & acd \\ c^2 & c & 1 & abd \\ d^2 & d & 1 & abc \end{vmatrix} \begin{matrix} \text{Inter change} \\ C_3 \text{ \& } C_1 \\ \end{matrix} = (-1) \begin{vmatrix} 1 & a & a^2 & bcd \\ 1 & b & b^2 & acd \\ 1 & c & c^2 & abd \\ 1 & d & d^2 & abc \end{vmatrix} \begin{matrix} R_2 - R_1 \\ R_3 - R_1 \\ R_4 - R_1 \end{matrix}$$

$$\begin{aligned}
 &= (-1) \begin{vmatrix} 1 & a & a^2 & bcd \\ 0 & b-a & b^2-a^2 & -cd(b-a) \\ 0 & c-a & c^2-a^2 & -bd(c-a) \\ 0 & d-a & d^2-a^2 & -bc(d-a) \end{vmatrix} = (b-a)(c-a)(d-a) \begin{vmatrix} 1 & a & a^2 & bcd \\ 0 & 1 & b+a & -cd \\ 0 & 1 & c+a & -bd \\ 0 & 1 & d+a & -bc \end{vmatrix} \begin{matrix} \text{Taking common} \\ (b-a) \text{ from } R_2 \\ (c-a) \text{ from } R_3 \\ (d-a) \text{ from } R_4 \end{matrix}
 \end{aligned}$$

$$\text{Expanding by } C_1, \text{ we get } = (-1)(-1)^3 (b-a)(c-a)(d-a) \begin{vmatrix} 1 & b+a & cd \\ 1 & c+a & bd \\ 1 & d+a & bc \end{vmatrix} \begin{matrix} R_2 - R_1 \\ R_3 - R_1 \end{matrix}$$

$$\begin{aligned}
 &= +(b-a)(c-a)(d-a) \begin{vmatrix} 1 & b+a & cd \\ 0 & c-b & -d(c-b) \\ 0 & d-b & -c(d-b) \end{vmatrix} \begin{matrix} \text{Taking common} \\ (c-b) \text{ from } R_2 \\ (d-b) \text{ from } R_3 \end{matrix}
 \end{aligned}$$

$$\begin{aligned}
 &= (b-a)(c-a)(d-a)(c-b)(d-b) \begin{vmatrix} 1 & b+a & cd \\ 0 & 1 & -d \\ 0 & 1 & -c \end{vmatrix} \text{ Expanding by } C_1, \text{ we get}
 \end{aligned}$$

$$\begin{aligned}
 &= (b-a)(c-a)(d-a)(c-b)(d-b)(d-c) \\
 &= [-(a-b)][-(a-c)][-(a-d)][-(b-c)][-(b-d)][-(c-d)] \\
 &= (-1)^6 (a-b)(a-c)(a-d)(b-c)(b-d)(c-d) \\
 &= (a-b)(a-c)(a-d)(b-c)(b-d)(c-d)
 \end{aligned}$$

Singular and Non-Singular Matrices

Matrix A is called *singular matrix* if $|A| = 0$ otherwise it is non-singular matrix.

Example 18: If the matrix $A = \begin{bmatrix} 2 & -3 & 5 \\ 1/2 & 4 & 6 \\ m & 0 & 8 \end{bmatrix}$ is singular find m.

Solution: The determinant of given matrix A is

$$|A| = 2 \begin{vmatrix} 4 & 6 \\ 0 & 8 \end{vmatrix} - (-3) \begin{vmatrix} 1/2 & 6 \\ m & 8 \end{vmatrix} + 5 \begin{vmatrix} 1/2 & 4 \\ m & 0 \end{vmatrix} = 2(32-0) + 3(4-6m) + 5(0-4m)$$

$= 64 + 12 - 18m - 20m = 76 - 38m$. Now, A singular therefore $|A| = 76 - 38m = 0 \Rightarrow A = 2$.

MINORS and COFACTORS

Let A be a square matrix of order n. The determinant of order $(n - 1)$ obtained from A by deleting its i^{th} row and j^{th} column is called a *Minor* corresponding to element a_{ij} . For instance, consider the matrix of order 3,

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$$

Then, $M_{11} = \begin{vmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{vmatrix}$, $M_{21} = \begin{vmatrix} a_{12} & a_{13} \\ a_{32} & a_{33} \end{vmatrix}$, $M_{31} = \begin{vmatrix} a_{12} & a_{13} \\ a_{22} & a_{23} \end{vmatrix}$, etc; are the minors corresponding

to elements a_{11} , a_{21} , and a_{31} respectively.

If A is a square matrix of order n, the *Cofactor* of an element a_{ij} is denoted by A_{ij} and is defined as:

$$A_{ij} = (-1)^{i+j} M_{ij}, \text{ where } M_{ij} \text{ is the corresponding minor.}$$

Example 01: Find minors and cofactors of each element of the following matrix

$$A = \begin{bmatrix} 1 & 2 & 0 \\ -2 & 3 & -1 \\ 0 & 0 & 5 \end{bmatrix}$$

Solution: Let M_{ij} denote the minor of the element a_{ij} . Then,

$$\begin{aligned} M_{11} &= \begin{vmatrix} 3 & -1 \\ 0 & 5 \end{vmatrix} = 15 - 0 = 15, & M_{12} &= \begin{vmatrix} -2 & -1 \\ 0 & 5 \end{vmatrix} = -10 - 0 = -10, & M_{13} &= \begin{vmatrix} -2 & 3 \\ 0 & 0 \end{vmatrix} = 0 - 0 = 0 \\ M_{21} &= \begin{vmatrix} 2 & 0 \\ 0 & 5 \end{vmatrix} = 10 - 0 = 10, & M_{22} &= \begin{vmatrix} 1 & 0 \\ 0 & 5 \end{vmatrix} = 5 - 0 = 5, & M_{23} &= \begin{vmatrix} 1 & 2 \\ 0 & 0 \end{vmatrix} = 0 - 0 = 0 \\ M_{31} &= \begin{vmatrix} 2 & 0 \\ 3 & -1 \end{vmatrix} = -2 - 0 = -2, & M_{32} &= \begin{vmatrix} 1 & 0 \\ -2 & -1 \end{vmatrix} = -1 - 0 = -1, & M_{33} &= \begin{vmatrix} 1 & 2 \\ -2 & 3 \end{vmatrix} = 3 + 4 = 7 \end{aligned}$$

Now using the definition of cofactors, we have

$$\begin{aligned} A_{11} &= (-1)^{1+1} M_{11} = 15, & A_{12} &= (-1)^{1+2} M_{12} = 10, & A_{13} &= (-1)^{1+3} M_{13} = 0 \\ A_{21} &= (-1)^{2+1} M_{21} = -10, & A_{22} &= (-1)^{2+2} M_{22} = 5, & A_{23} &= (-1)^{2+3} M_{23} = 1 \\ A_{31} &= (-1)^{3+1} M_{31} = -2, & A_{32} &= (-1)^{3+2} M_{32} = -1, & A_{33} &= (-1)^{3+3} M_{33} = 7 \end{aligned}$$

Definition: Let A be a matrix of order n given by

$$A = \begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \dots & \dots & \dots & \dots \\ a_{n1} & a_{n2} & \dots & a_{nn} \end{bmatrix}. \text{ We define determinant of matrix A by}$$

$$\det(A) = a_{i1}A_{i1} + a_{i2}A_{i2} + \dots + a_{in}A_{in} = \sum_{j=1}^n a_{ij}A_{ij} \quad \forall i, j \quad (1)$$

The expression on the right hand side of (1) is called an expansion of $\det(A)$ by cofactors. It is also known as an *Axiomatic definition of determinant* or *Laplacian Expansion*.

Example 02: Compute the determinant of matrix given below by Laplacian Expansion.

$$A = \begin{bmatrix} 2 & -1 & 3 \\ 5 & -7 & 4 \\ 1 & 6 & -2 \end{bmatrix}.$$

Solution: We expand the matrix by the second row. Then

$$|A| = a_{21}A_{21} + a_{22}A_{22} + a_{23}A_{23} = 5A_{21} + (-7)A_{22} + 4A_{23}. \text{ Now,}$$

$$A_{21} = (-1)^{2+1}M_{21} = - \begin{vmatrix} -1 & 3 \\ 6 & -2 \end{vmatrix} = -(2-18) = 16, \quad A_{22} = (-1)^{2+2}M_{22} = + \begin{vmatrix} 2 & 3 \\ 1 & -2 \end{vmatrix} = -4-3 = -7,$$

$$A_{23} = (-1)^{2+3}M_{23} = - \begin{vmatrix} 2 & -1 \\ 1 & 6 \end{vmatrix} = -(12+1) = -13. \text{ Thus, } |A| = 5(16) + (-7)(-7) + 4(-13) = 80 + 49 - 52 = 77.$$

Remark: (i) You may find the determinant of square matrix using Laplace Expansion method using any row of cofactor matrix using formula in (1).

(ii) If elements of any row are multiplied by the cofactors of the corresponding elements of a different row and these products are summed, the sum will be zero.

Adjoint Matrix

The adjoint of a square matrix A , denoted by $\text{adj}(A)$ is the transpose of matrix of cofactors. Symbolically,

$$\text{Adj}(A) = [A_{ij}]^t = [A_{ji}]$$

For example,

$$A = \begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \dots & \dots & \dots & \dots \\ a_{n1} & a_{n2} & \dots & a_{nn} \end{bmatrix}$$

The matrix of cofactors of the above matrix becomes

$$[A_{ij}] = \begin{bmatrix} A_{11} & A_{12} & \dots & A_{1n} \\ A_{21} & A_{22} & \dots & A_{2n} \\ \dots & \dots & \dots & \dots \\ A_{n1} & A_{n2} & \dots & A_{nn} \end{bmatrix}$$

or,

$$\text{Adj}(A) = [A_{ij}]^t = [A_{ji}] = \begin{bmatrix} A_{11} & A_{21} & \dots & A_{n1} \\ A_{12} & A_{22} & \dots & A_{n2} \\ \dots & \dots & \dots & \dots \\ A_{1n} & A_{2n} & \dots & A_{nn} \end{bmatrix}$$

Example 03: Find the adjoint of the following square matrices

$$(a) A = \begin{bmatrix} 1 & 1 \\ 3 & 5 \end{bmatrix} \quad (b) A = \begin{bmatrix} 1 & 3 & 5 \\ 2 & 4 & 7 \\ 9 & 8 & 6 \end{bmatrix}$$

$$\text{Solution: (a) } \text{adj}(A) = \begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{bmatrix}^t = \begin{bmatrix} 5 & -3 \\ -1 & 1 \end{bmatrix}^t = \begin{bmatrix} 5 & -1 \\ -3 & 1 \end{bmatrix}$$

Remark: It is easy to find the adjoint of square matrix of order 2. The process is to (i) Interchange the main diagonal elements. (ii) Change the signs of other two elements without changing their positions.

(b) Given that $A = \begin{bmatrix} 1 & 3 & 5 \\ 2 & 4 & 7 \\ 9 & 8 & 6 \end{bmatrix}$

$$\begin{aligned} A_{11} &= (-1)^{1+1} M_{11} = + \begin{vmatrix} 4 & 7 \\ 8 & 6 \end{vmatrix} = 24 - 56 = -32, & A_{12} &= (-1)^{1+2} M_{12} = - \begin{vmatrix} 2 & 7 \\ 9 & 6 \end{vmatrix} = (12 - 63) = 51, \\ A_{13} &= (-1)^{1+3} M_{13} = + \begin{vmatrix} 2 & 4 \\ 9 & 8 \end{vmatrix} = 16 - 36 = -20, & A_{21} &= (-1)^{2+1} M_{21} = - \begin{vmatrix} 3 & 5 \\ 8 & 6 \end{vmatrix} = -(18 - 40) = 22, \\ A_{22} &= (-1)^{2+2} M_{22} = + \begin{vmatrix} 1 & 5 \\ 9 & 6 \end{vmatrix} = 6 - 45 = -39, & A_{23} &= (-1)^{2+3} M_{23} = - \begin{vmatrix} 1 & 3 \\ 9 & 8 \end{vmatrix} = -(8 - 27) = 19, \\ A_{31} &= (-1)^{3+1} M_{31} = + \begin{vmatrix} 3 & 5 \\ 4 & 7 \end{vmatrix} = 21 - 20 = 1, & A_{32} &= (-1)^{3+2} M_{32} = - \begin{vmatrix} 1 & 5 \\ 2 & 7 \end{vmatrix} = -(7 - 10) = 3, \\ A_{33} &= (-1)^{3+3} M_{33} = + \begin{vmatrix} 1 & 3 \\ 2 & 4 \end{vmatrix} = 4 - 6 = -2. \text{ Now,} \end{aligned}$$

$$\text{Adj}(A) = \begin{bmatrix} A_{11} & A_{12} & A_{13} \\ A_{21} & A_{22} & A_{23} \\ A_{31} & A_{32} & A_{33} \end{bmatrix}^t = \begin{bmatrix} -32 & 51 & -20 \\ 22 & -39 & 19 \\ 1 & 3 & -2 \end{bmatrix}^t = \begin{bmatrix} -32 & 22 & 1 \\ 51 & -39 & 3 \\ -20 & 19 & -2 \end{bmatrix}$$

Multiplicative Inverse of a Square Matrix by Adjoint Matrix Method

We know that if matrices A and B are square matrices of order n and $AB = BA = I_n$ then matrix A is called inverse of matrix B and matrix B is called an inverse of matrix A, that is $A = B^{-1}$ and $B = A^{-1}$.

Theorem: If A is non-singular matrix, then inverse of A is given by $A^{-1} = \text{Adj}(A)/|A|$, $|A| \neq 0$.

Proof: For sake of simplicity, let us consider matrix A of order 3. Then

$$A \cdot \text{adj}(A) = \begin{bmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{bmatrix} \begin{bmatrix} A_1 & A_2 & A_3 \\ B_1 & B_2 & B_3 \\ C_1 & C_2 & C_3 \end{bmatrix}$$

Remark: The second matrix is a matrix of cofactors with transpose.

$$\begin{aligned} &= \begin{bmatrix} a_1 A_1 + b_1 B_1 + c_1 C_1 & a_1 A_2 + b_1 B_2 + c_1 C_2 & a_1 A_3 + b_1 B_3 + c_1 C_3 \\ a_2 A_1 + b_2 B_1 + c_2 C_1 & a_2 A_2 + b_2 B_2 + c_2 C_2 & a_2 A_3 + b_2 B_3 + c_2 C_3 \\ a_3 A_1 + b_3 B_1 + c_3 C_1 & a_3 A_2 + b_3 B_2 + c_3 C_2 & a_3 A_3 + b_3 B_3 + c_3 C_3 \end{bmatrix} \\ &= \begin{bmatrix} |A| & 0 & 0 \\ 0 & |A| & 0 \\ 0 & 0 & |A| \end{bmatrix}. \text{ This is by using the Laplace/axiomatic definition of the determinant.} \end{aligned}$$

Therefore, $A \cdot \text{adj}(A) = |A| \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = |A| \cdot I \text{ or } I = \frac{A \cdot \text{adj}(A)}{|A|}$

Pre-multiplying both sides by A^{-1} , we get

$$A^{-1} = \frac{\text{adj}(A)}{|A|} \text{ provided } |A| \neq 0.$$

Example 04: Find by adjoint method the inverse of the matrix $A = \begin{bmatrix} 1 & -1 & 2 \\ 2 & 1 & -1 \\ 3 & 2 & -2 \end{bmatrix}$.

Solution: Here $|A| = \begin{vmatrix} 1 & -1 & 2 \\ 2 & 1 & -1 \\ 3 & 2 & -2 \end{vmatrix} = \begin{vmatrix} 1 & -1 & 2 \\ 0 & 3 & -5 \\ 0 & 5 & -8 \end{vmatrix} \begin{matrix} R_2 - 2R_1 \\ R_3 - 3R_1 \end{matrix}$

Expanding by first column, we have $|A| = \begin{vmatrix} 3 & -5 \\ 5 & -8 \end{vmatrix} = -24 + 25 = 1 \neq 0$.

Hence A^{-1} exists. Now,

$$A_{11} = (-1)^{1+1} M_{11} = + \begin{vmatrix} 2 & -1 \\ 3 & -2 \end{vmatrix} = -2 + 2 = 0, \quad A_{12} = (-1)^{1+2} M_{12} = - \begin{vmatrix} 2 & -1 \\ 3 & -2 \end{vmatrix} = -(-4 + 3) = 1,$$

$$A_{13} = (-1)^{1+3} M_{13} = + \begin{vmatrix} 2 & 1 \\ 3 & 2 \end{vmatrix} = 4 - 3 = 1, \quad A_{21} = (-1)^{2+1} M_{21} = - \begin{vmatrix} -1 & 2 \\ 2 & -2 \end{vmatrix} = -(2 - 4) = 2,$$

$$A_{22} = (-1)^{2+2} M_{22} = + \begin{vmatrix} 1 & 2 \\ 3 & -2 \end{vmatrix} = -2 - 6 = -8, \quad A_{23} = (-1)^{2+3} M_{23} = - \begin{vmatrix} 1 & -1 \\ 3 & 2 \end{vmatrix} = -(2 + 3) = -5,$$

$$A_{31} = (-1)^{3+1} M_{31} = + \begin{vmatrix} -1 & 2 \\ 1 & -1 \end{vmatrix} = 1 - 2 = -1, \quad A_{32} = (-1)^{3+2} M_{32} = - \begin{vmatrix} 1 & 2 \\ 2 & -1 \end{vmatrix} = -(-1 - 4) = 5,$$

$$A_{33} = (-1)^{3+3} M_{33} = + \begin{vmatrix} 1 & -1 \\ 2 & 1 \end{vmatrix} = 1 + 2 = 3$$

$$\therefore \text{adj}A = \begin{bmatrix} 0 & 1 & 1 \\ 2 & -8 & -5 \\ -1 & 5 & 3 \end{bmatrix}^t = \begin{bmatrix} 0 & 2 & -1 \\ 1 & -8 & 5 \\ 1 & -5 & 3 \end{bmatrix} \Rightarrow A^{-1} = \frac{\text{adj}A}{|A|} = \begin{bmatrix} 0 & 2 & -1 \\ 1 & -8 & 5 \\ 1 & -5 & 3 \end{bmatrix}, \quad [|A| = 1]$$

CRAMER'S RULE

Gabriel Cramer (1704 - 1752) was a Swiss mathematician who published this rule in 1750. This rule is useful for finding the solution of n linear equations in n unknowns. Although the method to solve the system of non-homogeneous linear equations $Ax = b$, presented by Cramer is straight forward, nevertheless it is not recommended if the system contains more than three equations.

According to Cramer's rule the system of 2 equations in 2 variables is

$$\begin{aligned} a_1x + b_1y &= c_1 \\ a_2x + b_2y &= c_2 \end{aligned} \quad \text{has a solution if } \Delta = \begin{vmatrix} a_1 & b_1 \\ a_2 & b_2 \end{vmatrix} \neq 0, \text{ and the solution is given by}$$

$$x = \Delta_1 / \Delta, \quad y = \Delta_2 / \Delta. \quad \text{Here, } \Delta_1 = \begin{vmatrix} c_1 & b_1 \\ c_2 & b_2 \end{vmatrix} \text{ and } \Delta_2 = \begin{vmatrix} a_1 & c_1 \\ a_2 & c_2 \end{vmatrix}.$$

Similarly, consider the system of three linear equations in three variables

$$a_1x + b_1y + c_1z = d_1$$

$$a_2x + b_2y + c_2z = d_2$$

$$a_3x + b_3y + c_3z = d_3$$

The unique solution exists if $\Delta = \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix} \neq 0$ and the solution is given by

$$x = \Delta_1 / \Delta, \quad y = \Delta_2 / \Delta, \quad z = \Delta_3 / \Delta$$

where

$$\Delta_1 = \begin{vmatrix} d_1 & b_1 & c_1 \\ d_2 & b_2 & c_2 \\ d_3 & b_3 & c_3 \end{vmatrix}, \Delta_2 = \begin{vmatrix} a_1 & d_1 & c_1 \\ a_2 & d_2 & c_2 \\ a_3 & d_3 & c_3 \end{vmatrix}, \text{ and } \Delta_3 = \begin{vmatrix} a_1 & b_1 & d_1 \\ a_2 & b_2 & d_2 \\ a_3 & b_3 & d_3 \end{vmatrix}$$

Similarly, the above procedure can be extended to a system of n linear equations in n unknowns/variables.

Note: Sometimes we use D in place of Δ . Thus, if $\det(A) = |A| = D$.

Example 01: Solve the following system of linear equations by Cramer's Rule.

(i) $3x - 5y = 2$

$2x - 4y = 3$

(ii) $x_1 + x_2 + x_3 = 1, 2x_1 + 3x_2 + 4x_3 = 3, 4x_1 + 9x_2 + 10x_3 = 11$

(iii) $x_1 + x_2 + x_3 + x_4 = 6, 2x_1 - x_3 - x_4 = 4, 3x_3 + 6x_4 = 3, x_1 - x_4 = 5$

Solution: (i) Given equations are $3x - 5y = 2$ and $2x - 4y = 3$.

Let $A = \begin{bmatrix} 3 & -5 \\ 2 & -4 \end{bmatrix}$. Then, $\det(A) = D = \begin{vmatrix} 3 & -5 \\ 2 & -4 \end{vmatrix} = -12 + 10 = -2 \neq 0$

$D_1 = \begin{vmatrix} 2 & -5 \\ 3 & -4 \end{vmatrix} = -8 + 15 = 7$ and $D_2 = \begin{vmatrix} 3 & 2 \\ 2 & 3 \end{vmatrix} = 9 - 4 = 5$

Now, $x = \frac{D_1}{D} = \frac{7}{-2} = -\frac{7}{2}$ and $y = \frac{D_2}{D} = \frac{5}{-2} = -\frac{5}{2}$. Thus, $\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} -7/2 \\ -5/2 \end{bmatrix}$

(ii) System of linear equation is: $x_1 + x_2 + x_3 = 1, 2x_1 + 3x_2 + 4x_3 = 3, 4x_1 + 9x_2 + 10x_3 = 11$

$D = \begin{vmatrix} 1 & 1 & 1 \\ 2 & 3 & 4 \\ 4 & 9 & 10 \end{vmatrix} \xrightarrow{R_2 + (-2)R_1, R_3 + (-4)R_1} \begin{vmatrix} 1 & 1 & 1 \\ 0 & 1 & 2 \\ 0 & 5 & 6 \end{vmatrix} = \begin{vmatrix} 1 & 2 \\ 5 & 6 \end{vmatrix} = 6 - 10 = -4$. Also,

$D_1 = \begin{vmatrix} 1 & 1 & 1 \\ 3 & 3 & 4 \\ 11 & 9 & 10 \end{vmatrix} = 2, D_2 = \begin{vmatrix} 1 & 1 & 1 \\ 2 & 3 & 4 \\ 4 & 11 & 10 \end{vmatrix} = -8, D_3 = \begin{vmatrix} 1 & 1 & 1 \\ 2 & 3 & 3 \\ 4 & 9 & 11 \end{vmatrix} = 2$

$\therefore x_1 = \frac{D_1}{D} = \frac{2}{-4} = -\frac{1}{2}, x_2 = \frac{D_2}{D} = \frac{-8}{-4} = 2, x_3 = \frac{D_3}{D} = \frac{2}{-4} = -\frac{1}{2}$. So, $\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} -1/2 \\ 2 \\ -1/2 \end{bmatrix}$

(iii) Given system of linear equation is

$x_1 + x_2 + x_3 + x_4 = 6, 2x_1 - x_3 - x_4 = 4, 3x_3 + 6x_4 = 3, x_1 - x_4 = 5$

Here, $D = \begin{vmatrix} 1 & 1 & 1 & 1 \\ 2 & 0 & -1 & -1 \\ 0 & 0 & 3 & 6 \\ 1 & 0 & 0 & -1 \end{vmatrix} = (-1) \begin{vmatrix} 2 & -1 & -1 \\ 0 & 3 & 6 \\ 1 & 0 & -1 \end{vmatrix}$ (Expanding by C_2). Now, take 3 common from R_2

$D = -3(-1) \begin{vmatrix} 1 & 0 & -1 \\ 0 & 1 & 2 \\ 2 & -1 & -1 \end{vmatrix} \xrightarrow{R_3 + (-2)R_1} \begin{vmatrix} 1 & 0 & -1 \\ 0 & 1 & 2 \\ 0 & -1 & 1 \end{vmatrix} = 3 \begin{vmatrix} 1 & 2 \\ -1 & 1 \end{vmatrix} = 3(1 + 2) = 9$

$D_1 = \begin{vmatrix} 6 & 1 & 1 & 1 \\ 4 & 0 & -1 & -1 \\ 3 & 0 & 3 & 6 \\ 5 & 0 & 0 & -1 \end{vmatrix} = 30, D_2 = \begin{vmatrix} 1 & 6 & 1 & 1 \\ 2 & 4 & -1 & -1 \\ 0 & 3 & 3 & 6 \\ 1 & 5 & 0 & -1 \end{vmatrix} = 0,$

$$D_3 = \begin{vmatrix} 1 & 1 & 6 & 1 \\ 2 & 0 & 4 & -1 \\ 0 & 0 & 3 & 6 \\ 1 & 0 & 5 & -1 \end{vmatrix} = 39, \quad D_4 = \begin{vmatrix} 1 & 1 & 1 & 6 \\ 2 & 0 & -1 & 4 \\ 0 & 0 & 3 & 3 \\ 1 & 0 & 0 & 5 \end{vmatrix} = -15$$

$$\text{Now, } x_1 = \frac{D_1}{D} = \frac{30}{9} = \frac{10}{3}, \quad x_2 = \frac{D_2}{D} = \frac{0}{9} = 0, \quad x_3 = \frac{D_3}{D} = \frac{39}{9} = \frac{13}{3} \quad \text{and} \quad x_4 = \frac{D_4}{D} = \frac{-15}{9} = -\frac{5}{3}.$$

Hence, the required solution of the above system of equations is:

$$x_1 = 10/3, x_2 = 0, x_3 = 13/3, x_4 = -5/3.$$

Example 02: The Sum of three numbers is 6. If we multiply the third number by 2 and add the first number to the result, we get 7. By adding second and third numbers to three times the first number we get 12. Use determinants to find the numbers?

Solution: Let the three numbers be x, y and z. Then, from the given conditions, we have

$$x + y + z = 6, \quad x + 2z = 7, \quad 3x + y + z = 12. \text{ Now}$$

$$D = \begin{vmatrix} 1 & 1 & 1 \\ 1 & 0 & 2 \\ 3 & 1 & 1 \end{vmatrix} = 1 \begin{vmatrix} 0 & 2 \\ 1 & 1 \end{vmatrix} - 1 \begin{vmatrix} 1 & 2 \\ 3 & 1 \end{vmatrix} + 1 \begin{vmatrix} 1 & 0 \\ 3 & 1 \end{vmatrix} = (0-2) - (1-6) + (1-0) = 4.$$

$$D_{11} = \begin{vmatrix} 6 & 1 & 1 \\ 7 & 0 & 2 \\ 12 & 1 & 1 \end{vmatrix} = 6 \begin{vmatrix} 0 & 2 \\ 1 & 1 \end{vmatrix} - 1 \begin{vmatrix} 7 & 2 \\ 12 & 1 \end{vmatrix} + 1 \begin{vmatrix} 7 & 0 \\ 12 & 1 \end{vmatrix} = 6(0-2) - (7-24) + (7-0) = 12$$

$$D_2 = \begin{vmatrix} 1 & 6 & 1 \\ 1 & 7 & 2 \\ 3 & 12 & 1 \end{vmatrix} = 1 \begin{vmatrix} 7 & 2 \\ 12 & 1 \end{vmatrix} - 6 \begin{vmatrix} 1 & 2 \\ 3 & 1 \end{vmatrix} + 1 \begin{vmatrix} 1 & 7 \\ 3 & 12 \end{vmatrix} = 1(7-24) - 6(1-6) + 1(12-21) = 4$$

$$D_3 = \begin{vmatrix} 1 & 1 & 6 \\ 1 & 0 & 7 \\ 3 & 1 & 12 \end{vmatrix} = 1 \begin{vmatrix} 0 & 7 \\ 1 & 12 \end{vmatrix} - 1 \begin{vmatrix} 1 & 7 \\ 3 & 12 \end{vmatrix} + 6 \begin{vmatrix} 1 & 0 \\ 3 & 1 \end{vmatrix} = 1(0-7) - 1(12-21) + 6(1-0) = 8$$

$$\text{Thus, } x = \frac{D_1}{D} = \frac{12}{4} = 3, \quad y = \frac{D_2}{D} = \frac{4}{4} = 1, \quad z = \frac{D_3}{D} = \frac{8}{4} = 2$$

Hence the three numbers are: 3, 1 and 2.

Solution of Determinant Equation

We are familiar with solving different types of equations. For example, linear equation, quadratic equation, etc. In this section, we shall study the method of solving determinant equations.

Example 03: Solve the following equation for x

$$\begin{vmatrix} 1+x & 1 & 1 & 1 \\ 1 & 1+x & 1 & 1 \\ 1 & 1 & 1+x & 1 \\ 1 & 1 & 1 & 1+x \end{vmatrix} = 0.$$

Solution: Given that $\begin{vmatrix} 1+x & 1 & 1 & 1 \\ 1 & 1+x & 1 & 1 \\ 1 & 1 & 1+x & 1 \\ 1 & 1 & 1 & 1+x \end{vmatrix} = 0.$

Adding second, third and fourth row into first row, we get

$$\begin{vmatrix} x+4 & x+4 & x+4 & x+4 \\ 1 & 1+x & 1 & 1 \\ 1 & 1 & 1+x & 1 \\ 1 & 1 & 1 & 1+x \end{vmatrix} = 0. \text{ Taking } (x+4) \text{ common from first row, we get}$$

$$(x+4) \begin{vmatrix} 1 & 1 & 1 & 1 \\ 1 & 1+x & 1 & 1 \\ 1 & 1 & 1+x & 1 \\ 1 & 1 & 1 & 1+x \end{vmatrix} = 0. \text{ Subtracting first row from second, third and fourth row, we get}$$

$$(x+4) \begin{vmatrix} 1 & 1 & 1 & 1 \\ 0 & x & 0 & 0 \\ 0 & 0 & x & 0 \\ 0 & 0 & 0 & x \end{vmatrix} = 0. \text{ Expanding by } C_1, \text{ we get } (x+4) \begin{vmatrix} x & 0 & 0 \\ 0 & x & 0 \\ 0 & 0 & x \end{vmatrix} = 0 \Rightarrow x^3(x-4) = 0$$

This gives $x = 0$ or $x = 4$.

WORKSHEET 03

1. Evaluate:

$$\begin{array}{lll} \text{(a)} \begin{vmatrix} 3 & -1 \\ 2 & 4 \end{vmatrix} & \text{(b)} \begin{vmatrix} 0 & 3 \\ 5 & 7 \end{vmatrix} & \text{(c)} \begin{vmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{vmatrix} \end{array}$$

$$\begin{array}{lll} \text{(d)} \begin{vmatrix} 16 & 89 & 76 \\ 0 & 10 & 37 \\ 0 & 0 & 25 \end{vmatrix} & \text{(e)} \begin{vmatrix} 13 & 24 & 12 \\ 0 & 6 & 0 \\ 12 & 81 & 5 \end{vmatrix} & \text{(f)} \begin{vmatrix} 17 & 2 & 4 \\ 1 & 3 & 8 \\ 16 & -1 & -4 \end{vmatrix} \end{array}$$

$$\begin{array}{lll} \text{(g)} \begin{vmatrix} 6 & 13 & -2 \\ 5 & 37 & -5 \\ 1 & 13 & -2 \end{vmatrix} & \text{(h)} \begin{vmatrix} 0.5 & 0.1 & 0.8 \\ 1.5 & 0.3 & 0.6 \\ 1.0 & 0.4 & 0.2 \end{vmatrix} & \text{(i)} \begin{vmatrix} 1.6 & 0.4 & 1.2 \\ 2.2 & -0.3 & 2.5 \\ 0.4 & 0.2 & 0.2 \end{vmatrix} \end{array}$$

$$\begin{array}{lll} \text{(j)} \begin{vmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{vmatrix} & \text{(k)} \begin{vmatrix} 3 & 5 & -2 \\ 0 & 6 & 1 \\ 2 & 0 & -4 \end{vmatrix} & \text{(l)} \begin{vmatrix} 13 & 1 & 4 \\ 6 & 2 & 4 \\ 4 & 1 & 4 \end{vmatrix} \end{array}$$

$$\begin{array}{lll} \text{(m)} \begin{vmatrix} a & b & c \\ c & a & b \\ b & c & a \end{vmatrix} & \text{(n)} \begin{vmatrix} 1 & a & a^2 \\ 1 & b & b^2 \\ 1 & c & c^2 \end{vmatrix} & \text{(o)} \begin{vmatrix} b^2+c^2 & ab & ca \\ ab & c^2+a^2 & bc \\ ca & bc & a^2+b^2 \end{vmatrix} \end{array}$$

2. Without expanding show that

$$\begin{vmatrix} \sin^2 a & 1 & \cos^2 a \\ \sin^2 b & 1 & \cos^2 b \\ \sin^2 c & 1 & \cos^2 c \end{vmatrix} = 0 \quad \text{(b)} \begin{vmatrix} \sin^2 a & \cos 2a & \cos^2 a \\ \sin^2 b & \cos 2b & \cos^2 b \\ \sin^2 c & \cos 2c & \cos^2 c \end{vmatrix} = 0 \quad \text{(c)} \begin{vmatrix} \cos a & \sin a & \sin(a+d) \\ \cos b & \sin b & \sin(b+d) \\ \cos c & \sin c & \sin(c+d) \end{vmatrix} = 0$$

$$\begin{array}{lll} \text{(p)} \begin{vmatrix} 2 & 3 & -2 & 4 \\ 7 & 4 & -3 & 10 \\ 3 & 2 & 3 & 4 \\ -2 & 4 & 0 & 5 \end{vmatrix} & \text{(q)} \begin{vmatrix} 1 & -2 & 3 & -4 \\ 0 & 1 & -1 & 1 \\ 1 & 3 & 0 & -3 \\ 0 & -7 & 3 & 1 \end{vmatrix} & \text{(r)} \begin{vmatrix} 9 & 93 & 12 & -6 \\ 1 & 92 & 84 & -6 \\ 2 & 185 & 100 & -12 \\ 4 & 270 & 196 & -24 \end{vmatrix} \end{array}$$

$$\begin{array}{ll} \text{(s)} \begin{vmatrix} 1 & 1 & -1 & -1 & -1 \\ 1 & -1 & 1 & -1 & -1 \\ -1 & 1 & -1 & 1 & -1 \\ 1 & -1 & -1 & -1 & 1 \\ -1 & -1 & -1 & 1 & 1 \end{vmatrix} & \text{(t)} \begin{vmatrix} 2 & 0 & -4 & 6 \\ 4 & 5 & 1 & 0 \\ 0 & 2 & 6 & -1 \\ -3 & 8 & 9 & 1 \end{vmatrix} \end{array}$$

3. Without expansion, show that

$$(a) \begin{vmatrix} a & 1 & 1 & 1 \\ 1 & a & 1 & 1 \\ 1 & 1 & a & 1 \\ 1 & 1 & 1 & a \end{vmatrix} = (a-1)^3(a+3) \quad (b) \begin{vmatrix} \cos A & 1 & 0 \\ 0 & 2\cos A & 3 \\ 0 & 1 & 2\cos A \end{vmatrix} = \cos 3A$$

$$(c) \begin{vmatrix} \sin \alpha & \cos \alpha & 0 \\ -\sin \beta & \cos \beta & \sin \gamma \\ \cos \beta & \sin \beta & \cos \gamma \end{vmatrix} = \sin(\alpha + \beta + \gamma).$$

4. Find the inverse of the following matrices by adjoint method.

$$(a) A = \begin{bmatrix} 2 & 1 & 1 \\ 1 & 2 & 1 \\ 1 & 1 & 2 \end{bmatrix} \quad (b) B = \begin{bmatrix} -1 & 2 & -3 \\ 2 & 1 & 0 \\ 4 & -2 & 5 \end{bmatrix} \quad (c) C = \begin{bmatrix} 1 & 0 & 1 \\ 2 & 1 & 0 \\ 1 & -1 & 1 \end{bmatrix}$$

$$(d) D = \begin{bmatrix} 1 & -1 & 2 \\ 2 & 1 & 3 \\ 1 & -5 & 4 \end{bmatrix} \quad (e) E = \begin{bmatrix} 2 & -1 & 1 \\ 1 & 0 & -2 \\ 3 & 1 & 3 \end{bmatrix} \quad (f) F = \begin{bmatrix} 0 & -2 & -3 \\ 1 & 3 & 3 \\ -1 & -2 & -2 \end{bmatrix}.$$

5. Find the value of k such that the following matrices are singular:

$$(a) A = \begin{bmatrix} 2 & -3 & 5 \\ 1/2 & k & 6 \\ 3 & 2 & -1 \end{bmatrix} \quad (b) A = \begin{bmatrix} 8 & 0 & 9 \\ 5 & k & -3 \\ 2 & 1 & -5 \end{bmatrix}.$$

6. Solve for x each of the following equations:

$$(a) \begin{vmatrix} 1 & 2+x & 3 \\ 2 & 1 & 3+x \\ 3 & 2+x & 1 \end{vmatrix} = 0 \quad (b) \begin{vmatrix} 1 & 1 & 2 & 3 \\ 1 & 2-x^2 & 2 & 3 \\ 2 & 3 & 1 & 5 \\ 2 & 3 & 9 & 9-x^2 \end{vmatrix} = 0 \quad (c) \begin{vmatrix} 1 & x & x^2 & x^3 \\ 1 & 2 & 2^2 & 2^2 \\ 1 & 3 & 3^2 & 3^3 \\ 1 & 4 & 4^2 & 4^3 \end{vmatrix} = 0$$

$$(d) \begin{vmatrix} 1 & 2 & 3 & 4 & 5 \\ 1 & x+1 & 3 & 4 & 5 \\ 1 & 2 & x+1 & 4 & 5 \\ 1 & 2 & 3 & x+1 & 5 \\ 1 & 2 & 3 & 4 & x+1 \end{vmatrix} = 0 \quad (f) \begin{vmatrix} x & a & a & a & a \\ a & x & a & a & a \\ a & a & x & a & a \\ a & a & a & x & a \\ a & a & a & a & x \end{vmatrix} = 0$$

7. If possible, find the solution of the following systems of equations by using Cramer's Rule:

$$(a) \begin{cases} 5x + 3y = -1 \\ 9x + 7y = 4 \end{cases} \quad (b) \begin{cases} -4x + 5y = 3 \\ 4x - 5y = -3 \end{cases} \quad (c) \begin{cases} x + y + z = 1 \\ x - 2y - 3z = 1 \\ 3x + 2y + 4z = 5 \end{cases}$$

$$(d) \begin{cases} 3x + 2y - 5z = -1 \\ 2x - 3y - z = 11 \\ 5x - 2y + 7z = 9 \end{cases} \quad (e) \begin{cases} 4x + 2y - 9z = 2 \\ 3x + 4y + z = 5 \\ x - 3y + 2z = 8 \end{cases}$$

8. Show that if A is a non-singular matrix, then $\det(A^{-1}) = 1/\det(A)$.

9. Show that if k is a scalar and A is a square matrix of order n, then $\det(kA) = k^n \det(A)$.

CHAPTER FOUR

ANALYTICAL GEOMETRY IN R^3 EQUATION OF A LINE

4.1 INTRODUCTION

The subject of Geometry is almost as old as the human civilization. The famous Greek mathematician *Euclid* wrote many books on the subject of Geometry. Many developments in this subject occurred but in the seventeenth century, a French mathematician *Rene Descartes* (1596 – 1650 A.D) introduced the *Coordinate Geometry* that proved to be a revolution in the field of Geometry. He gave the concept of an entirely new science according to which coordinate systems and algebraic methods could be used to solve the problems of analytical geometry. He proved that all conic sections could be expressed in this new science of analytical geometry, by quadratic equations. In analytical geometry (also known as *coordinate geometry*), the geometric properties of curves and figures can be investigated by algebra. The analytical geometry can be defined as “A branch of Mathematics in which points are defined by means of a frame of reference and coordinates.”

In analytical geometry, the key step is to establish a correspondence between a real number and point on a line. This property is known as *Fundamental Theorem of Analytical Geometry*. This is done by arbitrarily designing one of the two directions along the line as a positive direction and the other as the negative direction. The positive direction is usually marked with an arrowhead as shown in figure 1; for horizontal lines the positive direction is generally taken to the right. The unit of measurement is then chosen and an arbitrary point, called the origin, is selected any where along the line. The line, the origin, the positive direction, and the unit of measurement define what is called a coordinate line or sometimes a real line. The real number corresponding to a point on the line is called the coordinate of the point. This real line is usually denoted by R_1 or simply by R .

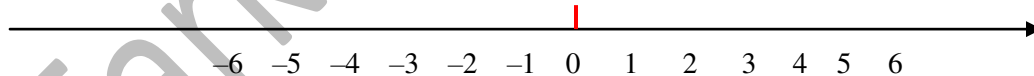


Fig. 1

If a point P is in the plane, we can locate its position with the help of two coordinates; one along the horizontal line (usually called x -axis) and one along the vertical line (usually called y -axis). These both lines intersect perpendicularly at the origin. To locate the position of the point in the plane, we first move say x units along the horizontal axis and then y units parallel to the vertical axis. The coordinates of the point P are then (x, y) .

This is shown in figure 2.

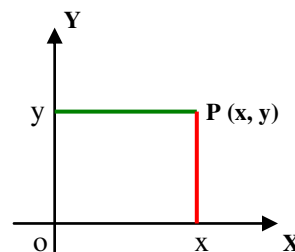


Fig. 2

The problem concerned with the geometric figures in the xy -plane is called coordinates or analytical geometry of two dimensions and is usually denoted by R^2 .

In this section we will discuss coordinate systems in three-dimensional space. In analytic geometry of two dimensions the underlying space is $R \times R = R^2$ and each point of R^2 is an ordered pair (x, y) of real numbers. In analytic geometry of three dimensions we consider the space $R \times R \times R$ or simply R^3 . Any element of R^3 is an ordered triple (x, y, z) where x, y, z are real numbers. A coordinate system in R^3 is defined as under:

We select a fixed point 'o' as an origin.

The x and y -axes are drawn mutually perpendicular through 'o' as in the case of plane analytic geometry. A line drawn through 'o' and perpendicular to both the x and y axes is taken as the z axis.

These three axes determine three mutually perpendicular planes. The plane determined by the x and y axes is the horizontal plane and is called xy -plane.

Similarly, the plane determined by the y and z axes is a vertical plane and is called yz -plane. The xz -plane is also vertical and is determined by x and z axes. These three planes are referred to as the coordinate planes (see figure 3).

Let P be any point in space. We draw through P three planes parallel to the coordinate planes. These three planes together with the coordinate planes form a box (a parallelepiped), shown in the figure 4.

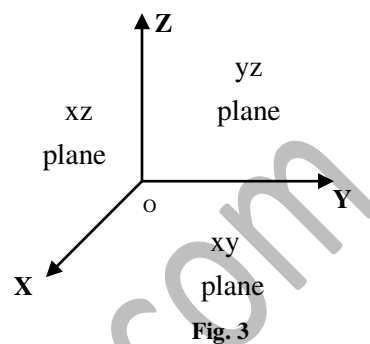


Fig. 3

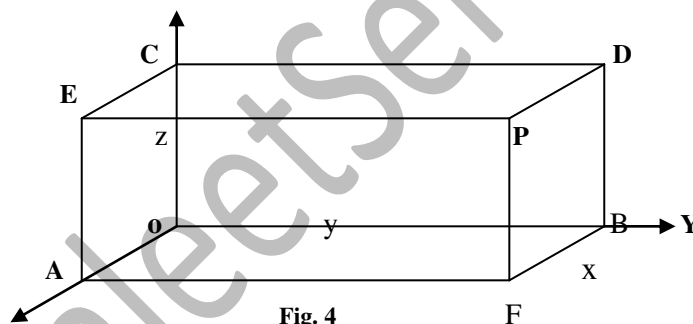


Fig. 4

The plane PEA F that is parallel to the yz plane meets the x -axis at A . Similarly the plane PDB F is parallel to the xz plane meets the y axis at B and the plane PECD is parallel the xy plane and meets the z axis at C . If the directed distance OA , OB and OC are x , y and z respectively then the point P is said to have rectangular (Cartesian) coordinates (x, y, z) and we write $P(x, y, z)$. Given an ordered triple (x, y, z) of real numbers there is a unique point P in space such that x is the directed distance of P from the yz -plane i.e $x = |DP|$, y is the directed distance of P from the xz -plane i.e., $y = |EP|$ and z is the directed distance of P from the xy -plane i.e., $z = |FP|$. Thus, there is a one-to-one correspondence between the set of points in space and the set $\{(x, y, z) : x, y, z \in R\}$ of ordered triples of real numbers. You may see the following about three-dimensional rectangular coordinate system.

Region	Description	Region	Description
xy -plane	Consists of all points of the form $(x, y, 0)$	x -axis	Consists of all points of the form $(x, 0, 0)$
xz -plane	Consists of all points of the form $(x, 0, z)$	y -axis	Consists of all points of the form $(0, y, 0)$
yz -plane	Consists of all points of the form $(0, y, z)$	z -axis	Consists of all points of the form $(0, 0, z)$

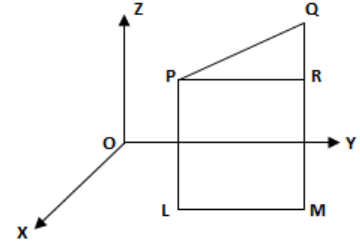
Distance between Two Points

Let $P(x_1, y_1, z_1)$ and $Q(x_2, y_2, z_2)$ be the two given points in R^3 .

Draw PL and QM perpendiculars from P and Q on the xy-plane.

Since L and M are in the xy-plane, their coordinates are (x_1, y_1) and (x_2, y_2) respectively. Then distance between two points L and M is

$$|LM| = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2} \text{ . Draw PR perpendicular on MQ.}$$



Clearly PLMR is a rectangle. Now $|RQ| = |MQ| - |MR| = |MQ| - |LP| = z_2 - z_1$

In the right-angled triangle PRQ: $|PQ|^2 = |PR|^2 + |RQ|^2$

or $|PQ|^2 = |LM|^2 + |RQ|^2$ since $|PR| = |LM|$ (opposite sides of the rectangle).

$$= (x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2$$

Hence,
$$|PQ| = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2}$$

Corollary: The distance of $P(x, y, z)$ from the origin $O(0, 0, 0)$ is $|OP| = \sqrt{x^2 + y^2 + z^2}$.

Example 01: Show that the three given points are the vertices of a right triangle, or the vertices of an isosceles triangle. (i) $A(1, 5, 0)$, $B(6, 6, 4)$, $C(0, 9, 5)$ (ii) $A(4, 9, 4)$, $B(0, 11, 2)$, $C(1, 0, 1)$

Solution: (i) Using distance formula, we have

$$|\overline{AB}| = \sqrt{(6-1)^2 + (6-5)^2 + (4-0)^2} = \sqrt{25+1+16} = \sqrt{42}$$

$$|\overline{BC}| = \sqrt{(0-6)^2 + (9-6)^2 + (5-4)^2} = \sqrt{36+9+1} = \sqrt{46}$$

$$|\overline{AC}| = \sqrt{(0-1)^2 + (9-5)^2 + (5-0)^2} = \sqrt{1+16+25} = \sqrt{42}$$

Since, $|AB| = |AC|$, therefore given vertices are the vertices of an isosceles triangle.

$$|\overline{AB}| = \sqrt{(0-4)^2 + (11-9)^2 + (2-4)^2} = \sqrt{16+4+4} = \sqrt{24}$$

(ii)
$$|\overline{BC}| = \sqrt{(1-0)^2 + (0-11)^2 + (1-2)^2} = \sqrt{1+121+1} = \sqrt{123}$$

$$|\overline{AC}| = \sqrt{(1-4)^2 + (0-9)^2 + (1-4)^2} = \sqrt{9+81+9} = \sqrt{99}$$

$$\therefore |\overline{BC}|^2 = |\overline{AB}|^2 + |\overline{AC}|^2, \text{ that is } 123 = 99 + 24 = 123$$

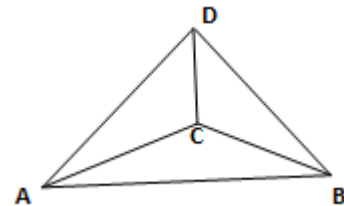
hence by Pythagoras theorem, the given vertices are the vertices of a right angled triangle having right angle at A.

Example 02: Show that the points $(1, 6, 1)$, $(1, 3, 4)$, $(4, 3, 1)$ and $(0, 2, 0)$ are the vertices of regular tetrahedron.

Solution: A solid figure bounded by four triangular faces is called a tetrahedron. A regular tetrahedron has four congruent equilateral triangles as its faces. Here we shall show that the six edges (sides) of the given tetrahedron are of equal length.

Let $A(1, 6, 1)$, $B(1, 3, 4)$, $C(4, 3, 1)$ and $D(0, 2, 0)$ then

$$|\overline{AB}| = \sqrt{(1-1)^2 + (3-6)^2 + (4-1)^2} = \sqrt{0+9+9} = \sqrt{18} = 3\sqrt{2}$$



$$|AC| = \sqrt{(4-1)^2 + (3-6)^2 + (1-1)^2} = \sqrt{9+9+0} = \sqrt{18} = 3\sqrt{2}$$

$$|AD| = \sqrt{(0-1)^2 + (2-6)^2 + (0-1)^2} = \sqrt{1+16+1} = \sqrt{18} = 3\sqrt{2}$$

$$|BC| = \sqrt{(4-1)^2 + (3-3)^2 + (1-4)^2} = \sqrt{9+0+9} = \sqrt{18} = 3\sqrt{2}$$

$$|BD| = \sqrt{(0-1)^2 + (2-3)^2 + (0-4)^2} = \sqrt{1+1+16} = \sqrt{18} = 3\sqrt{2}$$

$$|CD| = \sqrt{(0-4)^2 + (2-3)^2 + (0-1)^2} = \sqrt{16+1+1} = \sqrt{18} = 3\sqrt{2}$$

You may observe that $|AB| = |AC| = |AD| = |BC| = |BD| = |CD|$

Hence given points are the vertices of a regular tetrahedron

Coordinates of a Point Dividing a Line Segment in a Given Ratio

Let $A(x_1, y_1, z_1)$ and $B(x_2, y_2, z_2)$ be two distinct points on the line L in R^3 .

Let AB be a line segment and the point R having coordinates (x, y, z) divides the line join of AB in the ratio $m : n$. Let $O(0, 0, 0)$ be the origin, then OA , OB and OR are the position vectors which are denoted by $\mathbf{a}$, $\mathbf{b}$ and $\mathbf{c}$ respectively. It is given that

$$|AB| : |RB| = m : n \text{ or } |AR|/|BR| = m/n.$$

Using the head to tail rule of vectors, we have

$$OA + AR = OR \quad \text{This gives } \mathbf{a} + \mathbf{AR} = \mathbf{r} \quad (1)$$

The line segment AR is $m/(m+n)^{\text{th}}$ parts of AB .

$$\text{Substituting this in (1), we get } \mathbf{a} + [m/(m+n)] \mathbf{AB} = \mathbf{r} \quad (2)$$

Again using head to tail rule of vectors, we have

$$\mathbf{OA} + \mathbf{AB} = \mathbf{OB} \text{ or } \mathbf{AB} = \mathbf{b} - \mathbf{a}. \text{ Putting this value of } \mathbf{AB} \text{ in (2), we get}$$

Substituting this into (2) we get $\mathbf{a} + [m/(m+n)] (\mathbf{b} - \mathbf{a}) = \mathbf{r}$. Simplifying, we get

$$\text{Thus, } \mathbf{r} = (n\mathbf{a} + m\mathbf{b})/(m+n) \quad (3)$$

Now component-wise we can rewrite the above formula as below.

$$(x, y, z) = \left(\frac{mx_2 + nx_1}{m+n}, \frac{my_2 + ny_1}{m+n}, \frac{mz_2 + nz_1}{m+n} \right)$$

$$\text{Comparing, we have } x = \frac{mx_2 + nx_1}{m+n}, y = \frac{my_2 + ny_1}{m+n}, z = \frac{mz_2 + nz_1}{m+n}$$

These are the coordinates of the point R , that divide internally the line segment AB in the ratio $m : n$.

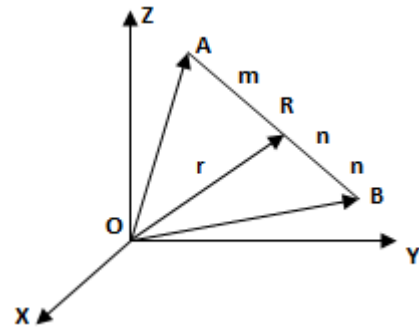
Remark:

- (i) If the point R divides the line segment AB externally in the ratio $m : n$, then

$$x = \frac{mx_2 - nx_1}{m-n}, y = \frac{my_2 - ny_1}{m-n}, z = \frac{mz_2 - nz_1}{m-n}$$

- (ii) If $R(x, y, z)$ is the midpoint of the line segment AB , then

$$(x, y, z) = \left(\frac{x_2 + x_1}{2}, \frac{y_2 + y_1}{2}, \frac{z_2 + z_1}{2} \right) \quad [\text{Since } m = n]$$



Example 04: Find the coordinates of the point dividing the join of A(-3, 1, 4) and B(5, -1, 6) internally in the ratio 3 : 5.

Solution: The coordinates of the point dividing the join of the points (x_1, y_1, z_1) and (x_2, y_2, z_2) in the ratio $m : n$ are $[x, y, z] = \left[\frac{mx_2 + nx_1}{m+n}, \frac{my_2 + ny_1}{m+n}, \frac{mz_2 + nz_1}{m+n} \right]$

Hence the required point dividing the join of given points in the ratio 3: 5 has coordinates

$$\left(\frac{5(-3) + 3(5)}{3+5}, \frac{3(-1) + 5(1)}{3+5}, \frac{3(4) + 5(6)}{3+5} \right) = \left(0, \frac{1}{4}, \frac{19}{4} \right)$$

Example 4.1.4 Find the ratio in which the yz -plane divides the segment joining the points B(3, -5, 8) and A(-2, 4, 7). Also find the coordinates of the point.

Solution: Let the yz -plane divide the join of the given points in $m : n$ ratio at point say R.

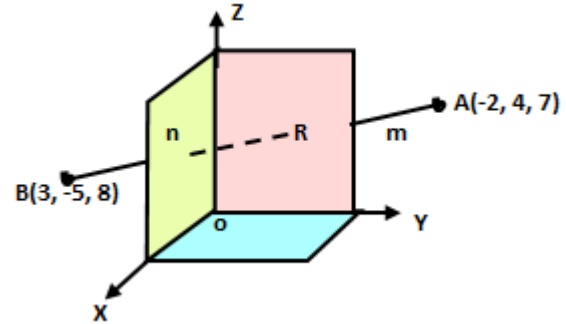
The x -coordinate of the point R, which divides the join of the given points, is $x = (3m - 2n) / (m + n)$.

As the yz -plane divides the line segment, its x -component must be zero. Hence,

$$3m - 2n = 0 \text{ giving } 3m = 2n \text{ or } m/n = 2/3 \text{ or } m : n = 2 : 3$$

This is the required ratio in which yz -plane divides the line segment. Hence the coordinates of the

$$\text{point R are: } \left(0, \frac{4(3) - 5(2)}{3+2}, \frac{7(3) + 8(2)}{3+2} \right) = \left(0, \frac{2}{5}, \frac{37}{5} \right)$$



STRAIGHT LINES in R^3

Readers are familiar with the equation of straight line in R^2 plane which is given by $y = mx + c$. We also know that if this line passes through two points (x_1, y_1) and (x_2, y_2) then the equation is

$$\frac{y - y_1}{y_2 - y_1} = \frac{x - x_1}{x_2 - x_1}$$

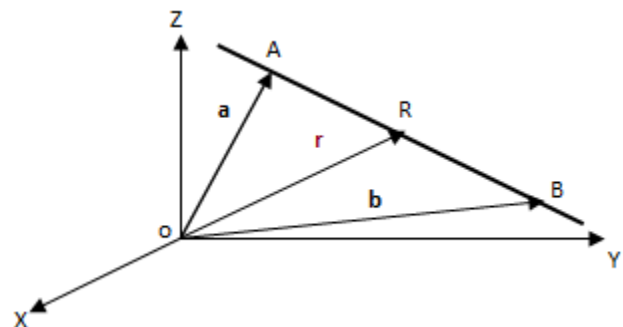
This form of equation of line is known as *Symmetric Form or Two Point Form* of straight line. In this section, we shall learn how to find an equation of a line in R^3 .

Definition: A straight line in R^3 is completely determined if it passes through two fixed points or it passes through a fixed point and is parallel to a given vector.

Let $A(x_1, y_1, z_1)$ and $B(x_2, y_2, z_2)$ be fixed points on the line L. In other words, AB is a line segment of L. Suppose $R(x, y, z)$ be any point on AB. Then,

$\mathbf{AR} = t \mathbf{AB}$, since vectors $\mathbf{AR}$ and $\mathbf{AB}$ are collinear, t being a real number called *parameter*. We let $\mathbf{a} = [a, b, c]$ and also suppose that vector $\mathbf{AB} = \mathbf{d}$. Then from the figure, using the vector law of addition, we have: $\mathbf{OR} = \mathbf{OA} + \mathbf{AR}$ or

$$\mathbf{r} = \mathbf{a} + t \mathbf{AB} = \mathbf{a} + t (\mathbf{b} - \mathbf{a}) = \mathbf{a} + t \mathbf{d} \quad (1)$$



Which; is equation of a line L in vector form.

Now $\mathbf{AR} = (x - x_1)\mathbf{i} + (y - y_1)\mathbf{j} + (z - z_1)\mathbf{k}$

Thus equation (1) becomes: $(x - x_1)\mathbf{i} + (y - y_1)\mathbf{j} + (z - z_1)\mathbf{k} = t(a\mathbf{i} + b\mathbf{j} + c\mathbf{k})$

Equating the components of $\mathbf{i}$, $\mathbf{j}$ and $\mathbf{k}$, we get

$$\left. \begin{aligned} x - x_1 &= at \\ y - y_1 &= bt \\ z - z_1 &= ct \end{aligned} \right\} \quad (2)$$

or $x = x_1 + at, y = y_1 + bt, z = z_1 + ct$

These are known as *Parametric Equations* of the line L where t is a parameter.

Now from (2) we have

$$\frac{x - x_1}{a} = t, \frac{y - y_1}{b} = t, \frac{z - z_1}{c} = t \quad \text{OR} \quad \frac{x - x_1}{a} = \frac{y - y_1}{b} = \frac{z - z_1}{c} = t$$

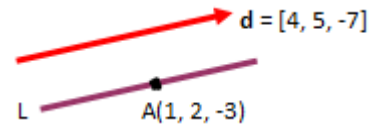
This is known as *symmetric form* of equation of line. The components of vector $\mathbf{d} = \mathbf{AB} = [a, b, c]$ are called *direction numbers* or *direction ratios* of line L .

Example 01: Find the parametric equations for the straight line through the point $A(1, 2, -3)$ and parallel to the vector $\mathbf{d} = [4, 5, -7]$.

Solution: Here $x_1 = 1, y_1 = 2$ and $z_1 = -3$ and $a = 4, b = 5, c = -7$

Using the formula: $x = x_1 + at, y = y_1 + bt, z = z_1 + ct$

the equation of straight line is: $x = 1 + 4t, y = 2 + 5t, z = -3 - 7t$.

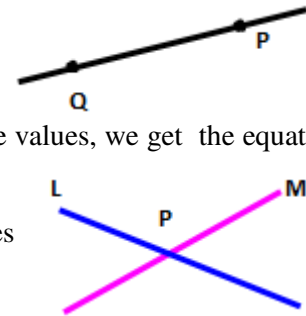


Example 02: Find the parametric equations of the straight line through the points $P(1, -5, 1)$ and $Q(4, -5, 4)$.

Solution: (First Method)

Here $\mathbf{PQ} = \mathbf{d} = [4 - 1, -5 + 5, 4 - 1] = [3, 0, 3]$. Using the formula

$x = x_1 + at, y = y_1 + bt, z = z_1 + ct$, through point P and substituting the values, we get the equation of line: $x = 1 + 3t, y = -5 + 0t, z = 1 + 3t$.



Example 03: Find the point of intersection (if it exists) of the pair of lines L and M given by

L: Through $A(2, -1, 0)$ and parallel to the vector $\mathbf{b} = [4, 3, -2]$

M: Through $B(-1, 3, 5)$ and parallel to the vector $\mathbf{c} = [1, 7, 3]$

Solution: Parametric equations of the lines are

$$L: \begin{cases} x = 2 + 4t \\ y = -1 + 3t \\ z = -2t \end{cases} \quad \text{and} \quad M: \begin{cases} x = -1 + s \\ y = 3 + 7s \\ z = 5 + 3s \end{cases}$$

The condition for L and M to intersect at P is that equations

$$2 + 4t = -1 + s, -1 + 3t = 3 + 7s, -2t = 5 + 3s$$

$$\text{Or } 4t - s + 3 = 0 \quad (i) \quad 3t - 7s - 4 = 0 \quad (ii) \quad 2t + 3s + 5 = 0 \quad (iii)$$

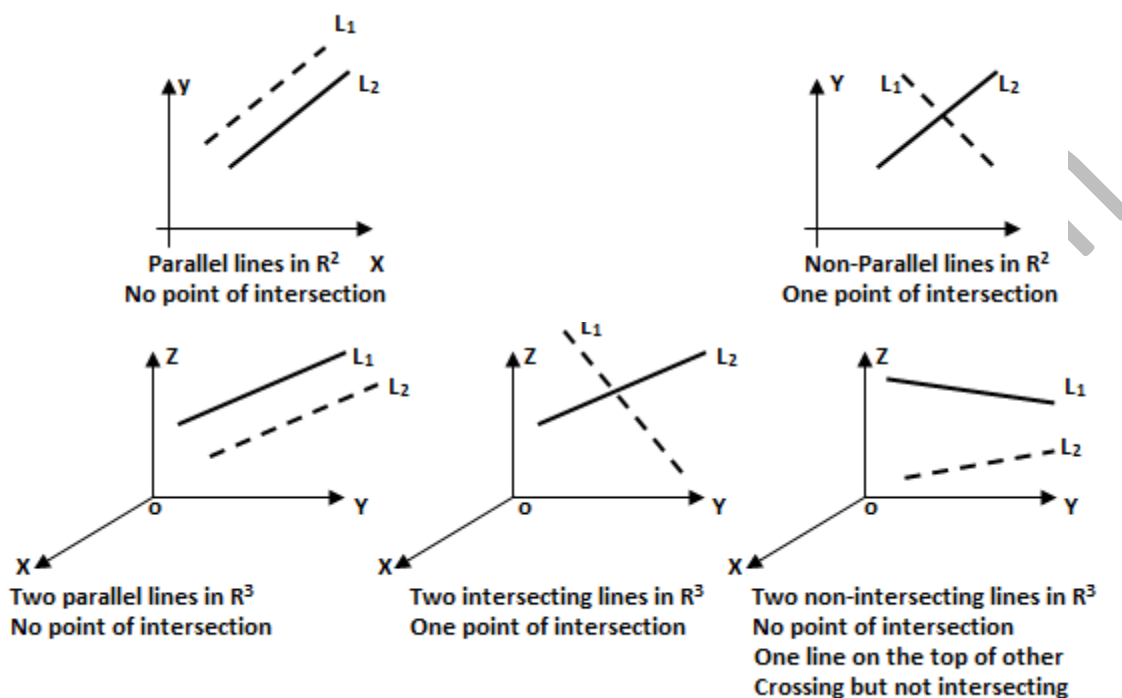
must have a unique solution. Now multiplying (i) by 3 and adding into (iii), we get

$$12t - 3s + 9 = 0 \quad \& \quad 2t + 3s + 5 = 0. \text{ Solving, we get } 14t + 14 = 0 \text{ or } t = -1. \text{ From, (1), we get } s = -1.$$

Since these values of t and s also satisfy equation (ii), therefore L and M intersect. Now substituting $t = -1$ in the equation of line L , we get $(x, y, z) = P(-2, -4, 2)$.

The point P can also be obtained from the equation of line M by substituting $s = -1$ in the equations for line M.

Remark: In R^2 if lines do not intersect then they must be parallel. But in R^3 if two lines do not intersect they may or may not be parallel. If the lines do not intersect and are not parallel they are known as *Skew Lines*. This is shown in figures below.



Example 04: Let L_1 and L_2 be the lines

$$\left. \begin{aligned} L_1 : x = 1 + 4t, y = 5 - 4t, z = -1 + 5t \\ L_2 : x = 2 + 8s, y = 4 - 3s, z = 5 + s \end{aligned} \right\} \text{ where } t \text{ \& } s \text{ are parameters of lines } L_1 \text{ \& } L_2.$$

(a) Are the lines parallel? (b) Do the lines intersect?

Solution: (a) The direction ratios of line L_1 are $[4, -4, 5]$ and that of line L_2 are $[8, -3, 1]$. These vectors are not equal nor multiple of each other. Hence, two lines are not parallel.

(b) If the lines L_1 and L_2 intersect each other then we must have

$$1 + 4t = 2 + 8s \Rightarrow 4t - 8s = 1 \quad (i)$$

$$5 - 4t = 4 - 3s \Rightarrow -4t + 3s = -1 \quad (ii)$$

$$-1 + 5t = 5 + s \Rightarrow 5t - s = 6 \quad (iii)$$

Adding first two equations yields $-5s = 0$ giving $s = 0$. Substituting this value of s in (i), we get $t = 1/4$. Now, put $s = 0$ and $t = 1/4$ in (iii) we get $5 = 6$ which is not true. Hence the two lines do not intersect each other. Hence the lines are neither parallel nor intersecting, which means that the given lines are skew lines.

Direction Ratios and Direction Cosines

In this section we shall discuss two important concepts of analytical geometry of R^3 . These are (i) Direction Ratios (ii) Direction Cosines.

Direction Ratios

The components of a vector which is parallel to line L or acting along the line are called direction ratios of line L.

Direction Cosines

The components of unit vector parallel to line L or acting along the line are called direction cosines of line L.

Remarks: (i) Unit vector is that whose magnitude is 1. (ii) If a vector is divided by its own magnitude a new vector is so formed is called unit vector. (iii) Direction cosines of x-axis, y-axis and z-axis respectively are $\mathbf{i} = [1, 0, 0]$, $\mathbf{j} = [0, 1, 0]$, $\mathbf{k} = [0, 0, 1]$ respectively.

Now why the components of unit vector are called direction cosines? Let us see this.

Consider a line L whose direction cosines are l, m and n . Let $\mathbf{u} = [l, m, n]$ be a unit vector. If α is the angle that line L makes with positive direction of x-axis then

$$\mathbf{u} \cdot \mathbf{i} = 1 \cdot \cos \alpha \text{ or } [l, m, n] \cdot [1, 0, 0] = \cos \alpha \text{ or } l(1) + m(0) + n(0) = \cos \alpha \text{ or } \cos \alpha = l.$$

If β is the angle that line L makes with positive direction of y-axis then

$$\mathbf{u} \cdot \mathbf{j} = 1 \cdot \cos \beta \text{ or } [l, m, n] \cdot [0, 1, 0] = \cos \beta \text{ or } l(0) + m(1) + n(0) = \cos \beta \text{ or } \cos \beta = m.$$

Similarly, if L makes an angle γ with positive direction of z-axis then

$$\mathbf{u} \cdot \mathbf{k} = 1 \cdot \cos \gamma \text{ or } [l, m, n] \cdot [0, 0, 1] = \cos \gamma \text{ or } l(0) + m(0) + n(1) = \cos \gamma \text{ or } \cos \gamma = n.$$

$$\text{Thus, } \mathbf{u} = [l, m, n] = [\cos \alpha, \cos \beta, \cos \gamma]$$

Now $\mathbf{u}$ is a unit vector hence $|\mathbf{u}| = 1$. This implies that

$$l^2 + m^2 + n^2 = 1 \text{ or } \cos^2 \alpha + \cos^2 \beta + \cos^2 \gamma = 1$$

The angles α, β and γ are called *direction angles* of line L.

Remark: The dot product between two vectors $\mathbf{a} = [a_1, a_2, a_3]$ and $\mathbf{b} = [b_1, b_2, b_3]$ is given by:

$$\mathbf{a} \cdot \mathbf{b} = a_1 b_1 + a_2 b_2 + a_3 b_3.$$

Example 05: What are the direction cosines of the coordinate axes?

Solution: Angle of x – axis along itself is 0° , therefore $\cos \alpha = \cos 0^\circ = 1$. Thus, its direction cosines are $[1, 0, 0]$.

Similarly, the direction cosines of y-axis and z – axis are $[0, 1, 0]$ and $[0, 0, 1]$ respectively.

Example 06: Find the direction ratios, direction cosines and measures of the direction angles of the straight line through P(1, -2, 0) and Q(5, -10, 1).

Solution: Let PQ be the segment of the line L then direction ratios of line L are $[5 - 1, -10 + 2, 1 - 0] = [4, -8, 1] = \mathbf{d}$. Now, $|\mathbf{d}| = \sqrt{(4)^2 + (-8)^2 + (1)^2} = 9$. Hence,

$$\cos \alpha = \frac{4}{\sqrt{(4)^2 + (-8)^2 + (1)^2}} = \frac{4}{9}; \cos \beta = \frac{-8}{\sqrt{(4)^2 + (-8)^2 + (1)^2}} = -\frac{8}{9}; \cos \gamma = \frac{1}{\sqrt{(4)^2 + (-8)^2 + (1)^2}} = \frac{1}{9}$$

Therefore, the direction cosines are: $\cos \alpha = 4/9$, $\cos \beta = -8/9$, $\cos \gamma = 1/9$. Now,

$$\cos \alpha = 4/9 \text{ or } \alpha = \cos^{-1}(4/9) = 63.6^\circ; \cos \beta = -8/9 \text{ or } \beta = \cos^{-1}(-8/9) = 152.74^\circ,$$

$$\cos \gamma = 1/9 \text{ or } \gamma = \cos^{-1}(1/9) = 83.62^\circ. \text{ Thus, } \alpha = 63.6^\circ, \beta = 152.74^\circ \text{ and } \gamma = 83.62^\circ$$

These are the direction angles of the line L.

Remark: A line has infinite sets of direction ratios as it passes through infinite number of different point, but these direction ratios are multiple of each other.

Example 07: Prove that if α , β and γ are the direction angles of a straight line, then $\sin^2 \alpha + \sin^2 \beta + \sin^2 \gamma = 2$.

Solution: We know that if α , β and γ are the direction angles of a line, then,

$$\cos^2 \alpha + \cos^2 \beta + \cos^2 \gamma = 1$$

This implies that, $1 - \sin^2 \alpha + 1 - \sin^2 \beta + 1 - \sin^2 \gamma = 1$ or $\sin^2 \alpha + \sin^2 \beta + \sin^2 \gamma = 3 - 1 = 2$.

Example 08: If two of the direction angles of a straight line are 45° and 60° , find the third direction angle?

Solution: Let $\alpha = 45^\circ$ and $\beta = 60^\circ$. Since α , β and γ are the direction angles of a straight line, we have:

$$\cos^2 \alpha + \cos^2 \beta + \cos^2 \gamma = 1 \quad \text{or} \quad \cos^2 45^\circ + \cos^2 60^\circ + \cos^2 \gamma = 1$$

Or, $\cos^2 \gamma = 1 - (1/2)^2 - (\sqrt{3}/2)^2 = 1 - 1/2 - 3/4 = 1/4$ or $\gamma = \cos^{-1}(1/2) = 60^\circ$ or $\gamma = 120^\circ$.

Parallel and Perpendicular Two Lines

Let L_1 and L_2 be two lines with equations

$$L: \frac{x-x_1}{c_1} = \frac{y-y_1}{c_2} = \frac{z-z_1}{c_3} \quad \text{and} \quad M: \frac{x-x_1}{d_1} = \frac{y-y_1}{d_2} = \frac{z-z_1}{d_3}$$

Here $\mathbf{c} = [c_1, c_2, c_3]$ and $\mathbf{d} = [d_1, d_2, d_3]$. If L_1 and L_2 are parallel then the ratio between their direction ratios/direction cosines must be same. That is,

$$\frac{c_1}{d_1} = \frac{c_2}{d_2} = \frac{c_3}{d_3} = \frac{l_1}{l_2} = \frac{m_1}{m_2} = \frac{n_1}{n_2} = k$$

Above equations can also be expressed as $\mathbf{c} = k\mathbf{d}$ where k is some constant/scalar. The condition that two lines are parallel is also true if $\mathbf{c} \times \mathbf{d} = \mathbf{0}$. If the lines L_1 and L_2 are perpendicular to each other then

$$\mathbf{c} \cdot \mathbf{d} = 0 \quad \text{or} \quad c_1 d_1 + c_2 d_2 + c_3 d_3 = 0 \quad \text{or} \quad l_1 l_2 + m_1 m_2 + n_1 n_2 = 0$$

Example 09: Are the straight lines

$$\frac{x-1}{2} = \frac{y+1}{3} = \frac{z}{-5} \quad \text{and} \quad \frac{x+3}{-1} = \frac{y-2}{4} = \frac{z-1}{2}$$

(i) Parallel? (ii) Perpendicular?

Solution: The direction ratios of given lines are $\mathbf{c} = [2, 3, -5]$ and $\mathbf{d} = [-1, 4, 2]$.

Since $\mathbf{c}$ and $\mathbf{d}$ are not multiple of each other hence given lines are not parallel.

Now, $\mathbf{c} \cdot \mathbf{d} = 2(-1) + 3(4) + (-5)(2) = -2 + 12 - 10 = 0$

Since $\mathbf{c} \cdot \mathbf{d} = 0$, this shows that the two lines L_1 and L_2 are perpendicular.

Co-linearity of Three Points

We know that the equation of a line passing through two points (x_1, y_1, z_1) and (x_2, y_2, z_2) is given by

$$\left. \frac{x-x_1}{x_2-x_1} = \frac{y-y_1}{y_2-y_1} = \frac{z-z_1}{z_2-z_1} = t \right\}$$

$$\text{or} \quad x = x_1 + t(x_2 - x_1), y = y_1 + t(y_2 - y_1), z = z_1 + t(z_2 - z_1) \quad (1)$$

Now if this line also passes through the point (x_3, y_3, z_3) , we must have

$$x_3 = x_1 + t(x_2 - x_1), y_3 = y_1 + t(y_2 - y_1), z_3 = z_1 + t(z_2 - z_1) \quad (2)$$

Eliminating t between (1) and (2), we get

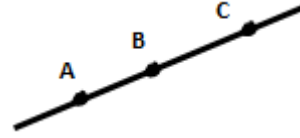
$$\begin{vmatrix} x_1 & y_1 & z_1 \\ x_2 & y_2 & z_2 \\ x_3 & y_3 & z_3 \end{vmatrix} = 0$$

This is the condition that three points are collinear, that is; the points lie on the same line.

Example 10: Show that the points A(3, 2, 4), B(1, 3, 2) and C(5, 1, 6) are collinear.

Solution: Method-I. We know that if three points are collinear, then

$$\begin{vmatrix} x_1 & y_1 & z_1 \\ x_2 & y_2 & z_2 \\ x_3 & y_3 & z_3 \end{vmatrix} = 0.$$



$$\text{Now, } \begin{vmatrix} x_1 & y_1 & z_1 \\ x_2 & y_2 & z_2 \\ x_3 & y_3 & z_3 \end{vmatrix} = \begin{vmatrix} 3 & 2 & 4 \\ 1 & 3 & 2 \\ 5 & 1 & 6 \end{vmatrix} = 3(18-2) - 2(6-10) + 4(1-15) = 48 + 8 - 56 = 0$$

Since given condition is satisfied, hence the points are collinear.

Method II: The direction ratios of the line segment AB are $[1-3, 3-2, 2-4] = [-2, 1, -2] = \mathbf{c}$. Similarly, the direction ratios of the line segment BC are $[5-1, 1-3, 6-2] = [4, -2, 4] = \mathbf{d}$. You may observe that $\mathbf{d} = -2\mathbf{c}$, hence the points A, B and C are collinear.

Angle between Two Straight Lines

If two lines are coplanar but are not parallel, they will intersect at a certain angle. Let l_1, m_1, n_1 be the direction cosines of a straight line L and l_2, m_2, n_2 be the direction cosines of a straight line M. if $\theta \in [0, \pi]$ be the measure of angle between lines L and M then it is also the measure of the angle between the direction vectors $\mathbf{c}$ and $\mathbf{d}$. Thus, $\mathbf{c} = l_1\mathbf{i} + m_1\mathbf{j} + n_1\mathbf{k}$ and $\mathbf{d} = l_2\mathbf{i} + m_2\mathbf{j} + n_2\mathbf{k}$, then

$$\mathbf{c} \cdot \mathbf{d} = |\mathbf{c}| |\mathbf{d}| \cos \theta \text{ or } \mathbf{c} \cdot \mathbf{d} = \cos \theta$$

NOTE: Vectors $\mathbf{c}$ and $\mathbf{d}$ are unit vectors, hence $|\mathbf{c}| = 1 = |\mathbf{d}|$.

$$\theta = \cos^{-1}(\mathbf{c} \cdot \mathbf{d}) = \cos^{-1}(l_1 l_2 + m_1 m_2 + n_1 n_2) \quad (1)$$

If c_1, c_2, c_3 and d_1, d_2, d_3 are the direction ratios of the line L and M respectively, then

$$\theta = \cos^{-1} \left(\frac{c_1 d_1 + c_2 d_2 + c_3 d_3}{\sqrt{c_1^2 + c_2^2 + c_3^2} \sqrt{d_1^2 + d_2^2 + d_3^2}} \right) \quad (2)$$

Example 11: Find the angle between straight lines

$$L: \frac{x-2}{3} = \frac{y+1}{-2} = \frac{z}{2} \text{ and } M: \frac{x-3}{-2} = \frac{y-6}{2} = \frac{z-4}{-1}.$$

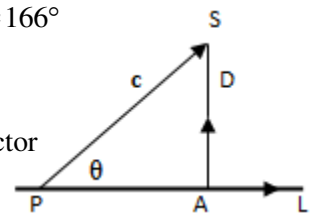
Solution: Here $\mathbf{c} = [3, -2, 2]$ and $\mathbf{d} = [-2, 2, -1]$ are the direction vectors of lines L and M respectively. If θ is an angle between them then using the formula (2), we have

$$\theta = \cos^{-1} \left[\frac{(3)(-2) + (-2)(2) + (2)(-1)}{\sqrt{9+4+4} \sqrt{4+4+1}} \right] = \cos^{-1} \left[\frac{-12}{3\sqrt{17}} \right] \approx 166^\circ$$

Distance from a Point to a Line

Consider a line L that passes through the point $P(x_1, y_1, z_1)$ with direction vector

$\mathbf{c} = [c_1, c_2, c_3]$. Let S (x_0, y_0, z_0) be the point from where we have to find the



length of perpendicular $|SA| = D$ to the line (see the adjacent figure).

$$\text{Now, } \sin \theta / |AS| / |PS| \text{ or } |AS| = D = |PS| \sin \theta = \frac{|PS| |c| \sin \theta}{|c|} = \frac{|PS \times c|}{|c|}$$

This is formula of distance between point S and the line L.

Example 12: Find the distance from the point S(1, 1, 5) to the line L: $x = 1 + t$, $y = 3 - t$, $z = 2t$.

Solution: From the equations of line L we observe that L passes through P(1, 3, 0) and is parallel to $c = [1, -1, 2]$. Thus $PS = [1 - 1, 1 - 3, 5 - 0] = [0 - 2, 5]$. Now,

$$\overline{PS} \times c = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 0 & -2 & 5 \\ 1 & -1 & 2 \end{vmatrix} = (-4 + 5)\hat{i} - (0 - 5)\hat{j} + (0 + 2)\hat{k} = \hat{i} + 5\hat{j} + 2\hat{k}$$

$$\text{Now, } D = \frac{|\overline{PS} \times c|}{|c|} = \frac{|\hat{i} + 5\hat{j} + 2\hat{k}|}{|\hat{i} - \hat{j} + 2\hat{k}|} = \frac{\sqrt{(1)^2 + (5)^2 + (2)^2}}{\sqrt{(1)^2 + (-1)^2 + (2)^2}} = \frac{\sqrt{1+25+4}}{\sqrt{1+1+4}} = \frac{\sqrt{30}}{\sqrt{6}} = \sqrt{5} \text{ units.}$$

Example 13: Find the equation of the straight line passing through the point (-2, 3, 1) and parallel to the straight line joining the points (3, -2, 4) and (1, 6, -7).

Solution: Let c_1 , c_2 and c_3 be the direction ratios of a straight line passing through P(3, -2, 4) and Q(1, 6, -7). Then, $c_1 = 1 - 3 = -2$, $c_2 = 6 + 2 = 8$ and $c_3 = -7 - 4 = -11$

Since the required line is parallel to the line that passes through the points (3, -2, 4) and (1, 6, -7) hence its direction ratios are $[c_1, c_2, c_3] = \overline{PQ} = [-2, 8, -11]$. Hence the equation of required line passing through (-2, 3, 1) with direction ratios $[-2, 8, -11]$ is given by

$$x = -2 - 2t, y = 3 + 8t, z = 1 - 11t$$

Example 14: Find the equation of the straight lines L_1 and L_2 in symmetric forms. Determine whether or not the pairs of lines intersect. Find the point of intersection if it exists.

L_1 : Through A (2, 1, 3) and B (-1, 2, -4); L_2 : Through P (5, 1, -2) and Q (0, 4, 3)

Solution: Direction ratios of the line L_1 are $c = [-1 - 2, 2 - 1, -4 - 3] = [-3, 1, -7]$

Direction ratios of the line L_2 are $d = [0 - 5, 4 - 1, 3 - (-2)] = [-5, 3, 5]$

Thus equations of line L_1 with direction ratios $[-3, 1, -7]$ and passing through the point say A (2, 1, 3)

in symmetric form are: $\frac{x-2}{-3} = \frac{y-1}{1} = \frac{z-3}{-7} = t$ (i)

Similarly, the equations of line L_2 with direction ratios $[-5, 3, 5]$ and passing through the point say

P (5, 1, -2) in symmetric form are: $\frac{x-5}{-5} = \frac{y-1}{3} = \frac{z+2}{5} = s$ (ii)

Now if two lines intersect, they must have a common point of intersection.

From (i) and (ii) we have: $x = -3t + 2$ $y = t + 1$ $z = -7t + 3$

and $x = -5s + 5$ $y = 3s + 1$ $z = 5s - 2$

This implies that $-3t + 2 = -5s + 5$ $t + 1 = 3s + 1$ $-7t + 3 = 5s - 2$

Or $-3t + 5s = 3$ (I) $t - 3s = 0$ (II) $-7t - 5s = -5$ (III)

Adding (I) and (III), we have $-10t = -2$ or $t = 1/5$. Put this in equation (I), we get $s = 18/25$.

Now substituting these values of t and s in equation (II), we see that $1/5 - 54/25 \neq 0$. Since equation (II) is not satisfied, there is no point of intersection between the lines L_1 and L_2 .

Example 15: If the edges of a rectangular parallelepiped are a , b , and c , show that the angles between the four diagonals are given by:

$$\cos^{-1} \left(\frac{\pm a^2 \pm b^2 \pm c^2}{a^2 + b^2 + c^2} \right).$$

Solution: Let OABCDEFG be a rectangular parallelepiped with O as origin and A, B and C are the points on the three axes as shown in the figure. If we assumed that $|OA| = a$, $|OB| = b$ and $|OC| = c$ then, A $(a, 0, 0)$, B $(0, b, 0)$ and C $(0, 0, c)$.

The coordinates of the other corner points are; D $(a, 0, c)$, E $(a, b, 0)$, F $(0, b, c)$ and G (a, b, c) .

The direction ratios of the diagonals; OG, AF, BD and

CF are $[a, b, c]$, $[-a, b, c]$, $[a, -b, c]$ and $[a, b, -c]$ respectively.

Using the formula for the angle between two lines,

with direction ratios $[a_1, a_2, a_3]$ and $[b_1, b_2, b_3]$, that is,

$$\theta = \cos^{-1} \left(\frac{a_1 b_1 + a_2 b_2 + a_3 b_3}{\sqrt{a_1^2 + a_2^2 + a_3^2} \sqrt{b_1^2 + b_2^2 + b_3^2}} \right).$$

Thus the angle between the diagonals OG and AF is: $\cos^{-1} \left(\frac{-a^2 + b^2 + c^2}{a^2 + b^2 + c^2} \right)$

The angle between the diagonals OG and BD is: $\cos^{-1} \left(\frac{a^2 - b^2 + c^2}{a^2 + b^2 + c^2} \right)$

The angle between the diagonals OG and CE is: $\cos^{-1} \left(\frac{a^2 + b^2 - c^2}{a^2 + b^2 + c^2} \right)$

The angle between the diagonals AF and BD is: $\cos^{-1} \left(\frac{-a^2 - b^2 + c^2}{a^2 + b^2 + c^2} \right)$

The angle between the diagonals AF and CE is: $\cos^{-1} \left(\frac{-a^2 + b^2 - c^2}{a^2 + b^2 + c^2} \right)$

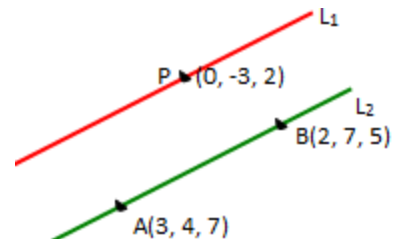
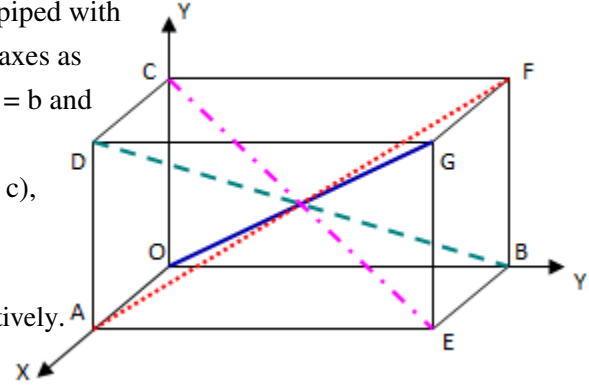
The angle between the diagonals BD and CE is: $\cos^{-1} \left(\frac{a^2 - b^2 - c^2}{a^2 + b^2 + c^2} \right)$

Combining the above results, we see that the angles between the four diagonals are:

$$\cos^{-1} \left(\frac{\pm a^2 \pm b^2 \pm c^2}{a^2 + b^2 + c^2} \right)$$

Example 16: Find the equations of the straight line that passes through the point P $(0, -3, 2)$ and is parallel to the straight line passing through the points A $(3, 4, 7)$ and B $(2, 7, 5)$.

Solution: Let L_1 be the required line. As this line is parallel to



the line L_2 that passes through the points A and B hence: Direction ratios of the line L_1 = Direction ratios of the line L_2 .

Now direction ratios of line L_2 are: $\mathbf{c} = [2 - 3, 7 - 4, 5 - 7]$

$= \mathbf{c} = [-1, 3, -2]$. Thus, equation of line L_1 passing through the point $P(0, -3, 2)$ with direction ratios $[-1, 3, -2]$ is:

$$\frac{x-0}{-1} = \frac{y+3}{3} = \frac{z-2}{-2}$$

Example 17: Find the equations of the straight line that passes through the point $P(2, 0, -2)$ and is perpendicular to each of the straight lines

$$L_1 \equiv \frac{x-3}{2} = \frac{y}{2} = \frac{z+1}{2} \text{ and } L_2 \equiv \frac{x}{3} = \frac{y+1}{-1} = \frac{z+2}{2}.$$

Solution: Let the direction ratios of the line L be $\mathbf{c} = [a, b, c]$.

Since, direction ratios of lines L_1 and L_2 are $\mathbf{d} = [2, 2, 2]$ and $\mathbf{e} = [3, -1, 2]$

respectively and line L is perpendicular on both L_1 and L_2 therefore,

$\mathbf{c} \cdot \mathbf{d} = 0$ and $\mathbf{c} \cdot \mathbf{e} = 0$. This gives:

$$2a + 2b + 2c = 0$$

$$\text{or } a + b + c = 0$$

And

$$3a + (-1)b + 2c = 0$$

$$\text{or } 3a - b + 2c = 0$$

Note: When you have two equations with three variables, the following method is used to find the unknowns.

$$\frac{a}{\begin{vmatrix} 1 & 1 \\ -1 & 2 \end{vmatrix}} = \frac{-b}{\begin{vmatrix} 1 & 1 \\ 3 & 2 \end{vmatrix}} = \frac{c}{\begin{vmatrix} 1 & 1 \\ 3 & -1 \end{vmatrix}} \quad \text{or} \quad \frac{a}{3} = \frac{b}{1} = \frac{c}{-4}$$

Thus equation of the required straight line L with direction ratios $[3, 1, -4]$ and passing through the point $P(2, 0, -2)$ is given by

$$\frac{x-2}{3} = \frac{y}{1} = \frac{z+2}{-4}.$$

Example.19: Find the equations of the straight line through the point $A(11, 4, -6)$ and perpendicular to the straight line $M: x = 4 - t, y = 7 + 2t, z = -1 + t$. Find also the coordinates of the foot of the perpendicular of the line L on the line M .

Solution: Let $F(x, y, z)$ be the foot of the perpendicular of the line

L on the given line M . Since line L passes through the point A ,

hence its direction ratios are: $\mathbf{c} = |\mathbf{AF}| = [x - 11, y - 4, z + 6]$.

The direction ratios of the line M are $\mathbf{d} = [-1, 2, 1]$.

Since, line $L \perp M$, hence

$$\mathbf{c} \cdot \mathbf{d} = 0 \text{ or } -1(x - 11) + 2(y - 4) + 1(z + 6) = 0$$

$$\text{Thus, } -x + 2y + z + 9 = 0 \quad (i)$$

$$\text{But, } x = 4 - t, y = 7 + 2t, z = -1 + t. \quad (ii)$$

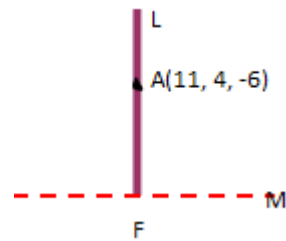
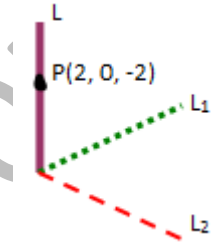
Substituting the values of x, y and z in equation (i), we obtain

$$-(4 + t) + (14 + 4t) + (-1 + t) + 9 = 0 \quad \text{or } 6t = -18 \quad \text{or } t = -3$$

Put this value of t in equations (ii) we get: $x = 7, y = 1$ and $z = -4$.

Therefore, $F = (7, 1, -4)$ which are the coordinates of the foot of the perpendicular of the line L on the line M .

The direction ratios of line L are therefore, $\mathbf{c} = [7 - 11, 1 - 4, -4 + 6] = [-4, -3, 2]$. Hence the equations of the line L passing through the point A and having direction ratios $[-4, -3, 2]$ are



$$\frac{x-11}{-4} = \frac{y-4}{-3} = \frac{z+6}{2}$$

Example 20: A straight line makes angles of measure α , β , γ , and δ with four diagonals of a cube. Prove that

$$\cos^2 \alpha + \cos^2 \beta + \cos^2 \gamma + \cos^2 \delta = 4/3.$$

Solution: Let OABCDEFG be a cube with each side equal to say 'a' and O as the origin (see the figure).

Then, coordinates of different vertices are;

O (0, 0, 0), A (a, 0, 0), B (0, a, 0), C (0, 0, a), D (a, 0, a), E (a, a, 0), F (0, a, a), and G (a, a, a). Thus direction ratios of diagonals: **OG**: [a, a, a], **AF**: [-a, a, a], **BD**: [a, -a, a] and **CE**: [a, a, -a].

Then their magnitudes are; $|\mathbf{OG}| = |\mathbf{AF}| = |\mathbf{BD}| = |\mathbf{CE}| = \sqrt{3} a$.

Now, direction cosines of diagonal **OG** are $[a/\sqrt{3} a, a/\sqrt{3} a, a/\sqrt{3} a] = [1/\sqrt{3}, 1/\sqrt{3}, 1/\sqrt{3}]$.

Similarly, direction cosines of diagonal **AF** are $[-1/\sqrt{3}, 1/\sqrt{3}, 1/\sqrt{3}]$. Direction cosines of diagonal **BD** are $[1/\sqrt{3}, -1/\sqrt{3}, 1/\sqrt{3}]$. Finally, the direction cosines of diagonal **CE** are $[1/\sqrt{3}, 1/\sqrt{3}, -1/\sqrt{3}]$.

Now, let L be a line with direction cosines l, m, n. If L makes angle α , with the diagonal **OG**, then $\cos \alpha = l \cdot (1/\sqrt{3}) + m \cdot (1/\sqrt{3}) + n \cdot (1/\sqrt{3})$. Squaring both sides, we have

$$\cos^2 \alpha = \frac{l^2}{3} + \frac{m^2}{3} + \frac{n^2}{3} + \frac{2lm}{3} + \frac{2mn}{3} + \frac{2nl}{3}.$$

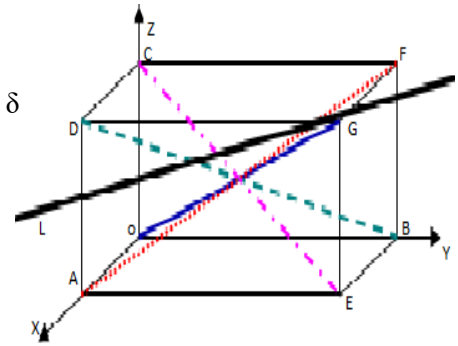
Similarly, if L makes an angles β , γ , and δ with the diagonals **AF**, **BD** and **CE** respectively, we have

$$\cos^2 \beta = \frac{l^2}{3} + \frac{m^2}{3} + \frac{n^2}{3} - \frac{2lm}{3} + \frac{2mn}{3} - \frac{2nl}{3}, \quad \cos^2 \gamma = \frac{l^2}{3} + \frac{m^2}{3} + \frac{n^2}{3} - \frac{2lm}{3} - \frac{2mn}{3} + \frac{2nl}{3}$$

$$\cos^2 \delta = \frac{l^2}{3} + \frac{m^2}{3} + \frac{n^2}{3} + \frac{2lm}{3} - \frac{2mn}{3} - \frac{2nl}{3}.$$

Adding the above results, and using the fact that $l^2 + m^2 + n^2 = 1$, we get

$$\cos^2 \alpha + \cos^2 \beta + \cos^2 \gamma + \cos^2 \delta = \frac{4l^2}{3} + \frac{4m^2}{3} + \frac{4n^2}{3} = \frac{4}{3}(l^2 + m^2 + n^2) = \frac{4}{3}.$$



WORKSHEET 04

1. Show that the points (4, 9, 4), (0, 11, 2) and (1, 0, 1) are the vertices of a right triangle?
2. Show that the point (1, 0, 2), (4, 3, 2) and (0, 7, 6) are the vertices of an isosceles triangle?
3. Show that the point (3, -1, 3), (1, -1, 2) (2, 1, 0) and (4, 1, 1) are the vertices of a rectangle?
4. Show that the ratio in which yz-plane divides the join of (-2, 4, 7) and (3, -5, 9) is 2 : 3.
5. Find the direction ratios, direction cosines, the direction angles and the parametric equations of a line passing through the points (1, -2, 0) and (5, -10, 1).
6. Are the points (1, 4, 3), (2, 6, 1) and (4, 3, 2) collinear?
7. Find the angle between the straight lines

$$L: \frac{x-4}{1} = \frac{y+3}{-4} = \frac{z+1}{7} \text{ and } M: \frac{x-1}{2} = \frac{y+1}{-3} = \frac{z+10}{8}$$

8. Find the direction cosines of two straight lines that are related by an equations: $l + m + n = 0$, $l^2 + m^2 - n^2 = 0$. Also find the measure of an angle between them.

9. Find the distance of the given point S from the given line L: $S = (0, -2, 1)$, $L: \frac{x-1}{4} = \frac{y+3}{-2} = \frac{z+1}{5}$.

10. Find the equations of the straight lines L and M in symmetric form. Determine if the lines intersect. Find the point of intersection if it exists.

L: through $(2, -1, 0)$ and parallel to $[1, 3, 4]$

M: through $(1, 2, 1)$ and parallel to $[2, 1, 5]$

11. Find the equation of straight line through the point $(5, -4, 4)$ and perpendicular to the straight line $x = -t$, $y = 1 + t$, $z = -2t$.

12. Find the equation of straight line through the point $(2, -1, 3)$ and perpendicular to the straight lines L: $x = -t$, $y = 1 + t$, $z = -2t$ and M: $x = 2s$, $y = 3s$ and $z = s$.

13. Show that the join of $(0, -1, 0)$ and $(2, 4, -1)$ intersects the join of $(1, 1, 1)$ and $(3, 3, 9)$.

14. Show that the following lines are parallel.

L: $x + 2y - z - 7 = 0 = -2x + y + z - 6$ and M: $3x + 6y - 3z - 8 = 0 = 2x - y - z$.

15. Show that following lines are perpendicular.

L: $x + 2y - 1 = 0 = 2y - z - 1$ and M: $x - y - 1 = 0 = x - 2y - z$.

CHAPTER FIVE

EQUATION OF PLANE IN R^3

INTRODUCTION

A plane in R^3 can uniquely be determined by specifying a point in the plane and a vector perpendicular to the plane. A vector perpendicular to a plane is called a *normal* to the plane.

In other words, the computation of equation of a plane requires a fixed point in the plane and a vector perpendicular/normal to this plane.

Derivation of Equation of a Plane

Let $P_0(x_0, y_0, z_0)$ be a point in the plane and $\mathbf{n} = [a, b, c]$ be a normal to the plane. Now let $P(x, y, z)$ be any point in the plane. Then

$$\mathbf{N} \cdot \mathbf{P_0P} = 0 \quad (1)$$

$$\text{Or} \quad [a, b, c] \cdot [x - x_0, y - y_0, z - z_0] = 0$$

$$\text{or} \quad a(x - x_0) + b(y - y_0) + c(z - z_0) = 0 \quad (2)$$

This is an equation of the plane through $P_0(x_0, y_0, z_0)$ with normal vector $\mathbf{n} = [a, b, c]$. This equation is known as *point-normal form* of plane. Formula (1) is the *vector form* of the plane. Equation (2) can be written as

$$ax + by + cz = ax_0 + by_0 + cz_0$$

Let d be a real number such that $ax_0 + by_0 + cz_0 = -d$

$$\text{then equation (2) becomes} \quad ax + by + cz + d = 0 \quad (3)$$

This is known as *general form* of the equation of plane.

Example 01: Find the equation of the plane passing through the point $(3, -1, 7)$ and perpendicular to the $\mathbf{n} = [4, 2, -5]$.

Solution: Here $x_0 = 3$, $y_0 = -1$, $z_0 = 7$ and $a = 4$, $b = 2$ and $c = -5$. Using point-normal form of the equation, we get $4(x - 3) + 2(y - 1) - 5(z - 7) = 0$. Simplifying, we get: $4x + 2y - 5z + 25 = 0$ which is equation of required plane.

Remark: In the following sections we are deriving the equation of plane under different conditions imposed on the plane.

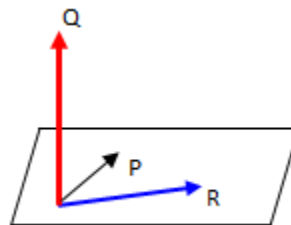
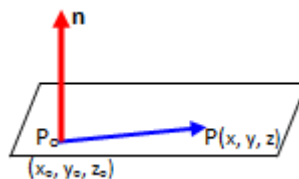
Plane Through Three Non-Collinear Points

Let $P(x_0, y_0, z_0)$, $Q(x_2, y_2, z_2)$ and $R(x_3, y_3, z_3)$ be three non-collinear points lying in the plane. Now the equation of plane in general form is

$$ax + by + cz - d = 0 \quad (i)$$

Since P , Q and R lie in this plane they must satisfy the equation of plane.

This means we must have:



$$ax_1 + by_1 + cz_1 - d = 0 \quad (\text{ii})$$

$$ax_2 + by_2 + cz_2 - d = 0 \quad (\text{iii})$$

$$ax_3 + by_3 + cz_3 - d = 0 \quad (\text{iv})$$

Eliminating a, b, c and -d from (i) to (iv), we get

$$\begin{vmatrix} x & y & z & 1 \\ x_1 & y_1 & z_1 & 1 \\ x_2 & y_2 & z_2 & 1 \\ x_3 & y_3 & z_3 & 1 \end{vmatrix} = 0$$

This is the equation of plane through three non-collinear points.

Example 02: Find the equation of the plane through three points P(1, -1, 2), Q(-3, -2, 6) and R(6, 0, 1).

Solution: The equation of the plane through three points is:

$$\begin{vmatrix} x & y & z & 1 \\ 1 & -1 & 2 & 1 \\ -3 & -2 & 6 & 1 \\ 6 & 0 & 1 & 1 \end{vmatrix} = 0$$

Interchanging the first and fourth column, we get

$$\begin{vmatrix} 1 & y & z & x \\ 1 & -1 & 2 & 1 \\ -3 & -2 & 6 & -3 \\ 6 & 0 & 1 & 6 \end{vmatrix} = 0$$

Subtracting first, second and third row from fourth, we get

$$\begin{vmatrix} 0 & y & z-1 & x-6 \\ 0 & -1 & 1 & -5 \\ 0 & -2 & 5 & -9 \\ 1 & 0 & 1 & 6 \end{vmatrix} = 0$$

Expanding by the first column, we get

$$\begin{vmatrix} y & z-1 & x-6 \\ -1 & 1 & -5 \\ -2 & 5 & -9 \end{vmatrix} = 0$$

Multiplying R_2 and R_3 by -1, we obtain

$$\begin{vmatrix} y & z-1 & x-6 \\ 1 & -1 & 5 \\ 2 & -5 & 9 \end{vmatrix} = 0$$

Expanding by the first row, we get: $y(-9 + 25) - (z - 1)(-1) + (x - 6)(-3) = 0$ or $3x - 16y - z = 17$

This is equation of required plane.

Remark: Above method of finding the equation of plane is some what cumbersome as it involves evaluation of determinant of order four. The following method is some what easier than the above method.

Example 03: Find the equation of plane that passes through the points $P(1, -1, 2)$, $Q(-3, -2, 6)$ and $R(5, -8, -2)$.

Solution: $\overrightarrow{PQ} = [-3 - 1, -2 + 1, 6 - 2] = [-4, -1, 4]$, $\overrightarrow{PR} = [5 - 1, -8 + 1, -2 - 2] = [4, -7, -4]$.

Now normal to the plane is a vector

$$\overrightarrow{PQ} \times \overrightarrow{PR} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ -4 & -1 & 4 \\ 4 & -7 & -4 \end{vmatrix} = \hat{i}(4 + 28) - \hat{j}(16 - 16) + \hat{k}(28 + 4) = 32\hat{i} + 32\hat{k} = [32, 0, 32] = \mathbf{n}$$

Hence equation of the plane with normal vector $\mathbf{n} = [32, 0, 32]$ and passing through the point $P(1, -1, 2)$ is $2(x - 1) + 0(y + 1) + 32(z - 2) = 0$ or $x + z - 3 = 0$

Intercept Form of Equation of Plane

Let the equation of a plane be $ax + by + cz = d$. Suppose that this plane intercepts three axes at $P(a, 0, 0)$, $Q(0, b, 0)$ and $R(0, 0, 1)$ respectively.

This means that the plane contains three non-collinear point P , Q and R .

Using the determinant formula for the equation of plane,

we get

$$\begin{vmatrix} x & y & z & 1 \\ a & 0 & 0 & 1 \\ 0 & b & 0 & 1 \\ 0 & 0 & c & 1 \end{vmatrix} = 0$$

Evaluating the determinant, we get: $bcx + acy + abz - abc = 0$

Dividing both sides by abc , we obtain: $\frac{x}{a} + \frac{y}{b} + \frac{z}{c} = 1$

This is equation of plane in intercept form.

Normal Form of Equation of Plane

Let l, m, n be the direction cosines of a normal from the origin to the plane and let p be the length of this normal. Let A be the foot of the perpendicular from the origin to the plane. Since the direction cosines of $\mathbf{OA}$ are l, m, n and $|\mathbf{OA}| = p$ so that coordinates of A are (lp, mp, np) .

Explanation:

$\cos \alpha = OB/OA$ or $OB = OA \cdot \cos \alpha = pl$. This is x -component of point A .

Similarly, the y and z components of A are pm and pn . Thus $A = (lp, mp, np)$

Now, let $P(x, y, z)$ be any point on the plane.

Then, $\mathbf{AP} = [x - lp, y - mp, z - np]$. Now $\mathbf{OA}$ is perpendicular to the plane. This implies that $\mathbf{OA}$ is perpendicular to $\mathbf{AP}$ thus,

$$\mathbf{OA} \cdot \mathbf{AP} = 0 \text{ or } [lp, mp, np] \cdot [x - lp, y - mp, z - np]$$

or $lp(x - lp) + mp(y - mp) + np(z - np) = 0$. Dividing by p , we get

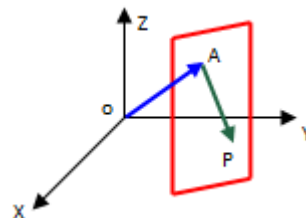
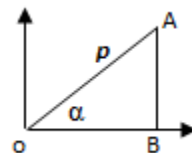
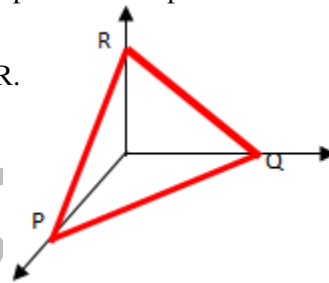
$$l(x - lp) + m(y - mp) + n(z - np) = 0. \text{ This gives,}$$

$$lx + my + nz - (l^2p + m^2p + n^2p) = 0$$

$$\text{or } lx + my + nz = (l^2p + m^2p + n^2p) = p(l^2 + m^2 + n^2)$$

$$\text{or } lx + my + nz = p$$

This is the equation of plane in normal form.



Steps to change the equation of plane from general to normal form:

- (i) Equation of plane in general form is $ax + by + cz = -d$
- (ii) Equation of plane in normal form is $lx + my + nz = p$

Comparing the two equations we get

$$l/a = m/b = n/c = -p/d = k \text{ or } l = ak, m = bk, n = ck \text{ and } p = -dk$$

Then, $l^2 + m^2 + n^2 = k^2(a^2 + b^2 + c^2)$. But $l^2 + m^2 + n^2 = 1$ therefore, $k = \pm 1/\sqrt{a^2 + b^2 + c^2}$

Now, we shall take that value of k which makes p a positive number. This is explained in the following example.

Example 04: Convert equation of the plane $2x - 3y - z + 5 = 0$ to

- (i) Intercept Form
- (ii) the Normal Form.

Solution: Intercept Form

Given equation of the plane is: $2x - 3y - z + 5 = 0$ or $2x - 3y - z = -5$

Dividing both sides by -5 , we have: $-\frac{2}{5}x + \frac{3}{5}y + \frac{1}{5}z = 1$ or $\frac{x}{-5/2} + \frac{y}{5/3} + \frac{z}{5} = 1$

This is the intercept form of plane where x -, y - and z intercepts are $-5/2$, $5/3$, and 5 .

Normal Form

Given equation of plane is: $2x - 3y - z = -5$

Comparing this equation with normal form: $lx + my + nz = p$.

We see that direction ratios of normal vector are $\mathbf{n} = [a, b, c] = [2, -3, -1]$. Hence,

$$k = \pm 1/\sqrt{a^2 + b^2 + c^2} = \pm \sqrt{2^2 + (-3)^2 + (-1)^2} = \pm \sqrt{14}.$$

Now we shall take that value of k which makes p a positive number. Now $p = -dk$. Here, $d = -5$ thus we take $k = -1/\sqrt{14}$.

Thus, $l = ak = -2/\sqrt{14}$, $m = bk = 3/\sqrt{14}$, $n = ck = 1/\sqrt{14}$ and $p = -dk = 5/\sqrt{14}$.

Thus normal form of given plane is: $\frac{-2}{\sqrt{14}}x + \frac{3}{\sqrt{14}}y + \frac{1}{\sqrt{14}}z = \frac{5}{\sqrt{14}}$.

Angle between Two Planes

Angle between two planes is defined as angle between their normal vectors. If equations of the two planes are

$$S_1: a_1x + b_1y + c_1z + d_1 = 0 \quad (i)$$

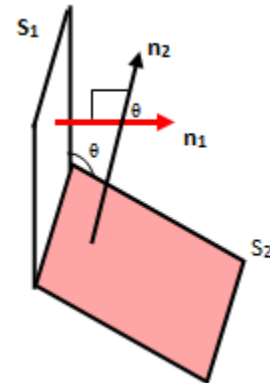
$$S_2: a_2x + b_2y + c_2z + d_2 = 0 \quad (ii)$$

then angle between them is the angle between their normal vectors.

Now the normal vectors of both planes are:

$\mathbf{n}_1 = [a_1, b_1, c_1]$ and $\mathbf{n}_2 = [a_2, b_2, c_2]$. Now, $\mathbf{n}_1 \cdot \mathbf{n}_2 = n_1 n_2 \cos \theta$ where θ is the angle between the vectors $\mathbf{n}_1$ and $\mathbf{n}_2$.

$$\text{Therefore, } \cos \theta = \left(\frac{\mathbf{n}_1 \cdot \mathbf{n}_2}{|\mathbf{n}_1| |\mathbf{n}_2|} \right) = \frac{a_1a_2 + b_1b_2 + c_1c_2}{\sqrt{a_1^2 + b_1^2 + c_1^2} \cdot \sqrt{a_2^2 + b_2^2 + c_2^2}}$$



If the planes are perpendicular then, $\theta = \pi/2$ so $\cos \theta = \cos \pi/2 = 0$. Thus, the condition for the planes to be perpendicular to each other is: $a_1a_2 + b_1b_2 + c_1c_2 = 0$.

Two planes are parallel if their normal vectors are parallel or their normal vectors are multiple of each other. Mathematically, if two vectors are parallel, we must have

$$\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2} = k$$

Example 05: Find the measure of angle between the planes $x - y + 2z = 1$ and $x + 2y - z = 6$.

Solution: Here $\mathbf{n}_1 = [1, -1, 2]$ and $\mathbf{n}_2 = [1, 2, -1]$. If θ is the angle between the planes, then

$$\cos \theta = \frac{\mathbf{n}_1 \cdot \mathbf{n}_2}{|\mathbf{n}_1| |\mathbf{n}_2|} = \frac{(1)(1) + (-1)(2) + (2)(-1)}{\sqrt{(1)^2 + (-1)^2 + (2)^2} \sqrt{(1)^2 + (2)^2 + (-1)^2}} = \frac{-3}{6} = -\frac{1}{2}$$

or $\theta = \cos^{-1}\left(-\frac{1}{2}\right) = \frac{2\pi}{3}$. So the acute angle between the given planes is $\theta = \frac{2\pi}{3}$.

Remark: Other angle is $\theta = \pi - 2\pi/3 = \pi/3$.

Example 06: Determine whether the planes $3x - 4y + 5z = 0$ and $-6x + 8y - 10z - 4 = 0$ are parallel.

Solution: If two planes are parallel their normal vectors are parallel. This means normal vectors are multiple of each other. Now,

Normal vector to the first plane is: $\mathbf{n}_1 = [3, -4, 5]$

Normal vector to the second plane is: $\mathbf{n}_2 = [-6, 8, -10]$

You may observe that $\mathbf{n}_2 = -2\mathbf{n}_1$. Hence normal vectors are parallel; therefore, two planes are parallel.

Distance between Point and Plane

Distance between point and a plane is defined as the length of perpendicular drawn from the point on the plane.

Statement: The (perpendicular) distance D of a point $A(x_1, y_1, z_1)$ from the plane

$$ax + by + cz + d = 0 \text{ is } |D| = \frac{|ax_1 + by_1 + cz_1 + d|}{\sqrt{a^2 + b^2 + c^2}}.$$

Proof: Let $P(x_0, y_0, z_0)$ be any point on the plane. The normal $\mathbf{n} = [a, b, c]$ is drawn in such a way that its initial point is at P .

The required distance D = length of the orthogonal projection of $\mathbf{PA}$ on $\mathbf{n}$. Thus,

$$|D| = (\mathbf{PA} \cdot \mathbf{n})/|\mathbf{n}| \quad (i)$$

Now,

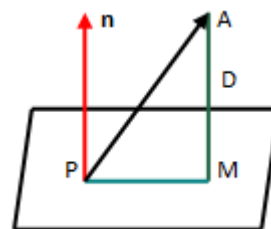
$$\mathbf{PA} = [x_1 - x_0, y_1 - y_0, z_1 - z_0]$$

Or $\mathbf{PA} = a(x_1 - x_0) + b(y_1 - y_0) + c(z_1 - z_0)$ and $|\mathbf{n}| = \sqrt{a^2 + b^2 + c^2}$.

$$\text{From (i) we have } |D| = \frac{|ax_1 + by_1 + cz_1 - (ax_0 + by_0 + cz_0)|}{\sqrt{a^2 + b^2 + c^2}} = \frac{|ax_1 + by_1 + cz_1 + d|}{\sqrt{a^2 + b^2 + c^2}}$$

Example 07: Find the distance between the point $A(3, -1, 2)$ and the plane $2x + y - z - 4 = 0$.

Solution: Here, $(x_1, y_1, z_1) = (3, -1, 2)$ and $[a, b, c] = [2, 1, -1]$. Now, we know that distance between the point and plane is given by



$$|D| = \frac{|ax_1 + by_1 + cz_1 + d|}{\sqrt{a^2 + b^2 + c^2}}$$

Substituting the values into this formula, we get

$$D = \frac{|(2)(3) + (1)(-1) + (-1)(2) + (-4)|}{\sqrt{(2)^2 + (1)^2 + (-1)^2}} = \frac{|6 - 1 - 2 - 4|}{\sqrt{6}} = \frac{1}{\sqrt{6}} \text{ units.}$$

Example 08: Find the equation of the plane through the point $(3, -2, 5)$ and perpendicular to the line $x = 2 + 3t$, $y = 1 - 6t$, $z = -2 + 2t$.

Solution: From the given equation of line, we see that the direction ratios of line are $\mathbf{c} = [a, b, c] = [3, -6, 2]$.

Since the plane is perpendicular to the given line, normal to the plane is parallel to the line, hence

The direction ratios of plane = The direction ratios of line. That is $\mathbf{n} = \mathbf{c} = [3, -6, 2]$.

Now Point – Normal form of equation of a plane is: $a(x - x_0) + b(y - y_0) + c(z - z_0) = 0$

Substituting the values, we get: $3(x - 3) - 6(y + 2) + 2(z - 5) = 0$ or $3x - 6y + 2z = 31$.

This is equation of required plane.

Example 09: Find the equation of plane through $(5, -1, 4)$ and perpendicular to each of the planes $x + y - 2z - 3 = 0$ and $2x - 3y + z = 0$.

Solution: A plane through $(5, -1, 4)$ is: $a(x - 5) + b(y + 1) + c(z - 4) = 0$ (1)

Here $\mathbf{n} = [a, b, c]$. Since required plane is perpendicular to each of the given planes,

we have: $a(1) + b(1) + c(-2) = 0$ or $a + b - 2c = 0$ (i) [since, $\mathbf{n} \cdot \mathbf{n}_1 = 0$]

and $a(2) + b(-3) + c(1) = 0$ or $2a - 3b + c = 0$ (ii) [since, $\mathbf{n} \cdot \mathbf{n}_2 = 0$]

Multiplying (i) by 3 and adding into (ii), we get $5a - 5c = 0$ or $a = c$.

Substituting $c = a$ into (i), we get $a + b - 2a = 0$ or $b - a = 0$ giving; $a = b$.

Hence, $a = b = c$ or $\frac{a}{1} = \frac{b}{1} = \frac{c}{1}$ that is $a : b : c = 1 : 1 : 1$

Thus, from (1) the required equation of the plane is: $x + y + z = 8$

Example 10: Find the equation of the plane which passes through the point $(3, 4, 5)$ and has x -intercept equal to -5 and is perpendicular to the plane $2x + 3y - z = 8$.

Solution: Equation of plane in intercept form is: $\frac{x}{a} + \frac{y}{b} + \frac{z}{c} = 1$

Here $a = -5$, hence we get: $\frac{x}{-5} + \frac{y}{b} + \frac{z}{c} = 1$ (i)

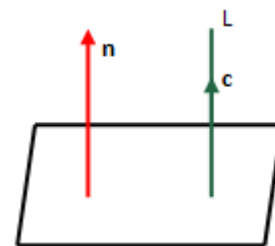
The normal to this plane is $\mathbf{n} = [-1/5, 1/b, 1/c]$

Since this plane is perpendicular to $2x + 3y - z = 8$, we have

$$2\left(-\frac{1}{5}\right) + 3\left(\frac{1}{b}\right) - \frac{1}{c} = 0 \text{ or } -\frac{2}{5} + \frac{3}{b} - \frac{1}{c} = 0 \text{ or } \frac{3}{b} - \frac{1}{c} = \frac{2}{5} \quad \text{(ii)}$$

Now the point $(3, 4, 5)$ lies on the plane (i), therefore, we have

$$\frac{3}{-5} + \frac{4}{b} + \frac{5}{c} = 1 \Rightarrow \frac{4}{b} + \frac{5}{c} = \frac{8}{5} \quad \text{(iii)}$$



Multiplying (ii) by 5 and adding into (iii), we have

$$\frac{15}{b} + \frac{4}{b} = 2 + \frac{8}{5} \quad \text{or} \quad \frac{19}{b} = \frac{18}{5} \quad \text{or} \quad \frac{b}{19} = \frac{5}{18} \quad \text{or} \quad b = \frac{95}{18}$$

Substituting this value into (ii), we get

$$\frac{4}{95/18} + \frac{5}{c} = \frac{8}{5} \quad \text{or} \quad \frac{72}{95} + \frac{5}{c} = \frac{8}{5} \quad \text{or} \quad \frac{5}{c} = \frac{8}{5} - \frac{72}{95} \quad \text{or} \quad \frac{5}{c} = \frac{80}{95} \quad \text{or} \quad \frac{c}{5} = \frac{95}{80} \quad \text{or} \quad c = \frac{95}{16}$$

Putting the values of b and c in equation (i), we obtain

$$\frac{x}{-5} + \frac{18y}{95} + \frac{16z}{95} = 1 \quad \text{or} \quad 19x - 18y - 16z + 95 = 0$$

This is equation of required plane.

Example 11: Show that the distance of the point $P(3, -4, 5)$ from the plane

$2x - 5y - 6z = 16$ measured parallel to the line $\frac{x}{2} = \frac{y}{1} = \frac{z}{-2}$ is $60/7$.

Solution: Direction ratios of the given line L are $[2, 1, -2]$.

Since the required line M is parallel to L, hence their direction ratios are same. Thus equation of the line M passing through the point $P(3, -4, 5)$ having direction ratios $[2, 1, -2]$ is:

$$\frac{x-3}{2} = \frac{y+4}{1} = \frac{z-5}{-2} = t \quad \text{or} \quad x = 2t + 3, \quad y = t - 4, \quad z = -2t + 5$$

Let these be the coordinates of the point Q, that is, $Q(2t + 3, t - 4, -2t + 5)$.

If this point Q lies on the given plane, it must satisfy the equation of the plane. Thus,

$$2(2t + 3) - 5(t - 4) - 6(-2t + 5) = 16.$$

Simplifying, we obtain $t = 20/7$. Hence, the coordinates of the point Q are $(61/7, -8/7, -5/7)$. Now using the distance formula between the two points, we get

$$|PQ| = \sqrt{\left(3 - \frac{61}{7}\right)^2 + \left(-4 + \frac{8}{7}\right)^2 + \left(5 + \frac{5}{7}\right)^2} = \sqrt{\left(\frac{40}{7}\right)^2 + \left(\frac{20}{7}\right)^2 + \left(\frac{40}{7}\right)^2} = \frac{60}{7}.$$

Example 12: If a, b, c are the intercepts on the coordinate axes and r is the distance of the origin from the plane, prove that $\frac{1}{r^2} = \frac{1}{a^2} + \frac{1}{b^2} + \frac{1}{c^2}$.

Solution: The intercept form of the required plane will be $\frac{x}{a} + \frac{y}{b} + \frac{z}{c} = 1$.

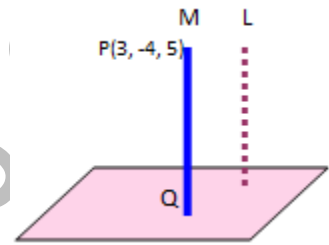
Now using the formula for distance between the point and the plane, the distance between this plane and the origin is:

$$r = \frac{|-1|}{\sqrt{(1/a^2) + (1/b^2) + 1/(c^2)}}. \text{ Squaring both sides and cross multiplying, we get}$$

$$\frac{1}{a^2} + \frac{1}{b^2} + \frac{1}{c^2} = \frac{1}{r^2} \quad \text{which is the required condition.}$$

Example 13: Find the equations of two planes whose distances from the origin are 3 units each and which are perpendicular to the line through the points $P(7, 3, 1)$ and $Q(6, 4, -1)$.

Solution: Direction ratios of the line joining the point P and Q are $[7 - 6, 3 - 4, 1 + 1] = [1, -1, 2]$.



Hence its direction cosines are $\frac{1}{\sqrt{6}}, \frac{-1}{\sqrt{6}}, \frac{2}{\sqrt{6}}$. Now using the equation of the plane in normal form,

the equations of the required planes are; $\frac{1}{\sqrt{6}}x - \frac{1}{\sqrt{6}}y + \frac{2}{\sqrt{6}}z = \pm 3$ or $x - y + 2z = \pm 3\sqrt{6}$.

Straight Line as Intersection of Two Planes

Consider two planes:

$$a_1 x + b_1 y + c_1 z = d_1$$

$$a_2 x + b_2 y + c_2 z = d_2$$

with normal vectors $\mathbf{n}_1 = [a_1, b_1, c_1]$ and $\mathbf{n}_2 = [a_2, b_2, c_2]$.

If these planes are parallel, then $\mathbf{n}_1$ and $\mathbf{n}_2$ must be collinear. The condition for this is:

$$\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2}.$$

If the planes do intersect, they form a line as shown in the figure. The equation of this line L can be found as follow:

Find any two points on the line L and find its direction ratios. This can be done by first equating say z equal to zero and then solving the two equations in x and y simultaneously. This gives the first point P_1 on the line L. Now choose say x equal to zero and solve for y and z. This results in two linear equations. Solve these and get another point P_2 on the line L. Now the direction vector of the line L would be $\mathbf{c} = \mathbf{P}_1\mathbf{P}_2$. With the help of $\mathbf{c}$ and the point say P_1 , we may find the equation of the straight line.

Example 14: Show that the planes $4x + 4y - 5z = 12$ and $8x + 12y - 13z = 32$ intersect each other. Hence find the equation of their line of intersection.

Solution: Here $\mathbf{n}_1 = [4, 4, -5]$ and $\mathbf{n}_2 = [8, 12, -13]$. We see that $\mathbf{n}_1$ and $\mathbf{n}_2$ are not multiple of each other, hence they are not collinear. This means that the planes intersect each other.

To find the line of their intersection, we put $z = 0$ and get:

$$4x + 4y = 12 \text{ and } 8x + 12y = 32 \quad \text{or} \quad x = 1 \text{ and } y = 2. \text{ Hence } P_1 = (1, 2, 0).$$

To get the second point on the line, we put $x = 0$. This gives:

$$4y - 5z = 12 \text{ and } 12y - 13z = 32 \quad \text{or} \quad z = -2 \text{ and } y = 1/2. \text{ Thus } P_2 = (0, 1/2, -2).$$

Now the direction vector of the line L is

$$\mathbf{c} = \mathbf{P}_1\mathbf{P}_2 = [0 - 1, (1/2) - 2, -2 - 0] = [-1, -3/2, -2] = [-2, -3, -4] = [2, 3, 4].$$

Hence the equation of required line with direction ratios 2, 3, 4 and passes through the point

$$P_1(1, 2, 0) \text{ is; } \frac{x-1}{2} = \frac{y-2}{3} = \frac{z}{4}.$$

PLANE AND STRAIGHT LINE

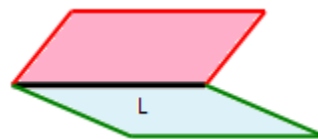
Consider a plane

$$S: ax + by + cz = d$$

and a straight line,

$$L: x = x_1 + c_1t, y = y_1 + c_2t, z = z_1 + c_3t$$

If there is only one point common between straight line and a plane, we say that the line and the plane intersect at a unique point (Figure 1). If a straight line and a plane have two or more points in common, we say that straight line lies in the plane (Figure 2).



If a straight line and a plane have no point in common, we say that the straight line and the plane are parallel (Figure 3)

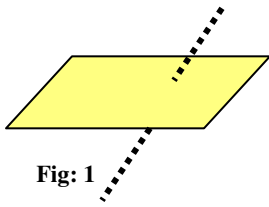


Fig: 1

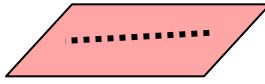


Fig: 2

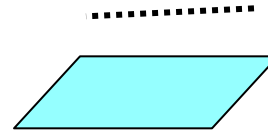


Fig: 3

Now the plane is:

$$S: ax + by + cz = d$$

and the line,

$$L: x = x_1 + c_1t, y = y_1 + c_2t, z = z_1 + c_3t$$

Let $P(x, y, z) = (x_1 + c_1t, y_1 + c_2t, z_1 + c_3t)$

Now substituting the coordinates of point P in the equation of plane:

- i. If we get only one value of t then the line L and plane S intersect each at one point.
- ii. If P does not satisfy the equation of plane we say that line does not lie in the plane. This means line and plane are parallel. In this case the normal to the plane will be perpendicular to the line and thus, we get

$$ac_1 + bc_2 + cc_3 = 0 \quad (1)$$

- iii. If the straight line lies in the plane, in addition to the above condition, the condition:

$$ax_1 + by_1 + cz_1 = d \quad (2)$$

should also be satisfied.

- iv. If the straight line L intersects the plane perpendicularly then normal to the plane will be parallel to the line and we have

$$\frac{a}{c_1} = \frac{b}{c_2} = \frac{c}{c_3} \quad (3)$$

- v. If the straight line neither satisfies the condition (1) nor condition (2) then the line intersects the plane in one point but not perpendicularly.

Example 01: Show that the straight line $\frac{x+3}{2} = \frac{y-4}{-7} = \frac{z}{3}$ is parallel to the plane $4x + 2y + 2z = 9$.

Solution: Here $\mathbf{c} = [c_1, c_2, c_3] = [2, -7, 3]$ and $\mathbf{n} = [a, b, c] = [4, 2, 2]$.

Now, if the line L and plane S are perpendicular then $\mathbf{c}$ and $\mathbf{n}$ are perpendicular to each other. This means $\mathbf{c} \cdot \mathbf{n} = 0$.

$$\text{or } ac_1 + bc_2 + cc_3 = (4)(2) + (2)(-7) + (2)(3) = 8 - 14 + 6 = 0.$$

This shows that line L is parallel to the plane S .

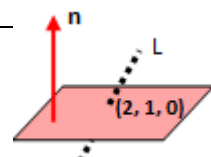
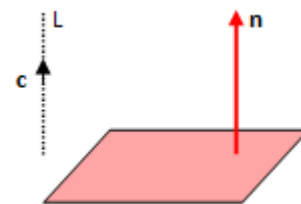
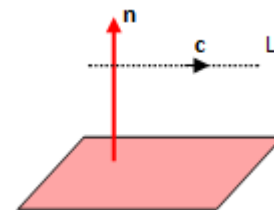
Example 02: Show that the straight line $\frac{x}{1} = \frac{y}{2} = \frac{z}{3}$ is

perpendicular to the plane $4x + 8y + 12z + 19 = 0$.

Solution: If the direction vector of the line is parallel to the normal vector of the plane then the line and the plane are said to be perpendicular to each other.

Here direction vector of the given line is $\mathbf{c} = [1, 2, 3]$ and normal vector of the plane is $\mathbf{n} = [4, 8, 12]$.

We see that $\mathbf{n} = 4\mathbf{c}$, therefore, the given line is perpendicular on the given plane.



Example 03: Determine the point, if any, common to the straight line

$x = 1 + t$, $y = t$, $z = -1 + t$ and the plane $x + y + z = 3$.

Solution: We know that line and plane intersect if and only if the dot product of direction vector and normal vector is non-zero.

Here the direction vector of a line L is $\mathbf{d} = [1, 1, 1]$ and the normal vector is $\mathbf{n} = [1, 1, 1]$.

Now $\mathbf{d} \cdot \mathbf{n} = 1.1 + 1.1 + 1.1 = 3 \neq 0$ which shows that line and plane intersect each other.

Now let $P(x, y, z) = (1 + t, t, -1 + t)$ be the point where the line and plane intersect each other.

Substituting this in equation of plane, we get: $1 + t + t - 1 + t = 3$ or $3t = 3$ giving $t = 1$.

Thus coordinates of point P are $P(2, 1, 0)$ which is the point of intersection of the given line and the plane.

Example 04: Prove that the straight lines $L: \frac{x-1}{2} = \frac{y+1}{-3} = \frac{z+10}{8}$ and $M: \frac{x-4}{1} = \frac{y+3}{-4} = \frac{z+1}{7}$

intersect. Also find their point of intersection and a plane through them.

Solution: The parametric equations for the above two given lines are:

$$L: \begin{cases} x = 1 + 2t \\ y = -1 - 3t \\ z = -10 + 8t \end{cases} \quad \text{and} \quad M: \begin{cases} x = 4 + s \\ y = -3 - 4s \\ z = -1 + 7s \end{cases}$$

$$1 + 2t = 4 + s \quad \text{or} \quad 2t - s - 3 = 0 \quad (\text{i})$$

$$\text{This gives,} \quad -1 - 3t = -3 - 4s \quad \text{or} \quad -3t + 4s + 2 = 0 \quad (\text{ii})$$

$$-10 + 8t = -1 + 7s \quad \text{or} \quad 8t - 7s - 9 = 0 \quad (\text{iii})$$

Multiply equation (i) by 4 and adding the result in equation (ii), we get, $t = 2$.

Putting this value of t in equation (ii), we get, $s = 1$. Now substituting the values of t and s in (iii), we see that:

$$8(2) - 7(1) - 9 = 16 - 7 - 9 = 16 - 16 = 0$$

Hence, both values of s and t satisfy the third equation, therefore, the lines L and M intersect.

Substituting the value of t for the line L , we get the required point of intersection as $(5, -7, 6)$.

Now, plane through these lines must contain the point of intersection $(5, -7, 6)$.

$$\text{Therefore,} \quad a(x-5) + b(y+7) + c(z-6) = 0 \quad (\text{iv})$$

The normal vector $\mathbf{n}$ of the plane is perpendicular to both lines, hence

$$\mathbf{n} \cdot \mathbf{c} = 0 \quad \text{and} \quad \mathbf{n} \cdot \mathbf{d} = 0.$$

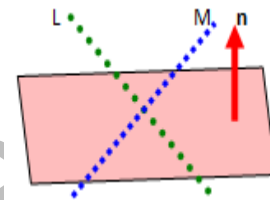
$$\text{or} \quad 2a - 3b + 8c = 0 \quad (\text{v})$$

$$a - 4b + 7c = 0 \quad (\text{vi})$$

Eliminating a , b , c from equations (iv), (v) and (vi), we get:

$$\begin{vmatrix} x-5 & y+7 & z-6 \\ 2 & -3 & 8 \\ 1 & -4 & 7 \end{vmatrix} = 0$$

Simplifying, we have: $11x - 6y - 5z - 67 = 0$, this is the equation of required plane through the given lines.



COPLANAR LINES

Consider the lines; $\frac{x-x_1}{c_1} = \frac{y-y_1}{c_2} = \frac{z-z_1}{c_3}$ and $\frac{x-x_2}{d_1} = \frac{y-y_2}{d_2} = \frac{z-z_2}{d_3}$

If the plane $ax + by + cz + d = 0$

contains the first line, we must have

$$a(x-x_1) + b(y-y_1) + c(z-z_1) = 0 \quad (i)$$

with

$$ac_1 + bc_2 + cc_3 = 0 \quad (ii)$$

And if the plane contains the second line, we must have

$$a(x_2-x_1) + b(y_2-y_1) + c(z_2-z_1) = 0 \quad (iii)$$

with

$$ad_1 + bd_2 + cd_3 = 0 \quad (iv)$$

Here, $[a, b, c]$ are the components of normal vector $\mathbf{n}$ to the plane.

Eliminating a, b and c from ((ii), (iii) and (iv), we get

$$\begin{vmatrix} x_2-x_1 & y_2-y_1 & z_2-z_1 \\ c_1 & c_2 & c_3 \\ d_1 & d_2 & d_3 \end{vmatrix} = 0 \quad (1)$$

This is the condition that two lines are coplanar.

In addition to this, if we wish to find the equation of the plane that contains the given lines, we must eliminate a, b and c from (i), (ii) and (iv) to get,

$$\begin{vmatrix} x-x_1 & y-y_1 & z-z_1 \\ c_1 & c_2 & c_3 \\ d_1 & d_2 & d_3 \end{vmatrix} = 0 \quad (2)$$

Example 01: Prove that the straight lines

$$\frac{x-1}{2} = \frac{y-2}{3} = \frac{z-3}{4} \quad \text{and} \quad \frac{x-2}{3} = \frac{y-3}{4} = \frac{z-4}{5}$$

are coplanar. Also find the equation of the plane that contains both lines.

Solution: Here $x_2-x_1 = 2-1=1$, $y_2-y_1 = 3-2=1$ and $z_2-z_1 = 4-3=1$

According to the given condition;

$$\begin{vmatrix} x_2-x_1 & y_2-y_1 & z_2-z_1 \\ c_1 & c_2 & c_3 \\ d_1 & d_2 & d_3 \end{vmatrix} = 0 \quad \text{or} \quad \begin{vmatrix} 1 & 1 & 1 \\ 2 & 3 & 4 \\ 3 & 4 & 5 \end{vmatrix} = 1(15-16) - 1(10-12) + 1(8-9) = 1+2-1=0$$

Since the required condition is satisfied, this proves that above lines are coplanar.

The equation of the plane that contains given lines is therefore given by the formula

$$\begin{vmatrix} x-x_1 & y-y_1 & z-z_1 \\ c_1 & c_2 & c_3 \\ d_1 & d_2 & d_3 \end{vmatrix} = 0$$

Substituting the given values, we get:

$$\begin{vmatrix} x-1 & y-2 & z-3 \\ 2 & 3 & 4 \\ 3 & 4 & 5 \end{vmatrix} = 0$$

Solving, we get:

$$x - 2y + z = 0.$$

Plane Through the Line of Intersection of Two Planes

Consider two planes:

$$a_1x + b_1y + c_1z = d_1$$

$$a_2x + b_2y + c_2z = d_2$$

with normal vectors $\mathbf{n}_1 = [a_1, b_1, c_1]$ and $\mathbf{n}_2 = [a_2, b_2, c_2]$.

If these planes intersect, they form a line.

The equation of the plane that passes through the line of the intersection of the given planes may be written as:

$$(a_1x + b_1y + c_1z - d_1) + k(a_2x + b_2y + c_2z - d_2) = 0 \quad (1)$$

Here k is some constant that is to be determined by imposing some condition on the required plane.

Example 02: Find the equation of the plane that passes through the line of intersection of the planes $2x - y + 3z = 0$ and $x + 2y - 2z - 3 = 0$ and perpendicular to the yz -plane.

Solution: Equation of the plane passing through the line of intersection of the two planes is:

$$(2x - y + 3z) + k(x + 2y - 2z - 3) = 0 \quad (i)$$

$$\text{or} \quad (2 + k)x + (-1 + 2k)y + (3 - 2k)z - 3k = 0 \quad (ii)$$

The normal to this plane is $\mathbf{n}_1 = [2 + k, -1 + 2k, 3 - 2k]$.

The equation of yz -plane is $x = 0$ with normal vector $\mathbf{n}_2 = [1, 0, 0]$.

Now $\mathbf{n}_1$ is perpendicular to $\mathbf{n}_2$, hence, $\mathbf{n}_1 \cdot \mathbf{n}_2 = 0$

Thus, $1(2 + k) + 0(-1 + 2k) + 0(3 - 2k) = 0$. Simplifying, we get $k = -2$.

Put this value of k in equation (ii), we have $5y - 7z = 6$. This is the equation of the required plane.

Example 03: Find the equation of the plane that passes through the line of intersection of the planes $2x - y + 3z = 0$ and $x + 2y - 2z - 3 = 0$ and is at unit distance from the origin.

Solution: The equation of the plane that passes through the line of intersection of the given planes is

$$(2x - y + 3z) + k(x + 2y - 2z - 3) = 0 \quad (i)$$

$$\text{or} \quad (2 + k)x + (-1 + 2k)y + (3 - 2k)z - 3k = 0 \quad (ii)$$

Using the distance formula between the point and a plane

$$D = \frac{|ax_1 + by_1 + cz_1 + d|}{\sqrt{a^2 + b^2 + c^2}}, \text{ we get}$$

$$D = \frac{|(2+k).0 + (-1+2k).0 + (3-2k).0 - 3k|}{\sqrt{(2+k)^2 + (-1+2k)^2 + (3-2k)^2}} = 1 = \frac{|-3k|}{\sqrt{(2+k)^2 + (-1+2k)^2 + (3-2k)^2}}$$

$$\Rightarrow |-3k| = \sqrt{(2+k)^2 + (-1+2k)^2 + (3-2k)^2}$$

Squaring both sides, we have

$$9k^2 = 4 + 4k + k^2 + 1 - 4k + 4k^2 + 9 - 12k + 4k^2$$

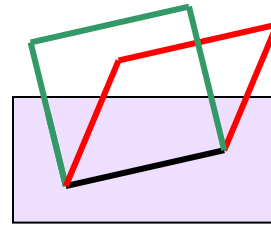
Simplifying, we get $-12k + 14 = 0$ or $k = 7/6$.

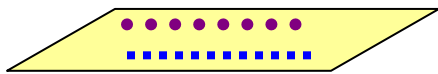
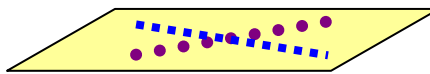
Putting this value of k in equation (i) and simplifying, we have: $19x + 8y + 4z = 21$.

This is the equation of the required plane.

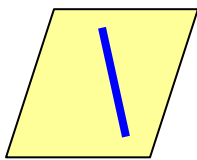
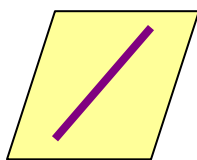
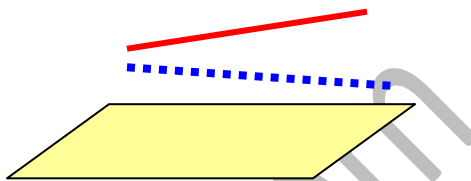
Skew Lines

Let L and M be two lines. If the lines are coplanar, then they either intersect each other at a common point or they are parallel. This is shown in the figure below:



**Parallel coplanar lines****Non-parallel coplanar lines**

Two non – coplanar straight lines neither intersect nor are they parallel. Such lines are called **skew lines**. It may be noted that any two skew lines may lie in parallel planes. These two facts are shown in the following figures.

**Skew Lines: Two lines lie in two different planes****The lines do not lie in the plane and they do not intersect each other**

Example 04 Show that the lines

$$L: \frac{x-3}{3} = \frac{y-8}{-1} = \frac{z-3}{1} \text{ and } M: \frac{x+3}{-3} = \frac{y+7}{2} = \frac{z-6}{4}$$

are skew lines.

Solution: First we check whether the lines are parallel or not.

Direction vectors of both the lines are: $\mathbf{d}_1 = [3, 1, 1]$ and $\mathbf{d}_2 = [-3, 2, 4]$.

Since the direction vectors are not parallel, therefore the lines are also not parallel.

Now let us see that whether the lines intersect or not.

Parametric equations of the lines are: $L: \begin{cases} x = 3 + 3t \\ y = 8 - t \\ z = 3 + t \end{cases}$ and $M: \begin{cases} x = -3 - 3s \\ y = -7 + 2s \\ z = 6 + 4s \end{cases}$

or $3 + 3t = -3 - 3s, 8 - t = -7 + 2s, 3 + t = 6 + 4s$

$$t + s + 2 = 0 \quad (\text{i})$$

or $-t - 2s + 15 = 0 \quad (\text{ii})$

$$t - 4s - 3 = 0 \quad (\text{iii})$$

Adding equations (i) and (ii), we get $-s + 17 = 0$ or $s = 17$

Put this value of s in (iii), we obtain $t - 68 - 3 = 0$ or $t = 71$

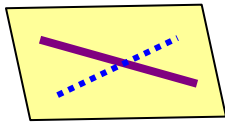
Substituting these values in equation (ii), we see that: $-71 - 34 + 15 \neq 0$

The values of t and s do not satisfy the first equation. Therefore, the lines do not intersect. It implies that lines are neither parallel nor intersecting. Hence the lines are skew lines.

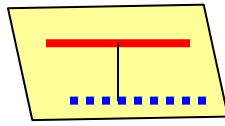
Shortest Distance between Two Straight Lines

If two lines L and M are coplanar then either they intersect or are parallel. In case they intersect, the shortest distance between them is zero. If they are parallel, the shortest distance between them is obtained by taking any point P on L and then finding the distance between this point P and the line M . If two lines are not coplanar (Skew lines), the shortest distance between them is the length of the line segment between them that is perpendicular to both line L and M .

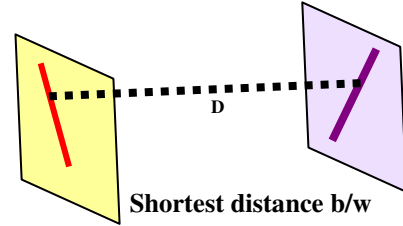
This is shown in the following figure.



**Shortest distance
b/w these line is zero**



**Shortest distance b/w
two parallel lines**



**Shortest distance b/w
two skew lines**

Example 05: Show that the two lines

$$L_1 = \begin{cases} x = 4 - t \\ y = -2 + 2t \\ z = 7 - 3t \end{cases} \quad \text{and} \quad L_2 = \begin{cases} x = 3 + 2s \\ y = -7 - 3s \\ z = 6 + 4s \end{cases}$$

are skew line. Also find the point on the lines such that the segment joining these points is perpendicular to both lines, and hence find the shortest distance between the given lines.

Solution: The direction ratios of L_1 are $-1, 2, -3$ and direction ratios of L_2 are $2, -3, 4$ respectively. We observe that these are not proportional to each other; therefore the given lines are not parallel. Now we find if the two lines intersect or not. Equating the two equations, we get

$$4 - t = 3 + 2s \quad \text{or} \quad -t - 2s + 1 = 0 \quad (\text{i})$$

$$-2 + 2t = -7 - 3s \quad \text{or} \quad 2t + 3s + 5 = 0 \quad (\text{ii})$$

$$7 - 3t = 6 + 4s \quad \text{or} \quad -3t - 4s + 1 = 0 \quad (\text{iii})$$

Solving equations (i) and (ii) we get $s = 7$ and $t = -1$. Substituting these values in (iii) we get

$$-3(-13) - 4(7) + 1 = 0 \quad \text{or} \quad 12 = 0 \quad \text{which is not true.}$$

This shows that the lines do not intersect. Now, if two lines are not parallel and non-intersecting then they are skew lines.

To find the shortest distance between these lines we proceed as under:

The general points on the lines L_1 and L_2 are $P(4 - t, -2 + 2t, 7 - 3t)$ and $Q(3 + 2s, -7 - 3s, 6 + 4s)$.

Thus, direction ratios of line PQ joining these points are:

$$3 + 2s - (4 - t), -7 - 3s - (-2 + 2t), 6 + 4s - (7 - 3t) \quad \text{or} \quad t + 2s - 1, -2t - 3s - 5, 3t + 4s - 1.$$

Now, PQ will be perpendicular to line L_1 if

$$\begin{aligned} &(-1)(t + 2s - 1) + (2)(-2t - 3s - 5) + (-3)(3t + 4s - 1) = 0 \\ &-t - 2s + 1 - 4t - 6s - 10 - 9t - 12s + 3 = 0 \Rightarrow -14t - 20s - 6 = 0 \\ &7t + 10s + 3 = 0 \end{aligned} \quad (\text{iv})$$

The line L_2 is perpendicular if

$$\begin{aligned} &(2)(t + 2s - 1) + (-3)(-2t - 3s - 5) + (4)(3t + 4s - 1) = 0 \\ &\text{or } 2t + 4s - 2 + 6t + 9s + 15 + 12t + 16s - 4 = 0 \\ &20t + 29s + 9 = 0 \end{aligned} \quad (\text{v})$$

Solving (iv) and (v), we get: $t = 1$ and $s = -1$

Therefore the required points are: $P(3, 0, 4)$ and $Q(1, -4, 2)$.

$$\text{The shortest distance} = \sqrt{(1-3)^2 + (-4-0)^2 + (2-4)^2} = 2\sqrt{6} \text{ units.}$$

Example 06: Find the shortest distance between the straight line joining the points $A(3, 2, -4)$ and $B(1, 6, -6)$ and the straight line joining the points $C(-1, 1, -2)$ and $D(-3, 1, -6)$. Also find equations of the line of shortest distance and coordinates of the feet of the common perpendiculars.

Solution: Let L denote the line joining the points $A(3, 2, -4)$ and $B(1, 6, -6)$ and M denote the line joining the points $C(-1, 1, -2)$ and $D(-3, 1, -6)$. Then their parametric equations are given by:

$$L: \begin{cases} x = 3 - 2t \\ y = 2 + 4t \\ z = -4 - 2t \end{cases} \quad \text{and} \quad M: \begin{cases} x = -1 - 2s \\ y = 1 \\ z = -2 - 4s \end{cases}$$

General points on the lines L and M are respectively:

$$P(3 - 2t, 2 + 4t, -4 - 2t) \quad \text{and} \quad Q(-1 - 2s, 1, -2 - 4s).$$

Direction ratios of the line PQ joining these points are:

$$[-1 - 2s - (3 - 2t), 1 - (2 + 4t), -2 - 4s - (-4 - 2t)] = [2t - 2s - 4, -4t - 1, 2t - 4s + 2]$$

PQ will be perpendicular to L if

$$(-2)(2t - 2s - 4) + (4)(-4t - 1) + (-2)(2t - 4s + 2) = 0 \quad \text{or} \quad 2t - s = 0 \quad (\text{i})$$

The same line PQ will be perpendicular to M if

$$(-2)(2t - 2s - 4) + (-4)(2t - 4s + 2) = 0 \quad \text{or} \quad 3t - 5s = 0 \quad (\text{ii})$$

Solving (i) and (ii), we get: $t = 0$ and $s = 0$.

Substituting these values in the equations of lines L and M , we get the required points (coordinates of the feet of the common perpendicular) as: $(3, 2, -4)$ and $(-1, 1, -2)$.

$$\text{The shortest distance} = \sqrt{(-1 - 3)^2 + (1 - 2)^2 + (-2 + 4)^2} = \sqrt{16 + 1 + 4} = \sqrt{21} \text{ units}$$

Equations of the line of shortest distance are:

$$\frac{x - 3}{-4} = \frac{y - 2}{-1} = \frac{z + 4}{2} \quad \text{or} \quad \frac{x + 1}{-4} = \frac{y - 1}{-1} = \frac{z + 2}{2}.$$

Example 07: Find the shortest distance between the straight lines

$$L: \frac{x - x_1}{c_1} = \frac{y - y_1}{c_2} = \frac{z - z_1}{c_3}; \quad M: \frac{x - x_2}{d_1} = \frac{y - y_2}{d_2} = \frac{z - z_2}{d_3}$$

Also find the equations of the line of shortest distance.

Solution: Let $A(x_1, y_1, z_1)$ and $B(x_2, y_2, z_2)$ be two points on the lines L and M respectively.

$$\text{Then } \mathbf{PQ} = [x_2 - x_1, y_2 - y_1, z_2 - z_1].$$

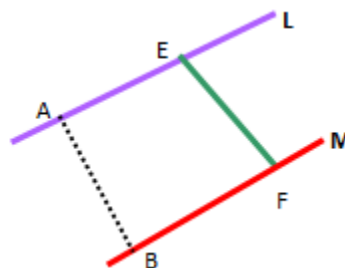
Let $\mathbf{EF}$ be perpendicular to both the straight lines L and M .

If l, m, n are the direction cosines of $\mathbf{EF}$ and $\mathbf{u}$ is a unit vector along $\mathbf{EF}$ then $\mathbf{u} = [l, m, n]$ and

$$\mathbf{EF} = \text{Projection of } \mathbf{AB} \text{ on } \mathbf{u} \text{ and is given by } \mathbf{P_u^{AB}} = \frac{|l(x_2 - x_1) + m(y_2 - y_1) + n(z_2 - z_1)|}{\sqrt{l^2 + m^2 + n^2}}$$

$$|l(x_2 - x_1) + m(y_2 - y_1) + n(z_2 - z_1)|, \text{ since } l^2 + m^2 + n^2 = 1.$$

Thus, the shortest distance between the lines is given by



$$|\mathbf{EF}| = |l(x_2 - x_1) + m(y_2 - y_1) + n(z_2 - z_1)| \quad (1)$$

Since $\mathbf{EF}$ and L intersect, therefore these two lines are coplanar. Equation of the plane containing these two lines is given by

$$\begin{vmatrix} x - x_1 & y - y_1 & z - z_1 \\ c_1 & c_2 & c_3 \\ 1 & m & n \end{vmatrix} = 0 \quad (i)$$

Again since $\mathbf{EF}$ and M intersect, therefore these two lines are coplanar. Equation of the plane containing these two lines is given by

$$\begin{vmatrix} x - x_2 & y - y_2 & z - z_2 \\ d_1 & d_2 & d_3 \\ 1 & m & n \end{vmatrix} = 0 \quad (ii)$$

Therefore, $\mathbf{EF}$ is the intersection of the planes (i) and (ii) whose equations non-symmetric form are given by

$$\begin{vmatrix} x - x_1 & y - y_1 & z - z_1 \\ c_1 & c_2 & c_3 \\ 1 & m & n \end{vmatrix} = 0 = \begin{vmatrix} x - x_2 & y - y_2 & z - z_2 \\ d_1 & d_2 & d_3 \\ 1 & m & n \end{vmatrix} \quad (2)$$

Example 08: Show that that the shortest distance between the straight lines

$$\frac{x-1}{2} = \frac{y-2}{3} = \frac{z-3}{4} \quad \text{and} \quad \frac{x-2}{3} = \frac{y-4}{4} = \frac{z-5}{5}$$

is $1/\sqrt{6}$ and the equations of the straight line perpendicular to both are

$$11x + 2y - 7z + 6 = 0 = 7x + y - 5z + 7$$

Solution: Let $\mathbf{n} = [a, b, c]$ be the direction vector of the common perpendicular to both given lines. Then

$$2a + 3b + 4c = 0 \quad (i)$$

$$3a + 4b + 5c = 0 \quad (ii)$$

Multiplying (i) by 3 and (ii) by 2 and subtracting, we have: $b + 2c = 0$ or $c = -b/2$

From (i), we have: $2a + 3b + 4(-1/2)b = 0$ or $2a + b = 0$ or $b = -2a$

Hence, $-2a = b = -2c$ or $\frac{a}{1} = \frac{b}{-2} = \frac{c}{1}$.

Thus, the shortest distance

$$= \frac{|a(x_2 - x_1) + b(y_2 - y_1) + c(z_2 - z_1)|}{\sqrt{a^2 + b^2 + c^2}} = \frac{|(1)(2-1) + (-2)(4-2) + (1)(5-3)|}{\sqrt{(1)^2 + (-2)^2 + (1)^2}} = \frac{1}{\sqrt{6}}$$

Equations of the shortest distance are:

$$\begin{vmatrix} x - x_1 & y - y_1 & z - z_1 \\ c_1 & c_2 & c_3 \\ 1 & m & n \end{vmatrix} = 0 = \begin{vmatrix} x - x_2 & y - y_2 & z - z_2 \\ d_1 & d_2 & d_3 \\ 1 & m & n \end{vmatrix}$$

Substituting the values, we get

$$\begin{vmatrix} x-1 & y-2 & z-3 \\ 2 & 3 & 4 \\ 1 & -2 & 1 \end{vmatrix} = 0 = \begin{vmatrix} x-2 & y-4 & z-5 \\ 3 & 4 & 5 \\ 1 & -2 & 1 \end{vmatrix}$$

$$(x-1)(11) - (y-2)(-2) + (z-3)(-7) = 0 = (x-2)(14) - (y-4)(-2) + (z-5)(-10)$$

$$11x - 11 + 2y - 4 - 7z + 21 = 0 = 14x - 28 + 2y - 8 - 10z + 50$$

$$11x + 2y - 7z + 6 = 0 = 7x + y - 5z + 7$$

WORKSHEET 05

- Find the equation of the plane passing through the points (2, 1, 1), (6, 3, 1) and (-2, 1, 2).
- Show that if two points (3, -1, 4) and (-1, 5, 0) are at equal distance from the point (x, y, z) then $2x - 3y + 2z = 0$.
- Transform the equations of the planes $3x - 4y + 5z = 0$ and $2x - y - 22z = 5$ to normal forms and hence find the angle between them.
- Find an equation of the plane through the points (1, 0, 1) and (2, 2, 1) and perpendicular to the plane $x - y - z + 4 = 0$.
- Find the equation of the plane that is perpendicular bisector of the line segment joining the points (3, 4, -1) and (5, 2, 7).
- Show that the join of (0, -1, 0) and (2, 4, -1) intersects the join of (1, 1, 1) and (3, 3, 9).
- Find the equation of the plane containing the line $x = 2t, y = 3t, z = 4t$ and intersection of the planes $x + y + z = 0$ and $2y - z = 0$.
- Find the family of the planes that have x-intercept 5, y-intercept 2 and a non-zero z-intercept such that they are perpendicular to the plane $3x - 2y + z - 4 = 0$.
- Find an equation of the plane passing through the point (2, -3, 1) and containing the line $x - 3 = 2y = 3z - 1$.
- Find the parametric equations of the line containing the point (2, 4, -3) and perpendicular to the plane $3x + 3y - 7z - 9 = 0$.
- Find the equation of the straight lines passing through the point (1, 2, 3) and parallel to the line $x - y + 2z - 5 = 0 = 3x + y + z + 6$.
- Find the equation of the plane passing through the point (2, 1, 4) and perpendicular to the vector $\mathbf{n} = [1, 4, 6]$.
- Find the equation of the plane through three points P(3, 2, 1), Q(4, 1, 0) and R(6, 0, 1).
- Convert the equation of plane $x + t - z + 3 = 0$ into (i) Intercept form and (ii) Normal Form.
- Find the measure of angle between the planes $2x - y + z = 1$ and $3x - 4y + z = 5$.
- Determine whether the planes $x - 3y + 4z = 1$ and $-2x + 6y + 5z = 2$ are perpendicular or not.
- Find the distance between the point (1, 3, 7) and the plane $x - y - z - 4 = 0$.
- Find the equation of the plane through the point (4, -2, 1) and perpendicular to the line $x = 3t, y = 6t, z = t$.
- Find the equation of the plane through (4, 1, 2) and perpendicular to each of the planes $2x + y - 2z = 3$ and $x - 3y + 4z = 7$.
- Find the equation of the plane which passes through the point (1, 2, 3) and has x-intercept equal to 6 and is perpendicular to the plane $x + y - z = 5$.

21. Find the distance of the point $P(1, -4, 1)$ from the plane $x - 3y - z = 1$ measured parallel to the line $\frac{x}{3} = \frac{y}{1} = \frac{z}{2}$.

22. Find the equations of two planes whose distances from the origin are 2 units each and which are perpendicular to the line through the points $(3, 3, 3)$ and $Q(6, 5, 1)$.

23. Show that the planes $x + 4y + 5z = 1$ and $2x + 2y - 3z = 3$ intersect each other. Hence find the equation of their line of intersection.

24. Show that the straight line $\frac{x+3}{4} = \frac{y+4}{-2} = \frac{z+1}{3}$ is parallel to the plane $2x - 2y + 4z = 9$.

25. Determine the point, if any, common to the straight line $x = 1 + 2t, y = 3t, z = 1 + t$ and the plane $2x + y - z = 5$.

26. Prove that the straight lines:

$$L: \frac{x+1}{1} = \frac{y-1}{3} = \frac{z-1}{2} \text{ and } M: \frac{x+4}{2} = \frac{y-3}{3} = \frac{z+1}{1}$$

intersect. Also find their point of intersection and a plane through them.

27. Prove that the straight lines:

$$\frac{x-2}{3} = \frac{y-3}{4} = \frac{z-4}{5} \text{ and } \frac{x-3}{4} = \frac{y-4}{5} = \frac{z-5}{6}$$

are coplanar. Also find the equation of the plane that contains both lines.

28. Find the equation of the plane that passes through the line of intersection of the planes $2x - y + 3z = 0$ and $x + 2y - 2z - 3 = 0$ and perpendicular to the xy -plane.

29. Find the equation of the plane that passes through the line of intersection of the planes $x - y + z = 1$ and $2x + y - z = 3$ and is at 2 units distance from the origin.

30. Are the following line skew?

$$L: \frac{x-4}{4} = \frac{y-7}{-1} = \frac{z-4}{2} \text{ and } M: \frac{x+2}{2} = \frac{y+3}{3} = \frac{z-2}{1}$$

Also find the shortest distance between the given lines.

31. Find the shortest distance between the straight line joining the points $A(1, 2, 3)$ and $B(3, 4, 5)$ and the straight line joining the points $C(1, 3, 2)$ and $D(3, 5, 6)$. Also find equations of the line of shortest distance and coordinates of the feet of the common perpendiculars.

32. Find the shortest distance between the lines.

$$\frac{x-1}{1} = \frac{y-2}{2} = \frac{z-3}{1} \text{ and } \frac{x-2}{4} = \frac{y-4}{7} = \frac{z-5}{1}$$

Also find the equation of the straight line perpendicular to both lines.

CHAPTER SIX

CYLINDRICAL AND SPHERICAL COORDINATES

INTRODUCTION

This section introduces two new coordinate systems for space $\mathbb{R}^3$; the cylindrical and the spherical coordinate systems. Cylindrical coordinates simplify the equations of cylinders whereas the spherical coordinates simplify the equations of spheres and cones. In fact, these coordinates arise in the study of planetary motion and the derivation of Kepler's laws in Physics and Astronomy.

Cylinder and Cylindrical Coordinates

It is frequently convenient to use cylindrical coordinates to locate a point in the space. We obtain cylindrical coordinates for space by combining polar coordinates in the xy -plane with z -axis. This assigns to every point in space one or more coordinate triples of the form (r, θ, z) , as shown in figure (1).

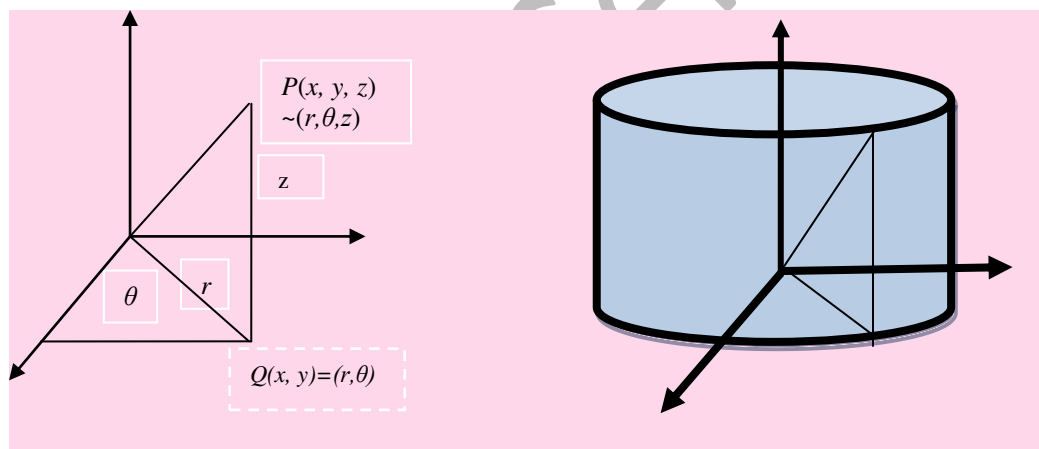


Fig. 1

Definition

Cylindrical coordinates represent a point P in space by ordered triples (r, θ, z) , in which

- (i) r and θ are polar coordinates for the vertical projection of P on the xy -plane.
- (ii) z is the rectangular vertical coordinate.

The relation between rectangular and cylindrical coordinates is given as under.

$$x = r \cos \theta, \quad y = r \sin \theta, \quad z = z, \quad \text{where } r = \sqrt{x^2 + y^2} \text{ and } \theta = \tan^{-1}(y/x)$$

Example 01: Find the cylindrical coordinates of the point whose rectangular coordinates are $\left(\frac{16}{5}, \frac{12}{5}, 3\right)$.

Solution: Given $x = 16/5$, $y = 12/5$ and $z = 3$. To find r and θ we have; by definition

$$r = \sqrt{x^2 + y^2} = \sqrt{\left(\frac{16}{5}\right)^2 + \left(\frac{12}{5}\right)^2} = \sqrt{\frac{400}{25}} = \sqrt{16} = 4.$$

Also,
$$\theta = \tan^{-1}\left(\frac{y}{x}\right) = \tan^{-1}\left(\frac{12/5}{16/5}\right) = \tan^{-1}\frac{3}{4} = 57.3^\circ, \quad \text{and } z = 3.$$

Therefore, the cylindrical coordinates are; $(r, \theta, z) = (4, 57.3^\circ, 3)$.

Example 02: Find the cylindrical coordinates of the point whose rectangular coordinates are $(-6/5, -2/5, 3)$.

Solution: Given, $x = -6/5$, $y = -2/5$, $z = 3$. To find r and θ we have; by definition

$$r = \sqrt{x^2 + y^2} = \sqrt{\left(\frac{-6}{5}\right)^2 + \left(\frac{-2}{5}\right)^2} = \sqrt{\frac{180}{25}} = 51.84.$$

$$\theta = \tan^{-1}\left(\frac{y}{x}\right) = \tan^{-1}\left(\frac{-2/5}{-6/5}\right) = \tan^{-1}(2) = 63.4^\circ \Rightarrow \theta = 180 + 63.4 = 243.4^\circ, \quad \text{and } z = 3.$$

It may be noted that since both x and y are negative, thus Since x and y both are negative, hence $\theta = 180 + 63.4 = 243.4^\circ$.

Therefore, required cylindrical coordinates are: $(r, \theta, z) = (4, 243.4^\circ, 3)$.

Example 03: Change the following cylindrical coordinates to rectangular coordinates: $(6, \pi/3, -5)$.

Solution: Given: $r = 6$, $\theta = \pi/3$, $z = -5$. To find x and y we have; by definition

$$x = r \cos \theta = 6 \cos(\pi/3) = 6(1/2) = 3.$$

Also,
$$y = r \sin \theta = 6 \sin(\pi/3) = 6(\sqrt{3}/2) = 3\sqrt{3}.$$

Thus, rectangular coordinates are: $(x, y, z) = (3, 3\sqrt{3}, -5)$.

Example 04: Express the equation $z = r^2 \cos 2\theta$ in rectangular coordinates.

Solution: Given that $z = r^2 \cos 2\theta$ or $z = r^2 (\cos^2 \theta - \sin^2 \theta) = r^2 \cos^2 \theta - r^2 \sin^2 \theta = x^2 - y^2$.

Thus the required equation in the rectangular coordinates is; $z = x^2 - y^2$.

Example 05: Express $x^2 - y^2 + z^2 = 1$ in cylindrical coordinates.

Solution: Given that $x^2 - y^2 + z^2 = 1$. Changing x and y into polar coordinates, we have

$$(r \cos \theta)^2 - (r \sin \theta)^2 - z^2 = 1 \quad \text{or} \quad r^2(\cos^2 \theta + \sin^2 \theta) - z^2 = 1 \quad \text{or} \quad r^2 - z^2 = 1$$

This is the required equation in cylindrical coordinates.

Spherical Coordinates

Spherical coordinates locate the given point $P(x, y, z)$ with two angles and a distance, as shown in figure 2.

The first coordinate $\rho = |\mathbf{OP}|$ is the distance from the origin to the point P . The second coordinate is the angle θ is measured in cylindrical coordinates. The third coordinate, ϕ is the angle which OP makes with the positive z - axis. It is customary to take $\rho \geq 0$, $0 \leq \theta \leq 2\pi$, $0 \leq \phi \leq \pi$. θ and ϕ are always measured in radians.

Definition

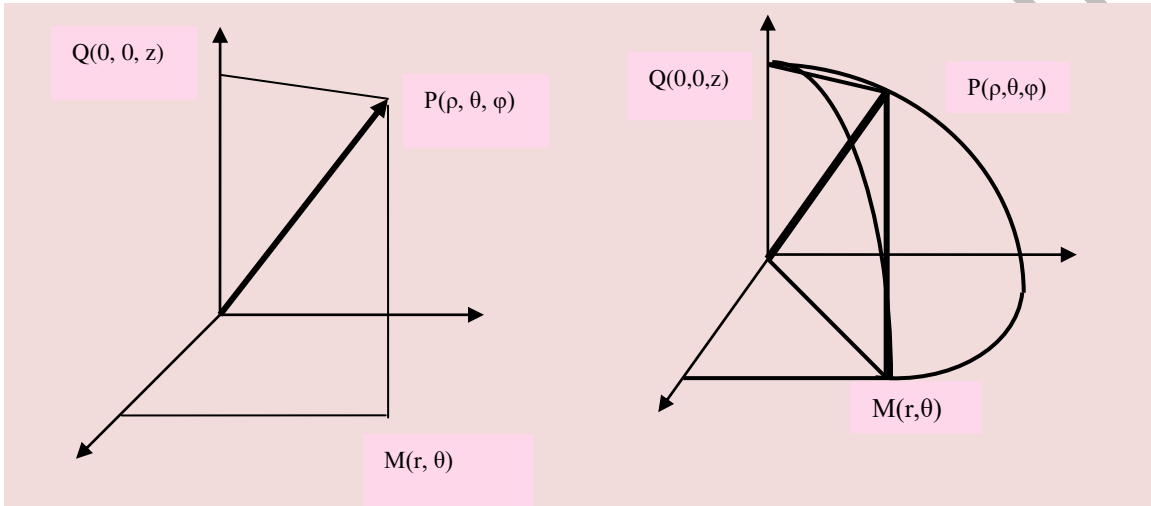
Spherical coordinates represent a point P in space by ordered triples (ρ, θ, φ) in which

- (i) ρ is the distance from P to the origin.
- (ii) θ is the angle and is same as in cylindrical coordinates.
- (iii) φ is the angle which **OP** makes with the positive z – axis.

It is clear from fig.2 that $r = |\mathbf{OM}| = |\mathbf{PQ}| = \rho \sin \varphi$, $|\mathbf{OQ}| = \rho \cos \varphi$ where as θ remains unchanged.

This give relationships between spherical and cylindrical coordinates of the point P. Since, $x = r \cos \theta$ and $y = r \sin \theta$ we have the following formulas to compute the rectangular coordinates of a point from its spherical coordinates.

$$x = \rho \sin \varphi \cos \theta, y = \rho \sin \varphi \sin \theta, z = \rho \cos \varphi$$

**Fig. 2**

Moreover, since ρ is the distance between the origin and P, we have

$$\rho = \sqrt{x^2 + y^2 + z^2} = \sqrt{r^2 + z^2}$$

Remarks:

- (i) The reason for the name spherical coordinates is that the surface $\rho = a$ (constant) represents a sphere.
- (ii) ρ is called the radial distance of P, angle θ the azimuth of P, and the angle φ the colatitudes of P. These three quantities are called spherical (or geographical) coordinates of P and we write $P(\rho, \theta, \varphi)$.

Example 06: Find the spherical coordinates of the point whose rectangular coordinates are $(4, -4\sqrt{3}, 6)$.

Solution: Given: $x = 4, y = -4\sqrt{3}, z = 6$. Required: $\rho = ?$, $\theta = ?$, $\varphi = ?$

We know that $\rho = \sqrt{x^2 + y^2 + z^2} = \sqrt{4^2 + (-4\sqrt{3})^2 + 6^2} = \sqrt{100} = 10$

Also, $\theta = \tan^{-1}\left(\frac{y}{x}\right) = \tan^{-1}\left(\frac{-4\sqrt{3}}{4}\right) = \tan^{-1}(-\sqrt{3}) = \frac{5\pi}{3} = 300^\circ$.

And, $\varphi = \cos^{-1}\left(\frac{z}{\rho}\right) = \cos^{-1}\left(\frac{6}{10}\right) = \cos^{-1}\left(\frac{3}{5}\right) = 53.1^\circ$

Thus, the required spherical coordinates are; $(\rho, \theta, \phi) = (10, 300^\circ, 53.1^\circ)$.

Example 07: Find the rectangular coordinates of point whose spherical coordinates are: $\left(4, \frac{\pi}{3}, \frac{2\pi}{3}\right)$.

Solution: Given: $\rho = 4, \theta = \frac{\pi}{3}, \phi = \frac{2\pi}{3}$. Required: $x = ?, y = ?, z = ?$

$$\text{Here } x = \rho \sin \phi \cos \theta = 4 \sin \frac{2\pi}{3} \cos \frac{\pi}{3} = 4 \cdot \frac{\sqrt{3}}{2} \cdot \frac{1}{2} = 2 \cdot \frac{\sqrt{3}}{2} = \sqrt{2} \sqrt{3} = \sqrt{6}$$

$$\text{Also, } y = \rho \sin \phi \sin \theta = 4 \sin \frac{2\pi}{3} \sin \frac{\pi}{3} = 4 \cdot \frac{\sqrt{3}}{2} \cdot \frac{\sqrt{3}}{2} = 4(3/4) = 3$$

$$\text{And, } z = \rho \cos \phi = 4 \cos 2\pi/3 = 4(-1/2) = -2$$

Thus, rectangular coordinates are: $(x, y, z) = (\sqrt{6}, 3, -2)$.

Example 08: Express equation $\rho = 2 \cos \theta \sin \phi$ in rectangular coordinates.

Solution: We have, $\rho = 2 \cos \theta \sin \phi$. Multiplying both sides by ρ , we get

$$\rho^2 = 2\rho \cos \theta \sin \phi \quad \text{or} \quad x^2 + y^2 + z^2 = 2x.$$

Example 09 Express equation $x^2 + y^2 + 2z = 6$ in spherical coordinates.

Solution: We have $x^2 + y^2 + 2z = 6$. Substituting the values for x, y, z , we get

$$(\rho \sin \phi \cos \theta)^2 + (\rho \sin \phi \sin \theta)^2 + 2(\rho \cos \phi) = 6 \Rightarrow \rho^2 \sin^2 \phi \cos^2 \theta + \rho^2 \sin^2 \phi \sin^2 \theta + 2\rho \cos \phi - 6 = 0$$

$$\rho^2 \sin^2 \phi (\cos^2 \theta + \sin^2 \theta) + 2\rho \cos \phi - 6 = 0 \Rightarrow \rho^2 \sin^2 \phi + 2\rho \cos \phi - 6 = 0$$

This is equation in spherical coordinates.

THE SPHERE

The locus of a point $P(x, y, z)$ such that its distance from some fixed point $C(a, b, c)$ always remains constant is called a sphere.

The fixed distance is called a radius and the fixed point C is called the center of the sphere.

Equation of Sphere

By definition $|CP| = r$. Using the distance formula, we get

$$\sqrt{(x-a)^2 + (y-b)^2 + (z-c)^2} = r$$

$$\text{or } (x-a)^2 + (y-b)^2 + (z-c)^2 = r^2 \quad (1)$$

Equation (1) is an equation of sphere.

If center lies at the origin, equation (1) becomes;

$$x^2 + y^2 + z^2 = r^2 \quad (2)$$

Now simplifying equation (1), we get

$$x^2 + y^2 + z^2 - 2ax - 2by - 2cz + a^2 + b^2 + c^2 - r^2 = 0 \quad (3)$$

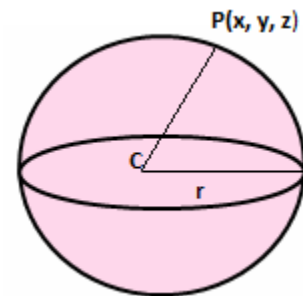
Let, $-2a = f, -2b = g, -2c = h$ and $a^2 + b^2 + c^2 - r^2 = d$.

Then equation (3) becomes: $x^2 + y^2 + z^2 + fx + gy + hz + d = 0 \quad (4)$

Equation (1) is known as “Standard Form” and equation (4) is known as “General Form” of equation of “Sphere”.

The center of the sphere is $(a, b, c) = (-f/2, -g/2, -h/2)$ and the radius r of the sphere is

$$r = \sqrt{a^2 + b^2 + c^2 - d}$$



Example 01: Show that in spherical coordinates, $\rho = c$ is an equation of sphere centered at origin with radius equal to c .

Solution: We know that in spherical coordinates, $\rho = \sqrt{x^2 + y^2 + z^2}$

Now $\rho = c$ or $\rho^2 = c^2$. This gives, $x^2 + y^2 + z^2 = c^2$.

This is equation of sphere centered at origin and radius equal to c .

Example 02: Find the equation of sphere whose center lies on y -axis and which passes through the points $(0, 2, 2)$ and $(4, 0, 0)$.

Solution: If center of sphere lies on the y -axis, then its coordinates are $(0, b, 0)$. Hence, equation of sphere will be

$$(x-0)^2 + (y-b)^2 + (z-0)^2 = r^2$$

Simplifying, we get: $x^2 + y^2 - 2xb + b^2 + z^2 = r^2$ (i)

Now this sphere passes through the points $(0, 2, 2)$ and $(4, 0, 0)$. Thus we have

$$0 + 4 - 0 + b^2 + 4 = r^2 \quad \text{or} \quad 8 + b^2 = r^2 \quad \text{(ii)}$$

$$\text{and} \quad 16 + 0 - 8b + b^2 = r^2 \quad \text{or} \quad 16 + b^2 - 8b = r^2 \quad \text{(iii)}$$

Subtracting (iii) from (ii), we obtain: $-16 + 8b = 0$ giving $b = -2$

Put this value of b in equation (ii), we get; giving $r^2 = 16$.

Now using equation (i) giving $x^2 + y^2 + z^2 + 4x = 12$ as equation of required sphere.

Example 03: Show that equation of sphere having the straight line joining the points (x_1, y_1, z_1) and (x_2, y_2, z_2) as a diameter is

$$(x - x_1)(x - x_2) + (y - y_1)(y - y_2) + (z - z_1)(z - z_2) = 0.$$

Solution: Let $A(x_1, y_1, z_1)$ and $B(x_2, y_2, z_2)$ be the end points of the diameter and let $P(x, y, z)$ be any point on the sphere. Then from geometry $\angle APB$ is a right angle. Thus AP and BP are perpendicular to each other. Now the direction ratios of vectors AP and BP are

$$\mathbf{AP} = [x - x_1, y - y_1, z - z_1] \text{ and } \mathbf{BP} = [x - x_2, y - y_2, z - z_2]$$

As $AP \perp BP$, hence $\mathbf{AP} \cdot \mathbf{BP} = 0$. Thus,

$$(x - x_1)(x - x_2) + (y - y_1)(y - y_2) + (z - z_1)(z - z_2) = 0.$$

This is the equation of the required sphere.

Example 04: Find the equation of the sphere which passes through $A(-3, 6, 0)$, $B(-2, -5, -1)$ and $C(1, 4, 2)$ and whose center lies on the hypotenuse of the triangle ABC .

Solution: Here, $|AB|^2 = (-2 + 3)^2 + (-5 - 6)^2 + (-1 - 0)^2 = 123$

$$|BC|^2 = (1 + 2)^2 + (4 + 5)^2 + (2 + 2)^2 = 99$$

$$|CA|^2 = (-3 - 1)^2 + (6 - 4)^2 + (0 - 2)^2 = 24$$

Observe that, $|AB|^2 = |BC|^2 + |CA|^2$ hence, triangle ABC is a right triangle, where AB is the hypotenuse.

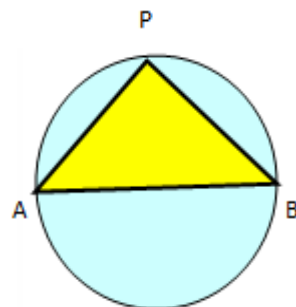
Now it is given that center of the sphere lies on hypotenuse side, that is, AB is a diameter.

Hence the equation of the sphere is given by using the formula:

$$(x - x_1)(x - x_2) + (y - y_1)(y - y_2) + (z - z_1)(z - z_2) = 0$$

$$\text{or} \quad (x + 3)(x + 2) + (y - 6)(y + 5) + (z - 0)(z + 1) = 0$$

Simplifying, we obtain $x^2 + y^2 + z^2 + 5x - y + z - 24 = 0$



Example 05: Find the center and radius of sphere $x^2 + y^2 + z^2 - 6x + 4z = 0$

Solution: Method I: Writing the equation as $(x^2 - 6x) + y^2 + (z^2 + 4z) = 0$

Completing the squares, we get: $(x^2 - 6x + 3^2) + y^2 + (z^2 + 4z + 2^2) = 0 + 3^2 + 2^2$

$$\text{or} \quad (x - 3)^2 + (y - 0)^2 + (z + 2)^2 = 13$$

This shows that center of given sphere is $(3, 0, -2)$ and radius is $\sqrt{13}$.

Method II: Comparing the given equation with general form of equation of sphere

$$x^2 + y^2 + z^2 + fx + gy + hz + d = 0$$

We see that $f = -6$, $g = 0$, $h = 4$ and $d = 0$. Now $a = -f/2 = 3$, $b = -g/2 = 0$, $c = -h/2 = -2$ and $d = 0$.

$$a^2 + b^2 + c^2 - r^2 = d = 0 \quad \text{or} \quad r^2 = a^2 + b^2 + c^2 = 9 + 0 + 4 = 13 \quad \text{giving } r = \sqrt{13}$$

Thus center of the sphere is $(a, b, c) = (3, 0, -2)$ and radius $r = \sqrt{13}$.

Example 06: Find the equation of sphere through the points $(0, 0, 0)$, $(0, -1, 1)$, $(-1, 2, 0)$ and $(1, 2, 3)$. Also find its center and radius.

Solution: Let the equation of sphere be

$$x^2 + y^2 + z^2 + fx + gy + hz + d = 0 \quad (1)$$

Now this sphere passes through the points $(0, 0, 0)$, $(0, -1, 1)$, $(-1, 2, 0)$ and $(1, 2, 3)$, hence these point must satisfy its equation. That is,

$$0 + 0 + 0 + 0 + 0 + 0 + d = 0. \quad \text{This implies that:} \quad d = 0 \quad (i)$$

$$0 + 1 + 1 + 0f - g + h + d = 0 \quad \text{or} \quad -g + h = -2 \quad (ii)$$

$$1 + 4 + 0 - f + 2g + 0h + d = 0 \quad \text{or} \quad -f + 2g = -5 \quad (iii)$$

$$1 + 4 + 9 + f + 2g + 3h + d = 0 \quad \text{or} \quad f + 2g + 3h = -14 \quad (iv)$$

$$\text{Adding (iii) and (iv), we get:} \quad 4g + 3h = -19 \quad (v)$$

$$\text{From (ii)} \quad h = g - 2 \quad (vi)$$

Put this value of h in equation (v), we get: $4g + 3(g - 2) = -19$ or $g = -13/7$.

Put this in equation (vi), we get $h = -27/7$.

Now put this value of g in equation (iii) and simplifying, we get $f = 2g + 5 = 9/7$. Thus equation of sphere (1) becomes:

$$x^2 + y^2 + z^2 + (9/7)x - (13/7)y - (27/7)z = 0. \quad \text{Multiplying by 7, we get}$$

$$7x^2 + 7y^2 + 7z^2 + 9x - 13y - 27z = 0$$

Example 07: Find the equation of sphere passing through the points $(0, -2, -4)$ and $(2, -1, -1)$ and having its center on the line $2x - 3y = 0 = 5y + 2z$.

Solution: (Note: $2x - 3y = 0$ and $5y + 2z = 0$ are planes. When they intersect they form a line). Let

$$\text{the equation of sphere be} \quad x^2 + y^2 + z^2 + fx + gy + hz + d = 0 \quad (1)$$

with center $(-f/2, -g/2, -h/2)$.

As this sphere passes through the points $(0, -2, -4)$ and $(2, -1, -1)$, we have:

$$0 + 4 + 16 + 0f - 2g - 4h + d = 0 \quad \text{or} \quad -2g - 4h + d = -20 \quad (i)$$

$$4 + 1 + 1 + 2f - g - h + d = 0 \quad \text{or} \quad 2f - g - h + d = -6 \quad (ii)$$

Since the center $(-f/2, -g/2, -h/2)$ lies on the line $2x - 3y = 0 = 5y + 2z$, we have

$$2(-f/2) - 3(-g/2) = 0 \quad \text{or} \quad 2f - 3g = 0 \quad (iii)$$

$$\text{and } 5(-g/2) + 2(-h/2) = 0 \quad \text{or} \quad 5g + 2h = 0 \quad (iv)$$

$$\text{Subtracting (ii) from (i), we get } -2f - g - 3h = -14 \quad \text{or} \quad g = 14 - 2f - 3h \quad (v)$$

Put this in (iii) and (iv), we get

$$2f - 42 + 6f + 9h = 0 \quad \text{or} \quad 8f + 9h = 42 \quad (\text{vi})$$

$$70 - 10f - 15h + 2h = 0 \quad \text{or} \quad 10f + 13h = 70 \quad (\text{vii})$$

Multiplying equation (vi) by 10 and (vii) by 8 and subtracting, we get:

$$80f + 90h = 420$$

$$80f + 104h = 280$$

Subtracting

$$\begin{array}{r} - \\ - \\ - \\ \hline -14h = -140 \end{array}$$

This gives, $h = 10$. Put this in equation (vi), we get $8f + 90 = 42$ or $f = -6$

Now put these values in equation (v), to obtain: $g = 14 - 2(-6) - 3(10) = -4$

Finally from equation (i), $d = 2g + 4h - 20 = 2(-4) + 4(10) - 20$ or $d = 12$

Thus equation (1) becomes: $x^2 + y^2 + z^2 - 6x - 4y + 10z + 12 = 0$

This is equation of required sphere.

Example 08: Find the equation of sphere that passes through the circle $x^2 + y^2 + z^2 = 9$, $2x + 3y + 4z = 5$ and the point $(1, 2, 3)$.

Solution: We know that $x^2 + y^2 + z^2 = 9$ is an equation of sphere and $2x + 3y + 4z = 5$ is a plane.

When they intersect each other at right angle, they form a circle. Now the equation of sphere that

contains the given circle is $(x^2 + y^2 + z^2 - 9) + k(2x + 3y + 4z - 5) = 0$ (1)

As this sphere passes through the point $(1, 2, 3)$, equation (1) gives

$$1 + 4 + 9 - 9 + k(2 + 6 + 12 - 5) = 0 \quad \text{or} \quad k = -5/15 = -1/3.$$

Put this value of k in equation (1) and simplify, we obtain

$$3x^2 + 3y^2 + 3z^2 - 2x - 3y - 4z + 5 = 0$$

This is the equation of required sphere.

Example 09: Find the equation of sphere that passes through the circle $x^2 + y^2 + z^2 = 1$, $2x + 4y + 5z = 6$ and touching the xy -plane.

Solution: The sphere through the given circle is

$$(x^2 + y^2 + z^2 - 1) + k(2x + 4y + 5z - 6) = 0$$

The center of this sphere is $(-k, -2k, -5k/2)$ and radius is

$$r = \sqrt{k^2 + 4k^2 + (25k^2/4) + 1 + 6k}$$

As the sphere touches the xy -plane, hence its radius is the z -component of its center.

Thus $-5k/2 = \sqrt{k^2 + 4k^2 + (25k^2/4) + 1 + 6k}$. Squaring both sides and simplifying, we get:

$$k = -1, \text{ and } k = -1/5$$

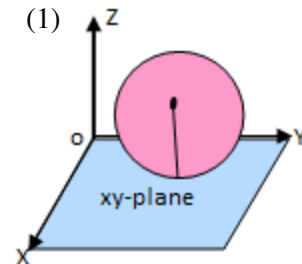
Put these values of k in equation (1), we get

$$x^2 + y^2 + z^2 - 2x - 4y - 5z + 5 = 0 \quad \text{and} \quad 5x^2 + 5y^2 + 5z^2 - 2x - 4y - z + 1 = 0$$

This shows that there are two such spheres that satisfy given conditions. In figure we have just shown one such circle. The other will be down the xy -plane that is difficult to show in 3-D plane.

Example 10: Show that the two circles $x^2 + y^2 + z^2 - 9 = 0 = x - 2y + 4z - 13$

and $x^2 + y^2 + z^2 + 6y - 6z + 21 = 0 = x + y + z + 2$



lie on the same sphere. Also find the equation of the sphere on which they lie.

Solution: The equation of two circles are

$$(x^2 + y^2 + z^2 - 9) + k(x - 2y + 4z - 13) = 0 \quad (i)$$

$$(x^2 + y^2 + z^2 + 6y - 6z + 21) + h(x + y + z + 2) = 0 \quad (ii)$$

If both circles lie on the same sphere then they must be identical. Thus by equating the coefficients of x , y and z terms, we get: $k = h$, $-2k = 6 + h$, $4k = -6 + h$ and $-9 - 13k = 21 + 2h$

All these equations yield: $k = h = -2$

Put $h = -k$ in (i), we obtain: $x^2 + y^2 + z^2 - 2x + 4y - 8z + 17 = 0$

This is the equation of required sphere.

Note: If we put $h = -2$ in (ii) we shall get the same result.

Example 11: Find the equation of sphere for which the following circle is a great circle.

$$x^2 + y^2 + z^2 + 7y - 2z + 2 = 0 = 2x + 3y + 4z - 8 = 0$$

Solution: The equation of required sphere is:

$$(x^2 + y^2 + z^2 + 7y - 2z + 2) + k(2x + 3y + 4z - 8) = 0 \quad (1)$$

The centre of this sphere is $(-k, -(7 + 3k)/2, 1 - 2k)$.

It may be noted that intersection of sphere and plane is a circle. This circle is a great circle if the plane $2x + 3y + 4z - 8 = 0$ must pass through the centre $(-k, -(7 + 3k)/2, 1 - 2k)$.

This implies: $2(-k) - 3(7 + 3k)/2 + 4(1 - 2k) - 8 = 0$. Simplifying, we get $k = -1$.

Put this value of k in equation (1), we get: $x^2 + y^2 + z^2 - 2x + 4y - 6z + 10 = 0$

This is equation of required sphere.

Example 12: The sphere is centered at $(2, -1, -1)$ and tangential to the plane $x - 2y + z + 7 = 0$. Find its equation.

Solution: Since the sphere is tangent to the plane, hence its radius is the distance between the plane and the center of the sphere. Using the formula of a distance between a plane and a point; i.e.,

$$D = \frac{|ax_1 + by_1 + cz_1 + d|}{\sqrt{a^2 + b^2 + c^2}} = \frac{|2 + 2 - 1 + 7|}{\sqrt{1 + 4 + 1}} = \frac{10}{\sqrt{6}} = r$$

Thus equation of sphere centered at $(2, -1, -1)$ and radius $r = 10/\sqrt{6}$ is given by using the formula:

$$(x - a)^2 + (y - b)^2 + (z - c)^2 = r^2 \text{ or } (x - 2)^2 + (y + 1)^2 + (z + 1)^2 = 100/6 = 50/3.$$

Simplifying, we obtain: $3x^2 + 3y^2 + 3z^2 - 12x + 6y + 6z - 32 = 0$

This is equation of required sphere.

Example 13: Given a sphere $3x^2 + 3y^2 + 3z^2 - 2x - 3y - 4z - 22 = 0$.

Find the equation of plane tangent to this sphere at the point $(1, 2, 3)$.

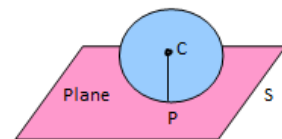
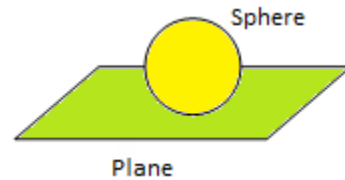
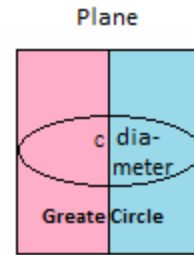
Solution: Refer to the figure shown here, we have to find equation of

the plane S . Now dividing given equation by 3, we get: $x^2 + y^2 + z^2 - 4x + 2y + 2z - 32/3 = 0$.

Center of this sphere is $C(2, -1, -1)$. Let $P(1, 2, 3)$ be the point where the plane touches the sphere.

Thus $\overrightarrow{PC}$ will be normal to this plane. Direction ratios of $\overrightarrow{PC}$ are $\mathbf{n} = [1 - 2, 2 + 1, 3 + 1] = [-1, 3, 4]$

Thus equation of the required plane passing through the point $P(1, 2, 3)$ with normal vector $\mathbf{n} = [-1, 3, 4]$ is: $-1(x - 1) + 3(y - 2) + 4(z - 3) = 0$ or $x - 3y - 4z + 17 = 0$.



WORKSHEET 06

1. Find the cylindrical coordinates of the point whose rectangular coordinates are $(2\sqrt{3}, 2, -2)$.
2. Find the spherical coordinates of the point whose rectangular coordinates are $(1, 1, \sqrt{6})$, $(-2, 2\sqrt{3}, 4)$ and $(-\sqrt{3}, 1, -2)$.
3. Express the following equations into rectangular coordinates:
(i) $\rho \cos \phi = 2$ (ii) $\rho = 7 \sin \theta \sin \phi$ (iii) $\rho^2 \cos 2\theta = a^2$ (iv) $z = 1 + \sin \theta$
4. Express the given equations into cylindrical and spherical coordinates.
(i) $(x + y)^2 - z^2 + 4 = 0$ (ii) $3x + y - 4z = 12$
5. Find the equation of sphere whose center lies on x-axis and which passes through the points $(1, 3, 0)$ and $(0, 5, 0)$.
6. Find the equation of the sphere which passes through the points $A(4, 9, 4)$, $B(0, 11, 2)$ and $C(1, 0, 1)$ and whose center lies on the hypotenuse of the triangle ABC.
7. Find the center and radius of the sphere $x^2 + y^2 + z^2 - 8x + 4y - 2z = 8$.
8. Find the equation of sphere through the points $(0, 0, 0)$, $(0, -1, 1)$, $(-1, 0, 2)$ and $(1, 2, 0)$. Also find its center and radius.
9. Find the equation of sphere passing through the points $(0, -2, -4)$ and $(2, -1, -1)$ and having its center on the line $x - 2z = 0 = 5x + 2y$.
10. Find the equation of sphere that passes through the circle $x^2 + y^2 + z^2 = 9$, $x + y + z = 8$ and the point $(1, 1, 1)$.
11. Find the equation of the sphere through the circle $x^2 + y^2 + z^2 = 1$, $2x + 4y + 5z = 6$ and touching the xz-plane.
12. Find the equation of the sphere for which the circle:
$$x^2 + y^2 + z^2 + 2x - 9y - z + 2 = 0 = x + y + z + 4 = 0$$
 is a great circle.
13. Find the equation of sphere centered at $(2, 1, 1)$ and tangent to the plane $2x - y + 3z + 1 = 0$.
14. Find the equation of the plane tangent to the sphere $x^2 + y^2 + z^2 - 4x - y - z + 12 = 0$ at the point $(1, 2, 1)$.

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CHAPTER SEVEN

MULTIPLE INTEGRALS

INTRODUCTION

Let $z = f(x, y)$ be a function of two variables defined and bounded in a closed region D in xy -plane. We divide D into small rectangles of length Δx and width Δy so that the area of this rectangle be $\Delta A = \Delta x \cdot \Delta y$. Let $P(x, y)$ be a point in this rectangle. Form a sum down as under.

$$S = \sum_{i=1}^n f(x_i, y_i) \Delta A_i$$

If the limit of S exists as n tends to infinity (∞) and the largest diagonal of the rectangle tends to zero, then this sum is called double integral and is usually denoted by;

$$\iint_D f(x, y) dA = \iint_D f(x, y) dx dy = \lim_{n \rightarrow \infty} \sum_{i=1}^n f(x_i, y_i) \Delta A_i$$

Double Integral as Volume

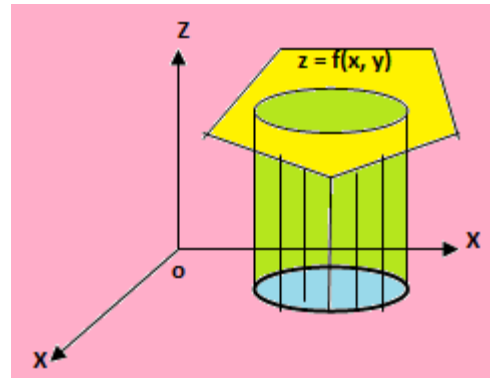
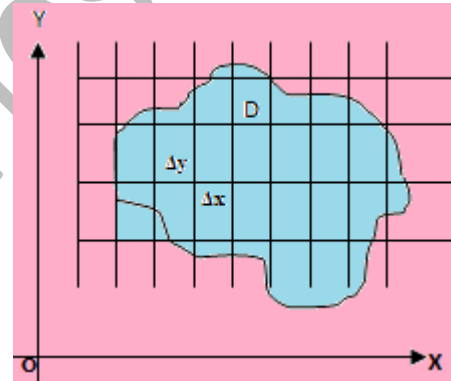
If $f(x, y) \geq 0$ in the region D then each term $f(x_i, y_i) \cdot \Delta A_i$ represents the volume of a parallelepiped with base ΔA_i and height $f(x_i, y_i)$ parallel to the z -axis. The sum of these volumes can be considered as an approximation to the volume of the solid below the surface $z = f(x, y)$ and above the region D in the xy -plane. The double integral then represents the volume below the graph of $z = f(x, y)$ and above the region D in xy -plane. This is shown in the figure.

Evaluation of Double Integral

Double integrals are calculated by using the techniques of evaluation of integration of function of one variable. This is done by expressing the double integral as a repeated integral. Let us suppose that the region D in the xy -plane be bounded on the top by the graph $y = g_1(x)$ and on the bottom by the graph $y = g_2(x)$. Also supposing that it extends from $x = a$ and $x = b$ on the right. Then

$$I = \iint_D f(x, y) dA = \int_a^b \int_{g_2(x)}^{g_1(x)} f(x, y) dy dx$$

This is shown in the following figure.



Double Integral in Polar Coordinates

When the region D is bounded by the straight lines $\theta = \alpha, \theta = \beta$ and the equations of the curves are in polar form $r = f_1(\theta)$ and $r = f_2(\theta)$, then double integral of the function $f(r, \theta)$ over D is given by $\iint_D f(r, \theta) dA$,

Here $dA = dx dy = r dr d\theta$ and $dx dy$ and $r dr d\theta$ are known as elements of area in Cartesian and Polar coordinates respectively.

Theorem: Prove that an element of area in Cartesian and polar coordinates is given by $dx dy = r dr d\theta$.

Proof: The element of area dA which is equal to $dx dy$ in Cartesian coordinates. It represents the area of a small rectangle when the region between two curves is divided into small rectangle meshes. Let us divide the area between two curves $y = g_1(x)$ and $y = g_2(x)$ in polar form. This is shown in the figure.

Let $|OP| = r$, $|OQ| = r + dr$, thus $|PQ| = dr$. Also let $\angle POR = d\theta$ and arc length $|RP| = s$. Then arc length $|RP| = s = r \cdot d\theta$. Now if the division is made infinitesimally small, then the figure PQSR will be a rectangle. Thus area of this small mesh will be

$$dA = |RP| \cdot |PQ| = r d\theta \cdot dr = r dr d\theta$$

Thus, $dA = dx dy = r dr d\theta$

Example 01: Evaluate the following double integrals

$$(i) \int_0^{23} \int_{10}^x (x+y) dx dy \quad (ii) \int_0^1 \int_1^2 (x+y) dy dx \quad (iii) \int_0^1 \int_{x^2}^x xy^2 dy dx$$

$$\begin{aligned} \text{Solution: } (i) \int_0^{23} \int_{10}^x (x+y) dx dy &= \int_0^{23} \left[\frac{x^2}{2} + xy \right]_{10}^x dy = \int_0^{23} \left(\frac{x^2}{2} + 3y \right) dy = \left[\frac{9y}{2} + \frac{3y^2}{2} \right]_1^2 \\ &= \left(\frac{9(2-1)}{2} + \frac{3(4-1)}{2} \right) = \frac{9}{2} + \frac{9}{2} = 9 \end{aligned}$$

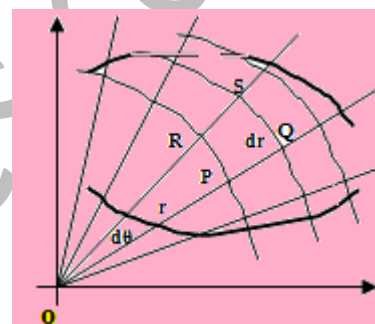
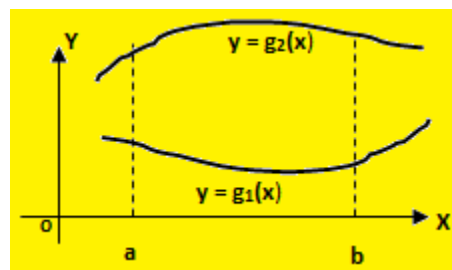
$$(ii) \text{ Similarly, } \int_0^1 \int_1^2 (x+y) dy dx = 9. \text{ Thus you may observe that if both limits of integration are constant}$$

and limits are interchanged and integration is done in reverse order, the result will be same. This is not the case when one of the limits of integration is variable.

$$\begin{aligned} (iii) \int_0^1 \int_{x^2}^x xy^2 dy dx &= \int_0^1 \left[x \frac{y^3}{3} \right]_{x^2}^x dx = \int_0^1 \left(\frac{x}{3} (x^3 - x^6) \right) dx = \int_0^1 \left(\frac{1}{3} (x^4 - x^7) \right) dx \\ &= \frac{1}{3} \left(\frac{x^5}{5} - \frac{x^7}{7} \right) \bigg|_0^1 = \frac{1}{3} \left(\frac{1}{5} - \frac{1}{7} \right) = \frac{1}{3} \times \frac{2}{35} = \frac{2}{105} \end{aligned}$$

$$\text{Example 02: Evaluate } \int_0^1 \int_0^1 \frac{1}{\sqrt{1+x^2} \sqrt{1+y^2}} dx dy$$

Solution: Both limits of integration are constants hence given integral can be rewritten as



$$\int_0^1 \int_0^1 \frac{1}{\sqrt{1-x^2} \sqrt{1-y^2}} dx dy = \int_0^1 \frac{dx}{\sqrt{1-x^2}} \cdot \int_0^1 \frac{dy}{\sqrt{1-y^2}} = \sin^{-1} x \Big|_0^1 \cdot \sin^{-1} y \Big|_0^1 = [\sin^{-1} 1 - \sin^{-1} 0]^2$$

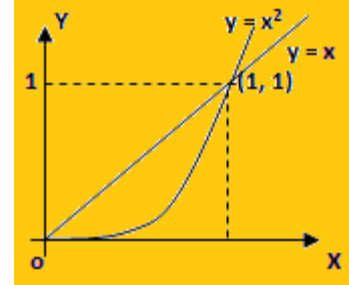
$$= (\pi/2)^2 = \frac{\pi^2}{4} \quad \left[\text{Note: } \sin^{-1} 0 = 0, \sin^{-1}(1) = \pi/2 \right]$$

Change of Order

Example 03: Evaluate $\iint_D xy(x+y) dy dx$ where D is the region

bounded by the line $y = x$ and parabola $y = x^2$.

Solution: Before the evaluation of integral we have to find the limits of integration. For this we have solve two equations simultaneously and get $x = 0$ and $x = 1$. Graphically this is show below.



Thus, $\iint_D xy(x+y) dy dx = \int_0^1 \int_{y=x^2}^x xy(x+y) dy dx$

$$= \int_0^1 \left\{ x^2 \left[\frac{y^2}{2} \right]_{x^2}^x + x \left[\frac{y^3}{3} \right]_{x^2}^x \right\} dx = \int_0^1 \left\{ x^2 \left[\frac{x^2 - x^4}{2} \right] + x \left[\frac{x^3 - x^6}{3} \right] \right\} dx = \int_0^1 \left\{ \frac{x^4 - x^6}{2} + \frac{x^4 - x^7}{3} \right\} dx$$

$$\frac{1}{2} \left[\frac{x^5}{5} - \frac{x^7}{7} \right] + \frac{1}{3} \left[\frac{x^5}{5} - \frac{x^8}{8} \right] \Big|_0^1 = \frac{1}{10} - \frac{1}{14} + \frac{1}{15} - \frac{1}{24} = \frac{3}{56}$$

Now, $y = x$ then $x = y$. Also, $y = x^2$ then, $x = \sqrt{y}$. Thus if the above integral is evaluated as

$$\iint_D xy(x+y) dx dy \text{ then } \iint_D xy(x+y) dx dy = \int_0^1 \int_{x=\sqrt{y}}^y xy(x+y) dx dy = \frac{3}{56}. \text{ (You may try this)}$$

Thus if you change the order of integration the value of integral will remain same but you have to take care of limits only. Here we produce two more problems that will make you to understand why change of integration is important.

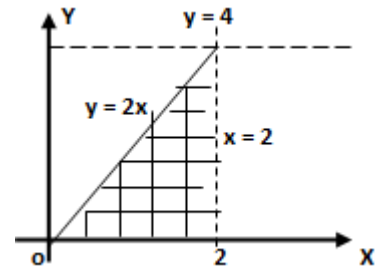
Example 04: Evaluate $\int_0^4 \int_{y/2}^2 e^{x^2} dx dy$

Solution: We see that this integral can not be evaluated in the given order, that is, to integrate first w.r.t x and then w.r.t y because we don't know the integration of the function $(e^x)^2$. Thus before evaluation of the integral, we must change its order. To do this, let us see the limits of x and y .

Limits of y : When $x = y/2$ and $x = 2$ then $y = 2x$ and $y = 0$ as the line $y = 2x$ begins from the origin.

Limits of x : When $y = 0$ and $y = 4$ then $x = 0$ and $x = 2$ respectively.

The graphs of these curves are shown in the above figure. Under the new set up, we see that, limits of x are: $x = 0$ and $x = 2$ and limits of y are: $y = 0$ and $y = 2x$. Thus,



$$I = \int_0^4 \int_{y/2}^2 e^{x^2} dx dy = \int_0^2 \int_0^{2x} e^{x^2} dy dx$$

The above procedure is known as "Change or Orders" of an integral. Now we first integrate w.r.t y to get

$$I = \int_0^2 \int_0^{2x} e^{x^2} dx = \int_0^2 e^{x^2} 2x dx$$

Putting $z = x^2$, we get $dz = 2x dx$. Now if $x = 0$ then $z = 0$ and, if $x = 2$ then $z = 4$. Thus

$$I = \int_0^4 e^z dz = e^z \Big|_0^4 = e^4 - e^0 = e^4 - 1$$

Example 05: Evaluate $\int_0^2 \int_{y^2}^4 y \cos x^2 dx dy$

Solution: We see that this integral can not be evaluated in the given order, that is, to integrate first w.r.t x and then w.r.t y because we don't know the integration of the function $\cos x^2$. Thus before evaluation of the integral, we must change its order. To do this, let us see the limits of x and y .

Limits of y : When $x = y^2$ and $x = 4$ then $y = \sqrt{x}$ and $y = 2$ respectively.

Limits of x are: When $y = 0$ and $y = 2$ then $x = 0$ and 4 respectively.

The graphs of these curves are shown in the above figure. Under the new set up, we see that:

Limits of x are $x = 0$ and $x = 4$ and limits of y are $y = 0$ and $y = \sqrt{x}$. Thus,

$$I = \int_0^2 \int_{y^2}^4 y \cos x^2 dx dy = \int_0^4 \int_0^{\sqrt{x}} y \cos x^2 dx dy. \text{ Now we first integrate w.r.t } y \text{ to get}$$

$$I = \int_0^4 y^2 / 2 \Big|_0^{\sqrt{x}} \cos x^2 dx = \frac{1}{4} \int_0^4 2x \cos x^2 dx. \text{ Putting } z = x^2, \text{ we get } dz = 2x dx.$$

Now, if $x = 0$ then $z = 0$ and, if $x = 4$ then $z = 16$. Thus,

$$I = \frac{1}{4} \int_0^{16} \cos z dz = \frac{1}{4} [\sin z]_0^{16} = \frac{1}{4} (\sin 16 - \sin 0) = \frac{1}{4} \sin 16.$$

Example 06: Evaluate $\iint_D e^{-(x^2+y^2)} dx dy$, where D is the closed disk of radius 'a' with center at origin.

Note: Whenever we have a function of the form $(x^2 + y^2)$, we put: $x = r \cos \theta$ and $y = r \sin \theta$.

Solution: Putting $x = r \cos \theta$ and $y = r \sin \theta$, we get $x^2 + y^2 = r^2$ and $dx dy = r dr d\theta$. (This has been shown above). Moreover, the limits of r are: $r = 0$ to $r = a$, and limits of θ are: $\theta = 0$ to $\theta = 2\pi$.

$$\text{Thus, } \iint_D e^{-(x^2+y^2)} dx dy = \int_0^{2\pi} \int_0^a e^{-r^2} r dr d\theta = \frac{1}{2} \int_0^{2\pi} \int_0^a e^{-r^2} 2r dr d\theta = \frac{1}{2} \int_0^{2\pi} e^{-r^2} 2r dr \int_0^{2\pi} d\theta$$

Now putting $z = r^2$, giving $dz = 2r dr$. If $r = 0$ then $z = 0$ and if $r = a$ then $z = a^2$.

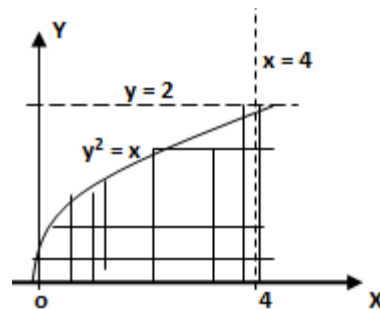
$$\text{Hence } I = \left(\frac{1}{2} \int_0^{a^2} e^{-z} dz \right) [\theta]_0^{2\pi} = \frac{2\pi}{2} [-e^{-z}]_0^{a^2} = -\pi(e^{-a^2} - 1) = \pi(1 - e^{-a^2})$$

Example 07: Evaluate $\iint_D \frac{1}{1+x^2+y^2} dx dy$, where D is the closed disc with radius a .

Solution: Let $x = r \cos \theta$ and $y = r \sin \theta$, giving $x^2 + y^2 = r^2$. Moreover, $dx dy = r dr d\theta$.

Also, the limits for r are: $r = 0$ to $r = a$, and limits for θ are: $\theta = 0$ to $\theta = 2\pi$. Thus,

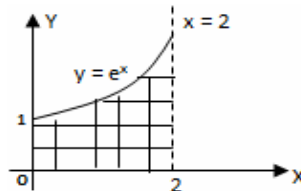
$$I = \int_0^{2\pi} \int_0^a \frac{1}{1+r^2} r dr d\theta = \frac{1}{2} \int_0^{2\pi} \frac{2r}{1+r^2} dr \int_0^{2\pi} d\theta = \frac{1}{2} \ln[1+r^2]_0^a [\theta]_0^{2\pi} = \frac{1}{2} 2\pi [\ln(1+a^2) - \ln 1] = \pi [\ln(1+a^2)] \quad (\ln 1 = 0)$$



Example 08: Evaluate $\iint_D dy dx$ and $\iint_D dx dy$, where D is the region

bounded by the y-axis, the line $x = 2$ and the curve $y = e^x$.

Solution: Before we evaluate the given integrals, let us draw the given curves. These are shown in the figure. Now the first integral has to be evaluated w.r.t y and then integral w.r.t x. It may be noted that when $x = 0$ then $y = e^0 = 1$. Hence the limits for y are, $y = 1$ to $y = e^x$. Thus,



$$\iint_D dy dx = \int_0^2 \int_1^{e^x} dy dx = \int_0^2 [y]_1^{e^x} dx = \int_0^2 [e^x - 1] dx = [e^x - x]_0^2 = [e^2 - e^0] - [2 - 0] = e^2 - 3$$

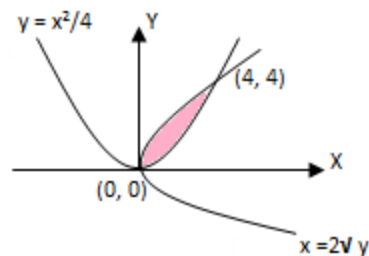
Let us evaluate the second integral. Here we first integrate w.r.t x and then w.r.t y. Now if $x = 0$ then $y = e^0 = 1$ and if $x = 2$ then $y = e^2$. Thus limits for y are 1 to e^2 . The limits for x are found by solving the equation $y = e^x$ giving $x = \ln y$. This is the lower limit of x. The upper limit for $x = 2$. Thus,

$$\begin{aligned} \iint_D dx dy &= \int_1^{e^2} \int_{\ln y}^2 dx dy = \int_1^{e^2} [x]_{\ln y}^2 dy = \int_1^{e^2} [2 - \ln y] dy = [2y - \{y \ln y - y\}]_1^{e^2} = [3y - y \ln y]_1^{e^2} \\ &= [3e^2 - 3] - [e^2 \ln e^2 - 1 \ln 1] = 3e^2 - 3 - 2e^2 - 0 = e^2 - 3. \end{aligned}$$

Note: i. $\ln 1 = 0$ ii. $\ln e^2 = 2 \ln e = 2 \cdot 1 = 2$ iii. $\int \ln y dy = y \ln y - y$

Example 09: Evaluate $\iint_D y dx dy$ where D is the region bounded by parabolas $y^2 = 4x$ and $x^2 = 4y$.

Solution: Solving given equations simultaneously, we see that points of intersection between two curves are (0, 0) and (4, 4). Thus limits for x are 0 to 4. Similarly, the limits for y are $y = 2\sqrt{x}$ to $x^2/4$.



Graphically the region is shown below.

Thus,

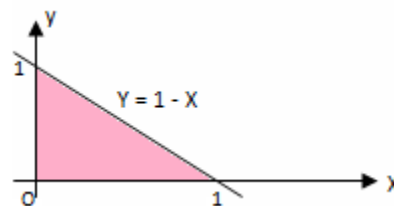
$$\iint_D y dx dy = \int_0^4 \int_{x^2/4}^{2\sqrt{x}} y dx dy = \int_0^4 \left[\frac{y^2}{2} \right]_{x^2/4}^{2\sqrt{x}} dx = \frac{1}{2} \int_0^4 \left[4x - \frac{x^4}{16} \right] dx = \frac{1}{2} \left[2x^2 - \frac{x^5}{80} \right]_0^4 = \frac{1}{2} \left[32 - \frac{1024}{80} \right] = \frac{48}{5}$$

Example 10: Evaluate $\iint_D e^{2x+3y} dx dy$ where D is the region bounded by the lines $x = 0$,

$y = 0$ and $x + y = 1$.

Solution: $x = 0$ is the equation of y-axis, $y = 0$ is the equation of x-axis. Thus area shown graphically is as under.

If $x = 0$ then $y = 1$ and if $y = 0$ then $x = 1$. Thus, points of intersection are (0, 1) and (1, 0). Hence limits for x are 0 to 1 and limits for y are 0 to $1 - x$.



$$\text{Hence, } \iint_D e^{2x+3y} dx dy = \int_0^1 \int_0^{1-x} e^{2x+3y} dx dy$$

$$= \int_0^1 e^{2x} \int_0^{1-x} e^{3y} dy dx = \int_0^1 e^{2x} \left[\frac{e^{3y}}{3} \right]_0^{1-x} dx = \frac{1}{3} \int_0^1 e^{2x} (e^{3-3x} - 1) dx \quad (\because e^0 = 1)$$

$$= \frac{1}{3} \int_0^1 (e^{3-x} - e^{2x}) dx = \frac{1}{3} \left[-e^{3-x} - \frac{e^{2x}}{2} \right]_0^1 = \frac{1}{6} (2e^3 - 3e^2 + 1)$$

APPLICATIONS OF DOUBLE INTEGRALS

In this section we shall discuss various applications of double integration that are useful in solving many engineering problems.

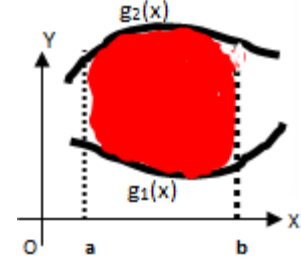
Area and Volume by Double Integrals

In the above section we have studied integrals of the form $\iint_D f(x, y) dx dy$.

If $f(x, y) = 1$ throughout D , then above integral becomes

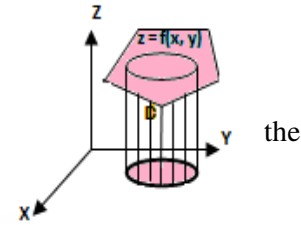
$$I = \iint_D dx dy = \int_a^b \int_{g_1(x)}^{g_2(x)} dy dx$$

This will provide the area bounded by two curves $y = g_1(x)$, $y = g_2(x)$ and the lines $x = a$, $x = b$. The shaded region shows this area.



If $f(x, y) \neq 1$, then we have already discussed that integral $\iint_D f(x, y) dx dy$

provides the volume of the solid below the surface $z = f(x, y)$ and above region D in the xy -plane. This is shown in the adjacent figure.



Mass of a Body

We know that mass of a body is defines as: $\text{Mass} = \text{Density} \times \text{Area}$.

Thus, $\text{Element of Mass} = \text{Density} \times \text{Element of Area}$

or $dm = \delta(x, y) \times dA = \delta(x, y) \times dx dy$

Here it is assumed that the mass of a body is continuously distributed over some region D in the xy -plane and $\delta(x, y)$ is the density at point (x, y) of D . In such case, the mass is found by using the

formula: $M = \iint_D \delta(x, y) dx dy$

Center of Gravity

In terms of double integral, the “Center of Gravity” $G(\bar{x}, \bar{y})$ of a solid whose mass is distributed continuously over the region D is given by;

$$\bar{x} = \frac{\iint_D x \delta(x, y) dx dy}{\iint_D \delta(x, y) dx dy} \quad \bar{y} = \frac{\iint_D y \delta(x, y) dx dy}{\iint_D \delta(x, y) dx dy}$$

Moment of Inertia

(a) Moment of Inertia about the x -axis: $I_x = \iint_D y^2 \delta(x, y) dx dy$

(b) Moment of Inertia about the y -axis: $I_y = \iint_D x^2 \delta(x, y) dx dy$

(c) Moment of Inertia about the origin: $I_0 = \iint_D (x^2 + y^2) \delta(x, y) dx dy$

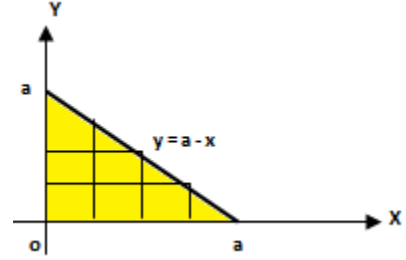
Example 01: By means of double integration, find the area of the region bounded by

(a) The coordinate axes and the straight line $x + y = a$.

Solution: The required area is shown in the figure.

Now area by double integral is given by: $\iint_D dy dx$

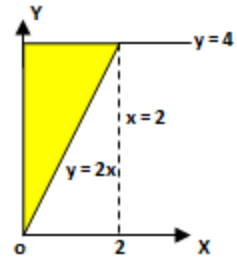
Now the limits for x are: $x = 0$ to $x = a$, and the limits for y are from $y = 0$ to $y = a - x$. Thus required area is;



$$A = \iint_D dy dx = \int_0^a \int_0^{a-x} dy dx = \int_0^a [y]_0^{a-x} dx = \int_0^a (a-x) dx = \left[ax - x^2/2 \right]_0^a = a^2 - a^2/2 = a^2/2$$

(b) The y -axis, the straight line $y = 2x$ and the straight line $y = 4$.

Solution: The required area is shown in the figure. Now area by double integral is given by $\iint_D dy dx$. Solving the two equations $y = 4$ and $y = 2x$, we see that



$2x = 4$ or $x = 2$. Thus limits for x are $x = 0$ and $x = 2$. The limits for y are very clear, that is; $y = 0$ and $y = 2x$. Thus,

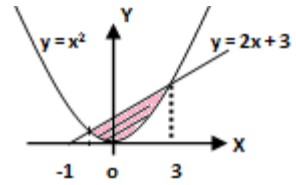
$$A = \iint_D dy dx = \int_0^2 \int_0^{2x} dy dx = \int_0^2 [y]_0^{2x} dx = \int_0^2 2x dx = \left[2x^2/2 \right]_0^2 = 2^2 - 0^2 = 4$$

Example 02: Find the area bounded by the parabola $y = x^2$ and the line $y = 2x + 3$.

Solution: The graphs of the both curve are shown. Solving the two equations simultaneously, we see that:

$$x^2 = 2x + 3 \text{ or } x^2 - 2x - 3 = 0 \text{ giving } x = 3 \text{ or } x = -1,$$

Thus, required area is given by



$$A = \iint_D dy dx = \int_{-1}^3 \int_{x^2}^{2x+3} dy dx = \int_{-1}^3 [y]_{x^2}^{2x+3} dx = \int_{-1}^3 (2x + 3 - x^2) dx = \left[\frac{2x^2}{2} + 3x - \frac{x^3}{3} \right]_{-1}^3 = (9 + 9 - 9) - (1 - 3 + 1/3) = 9 + 2 - 1/3 = 11 - 1/3 = 32/3$$

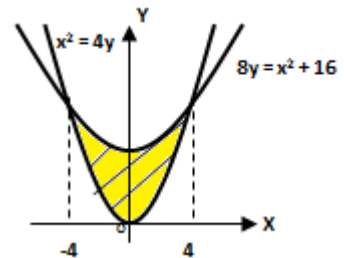
Example 03: Find the area bounded by the parabolas $x^2 = 4y$ and $8y = x^2 + 16$.

Solution: The graphs of both parabolas are shown in the figure.

Solving the given equations simultaneously, we get: $(x^2 + 16)/8 = x^2/4$

$$\text{or } x^2 + 16 = 2x^2 \text{ or } x^2 = 16 \text{ giving } x = -4 \text{ or } x = 4.$$

Thus required area is given by



$$A = \iint_D dy dx = \int_{-4}^4 \int_{x^2/4}^{(x^2+16)/8} dy dx = \int_{-4}^4 [y]_{x^2/4}^{(x^2+16)/8} dx = \int_{-4}^4 \left[\frac{x^2+16}{8} - \frac{x^2}{4} \right] dx = \int_{-4}^4 \left[\frac{16-x^2}{8} \right] dx = \frac{1}{8} \left[16x - \frac{x^3}{3} \right]_{-4}^4 = \frac{1}{8} \left[(64 - \frac{64}{3}) - (-64 + \frac{64}{3}) \right] = \frac{1}{8} (128 - 128/3) = 32/3$$

Example 04: Find the area outside the circle $r = 3$ and inside the cardioids $r = 2(1 + \cos \theta)$.

Solution: We know that an equation of a circle centered at origin and radius 3 is given by $x^2 + y^2 = 3^2$. In polar coordinates where $x = r \cos \theta$ and $y = r \sin \theta$ this becomes: $r^2 \cos^2 \theta + r^2 \sin^2 \theta = 3^2$ or $r^2 = 3^2$ giving $r = 3$. Thus $r = 3$ is equation of circle centered at origin with $r = 3$.

The curve $r = 2(1 + \cos \theta)$ is known as “Cardioids”.

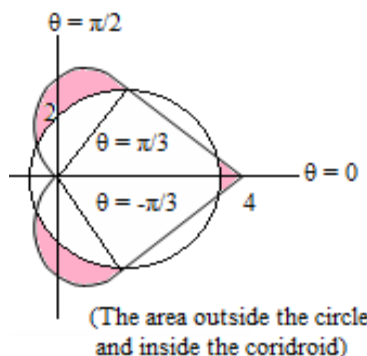
Let us plot its graph. Taking different values of θ

θ	0	$\pi/3$	$\pi/2$	$2\pi/3$	π
R	4	3	2	1	0

It may further be noted that in the polar coordinates, if θ is Changed with $-\theta$ and the equation does not change, then the curve is symmetric about the initial axis. Here if we change θ by $-\theta$, we get:

$$r = 2(1 + \cos(-\theta)) = 2(1 + \cos \theta) \quad [\text{since } \cos(-\theta) = \cos \theta]$$

We see that there is no change in the equation of the curve; hence the given curve is symmetric about the initial line. Now if we plot the circle and cardioids on the same plane, we will have the figure shown above. The shaded region is the required region whose area is to be found. Since the shaded area above and below the initial axis is same, hence required area is:



$$A = 2 \iint_D r \, dr \, d\theta \quad [\text{Note: } dx \, dy = r \, dr \, d\theta]$$

Now the circle $r = 3$ and the cardioids $r = 2(1 + \cos \theta)$ intersect each other, hence $2(1 + \cos \theta) = 3$ or $1 + \cos \theta = 3/2$ or $\cos \theta = 1/2$ giving $\theta = \cos^{-1}(1/2) = 60^\circ = \pi/3$. Thus limits for θ are 0 to $\pi/3$. Moreover, limits for r are: $r = 3$ to $r = 2(1 + \cos \theta)$. Thus,

$$\begin{aligned} A &= \iint_D r \, dr \, d\theta = 2 \int_0^{\pi/3} \int_3^{2(1+\cos\theta)} r \, dr \, d\theta = 2 \int_0^{\pi/3} \left[\frac{r^2}{2} \right]_3^{2(1+\cos\theta)} d\theta = \int_0^{\pi/3} [4(1+\cos\theta)^2 - 9] d\theta \\ &= \int_0^{\pi/3} (-5 + 8\cos\theta + 4\cos^2\theta) d\theta = \int_0^{\pi/3} (-5 + 8\cos\theta + 4 \frac{1+\cos 2\theta}{2}) d\theta \\ &= \int_0^{\pi/3} (-5 + 8\cos\theta + 2 + 2\cos 2\theta) d\theta = \int_0^{\pi/3} (-3 + 8\cos\theta + 2\cos 2\theta) d\theta \\ \therefore A &= \left[-3\theta + 8\sin\theta + \frac{\sin 2\theta}{2} \right]_0^{\pi/3} = \left[-3\pi/3 + 8\sin(\pi/3) + \frac{\sin(2\pi/3)}{2} \right] = -\pi + 8 \frac{\sqrt{3}}{2} + \frac{\sqrt{3}}{2} \end{aligned}$$

$$\text{Thus required area } A = \frac{9\sqrt{3}}{2} - \pi$$

Example 05: Find the area inside the circle $r = 4 \sin \theta$ and outside the lemniscates $r^2 = 8 \cos 2\theta$.

Solution: The shape of lemniscates is shown in the figure. The required area is also shown. Since, $r = 4 \sin \theta$ or $r^2 = 16 \sin^2 \theta$. Also $r^2 = 8 \cos 2\theta$. Now, equating both equations, we get,

$$16 \sin^2 \theta = 8 \cos 2\theta \text{ or } 2 \sin^2 \theta = (1 - 2 \sin^2 \theta) \text{ or } 4 \sin^2 \theta = 1 \text{ or } \sin \theta = \pm 1/2 \text{ giving } \theta = \pm \pi/6.$$

$$[\text{Note: } \cos 2\theta = 1 - 2 \sin^2 \theta]$$

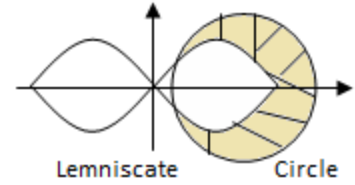
This is clear from the figure that the circle and lemniscates intersect each other at $\theta = \pm \pi/6$.

Moreover, when $\theta = \pi/4$, then $r^2 = 8 \cos 2\theta = 8 \cos \pi/2 = 0$ giving $r = 0$.

Thus limits for θ are: $\theta = \pi/6$ and $\theta = \pi/4$.

The limits for r are: $r = \sqrt{8\cos 2\theta}$ and $r = 4\sin \theta$.

Further to notice that since the curves is symmetric about the initial line, hence required area is:



$$\begin{aligned} A &= \iint_D r \, dr \, d\theta = 2 \int_{\pi/6}^{\pi/4} \int_{\sqrt{8\cos 2\theta}}^{4\sin \theta} r \, dr \, d\theta = 2 \int_{\pi/6}^{\pi/4} \left[\frac{r^2}{2} \right]_{\sqrt{8\cos 2\theta}}^{4\sin \theta} d\theta = \int_{\pi/6}^{\pi/4} [16\sin^2 \theta - 8\cos 2\theta] d\theta \\ &= \int_{\pi/6}^{\pi/4} [16 \frac{1 - \cos 2\theta}{2} - 8\cos 2\theta] d\theta = \int_{\pi/6}^{\pi/4} (8 - 8\cos 2\theta - 8\cos 2\theta) d\theta = \int_{\pi/6}^{\pi/4} (8 - 16\cos 2\theta) d\theta \\ &= \left[8\theta - 16 \frac{\sin 2\theta}{2} \right]_{\pi/6}^{\pi/4} = 8 \left\{ \left[\frac{\pi}{4} - \frac{\pi}{6} \right] - \left[\sin \frac{\pi}{2} - \sin \frac{\pi}{3} \right] \right\} = \frac{8\pi}{3} + 4(\sqrt{3} - 1) \end{aligned}$$

Example 06: Find the volume of the solid bounded by paraboloid $z = 4 - x^2 - y^2$ and the xy -plane.

Solution: In xy -plane, $z = 0$. Hence the given equation becomes

$$0 = 4 - x^2 - y^2 \text{ or } x^2 + y^2 = 4, \text{ which is a circle.}$$

The given solid is shown in the figure.

Since the lower part of the solid is circular; hence we use polar coordinates to get: $r^2 = x^2 + y^2$ and $dx \, dy = r \, dr \, d\theta$.

The limits for r are: $r = 0$ and $r = 2$ [The circle $x^2 + y^2 = 4$ has radius $r = 2$]

The limits for θ are: $\theta = 0$ and $\theta = 2\pi$. Thus the required volume is given by

$$\begin{aligned} V &= \iint_D z \, dy \, dx = \iint_D (4 - x^2 - y^2) \, dy \, dx = \int_0^{2\pi} \int_0^2 (4 - r^2) r \, dr \, d\theta = \int_0^{2\pi} 1 d\theta \left[\frac{4r^2}{2} - \frac{r^4}{4} \right]_0^2 \\ &= [\theta]_0^{2\pi} [8 - 4] = 4(2\pi) = 8\pi \end{aligned}$$

Example 07: Find the volume in the first octant between the planes $z = 0$, $z = x + y + 2$ and inside the cylinder $x^2 + y^2 = 16$.

Solution: The plane $z = 0$ is the xy -plane. On this plane, z is always zero.

Now $x^2 + y^2 = 16$ is circle centered at origin with radius 4. Finally the equation $z = x + y + 2$ is the equation of plane. The figure is shown here.

Now the required volume is given by

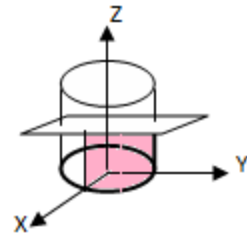
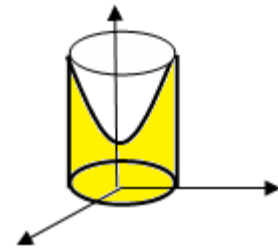
$$V = \iint_D z \, dy \, dx$$

To find the limits of x and y , we proceed as under:

Since $x^2 + y^2 = 16$ or $y = \sqrt{16 - x^2}$. In the first "Octant" $y = 0$ and $y = \sqrt{16 - x^2}$.

If $y = 0$, then $x^2 = 16$ giving $x = 4$ as the volume is to be found in the first "Octant", hence we take the positive value of x . Thus limits of x are: $x = 0$ to $x = 4$. Thus,

$$V = \iint_D z \, dy \, dx = \int_0^4 \int_0^{\sqrt{16-x^2}} (x + y + 2) \, dy \, dx = \int_0^4 \left[xy + \frac{y^2}{2} + 2y \right]_0^{\sqrt{16-x^2}} dx$$



$$= \int_0^4 \left[x\sqrt{16-x^2} + \frac{16-x^2}{2} + 2\sqrt{16-x^2} \right] dx \quad (1)$$

Now let us take each integral separately.

$$\int x\sqrt{16-x^2} dx = -\frac{1}{2} \int (16-x^2)^{1/2} (-2x) dx = -\frac{1}{2} \frac{(16-x^2)^{3/2}}{3/2} = -\frac{1}{3} (16-x^2)^{3/2}$$

Putting the limits, we get

$$\int_0^4 x\sqrt{16-x^2} dx = -\frac{1}{3} \left[(16-x^2)^{3/2} \right]_0^4 = -\frac{1}{3} \left[0 - (16)^{3/2} \right] = + (4)^3 / 3 = 64/3$$

$$\int_0^4 \left[\frac{16-x^2}{2} \right] dx = \left[8x - \frac{x^3}{6} \right]_0^4 = 32 - \frac{32}{3} = 64/3$$

Finally, using the formula $\int \sqrt{a^2-x^2} dx = \frac{x\sqrt{a^2-x^2}}{2} + \frac{a^2}{2} \sin^{-1}\left(\frac{x}{a}\right)$, we get

$$2 \int_0^4 \sqrt{16-x^2} dx = 2 \left[\frac{x\sqrt{16-x^2}}{2} \right]_0^4 + 2 \left[\frac{16}{2} \sin^{-1}\left(\frac{x}{4}\right) \right]_0^4 = 0 + 16(\sin^{-1}(1) - \sin^{-1}(0)) = 16 \frac{\pi}{2} = 8\pi$$

Now collecting all these integrals and put them in equation (1), we obtain

$$V = 64/3 + 64/3 + 8\pi = 128/3 + 8\pi$$

Example 08: Find the volume bounded by the cylinder $x^2 + y^2 = 4$ and the planes $y + z = 4$ and $z = 0$.

Solution: Since $x^2 + y^2 = 4$ or $y = \pm\sqrt{4-x^2}$ so limits for y are:

$y = -\sqrt{4-x^2}$ and $y = +\sqrt{4-x^2}$. Also limits for x are: $x = -2$ and $x = +2$. Thus the required volume is given by:

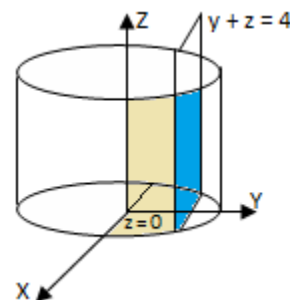
$$\begin{aligned} V &= \iint_D z dy dx = \int_{-2}^2 \int_{-\sqrt{4-x^2}}^{\sqrt{4-x^2}} (4-y) dy dx = \int_{-2}^2 \left[4y - \frac{y^2}{2} \right]_{-\sqrt{4-x^2}}^{\sqrt{4-x^2}} dx \\ &= \int_{-2}^2 \left[4(\sqrt{4-x^2} + \sqrt{4-x^2}) - \frac{1}{2} \{ (4-x^2) - (4-x^2) \} \right] dx = 8 \int_{-2}^2 \left[(\sqrt{4-x^2}) - \frac{1}{2} \{0\} \right] dx \\ &= 8 \left[\frac{x\sqrt{4-x^2}}{2} + \frac{4}{2} \sin^{-1}\left(\frac{x}{4}\right) \right]_{-2}^2 \end{aligned}$$

Substituting the limits and simplifying, we get: $V = 16\pi$

Formula: $\int \sqrt{a^2-x^2} dx = \frac{x\sqrt{a^2-x^2}}{2} + \frac{a^2}{2} \sin^{-1}\left(\frac{x}{a}\right)$. NOTE: $\sin^{-1} \frac{1}{2} = \frac{\pi}{6}$ and $\sin^{-1} \frac{-1}{2} = \frac{-\pi}{6}$

Example 09: Find the volume of the solid bounded by the curves $x^2 + y^2 = 4$, $z = \sqrt{16-x^2-y^2}$ and $z = 0$.

Solution: Since $x^2 + y^2 = 4$ we have, $y = \pm\sqrt{4-x^2}$. If $y = 0$ then $x = \pm 2$. Thus required volume is given by;



$$V = \iint_D z \, dy \, dx = \int_{-2}^2 \int_{-\sqrt{4-x^2}}^{\sqrt{4-x^2}} \sqrt{16-x^2-y^2} \, dy \, dx$$

Using polar coordinates: $x = r \cos \theta$ and $y = r \sin \theta$, we get $x^2 + y^2 = r^2$. Moreover, $dx \, dy = r \, dr \, d\theta$. The limits for θ are 0 to 2π . Limits for r are 0 to 2. Thus,

$$V = \iint_D z \, dy \, dx = \int_0^{2\pi} \int_0^2 \sqrt{16-r^2} \, r \, dr \, d\theta$$

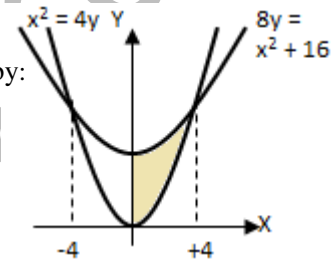
Let $u = 16 - r^2$ or $du = -2r \, dr$ giving $r \, dr = -du/2$. Also if $r = 0$ then $u = 16$ and if $r = 2$ then $u = 12$.

$$\text{Thus, } V = \frac{1}{2} \int_0^{2\pi} \int_{16}^{12} u^{1/2} \, du \, d\theta = -\frac{1}{2} \int_0^{2\pi} 1 \, d\theta \left[\frac{u^{3/2}}{3/2} \right]_{16}^{12} = \frac{-2}{3.2} [\theta]_0^{2\pi} [12^{3/2} - 16^{3/2}] = \frac{16\pi}{3} (8 - 3\sqrt{3})$$

Example 10: Find the center of gravity of the plane area of uniform density bounded by $x^2 = 4y$ and $8y = x^2 + 16$ in the first quadrant.

Solution: The coordinates of “Center of Gravity” of any solid body are given by:

$$\bar{x} = \frac{\iint_D x \rho(x, y) \, dA}{\iint_D \rho(x, y) \, dA}, \bar{y} = \frac{\iint_D y \rho(x, y) \, dA}{\iint_D \rho(x, y) \, dA}, \text{ where } dA = dy \, dx$$



The plane area has a uniform density, hence $\rho(x, y) = 1$.

$$\text{Thus center of gravity is given by: } \bar{x} = \frac{\iint_D x \, dA}{\iint_D dA}, \bar{y} = \frac{\iint_D y \, dA}{\iint_D dA}$$

Now from the given equations, $y = x^2/4$ and $y = (x^2 + 16)/8$ or $x^2/4 = (x^2 + 16)/8$. Solving, we get, $x = -4$ and $x = +4$. But we have to find the C.O.G of the area which is in the first quadrant. Hence limits of x are from $x = 0$ to $x = 4$. Now,

$$\iint_D y \, dA = \iint_D y \, dy \, dx = \int_0^4 \int_{x^2/4}^{2+x^2/8} y \, dy \, dx = \int_0^4 \left[\frac{y^2}{2} \right]_{x^2/4}^{2+x^2/8} dx$$

$$\begin{aligned} \iint_D dA &= \iint_D dy \, dx = \int_0^4 \int_{x^2/4}^{2+x^2/8} dy \, dx = \int_0^4 [y]_{x^2/4}^{2+x^2/8} dx = \int_0^4 \left(2 + \frac{x^2}{8} - \frac{x^2}{4} \right) dx = \int_0^4 \left(2 - \frac{x^2}{8} \right) dx \\ &= \left[2x - \frac{x^3}{24} \right]_0^4 = (8 - 8/3) - 0 = 16/3. \end{aligned}$$

$$\begin{aligned} \iint_D x \, dA &= \iint_D x \, dy \, dx = \int_0^4 \int_{x^2/4}^{2+x^2/8} x \, dy \, dx = \int_0^4 x [y]_{x^2/4}^{2+x^2/8} dx = \int_0^4 x \left(2 + \frac{x^2}{8} - \frac{x^2}{4} \right) dx \\ &= \int_0^4 \left(2x - \frac{x^3}{8} \right) dx = \left[x^2 - \frac{x^4}{32} \right]_0^4 = (16 - 8) - 0 = 8 \\ &= \frac{1}{2} \int_0^4 \left[\frac{x^2 + 16}{8} \right]^2 - \frac{x^4}{16} dx = \frac{1}{2} \cdot \frac{1}{64} \int_0^4 (265 + 32x^2 - 3x^4) dx \end{aligned}$$

$$= \frac{1}{128} \left[256x + \frac{32}{3}x^3 - 3\frac{x^5}{5} \right]_0^4 = \frac{1}{128} \left[256(4) + \frac{32}{3}(4^3) - 3\frac{4^5}{5} \right] = 128/15$$

Thus C.O.G of the plane in the first quadrant is given by:

$$\bar{x} = 8 \div \frac{16}{3} = \frac{3}{2} \text{ and } \bar{y} = \frac{128}{15} \div \frac{16}{3} = \frac{8}{5}. \text{ Thus, } (\bar{x}, \bar{y}) = \left(\frac{3}{2}, \frac{8}{5} \right)$$

Example 11: Find the mass of square plate of size 'a' if the density varies as the square of the distance from the vertex.

Solution: Consider the square OABC where O is at origin, A is on the x-axis and the C is on the y-axis.

Let P(x, y) be an element of area of this square. Then,

$|OP|^2 = x^2 + y^2$. Now the density of the element of area is proportional to $|OP|$. Thus

$$\rho = k(OP)^2 = k(x^2 + y^2)$$

The required mass is therefore,

$$M = \iint_D \rho \, dy \, dx = \int_0^a \int_0^a k(x^2 + y^2) \, dy \, dx = k \int_0^a \left[x^2 y + y^3 / 3 \right]_0^a \, dx = k \int_0^a (ax^2 + a^3 / 3) \, dx$$

$$\therefore M = \frac{k}{3} \left[ax^3 + a^3 x \right]_0^a = \frac{2a^4 k}{3}$$

Example 12: Find the mass of circular plate of radius 'a' if the density varies as the square of the distance from a point on the circumference to the center of the circle.

Solution: Consider a circle centered at origin O. Let P(x, y) be an element of area of the circle. Then $|OP|^2 = x^2 + y^2$. Now the density of the element of area is proportional to OP. Thus,

$$\rho = k(OP)^2 = k(x^2 + y^2)$$

The required mass is then given by using the polar equations, i.e.

$x = r \cos \theta$, $y = r \sin \theta$ giving $r^2 = x^2 + y^2$ and $dx \, dy = r \, dr \, d\theta$. Moreover, the limits for r are: $r = 0$ and $r = a$. Limits for θ are: $\theta = 0$ and $\theta = 2\pi$. Thus,

$$M = \iint_D \rho \, dy \, dx = \iint_D k(x^2 + y^2) \, dy \, dx = k \int_0^{2\pi} \int_0^a r^2 \cdot r \, dr \, d\theta = k \left[\frac{r^4}{4} \right]_0^a \int_0^{2\pi} 1 \, d\theta = \frac{ka^4}{4} \theta \Big|_0^{2\pi} = k\pi a^4 / 2$$

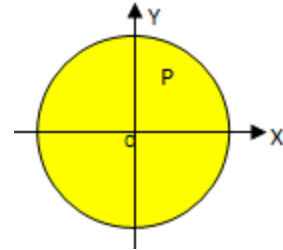
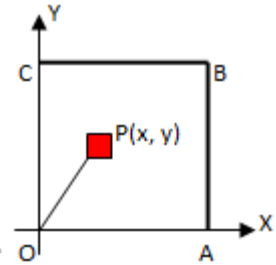
Example 13: Find the moments of inertia I_x , I_y and I_0 for the quarter disk $x^2 + y^2 \leq a^2$, $a > 0$ if the density is proportional to the distance from the origin.

Solution: Given that density is proportional to the distance from the origin, hence $\rho = kr$. Using the polar coordinates: $x = r \cos \theta$, $y = r \sin \theta$, $dx \, dy = r \, dr \, d\theta$, the limits for r are: $r = 0$ and $r = a$. The limits for θ are: $\theta = 0$ and $\theta = \pi/2$ (Due to quarter disk)

Now by definition:

$$I_x = \iint_D y^2 \rho(x, y) \, dy \, dx = \iint_D r^2 \sin^2 \theta \cdot kr \cdot r \, dr \, d\theta = k \int_0^{\pi/2} \int_0^a r^4 \sin^2 \theta \, dr \, d\theta$$

$$= k \left[\frac{r^5}{5} \right]_0^a \int_0^{\pi/2} \sin^2 \theta \, d\theta = k \frac{a^5}{5} \frac{1}{2} \int_0^{\pi/2} 1 \, d\theta = k \frac{a^5}{10} [\theta]_0^{\pi/2} = k \frac{\pi a^5}{20}$$



$$\begin{aligned}\text{Similarly, } I_y &= \iint_D x^2 \rho(x, y) dy dx = \iint_D r^2 \cos^2 \theta \cdot k r \cdot r dr d\theta = k \int_0^{\pi/2} \int_0^a r^4 \cos^2 \theta dr d\theta \\ &= k \left[\frac{r^5}{5} \right]_0^a \int_0^{\pi/2} \cos^2 \theta d\theta = k \frac{a^5}{5} \frac{1}{2} \int_0^{\pi/2} 1 d\theta = k \frac{a^5}{10} [\theta]_0^{\pi/2} = k \frac{\pi a^5}{20}\end{aligned}$$

$$\begin{aligned}\text{Finally, } I_0 &= \iint_D (x^2 + y^2) \rho(x, y) dy dx = \iint_D r^2 k r \cdot r dr d\theta = k \int_0^{\pi/2} \int_0^a r^4 dr d\theta \\ &= k \left[\frac{r^5}{5} \right]_0^a \int_0^{\pi/2} d\theta = k \frac{a^5}{5} \int_0^{\pi/2} 1 d\theta = k \frac{a^5}{5} [\theta]_0^{\pi/2} = k \frac{\pi a^5}{10}\end{aligned}$$

TRIPLE INTEGRALS AND THEIR APPLICATIONS

Consider a function $f(x, y, z)$ which is continuous at every point of a finite region V of three dimensional space. Divide the region V into n sub-regions of volumes $\Delta V_1, \Delta V_2, \dots, \Delta V_n$. Let (x_i, y_i, z_i) be an arbitrary point in the i^{th} sub-region V_i . Now consider the sum

$$S = \sum_{i=1}^n f(x_i, y_i, z_i) \Delta V_i$$

The limit of this sum as n tends to infinity and ΔV_i tends to zero, if exists, is called the triple integral of $f(x, y, z)$ over the region V and is denoted by:

$$\iiint_V f(x, y, z) dV = \int_{x_1}^{x_2} \int_{y_1}^{y_2} \int_{z_1}^{z_2} f(x, y, z) dz dy dx$$

Example 01: Evaluate $\int_0^1 \int_1^4 \int_2^6 (x + y + z) dz dy dx$

$$\begin{aligned}\text{Solution: } I &= \int_0^1 \int_1^4 \int_2^6 (x + y + z) dz dy dx = \int_0^1 \int_1^4 \left[xz + yz + \frac{z^2}{2} \right]_2^6 dy dx \\ &= \int_0^1 \int_1^4 [x(6-2) + y(6-2) + (36-4)/2] dy dx = \int_0^1 \int_1^4 (4x + 4y + 16) dy dx \\ &= \int_0^1 \left[4xy + 4 \frac{y^2}{2} + 16y \right]_1^4 dx = \int_0^1 (12x + 30 + 48) dx = \left[6x^2 + 78x \right]_0^1 = 84\end{aligned}$$

Example 02: Evaluate $\int_0^a \int_0^x \int_0^{x+y} e^{x+y+z} dz dy dx$

$$\begin{aligned}\text{Solution: } \int_0^a \int_0^x \int_0^{x+y} e^{x+y+z} dz dy dx &= \int_0^a \int_0^x \int_0^{x+y} e^{x+y} e^z dz dy dx = \int_0^a \int_0^x e^{x+y} \left[e^z \right]_0^{x+y} dy dx \\ &= \int_0^a \int_0^x e^{x+y} [e^{x+y} - 1] dy dx = \int_0^a \int_0^x (e^{2x+2y} - e^{x+y}) dy dx \\ &= \int_0^a e^{2x} \left[\frac{e^{2y}}{2} \right]_0^x - e^x [e^y]_0^x dx = \int_0^a \left[e^{2x} \left(\frac{e^{2x} - 1}{2} \right) - e^x (e^x - 1) \right] dx\end{aligned}$$

$$= \int_0^a \left[\frac{e^{4x}}{2} - \frac{3e^{2x}}{2} + e^x \right] dx = \left[\frac{e^{8x}}{8} - \frac{3e^{2x}}{4} + e^x \right]_0^a = \left[\frac{e^{4a} - 6e^{2a} + 8e^a - 3}{8} \right]$$

Example 03: Evaluate $\int_0^{\pi/2} \int_0^{a \sin \theta} \int_0^{(a^2-r^2)/a} r \, dz \, dr \, d\theta$

Solution:
$$\int_0^{\pi/2} \int_0^{a \sin \theta} \int_0^{(a^2-r^2)/a} r \, dz \, dr \, d\theta = \int_0^{\pi/2} \int_0^{a \sin \theta} r \left[z \right]_0^{(a^2-r^2)/a} dr \, d\theta = \int_0^{\pi/2} \int_0^{a \sin \theta} r \left(\frac{a^2-r^2}{a} \right) dr \, d\theta$$

$$= \frac{1}{a} \int_0^{\pi/2} \left(\frac{a^2 r^2}{2} - \frac{r^4}{4} \right) \Big|_0^{a \sin \theta} d\theta = \frac{a^3}{4} \left[\int_0^{\pi/2} 2 \sin^2 \theta - \sin^4 \theta \, d\theta \right]$$

Now using the reduction formula $\int_0^{\pi/2} \sin^n \theta \, d\theta = \frac{n-1}{n} \int_0^{\pi/2} \sin^{n-2} \theta \, d\theta$, we have:

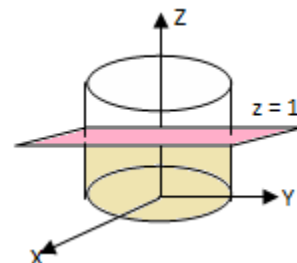
$$\int_0^{\pi/2} \sin^2 \theta \, d\theta = \frac{2-1}{2} \int_0^{\pi/2} \sin^{2-2} \theta \, d\theta = \frac{1}{2} \int_0^{\pi/2} 1 \, d\theta = \frac{1}{2} \cdot \frac{\pi}{2} = \frac{\pi}{4}$$

Similarly, $\int_0^{\pi/2} \sin^4 \theta \, d\theta = \frac{4-1}{4} \int_0^{\pi/2} \sin^2 \theta \, d\theta = \frac{3}{4} \cdot \frac{\pi}{4} = \frac{3\pi}{16}$

Thus,
$$\int_0^{\pi/2} \int_0^{a \sin \theta} \int_0^{(a^2-r^2)/a} r \, dz \, dr \, d\theta = \frac{a^3}{4} \left[2 \cdot \frac{\pi}{4} - \frac{3\pi}{16} \right] = \frac{a^3}{4} \left[\frac{8\pi - 3\pi}{16} \right] = \frac{5\pi a^3}{64}$$

Example 04: Evaluate the following by changing into cylindrical coordinates

$$\int_{-1}^1 \int_{-\sqrt{1-x^2}}^{\sqrt{1-x^2}} \int_{x^2+y^2}^1 z(x^2+y^2) \, dz \, dy \, dx$$



Solution: The solid is bounded above by $z = x^2 + y^2$ and below by $z = 1$.

Now $x^2 + y^2 = 1$ is a circle of radius 1, hence after changing into the cylindrical coordinates:

Limits for r are: $r = 0$ and $r = 1$. Limits for θ are: $\theta = 0$ and $\theta = 2\pi$.

Limits for z are: $z = x^2 + y^2$ and $z = 1$.

Moreover, $x^2 + y^2 = r^2$, $dx \, dy = r \, dr \, d\theta$, $dz = dz$.

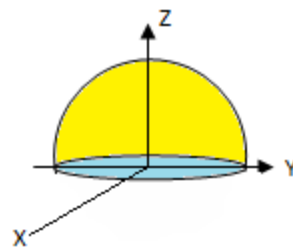
Thus
$$I = \int_0^1 \int_0^{2\pi} \int_{r^2}^1 z(r^2) r \, dz \, d\theta \, dr = \int_0^1 \left[\frac{z^2}{2} \right]_{r^2}^1 r^3 \, dr \int_0^{2\pi} 1 \, d\theta = \frac{1}{2} \int_0^1 (1-r^4) r^3 \, dr \cdot [\theta]_0^{2\pi}$$

$$= \frac{1}{2} \left[\frac{r^4}{4} - \frac{r^8}{8} \right]_0^1 (2\pi) = \frac{1}{2} \cdot \frac{1}{8} 2\pi = \frac{\pi}{8}$$

Example 05: Use cylindrical coordinates to evaluate $\iiint_S z \sqrt{x^2 + y^2} \, dV$,

where S is the hemisphere $x^2 + y^2 + z^2 \leq 4$ and $z \geq 0$.

Solution: The given integral is to be evaluated over the upper part of the sphere.



Using the equations $x = r \cos \theta$, $y = r \sin \theta$, we get $r^2 = x^2 + y^2$, and z remains unchanged.

Limits for r are: $r = 0$ and $r = 2$ because the radius of sphere $r = 2$.

Limits for θ are: $\theta = 0$ and $\theta = 2\pi$.

Limits for z are: $z = 0$ and $z = \sqrt{4 - x^2 - y^2} = \sqrt{4 - r^2}$. Moreover, $x^2 + y^2 = r^2$, $dx dy = r dr d\theta$, $dz = dz$.

$$\begin{aligned} \text{Thus, } I &= \int_0^2 \int_0^{2\pi} \int_0^{\sqrt{4-r^2}} z(r) r dz d\theta dr = \int_0^2 \left[\frac{z^2}{2} \right]_0^{\sqrt{4-r^2}} r^2 dr \int_0^{2\pi} 1 d\theta = \frac{1}{2} \int_0^2 (4-r^2) r^2 dr \cdot [\theta]_0^{2\pi} \\ &= \frac{1}{2} \left[\frac{4r^3}{3} - \frac{r^5}{5} \right]_0^2 (2\pi) = \frac{1}{2} \cdot \frac{64}{15} \cdot 2\pi = \frac{64\pi}{15} \end{aligned}$$

Example 06: Use spherical coordinates to evaluate $\iiint_S z \sqrt{x^2 + y^2 + z^2} dV$, where S is defined

$$\text{by } \sqrt{x^2 + y^2} \leq z \leq \sqrt{1 - x^2 - y^2}.$$

Solution: Here S is bounded below by the 'cone' $z^2 = x^2 + y^2$ and above by sphere $x^2 + y^2 + z^2 = 1$ of radius 1 centered at origin.

The figure is shown here. Using the spherical coordinates, we have:

$$x^2 + y^2 + z^2 = \rho^2, dV = dx dy dz = \rho^2 \sin \phi d\rho d\phi d\theta \text{ and } z = \rho \cos \phi.$$

The limits for ρ are: $\rho = 0$ and $\rho = 1$. The limits for θ are: $\theta = 0$ and $\theta = 2\pi$

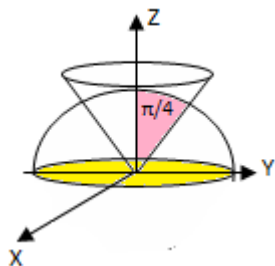
Limits for ϕ are found as follows:

The equation of cone is: $z^2 = x^2 + y^2$. Now in spherical coordinates, $x = \rho \sin \phi \cos \theta$, $y = \rho \sin \phi \sin \theta$, and $z = \rho \cos \phi$. Thus equation of cone becomes:

$(\rho \cos \phi)^2 = (\rho \sin \phi \cos \theta)^2 + (\rho \sin \phi \sin \theta)^2$ Or $\cos^2 \phi = \sin^2 \phi (\cos^2 \theta + \sin^2 \theta)$ or $\cos^2 \phi = \sin^2 \phi$

or $\tan^2 \phi = 1$ or $\tan \phi = \pm 1$ giving $\phi = \pm \pi/4$. Since integral is taken in the upper half of the sphere, hence $\phi = +\pi/4$. Thus limits for ϕ are 0 to $\pi/4$. Hence,

$$\begin{aligned} I &= \iiint_S z \sqrt{x^2 + y^2 + z^2} dV = \int_0^{2\pi} \int_0^{\pi/4} \int_0^1 (\rho \cos \phi) \rho (\rho^2 \sin \phi) d\rho d\phi d\theta \\ &= \frac{1}{2} \int_0^{2\pi} \int_0^{\pi/4} \int_0^1 \rho^4 (2 \cos \phi \sin \phi) d\rho d\phi d\theta = \frac{1}{2} \int_0^{2\pi} \int_0^{\pi/4} \left[\frac{\rho^5}{5} \right]_0^1 \sin 2\phi d\phi d\theta \\ &= \frac{1}{2} \cdot \frac{1}{5} \int_0^{2\pi} \left[\frac{\cos 2\phi}{2} \right]_0^{\pi/4} d\theta = \frac{1}{20} [\cos \pi/2 - \cos 0] [\theta]_0^{2\pi} = \frac{1}{20} [0 - 1] (2\pi) = -\frac{\pi}{10} \quad \{\text{Neglecting the -ve sign}\} \end{aligned}$$



WORKSHEET 07

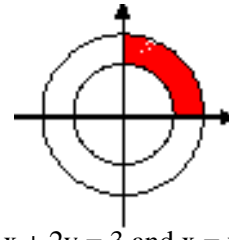
1. Evaluate the following Integral

- (i) $\int_0^1 \int_1^{3/2} xy(1+x+y) dx dy$ (ii) $\int_0^3 \int_0^x e^{y/x} dy dx$ (iii) $\int_1^4 \int_1^x \frac{1}{xy} dy dx$ (iv) $\int_0^2 \int_{x^2}^x (1+x^2+y) dy dx$
- (v) $\int_0^2 \int_y^{y^2+2} xy^2 dx dy$ (vi) $\int_0^4 \int_0^{\sqrt{1-x^2}} y dy dx$ (vii) $\int_0^3 \int_x^{\sqrt{x}} (x^2+y^2) dy dx$ (viii) $\int_0^4 \int_0^{\sqrt{16-y^2}} dx dy$

2. Evaluate $\iint_D e^{-(x^2+y^2)} dx dy$, where D is a region in the first

quadrant bounded by the circles $x^2 + y^2 = 1$ and $x^2 + y^2 = 4$. [See the figure]

NOTE: Here limits of r are from $r = 1$ to $r = 2$ and limits for θ are from $\theta = 0$ to $\theta = \pi/2$.



3. Evaluate $\iint_D (4 - 2x - 3y) dx dy$, where D is the region bounded by $y = 0$, $x + 2y = 3$ and $x = y^2$.

4. Evaluate $\iint_D xy dx dy$, where D is the region bounded by $y = x/4$ and $y^2 = x$.

5. Evaluate $\iint_D xy(x - y) dx dy$, where D is the region bounded by $y = x$ and $y = x^2$.

6. Evaluate $\iint_D \frac{xy}{\sqrt{1-y^2}} dx dy$, where D is the region bounded by upper half of circle $x^2 + y^2 = 1$.

8. Change the order of integration and hence evaluate the following integrals.

i. $\int_0^1 \int_{x^2}^{2-x} xy dx dy$

ii. $\int_0^\infty \int_x^\infty \frac{e^{-y}}{y} dx dy$

iii. $\int_0^1 \int_{x^2}^x (x^2 + y^2)^{-1/2} dy dx$

9. Use double integrals, find the area between:

i. The straight line $y = x$ and parabola $y = x^2$.

ii. The straight line $y = 2 - x$ and the parabola $y^2 = 2(2 - x)$

iii. The parabolas $y^2 = 4ax$ and $x^2 = 4ay$.

10. Find the COG, Moments of Inertia and the mass of the body for the curves in question 9 assuming the uniform density.

11. Evaluate the following triple integrals

i. $\int_0^a \int_0^a \int_0^a (xy + yz + zx) dz dy dx$

ii. $\int_{1/x}^3 \int_0^1 \int_0^{\sqrt{xy}} xyz dz dy dx$

iii. $\int_0^2 \int_0^x \int_0^{x+y} e^{x+y+z} dz dy dx$

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